Course Notes

Spacecraft Attitude Dynamics

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Part 1:
Attitude dynamics and kinematics

Note for the reader

These short notes are in support of the course “Spacecraft Attitude Dynamics”, they are not intended to replace any textbook. Interested readers are encouraged to consult also printed textbooks and archival papers.

Franco Bernelli Zazzera

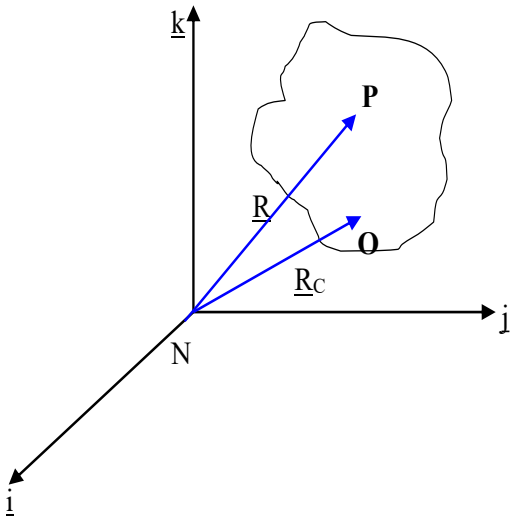
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Fundamental quantities

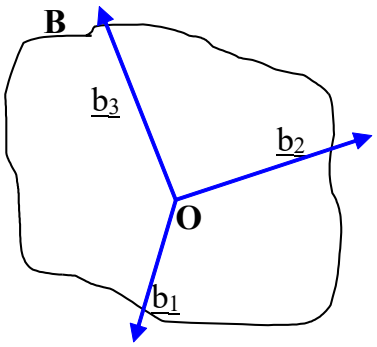
Inertial and body fixed frames

Here we will consider **N** = [i̲ :j̲ :k̲] to be a fixed inertial frame, where i̲, j̲, k̲ are unit vectors that form a basis of an orthonormal frame



The position of the centre of mass **O** of the rigid body $\underline{R}_C$ and a position vector $\underline{R}$ of an infinitesimal mass **P** of the rigid body can be expressed in the inertial frame, such that, $\underline{R} = x'\underline{\imath} + y'\underline{\jmath} + z'\underline{k}$.

We now introduce a body frame **B** = [$\underline{b}_1$: $\underline{b}_2$: $\underline{b}_3$] an orthonormal frame fixed in the body whose origin is positioned at the centre of mass O



In this way the orientation of a rigid body can be expressed as the relative attitude between the body fixed frame and an inertially fixed frame. Moreover, the attitude of the rigid body $A_{B/N}$ is defined as the relative orientation between the body fixed frame **B** and the inertial frame **N.** Since the orbital and attitude dynamics are generally de-coupled, in the analysis of a spacecraft's attitude we can ignore the translational component between the two frames and consider just the relative orientation such that:

$$\mathbf{B}=A_{B/N}\mathbf{N}$$

$A_{B/N}$ is an orthonormal matrix with the following properties:

$$A_{B/N}A_{B/N}^{T} = I_{3\times3}, det(A_{B/N}) = 1$$

The attitude matrix is also known as the Direct Cosine Matrix (DCM).
In addition, we denote the instantaneous angular velocity of each frame as $\underline{\omega}_N = \omega_1{}'\underline{\imath} + \omega_2{}'\underline{\jmath} + \omega_3{}'\underline{k}$and $\underline{\omega}_B = \omega_1\underline{b}_1 + \omega_2\underline{b}_2 + \omega_3\underline{b}_3$expressed in the basis of there repective frames. The angular velocity of the frame **N** can be expressed in the body frame as $\underline{\omega}_N^B = A_{B/N}\underline{\omega}_N$
The the relative angular velocity $\underline{\omega}_{B/N}$ expressed in body fixed coordinates associated with the relative attitude matrix $A_{B/N}$ is defined by

$$\underline{\omega}_{B/N} = \underline{\omega}_B - \underline{\omega}_N^B = \underline{\omega}_B - A_{B/N}\underline{\omega}_N$$

Since in this case *N* is a fixed inertial frame then $\underline{\omega}_N = \underline{0}$ and therefore $\underline{\omega}_{B/N} = \underline{\omega}_B$.
Note that later in the course we will need to define the attitude between the body frame and a moving frame.

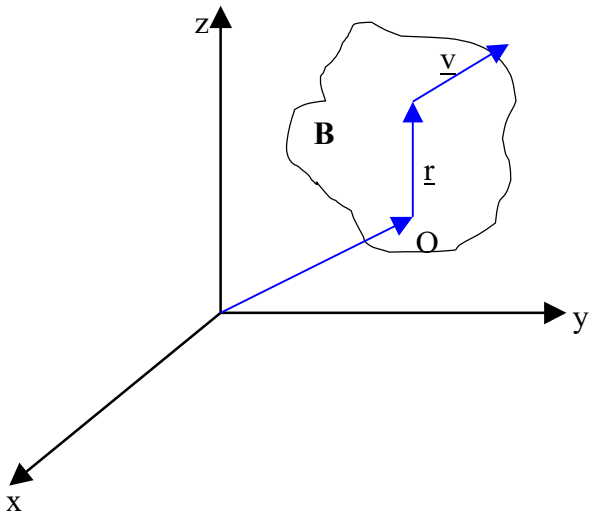
One of the theorems that is used throughout this course is the Transport Theorem. It relates the derivative of an arbitrary vector $\vec{x}$ taken with respect to one orthonormal frame, for example **N**, to a derivative taken with respect to another frame **B** where $\underline{\omega}_{B/N}$ is the relative angular velocity vector between the frames. The Transport Theorem is summarize in the equation:

$$\left[\frac{d\vec{x}}{dt}\right]_N = \left[\frac{d\vec{x}}{dt}\right]_B + \underline{\omega}_{B/N} \times \vec{x}$$

From here on when referiong to the angular velocity of the body frame with respect to the inertial frame expressed in body fixed coordinates we will simply write $\underline{\omega} = \underline{\omega}_{B/N}$.

Angular momentum of a rigid body

Let us start by defining the physical quantities that are fundamental and recurrent in the rotational motion of rigid spacecraft. These quantities are the angular momentum and the kinetic energy. The angular momentum of a rigid body "B" is defined referring to a reference frame attached to the rigid body and moving with it, with origin O:



Considering an infinitesimal point mass, it is defined as:

$$d\underline{h}_o = \underline{r} \wedge \underline{v} \cdot dm$$

Therefore, for the whole rigid body "B":

$$\underline{h}_o = \int_B \underline{r} \wedge \underline{v} \cdot dm$$

The velocity is expressed as:

$$\underline{v} = \underline{v}_o + \underline{\omega} \wedge \underline{r} + \underline{v}_{iB}$$

where $\underline{v}_o$ is the velocity of the origin of the reference system attached to the rigid body, $\underline{\omega}$ is the angular velocity of the rigid body and $\underline{v}_{iB}$ is the relative velocity between any point of the body and the origin of the reference system. For a rigid body, this relative velocity is always zero.
Evaluating the integral yields:

$$\underline{h}_o = \int_B \underline{r} \wedge (\underline{v}_o + \underline{\omega} \wedge \underline{r}) dm = -\underline{v}_o \wedge \int_B \underline{r} dm + \int_B \underline{r} \wedge (\underline{\omega} \wedge \underline{r}) dm$$

We must now evaluate the vector cross products:

$$\underline{\omega} = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} \qquad \underline{r} = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}$$

$$(\underline{\omega} \wedge \underline{r}) = \begin{Bmatrix} \omega_y z - \omega_z y \\ \omega_z x - \omega_x z \\ \omega_x y - \omega_y x \end{Bmatrix}$$

$$\underline{r} \wedge (\underline{\omega} \wedge \underline{r}) = \begin{Bmatrix} \omega_x y^2 - \omega_y xy + \omega_x z^2 - \omega_z xz \\ \omega_y z^2 - \omega_z yz + \omega_y x^2 - \omega_x xy \\ \omega_z x^2 - \omega_x xz + \omega_z y^2 - \omega_y yz \end{Bmatrix}$$

The first term of the integral can be expressed as a function of the first moment of mass, that is a measure of the distance between the origin of the reference system and the center of mass, while integrating the terms of the second vector term, since angular velocity is constant at any point of the rigid body, we can form the so called inertia matrix and write the angular momentum as:

$$\underline{h}_o = -\underline{v}_o \wedge S_o + I\underline{\omega}$$

where S_o is the first moment of mass and I is the inertia matrix that is expressed as:

$$S_o = \int_B \underline{r} dm$$

$$I = \begin{bmatrix} I_{xx} = \int_B (y^2 + z^2)dm & I_{xy} = \int_B -xydm & I_{xz} = \int_B -xzdm \\ I_{yx} = \int_B -yxdm & I_{yy} = \int_B (x^2 + z^2)dm & I_{yz} = \int_B -yzdm \\ I_{zx} = \int_B -zxdm & I_{zy} = \int_B -zydm & I_{zz} = \int_B (x^2 + y^2)dm \end{bmatrix}$$

Inertia matrix

The inertia matrix is symmetric and the terms along the diagonal are called inertia moments, while the off-diagonal terms are called inertia products. Inertia moments are always positive, while there is no general rule for the inertia products. The inertia matrix, by its definition, must respect precise constraints among its elements. It is possible to define 6 triangular inequalities that relate the inertia moments, which represent constraints on the admissible values of inertia moments. These constraints are on the sum of two inertia moments and on the difference between two inertia moments, for example:

$$I_{xx} + I_{yy} = \int (x^2 + y^2 + 2z^2)dm \geq \int (x^2 + y^2)dm = I_{zz}$$
$$I_{xx} - I_{yy} = \int (y^2 - x^2)dm \leq \int (x^2 + y^2)dm = I_{zz}$$

The remaining inequalities are obtained by simple rotation of the labels x, y and z.
There exist also 3 inequalities that relate the inertia products:

$$I_{xx} = \int (y^2 + z^2)dm = \int (y + z)^2 dm - \int 2zydm = A + 2I_{zy}$$
$$A \geq 0 \qquad I_{xx} \geq 0$$
$$I_{xx} \geq 2I_{zy}$$

Again, remaining inequalities are obtained by simple rotation of the labels x, y and z. Overall, 9 constraints relate the elements of the inertia matrix.

Rotation kinetic energy of a rigid body

Kinetic energy is defined as:

$$2T = \int_B \underline{v} \cdot \underline{v} dm = \int_B \left(\underline{v_o} + \underline{\omega} \wedge \underline{r}\right) \cdot \left(\underline{v_o} + \underline{\omega} \wedge \underline{r}\right) dm$$
$$= \int_B \left(\underline{v_o} \cdot \underline{v_o} + 2\underline{v_o} \cdot \left(\underline{\omega} \wedge \underline{r}\right) + \left(\underline{\omega} \wedge \underline{r}\right) \cdot \left(\underline{\omega} \wedge \underline{r}\right)\right) dm$$

The first term represents the kinetic energy due to the linear motion of the rigid body, not of interest in this case since we study only rotational motion. Therefore, the kinetic energy due to the angular motion is expressed as:

$$2T_{rot} = \int_B \left(2\underline{v_o} \cdot (\underline{\omega} \wedge \underline{r}) + (\underline{\omega} \wedge \underline{r}) \cdot (\underline{\omega} \wedge \underline{r})\right) dm = 2\underline{v_o} \cdot \underline{\omega} \wedge S_o + \int_B (\underline{\omega} \wedge \underline{r}) \cdot (\omega \wedge \underline{r}) dm$$

Expanding the term to be integrated we have:

$$(\underline{\omega} \wedge \underline{r}) \cdot (\underline{\omega} \wedge \underline{r}) = \omega_y^2 z^2 + \omega_z^2 y^2 + \omega_z^2 x^2 + \omega_x^2 z^2 + \omega_x^2 y^2 + \omega_y^2 x^2 - 2\omega_y \omega_z yz - 2\omega_z \omega_x xz - 2\omega_y \omega_x xy$$

That, once integrated, leads to:

$$\int_B (\underline{\omega} \wedge \underline{r}) \cdot (\underline{\omega} \wedge \underline{r}) dm = I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2 + 2I_{yz}\omega_y\omega_z + 2I_{xz}\omega_z\omega_x + 2I_{xy}\omega_y\omega_x$$

We can then write the kinetic energy using matrix notation, as:

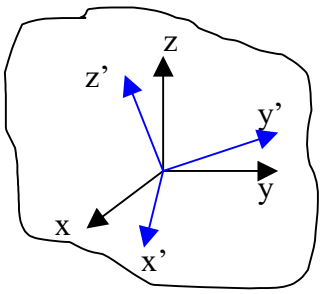
$$2T = 2\underline{v_o} \cdot \underline{\omega} \wedge S_o + \underline{\omega} \cdot I\underline{\omega}$$

By choosing the origin O coincident with the center of mass of the system, S_o vanishes and the angular momentum and kinetic energy become:

$$\underline{h} = I \cdot \underline{\omega}$$
$$2T = \underline{\omega} \cdot I \cdot \underline{\omega}$$

Principal axes and principal inertia moments

We can now try to find a further simplification of the expressions of the angular momentum and kinetic energy, trying to simplify the inertia matrix. In order to do so, we must change the reference system, keeping the origin coincident with the center of mass but such that the kinetic energy is a purely quadratic expression in the components of $\underline{\omega}$, thus reducing the inertia matrix to diagonal form.



Any vector can be transformed from one reference system to another by multiplication by the direction cosine matrix. This matrix contains the projections on the second reference frame of the axis defining the first reference frame.

$$\underline{\omega'} = a\underline{\omega}$$

a represents the direction cosines matrix, that is orthogonal ($a^T=a^{-1}$). Applying the multiplication rule to the expression of the kinetic energy leads to:

$$2T = \left(a^T \underline{\omega'}\right) \cdot I \cdot \left(a^T \underline{\omega'}\right)$$

and, in matrix notation[1]:

$$2T = \underline{\omega'}^T a I a^T \underline{\omega'} = \underline{\omega'}^T I' \underline{\omega'}$$

where I' is the inertia matrix in the new reference frame (a I a^T).
Coming back to the problem of finding a rotation capable of transforming the inertia matrix into diagonal form, we can write this problem as:

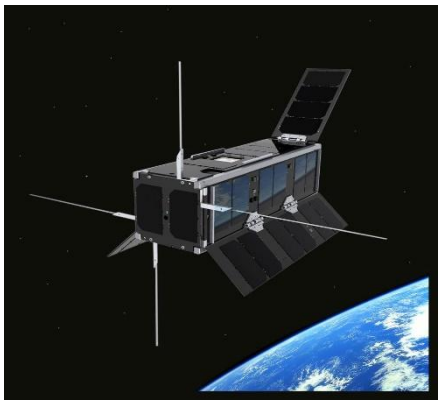
$$a I a^T = diag\left(I_x', I_y', I_z'\right)$$

This is the standard eigenvalue problem for matrix I. Once the eigenvalue problem is solved, the rotation matrix a is composed by the eigenvectors associated to the eigenvalues of I, that are I_x, I_y and I_z. Each row of rotation matrix a is one eigenvector. Eigenvectors correspond to the direction cosines of the new reference frame, called principal inertia frame, with respect to the starting reference frame. The principal inertia frame is therefore reference frame with origin in the center of mass of the rigid body and such that all inertia products are equal to zero. Calling I_x, I_y, I_z, the principal inertia moments, the angular momentum and kinetic energy become:

$$\underline{h} = I\underline{\omega} = [I_x\omega_x \quad I_y\omega_y \quad I_z\omega_z]^T$$
$$2\mathrm{T} = \underline{\omega} I \underline{\omega} = I_x\omega_x^2 + I_y\omega_y^2 + I_z\omega_z^2$$

It is pointed out that a plane of symmetry (in terms of mass distribution) includes 2 principal inertia axes, and then the third is uniquely defined as the axis orthogonal to the plane and passing through the center of mass. It is also pointed out that, normally, it is easy to evaluate the inertia matrix in a geometrically meaningful reference. Should this not be a principal axis frame, it can be calculated by solving the eigenvalue problem as described.
The principal inertia matrix includes always the maximum possible and the least possible inertia moment, no other reference can show along the diagonal greater or smaller inertia moments.

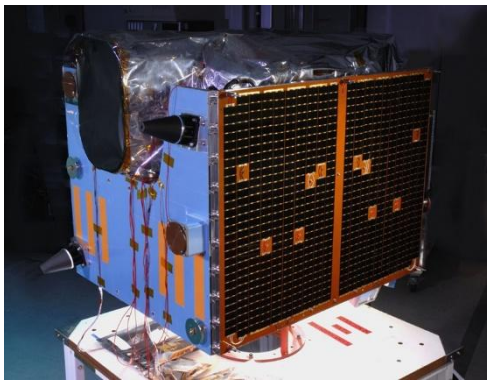
Typical values of the principal moments of inertia for a range of spacecraft are given below:



UKube-1 – 3U CubeSat
$I_1 = 0.0109 kgm^2, I_2 = 0.0504 kgm^2, I_3 = 0.055 kgm^2$

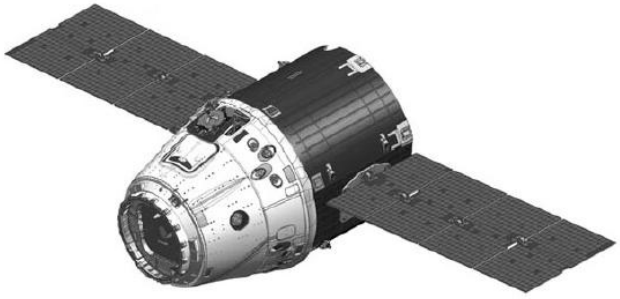
[1] Matrix notation for vector dot product (inner product or scalar product) consists in the following:

$$\underline{v}_1 \cdot \underline{v}_2 = v_1^T v_2$$



Rapid Eye - Micro-spacecraft
$I_1 = 19.5 kgm^2, I_2 = 19 kgm^2, I_3 = 12.6 kgm^2$

Space X 's unmanned 10 tonne spacecraft
$I_1 = 20,000 kgm^2, I_2 = 20,000 kgm^2, I_3 = 25,000 kgm^2$



Or, for example, using a simple cuboid geometry rotating about its centre of mass which is positioned as its centre of geometry then we have:

$$I_d = I_1 = \frac{m}{12}\left(w^2 + h^2\right)$$

$$I_w = I_2 = \frac{m}{12}\left(d^2 + h^2\right)$$

$$I_h = I_3 = \frac{m}{12}\left(w^2 + d^2\right)$$

Inertia ellipsoid, kinetic energy ellipsoid, angular momentum ellipsoid

Assuming that the angular velocity has a constant direction, we can write:

$$\underline{\omega} = \omega\hat{\eta}$$

where η is the unit vector defining the direction of $\underline{\omega}$. The kinetic energy can be written in the form:

$$2T = \underline{\omega} I \underline{\omega} = I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2 + 2\left(I_{xy}\omega_x\omega_y + I_{yz}\omega_y\omega_z + I_{xz}\omega_x\omega_z\right) = I_\eta \omega_\eta^2$$

The expression of I_η can now be evaluated as:

$$I_\eta = I_{xx}\left(\frac{\omega_x}{\omega}\right)^2 + I_{yy}\left(\frac{\omega_y}{\omega}\right)^2 + I_{zz}\left(\frac{\omega_z}{\omega}\right)^2 + 2\left(I_{xy}\left(\frac{\omega_x}{\omega}\right)\left(\frac{\omega_y}{\omega}\right) + I_{yz}\left(\frac{\omega_y}{\omega}\right)\left(\frac{\omega_z}{\omega}\right) + I_{xz}\left(\frac{\omega_x}{\omega}\right)\left(\frac{\omega_z}{\omega}\right)\right)$$

with:

$$\frac{\omega_x}{\omega} = \frac{\underline{\underline{\omega i}}}{\omega} = l_{\eta x}$$
$$\frac{\omega_y}{\omega} = \frac{\underline{\underline{\omega j}}}{\omega} = l_{\eta y}$$
$$\frac{\omega_z}{\omega} = \frac{\underline{\underline{\omega k}}}{\omega} = l_{\eta z}$$

defined as the direction cosines of unit vector η. In a principal inertia reference system we can write:

$$I_\eta = I_x l_{\eta x}^2 + I_y l_{\eta y}^2 + I_z l_{\eta z}^2$$

that can be transformed into:

$$\frac{l_{\eta x}^2/I_\eta}{1/I_x} + \frac{l_{\eta y}^2/I_\eta}{1/I_y} + \frac{l_{\eta z}^2/I_\eta}{1/I_z} = 1$$

This is an ellipsoid, called inertia ellipsoid, in which the coordinates of each point represent the components of the generic direction η divided by the square root of the inertia moment associated to that direction, while the semi-axes are inversely proportional to the square root of each principal inertia moment.

Semi-axes:	$\sqrt{1/I_x}, \sqrt{1/I_y}, \sqrt{1/I_z}$
Point coordinates	$l_{\eta x}/\sqrt{I_\eta}\,, l_{\eta y}/\sqrt{I_\eta}\,, l_{\eta z}/\sqrt{I_\eta}$

Representing this ellipsoid in a Cartesian space, in a principal inertia frame, it is possible to calculate the inertia moment associated to any direction. Tracing any direction, the intersection with the ellipsoid occurs at a distance from the origin equal to $1/\sqrt{I_\alpha}$, where $\underline{\alpha}$ is the unit vector of the given direction. In the same way, the intersection of the ellipsoid with the coordinate axes allows to calculate the principal inertia moments.

Now we can write the expression of the kinetic energy as a function of the generic inertia moment in the direction η:

$$2T = I_\eta \omega^2 = I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2$$

This can be written also as:

$$\frac{\omega_x^2}{2T/I_x} + \frac{\omega_y^2}{2T/I_y} + \frac{\omega_z^2}{2T/I_z} = 1$$

that represents another ellipsoid, called kinetic energy ellipsoid, in a space defined by angular velocities. This ellipsoid represents all possible angular velocities compatible with the given kinetic energy T. The intersection of the ellipsoid with any direction is at distance $\sqrt{2T/I_\alpha}$ from the origin. A similar representation holds for the angular momentum:

$$h^2 = \underline{h} \cdot \underline{h} = I_x^2\omega_x^2 + I_y^2\omega_y^2 + I_z^2\omega_z^2$$

that can be written as:

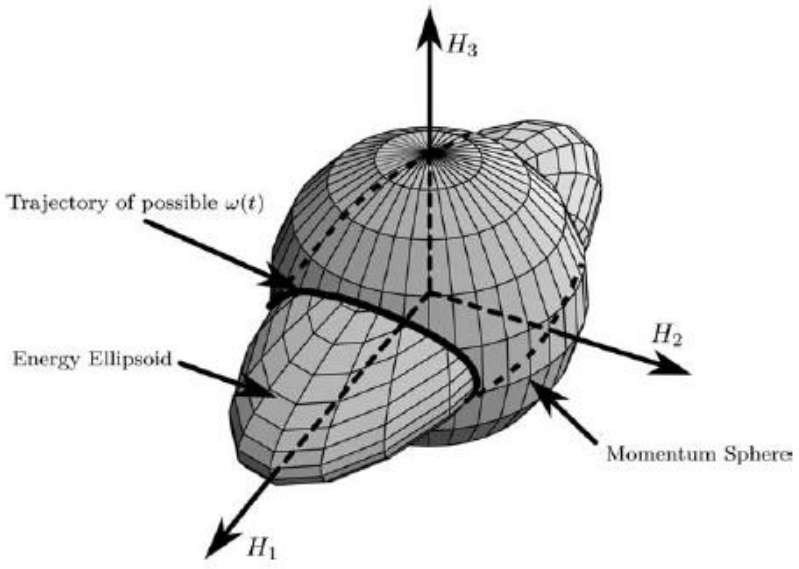
$$\frac{\omega_x^2}{h^2/I_x^2} + \frac{\omega_y^2}{h^2/I_y^2} + \frac{\omega_z^2}{h^2/I_z^2} = 1$$

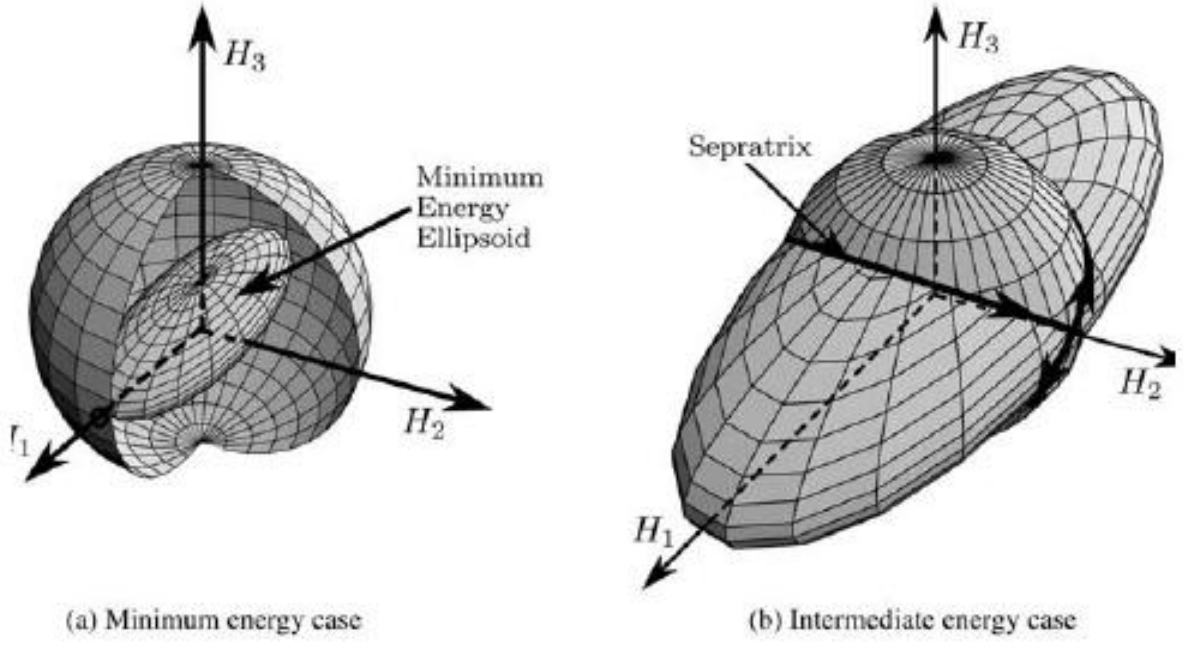
Finally, we have a third ellipsoid, called angular momentum ellipsoid, which represents the compatibility of the angular velocity with the given level of angular momentum.

Geometrical/graphical description of torque-free motion

Polhode and herpolhode

The inertia ellipsoid has no intersection with the kinetic energy ellipsoid, unless the two are coincident, while the angular momentum ellipsoid must intersect the kinetic energy ellipsoid at least in one point in order to represent a real motion. In fact, if no energy dissipation exists and no torque is applied, both kinetic energy and angular momentum must be constant. So, given T and h, angular motion must evolve in such a way that angular velocity vector identifies the intersection of the kinetic energy ellipsoid with the angular momentum ellipsoid. In fact, angular velocity must be compatible with the energy level and with the angular momentum defined by the initial conditions of motion. The intersection of the kinetic energy ellipsoid with the angular momentum ellipsoid generates two lines, called polhodes.





(a) Minimum energy case (b) Intermediate energy case

So far we have looked at the problem in a body frame, not fixed in inertial space. In an inertial frame, we will observe vector $\underline{h}$ as a fixed vector, if no torque is applied. Then, if no dissipation is present, we have:

$$T = \underline{\omega} \cdot \underline{h} = cos\, t$$
$$\underline{h} = cos\, t$$

We can observe that the projection of $\underline{\omega}$ over $\underline{h}$ is constant:

$$\omega_h = \frac{\underline{\omega} \cdot \underline{h}}{|\underline{h}|} = \frac{T}{|\underline{h}|} = cos\, t$$

Now, considering a plane orthogonal to $\underline{h}$, at a distance ω_h from the center of mass, the terminal point of vector $\underline{\omega}$ lies always on this plane due to the conservation of the projection of $\underline{\omega}$ over $\underline{h}$. This plane is called invariant plane. All this is a consequence of conservation of kinetic energy T with time:

$$\frac{d}{dt}(\underline{\omega} \cdot \underline{h}) = \frac{1}{dt}(d\underline{\omega} \cdot \underline{h} + \underline{\omega} \cdot d\underline{h}) = 0$$
$$d\underline{h} = 0$$
$$d\underline{\omega} \cdot \underline{h} = 0 \quad \Rightarrow \quad d\underline{\omega} \perp \underline{h}$$

The line traced by the angular velocity on the invariant plane is in general not closed, it is called herpolhode. The herpolhode is in any case a curve limited in space. In the body axes reference frame the derivative of the angular velocity vector is always tangent to the kinetic energy ellipsoid, while in the inertial space the kinetic energy ellipsoid rolls on the invariant plane as the angular velocity draws the herpolhode.
From the mathematical point of view, the intersection of the two ellipsoids is evaluated as:

$$\frac{\omega_x^2}{2T/I_x} + \frac{\omega_y^2}{2T/I_y} + \frac{\omega_z^2}{2T/I_z} = 1$$
$$\frac{\omega_x^2}{h^2/I_x^2} + \frac{\omega_y^2}{h^2/I_y^2} + \frac{\omega_z^2}{h^2/I_z^2} = 1$$

Equating the two left hand terms, equal to 1, we find the polhode:

$$\omega_x^2\left[I_x\left(\frac{I_x}{h^2}-\frac{1}{2T}\right)\right] + \omega_y^2\left[I_y\left(\frac{I_y}{h^2}-\frac{1}{2T}\right)\right] + \omega_z^2\left[I_z\left(\frac{I_z}{h^2}-\frac{1}{2T}\right)\right] = 0$$

In order to obtain a real solution, the three terms into brackets must show differences in sign. Multiplying each term by h^2 we have three terms like:

$$I - \frac{h^2}{2T}$$

Assuming $I_x > I_y > I_z$ we obtain the conditions for a sign change:

$$I_x > \frac{h^2}{2T} \qquad \text{and} \qquad I_z < \frac{h^2}{2T}$$

It is not necessary to impose conditions on I_y since the change in sign is already guaranteed by the previous conditions. The final constraint is then:

$$I_x > \frac{h^2}{2T} > I_z$$

that guarantees the possibility of having a real rotational motion.

Given the result so far obtained, it is difficult to make any consideration on the polhode as a line in space, while it is interesting to analyze its projections onto the three coordinate planes. We have:

$$\frac{I_x\omega_x^2}{2T} + \frac{I_y\omega_y^2}{2T} + \frac{I_z\omega_z^2}{2T} = 1$$
$$I_x\omega_x^2 + I_y\omega_y^2 + I_z\omega_z^2 = 2T$$

Looking at the projection onto the plane (x-y), we reduce to two dimensions by eliminating the coordinate ω_z:

$$I_z\omega_z^2 = 2T - I_x\omega_x^2 - I_y\omega_y^2$$

Substitution into the polhode equation gives:

$$I_x\omega_x^2\left(\frac{I_x}{h^2}-\frac{1}{2T}+\frac{1}{2T}-\frac{I_z}{h^2}\right) + I_y\omega_y^2\left(\frac{I_y}{h^2}-\frac{I_z}{h^2}\right) + 2T\left(\frac{I_z}{h^2}-\frac{1}{2T}\right) = 0$$
$$I_x\omega_x^2\left(\frac{I_x}{h^2}-\frac{I_z}{h^2}\right) + I_y\omega_y^2\left(\frac{I_y}{h^2}-\frac{I_z}{h^2}\right) + 2T\frac{I_z}{h^2} - 1 = 0$$

$$\omega_x^2\left(\frac{I_x(I_x-I_z)}{h^2}\right)+\omega_y^2\left(\frac{I_y(I_y-I_z)}{h^2}\right)+\frac{2TI_z-h^2}{h^2}=0$$
$$\omega_x^2\left(\frac{I_x(I_x-I_z)}{h^2-2TI_z}\right)+\omega_y^2\left(\frac{I_y(I_y-I_z)}{h^2-2TI_z}\right)=1$$

that represents a planar conic curve. Operating in the same way, we have the projections onto the (y-z) plane:

$$\omega_y^2\left(\frac{I_y(I_y-I_x)}{h^2-2TI_x}\right)+\omega_z^2\left(\frac{I_z(I_z-I_x)}{h^2-2TI_x}\right)=1$$

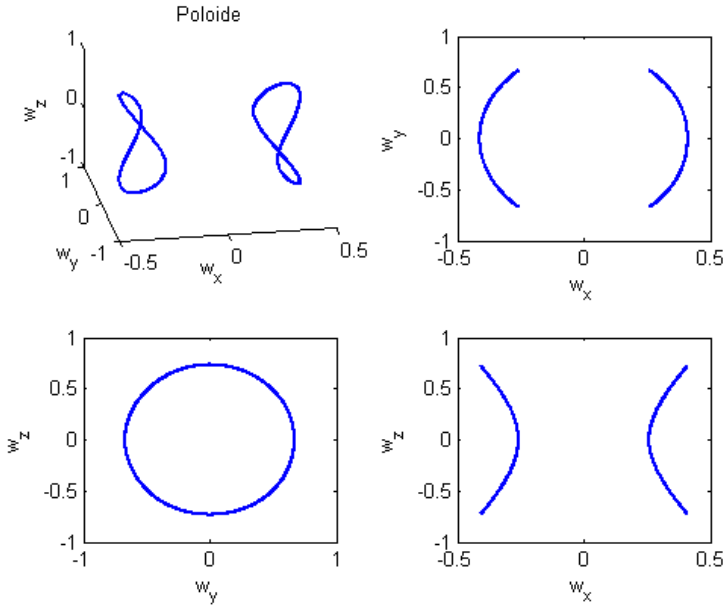
and onto the (x-z) plane:

$$\omega_x^2\left(\frac{I_x(I_x-I_y)}{h^2-2TI_y}\right)+\omega_z^2\left(\frac{I_z(I_z-I_y)}{h^2-2TI_y}\right)=1$$

Analyzing the planar conic curves and the signs of all coefficients, we have that:

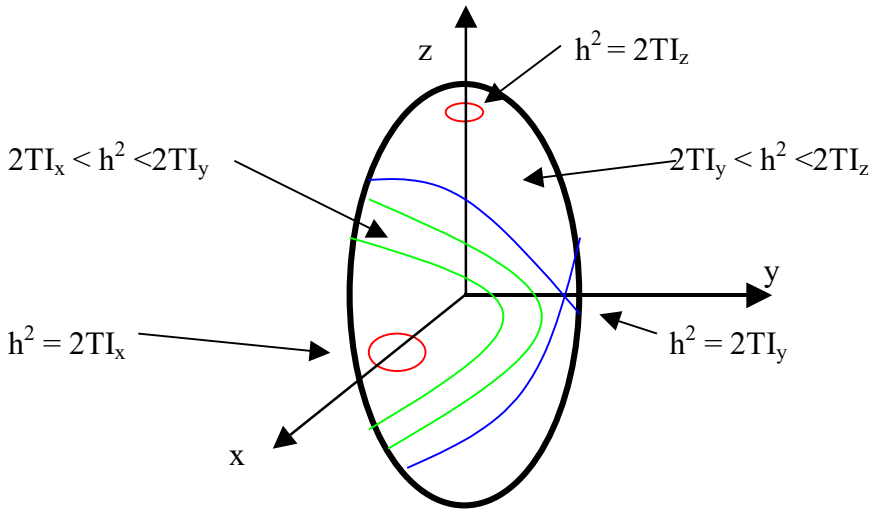
$$h^2-2TI_z>0$$
$$h^2-2TI_x<0$$
$$h^2-2TI_y?$$

Therefore, we know that the projections on planes (x-y) and (y-z) are ellipses, since coefficients are of the same sign, while projection on plane (x-z) is a hyperbola since the signs of the coefficients are different.



The fact that, looking at the polhode in the (x-z) plane, i.e. looking at it from the y axis, we see a hyperbola can be considered as a sort of instability. In fact, if the angular velocity is slightly shifted from the x-axis or from the z-axis it will remain confined in a region close to the axis, while a slight shift from the y-axis would cause a dramatic departure from the original condition. All this can be seen by looking at the kinetic energy ellipsoid and at the trajectories that are drawn on it by the angular

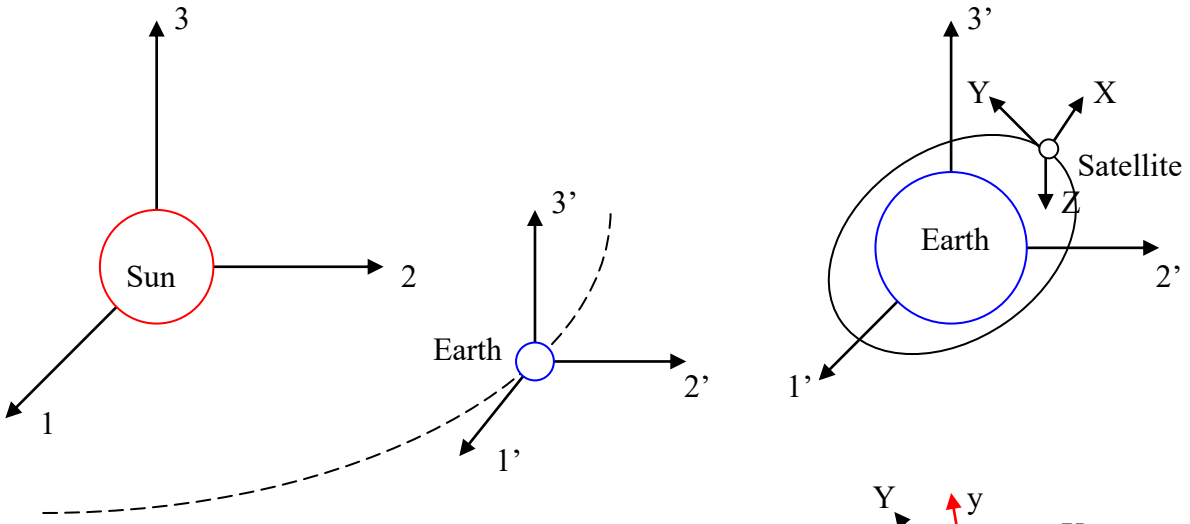
velocity vector. It is possible to divide the ellipsoid into areas corresponding to different initial conditions:

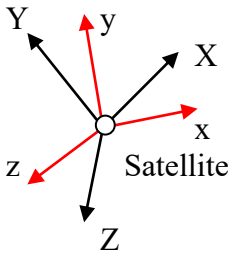


The inequalities hold for a given ellipsoid where the energy level has been fixed.

Attitude parameters

It is always necessary to be able to switch from a reference frame to another reference frame, such as the case of representing the principal inertia frame (body frame) as seen by an inertial frame or by an Earth-centered reference frame or even by a local-vertical-local-horizontal frame.





X is the satellite local vertical direction
Y is the satellite velocity direction
Z is the third direction, orthogonal to X and Y

x,y,z are the satellites' principal inertia axes

To define the orientation of one frame with respect to another, three parameters are the minimal set required. Often redundant parameters are used, i.e. more than three, either in order to improve the physical insight into the transformation or to simplify some numerical analysis.

Direction cosines

The attitude of the rigid-body $A_{B/N}$ is defined as the relative orientation between the body fixed frame **B** and the inertial frame **N** defined explcitly as:

$$\mathbf{B}=A_{B/N}\mathbf{N}$$

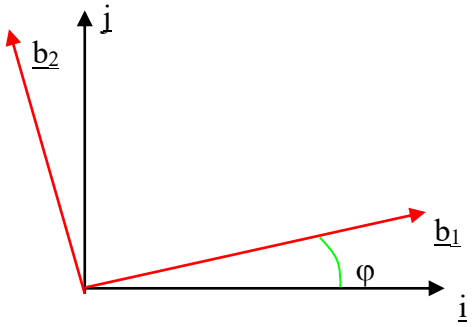
$A_{B/N}$ is an orthonormal matrix with the following properties:

$$A_{B/N}A_{B/N}^{T} = I_{3\times3}, det(A_{B/N}) = 1$$

Each axis of the reference frame that we target is defined in the current reference frame by the three components of its unit direction vector. Calling *u*, *v* and *w* the unit vectors of the new reference frame, assembling the components of the three axes we obtain the direction cosines matrix:

$$A_{B/N} = \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

in which each row represents one axis. This matrix allows switching from the representation of a vector in one reference to another reference. As example, let consider a planar rotation:



$$A_{B/N} = \begin{bmatrix} cos\,\phi & sin\,\phi & 0 \\ -sin\,\phi & cos\,\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The formal expression of the unit vector in the new reference ($\underline{a}_B$) is obtained by multiplying the original unit vector ($\underline{a}_N$) by the direction cosine matrix A:

$$\underline{a}_B = A_{B/N}\underline{a}_N$$

For the inverse rotation it is necessary to write the direction cosine matrix of axes 1,2,3 in the reference *u,v,w*; the result is the transpose of matrix A:

$$\underline{a}_N = A_{B/N}^T\underline{a}_B$$

This transformation does not affect the magnitude of the vectors and the relative angles between vectors. This attitude representation includes 9 parameters, 6 more than the minimal set, but it must be considered that the three axes are mutually orthogonal. So, there must be six constraints among direction cosines, three expressing mutual orthogonality and three expressing the invariability of magnitude of the unit vectors:

$$\begin{cases} |\underline{u}| = u_1^2 + u_2^2 + u_3^2 = 1 \\ |\underline{v}| = v_1^2 + v_2^2 + v_3^2 = 1 \\ |\underline{w}| = w_1^2 + w_2^2 + w_3^2 = 1 \end{cases} \qquad \text{Magnitude}$$

$$\begin{cases} \underline{u} \perp \underline{v} \quad \Rightarrow \quad \underline{u} \cdot \underline{v} = u_1v_1 + u_2v_2 + u_3v_3 = 0 \\ \underline{u} \perp \underline{w} \quad \Rightarrow \quad \underline{u} \cdot \underline{w} = u_1w_1 + u_2w_2 + u_3w_3 = 0 \\ \underline{v} \perp \underline{w} \quad \Rightarrow \quad \underline{v} \cdot \underline{w} = v_1w_1 + v_2w_2 + v_3w_3 = 0 \end{cases} \qquad \text{Orthogonality conditions}$$

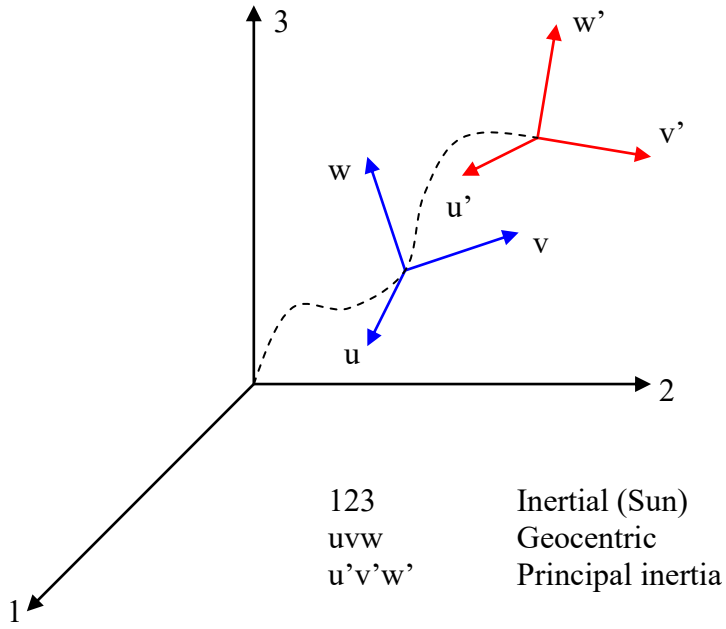
The six constraints among the direction cosines lead to the following property of the direction cosines matrix:

$$AA^T = I$$

Then, A is orthogonal so that the following holds:

$$A^T = A^{-1}$$

We can now see how to treat a sequence of two rotations:



The generic rotation of vector a from the reference 123 into reference uvw is expressed as:

$$a_{uvw} = A \cdot a_{123}$$

In the same way, the second rotation from reference uvw to reference u'v'w' is expressed as:

$$a_{u'v'w'} = A' \cdot a_{uvw}$$

Then, the overall rotation from reference 123 to reference u'v'w' becomes:

$$a_{u'v'w'} = A" \cdot a_{123} = A' \cdot a_{uvw} = A'A \cdot a_{123}$$

Concluding, in order to compose a sequence of two rotations it is sufficient to build the overall direction cosine matrix as the product of the two individual rotations (direction cosine matrix), taken in the reversed order compared to the sequence of rotations.

$$A" = A'A$$

Generalizing the concept, lets say we have two frames **L** and **N** and a vector expressed with respect to each frame. Then their orientation is related by:

$$\underline{a}_L = A_{L/N} \cdot \underline{a}_N$$

A second rotation could be performed in the same way between a frame **L** and a frame **B**:

$$\underline{a}_B = A_{B/L} \cdot \underline{a}_L$$

So, to rotate a vector expressed in the frame **N** to **B** we can write:

$$\underline{a}_B = A_{B/L} A_{L/N} \underline{a}_N$$

If we have two different moving frames expressed with respect to the same inertial frame we can write:

$$A_{B/N} = A_{B/L} A_{L/N} \Leftrightarrow A_{B/L} = A_{B/N} A_{L/N}^T$$

The relative rotation between two frames is often called the rotation error such that $A_{B/L} = A_e$, $A_{B/N}$ is the spaceraft body frame with respect to this frame while $A_{L/N} = A_d$ usually corresponds to some desired reference frame. If the spacecraft attitude matches the desired reference frame, then $A_e = I$ which corresponds to a zero error.

Euler axis and angle

From matrix algebra it is known that real, orthogonal matrices have one unit eigenvalue, to which we associate the eigenvector e:

$$A\underline{e} = \underline{e} \qquad \text{(the eigenvalue is 1)}$$

Therefore, vector e does not change due to the rotation represented by matrix A. This is possible only if the rotation occurs around axis e. This axis is called Euler axis, and the rotation amplitude around this axis is called Euler angle. We can now try to relate the direction cosines matrix A with vector e. If we consider elementary rotations around each coordinate axis 1, 2 and 3, we can notice that:

$$A_3(\phi) = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \underline{e} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A_2(\phi) = \begin{bmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{bmatrix} \qquad \underline{e} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$A_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \qquad \underline{e} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

In each case, the trace of matrix A is the same and equals:

$$tr(A) = 1 + 2\cos\phi$$

Then it is possible to infer the relation between Euler angle and rotation matrix A:

$$\cos\phi = \frac{1}{2}(tr(A) - 1)$$

Given matrix A it is possible to evaluate e and φ with simple operations.

Matrix A has the following form in relation to components of the Euler axis e and angle φ:

$$A = \begin{bmatrix} \cos\phi + e_1^2(1-\cos\phi) & e_1e_2(1-\cos\phi) + e_3\sin\phi & e_1e_3(1-\cos\phi) - e_2\sin\phi \\ e_1e_2(1-\cos\phi) - e_3\sin\phi & \cos\phi + e_2^2(1-\cos\phi) & e_2e_3(1-\cos\phi) + e_1\sin\phi \\ e_1e_3(1-\cos\phi) + e_2\sin\phi & e_2e_3(1-\cos\phi) - e_1\sin\phi & \cos\phi + e_3^2(1-\cos\phi) \end{bmatrix}$$

or:

$$A = I\cos\phi + (1-\cos\phi)\underline{e}\underline{e}^T - \sin\phi[e\wedge]$$

where:

$$[e\wedge] = \begin{bmatrix} 0 & -e_3 & e_2 \\ e_3 & 0 & -e_1 \\ -e_2 & e_1 & 0 \end{bmatrix}$$

represents cross product matrix operator.
If we know the attitude parameters expressed as Euler axis and angle, it is therefore straightforward to evaluate the direction cosine matrix A. The inverse operation is also possible, given the rotation matrix A evaluate Euler axis and angle, in fact:

$$\phi = cos^{-1}\left[\frac{1}{2}(tr(A)-1)\right]$$

$$\begin{cases} e_1 = \dfrac{(A_{23}-A_{32})}{2sin\phi} \\ e_2 = \dfrac{(A_{31}-A_{13})}{2sin\phi} \\ e_3 = \dfrac{(A_{12}-A_{21})}{2sin\phi} \end{cases}$$

It is remarked that the direct transformation, from Euler axis/angle to rotation matrix A, is always possible, while the inverse transformation, from rotation matrix A to Euler axis/angle, is not always defined. In fact, when sinφ becomes zero the Euler axis is undetermined. This occurs when φ = n π, that physically means that the axis is not uniquely determined, i.e., it is possible to reach the final configuration through at least two rotations of same amplitude but different axis. When the inverse transformation is not determined, the condition is known as a singularity in the attitude parameterization.

It is also remarked that the inverse trigonometric function arcos returns two possible values of the Euler angle ($\pm\phi$) that in turn provide two different axes ($\pm\underline{e}$). This is actually different from the singularity condition, since we are always looking at the same axis, once from the positive and once from the negative direction, and to each condition we associate the same rotation angle with opposite sign. This is physically the same condition.

Euler axis/angle parameters are useful since they are only 4 parameters (the constraint condition is associated to the fact that $\underline{e}$ is a unit vector), but unfortunately it is not possible to infer a relation for sequence of rotations.

Quaternion

A quaternion is a vector of four parameters, linked to the Euler axis/angle parameters through the relation:

$$\begin{cases} q_1 = e_1 sin\dfrac{\phi}{2} \\ q_2 = e_2 sin\dfrac{\phi}{2} \\ q_3 = e_3 sin\dfrac{\phi}{2} \\ q_4 = cos\dfrac{\phi}{2} \end{cases}$$

It can be noticed that the quaternion is normalized:

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1$$

In some cases it is convenient to divide the quaternion into a vector part $\underline{q}$ and a scalar part q_4:

$$\underline{q} = \begin{Bmatrix} q_1 \\ q_2 \\ q_3 \end{Bmatrix} \quad , \quad q_4$$

The direct transformation, given the quaternion evaluate the direction cosine matrix A, leads to:

$$A = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_2 - q_3q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix}$$

or:

$$A = \left(q_4^2 - \underline{q}^T\underline{q}\right) I + 2\underline{q}\underline{q}^T - 2q_4[q\wedge]$$

where:

$$[q\wedge] = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix}$$

The inverse transformation, from matrix A to quaternion, is represented as:

$$\begin{cases} q_1 = \dfrac{1}{4q_4}(A_{23} - A_{32}) \\ q_2 = \dfrac{1}{4q_4}(A_{31} - A_{13}) \\ q_3 = \dfrac{1}{4q_4}(A_{12} - A_{21}) \\ q_4 = \pm\dfrac{1}{2}(1 + A_{11} + A_{22} + A_{33})^{\frac{1}{2}} \end{cases}$$

The sign ambiguity has the same explanation given for the Euler axis/angle case. We are always looking at the same axis, once from the positive and once from the negative direction, and to each condition we associate the same rotation angle with opposite sign. This is physically the same condition. Quaternions have no singular condition even in the inverse transformation. In fact, should q_4 become zero it is always possible to evaluate one of the other components of the quaternion, which

will be surely different from zero due to the normalization constraint, and then evaluate the remaining three components. The inverse transformation has then the following three further sets of alternative equations:

$$q_1^2 = \pm\frac{1}{2}\sqrt{1 + A_{11} - A_{22} - A_{33}}$$
$$q_2^2 = \frac{1}{4q_1^2}(A_{12} + A_{21})$$
$$q_3^2 = \frac{1}{4q_1^2}(A_{13} + A_{31})$$
$$q_4^2 = \frac{1}{4q_1^2}(A_{23} - A_{32})$$

$$q_2^3 = \pm\frac{1}{2}\sqrt{1 - A_{11} + A_{22} - A_{33}}$$
$$q_1^3 = \frac{1}{4q_2^3}(A_{12} + A_{21})$$
$$q_3^3 = \frac{1}{4q_2^3}(A_{23} + A_{32})$$
$$q_4^3 = \frac{1}{4q_2^3}(A_{31} - A_{13})$$

$$q_3^4 = \pm\frac{1}{2}\sqrt{1 - A_{11} - A_{22} + A_{33}}$$
$$q_1^4 = \frac{1}{4q_3^4}(A_{13} + A_{31})$$
$$q_2^4 = \frac{1}{4q_3^4}(A_{23} + A_{32})$$
$$q_4^4 = \frac{1}{4q_3^4}(A_{12} - A_{21})$$

It is even possible to select the set of equations to adopt on the basis of the maximization of the square root term, for a reduced truncation error in the following operations. Unfortunately, quaternions have no physical meaning and therefore their use is not intuitive.
Adopting quaternions, it is possible to express a sequence of two consecutive rotations, combining quaternion components in the reversed order of rotations. Given a first rotation represented by quaternion q, a second rotation represented by quaternion q', the overall rotation is represented by quaternion q" computed as $q^{''} = q \otimes q^{'}$:

$$q" = \begin{bmatrix} q_4^{'} & q_3^{'} & -q_2^{'} & q_1^{'} \\ -q_3^{'} & q_4^{'} & q_1^{'} & q_2^{'} \\ q_2^{'} & -q_1^{'} & q_4^{'} & q_3^{'} \\ -q_1^{'} & -q_2^{'} & -q_3^{'} & q_4^{'} \end{bmatrix} q$$

Recalled the relative attitude error $A_e = A_{B/N}A_d^T$ this is equivalent to $\hat{q}_e = \hat{q}_d^T \otimes \hat{q}_{B/N}$ which is explicitly:

$$\begin{bmatrix} q_{1e} \\ q_{2e} \\ q_{3e} \\ q_{4e} \end{bmatrix} = \begin{bmatrix} q_{4c} & q_{3c} & -q_{2c} & -q_{1c} \\ -q_{3c} & q_{4c} & q_{1c} & -q_{2c} \\ q_{2c} & -q_{1c} & q_{4c} & -q_{3c} \\ q_{1c} & q_{2c} & q_{3c} & q_{4c} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Gibbs vector

Gibbs vector is a minimal attitude parameterization, since it includes only three parameters. It can also be seen as a non-normalized quaternion, and can be inferred from Euler axis/angle:

$$\begin{cases} g_1 = \dfrac{q_1}{q_4} = e_1 \, tan\dfrac{\phi}{2} \\ g_2 = \dfrac{q_2}{q_4} = e_2 \, tan\dfrac{\phi}{2} \\ g_3 = \dfrac{q_3}{q_4} = e_3 \, tan\dfrac{\phi}{2} \end{cases}$$

Also in this case it is possible to represent the direct transformation from Gibbs vector to rotation matrix A:

$$A = \frac{1}{1 + g_1^2 + g_2^2 + g_3^2} \begin{bmatrix} 1 + g_1^2 - g_2^2 - g_3^2 & 2(g_1 g_2 + g_3) & 2(g_1 g_3 - g_2) \\ 2(g_1 g_2 - g_3) & 1 - g_1^2 + g_2^2 - g_3^2 & 2(g_2 g_3 + g_1) \\ 2(g_1 g_3 + g_2) & 2(g_2 g_3 - g_1) & 1 - g_1^2 - g_2^2 + g_3^2 \end{bmatrix}$$

or:

$$A = \frac{\left(1 - \underline{g}^2\right) I + 2\underline{g}\underline{g}^T - 2[g \wedge]}{\left(1 + \underline{g}^2\right)}$$

where:

$$[g \wedge] = \begin{bmatrix} 0 & -g_3 & g_2 \\ g_3 & 0 & -g_1 \\ -g_2 & g_1 & 0 \end{bmatrix}$$

The inverse transformation is given by:

$$\begin{cases} g_1 = \dfrac{A_{23} - A_{32}}{1 + A_{11} + A_{22} + A_{33}} \\ g_2 = \dfrac{A_{31} - A_{13}}{1 + A_{11} + A_{22} + A_{33}} \\ g_3 = \dfrac{A_{12} - A_{21}}{1 + A_{11} + A_{22} + A_{33}} \end{cases}$$

The inverse transformation becomes singular when $\varphi = (2n + 1)\,\pi$.

Two consecutive rotations (first rotation expressed by vector g, second rotation represented by vector g') can be merged into a single Gibbs vector g" with the following rule:

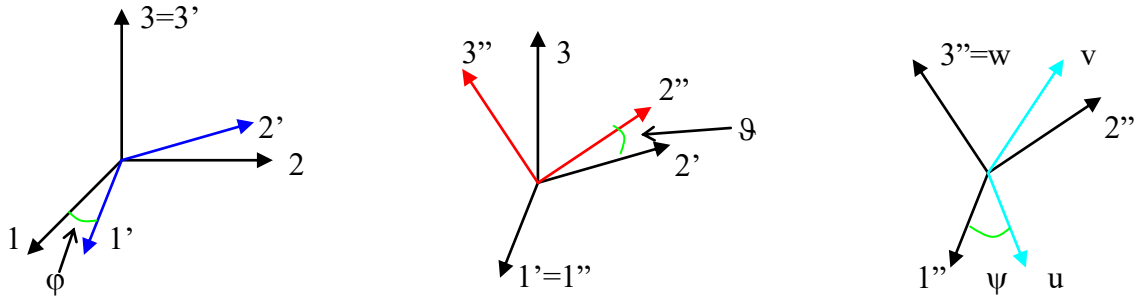
$$g'' = \frac{\underline{g} + \underline{g}' - \underline{g}' \wedge \underline{g}}{1 - \underline{g} \cdot \underline{g}'}$$

Also this parameterization has a poor physical meaning and furthermore it has singularity configuration in the inverse transformation, therefore part of the advantages of having only three parameters are lost.

Euler angles

The attitude can be represented by means of three Euler angles, which are a minimal representation and have a clear physical interpretation. The concept is based on the fact that it is always possible to make two orthogonal frames overlap by appropriate rotation of one of them three times around its reference axes.

Let us make an example to overlap systems 1,2,3 and u,v,w.
We start by rotating by angle φ around axis 3, that brings the triad in the new configuration 1',2',3'.
Now rotate by angle ϑ around axis 1', that brings the triad in the configuration 1",2",3".
Finally, rotate by angle ψ around axis 3" to superimpose the triad to the reference u,v,w.



The direction cosine matrix can be written by adopting the rule of consecutive rotations. In fact, it is possible to combine the elementary rotation matrices since each rotation is around one of the coordinate axes:

$$A_{313}(\phi, \vartheta, \psi) = A_3(\psi) \cdot A_1(\vartheta) \cdot A_3(\phi)$$

where:

$$A_3(\psi) = \begin{bmatrix} cos\,\psi & sin\psi & 0 \\ -sin\psi & cos\,\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A_1(\vartheta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\,\vartheta & sin\vartheta \\ 0 & -sin\vartheta & cos\,\vartheta \end{bmatrix}$$

$$A_3(\phi) = \begin{bmatrix} cos\,\phi & sin\phi & 0 \\ -sin\phi & cos\,\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The labels 1,2,3 of the rotation matrices do not indicate in this case the axis of rotation, but rather the rotation type, i.e., the family of axes around which the triad is rotating. The overall rotation matrix is

labeled according to the rotation sequence and the rotation angles associated to each individual rotation:

$$A_{313}(\phi, \vartheta, \psi)$$

In this case it is fundamental to remind that matrix products must be made in the reversed order as compared to the rotations. It is also possible to superimpose the two triads by a different set of rotation axes, and in this case the values of the rotation angles will also change radically.
This is true even if only the last rotation is changed, all angles will change their magnitude. Matrix A_{313} is then written as:

$$A_{313} = \begin{bmatrix} cos\,\psi\, cos\,\phi - sin\psi sin\phi\, cos\,\vartheta & cos\,\psi\, sin\phi + sin\psi\, cos\,\phi\, cos\,\vartheta & sin\psi sin\vartheta \\ -sin\psi\, cos\,\phi - cos\,\psi\, sin\phi\, cos\,\vartheta & -sin\psi sin\phi + cos\,\psi\, cos\,\phi\, cos\,\vartheta & cos\,\psi\, sin\vartheta \\ sin\phi sin\vartheta & -\,cos\,\phi\, sin\vartheta & cos\,\vartheta \end{bmatrix}$$

It can be noticed that the inverse transform, from matrix A to Euler angles, in this case takes the form:

$$\begin{cases} \vartheta = cos^{-1}(A_{33}) \\ \phi = -\,tan^{-1}\left(\dfrac{A_{31}}{A_{32}}\right) \\ \psi = tan^{-1}\left(\dfrac{A_{13}}{A_{23}}\right) \end{cases}$$

In this case we notice a singularity when sin ϑ tends to zero. This means that the first and last rotations are actually around the same physical direction in space, therefore it is not possible to distinguish them individually, but rather only the sum or difference of the two.

With Euler angles we have a different rotation matrix A for each combination of rotation axes. Two consecutive rotations cannot be around the same axis, they would become one single rotation, therefore the possible sequences are 12, out of which 6 have all indexes different and 6 are characterized by the same first and third index:

312,213,123,321,231,132	all different indexes
313,323,212,232,131,121	first and third index equal

The different sets of Euler angles are singular for different values of the second rotation angle ϑ, depending on the fact that the indexes are all different or the first and third index coincide. When all indexes are different, the singularity condition is ϑ = (2n+1)π/2, while for equal first and third index the singularity occurs when ϑ = nπ.
Here follow the symbolic expressions of the direction cosine matrix for the 12 possible sequences, assuming that the first rotation is always represented by angle φ, the second by angle ϑ and the third by angle ψ.

$$A_{123} = \begin{bmatrix} \cos\psi\cos\vartheta & \cos\psi\sin\vartheta\sin\phi + \sin\psi\cos\phi & -\cos\psi\sin\vartheta\, cos\,\phi + \sin\psi\sin\phi \\ -\sin\psi\cos\vartheta & -\sin\psi\sin\vartheta\sin\phi + \cos\psi\cos\phi & \sin\psi\sin\vartheta\, cos\,\phi + \cos\psi\sin\phi \\ \sin\vartheta & -\cos\vartheta\sin\phi & \cos\vartheta\,\cos\phi \end{bmatrix}$$

$$A_{132} = \begin{bmatrix} \cos\psi\cos\vartheta & \cos\psi\sin\vartheta\cos\phi + \sin\psi\sin\phi & \cos\psi\sin\vartheta\sin\phi - \sin\psi\cos\phi \\ -\sin\vartheta & \cos\phi\cos\vartheta & \cos\vartheta\sin\phi \\ \sin\psi\cos\vartheta & \sin\psi\sin\vartheta\cos\phi - \cos\psi\sin\phi & \sin\psi\sin\vartheta\sin\phi + \cos\psi\cos\phi \end{bmatrix}$$

$$A_{231}=\begin{bmatrix}\cos\vartheta\cos\phi & \sin\vartheta & -\cos\vartheta\sin\phi\\ -\cos\psi\sin\vartheta\cos\phi+\sin\psi\sin\phi & \cos\psi\cos\vartheta & \cos\psi\sin\vartheta\sin\phi+\sin\psi\cos\phi\\ \sin\psi\sin\vartheta\cos\phi+\cos\psi\sin\phi & -\sin\psi\cos\vartheta & -\sin\psi\sin\vartheta\sin\phi+\cos\psi\cos\phi\end{bmatrix}$$

$$A_{213}=\begin{bmatrix}\cos\psi\cos\phi+\sin\psi\sin\phi\sin\vartheta & \sin\psi\cos\vartheta & -\cos\psi\sin\phi+\sin\psi\cos\phi\sin\vartheta\\ -\sin\psi\cos\phi+\cos\psi\sin\phi\sin\vartheta & \cos\psi\cos\vartheta & \sin\psi\sin\phi+\cos\psi\cos\phi\sin\vartheta\\ \sin\phi\cos\vartheta & -\sin\vartheta & \cos\phi\cos\vartheta\end{bmatrix}$$

$$A_{312}=\begin{bmatrix}\cos\psi\cos\phi-\sin\psi\sin\phi\sin\vartheta & \cos\psi\sin\phi+\sin\psi\cos\phi\sin\vartheta & -\sin\psi\cos\vartheta\\ -\sin\phi\cos\vartheta & \cos\phi\cos\vartheta & \sin\vartheta\\ \sin\psi\cos\phi+\cos\psi\sin\phi\sin\vartheta & \sin\psi\sin\phi-\cos\psi\cos\phi\sin\vartheta & \cos\vartheta\cos\psi\end{bmatrix}$$

$$A_{321}=\begin{bmatrix}\cos\phi\cos\vartheta & \cos\vartheta\sin\phi & -\sin\vartheta\\ -\cos\psi\sin\phi+\sin\psi\cos\phi\sin\vartheta & \cos\psi\cos\phi+\sin\psi\sin\phi\sin\vartheta & \sin\psi\cos\vartheta\\ \sin\psi\sin\phi+\cos\psi\cos\phi\sin\vartheta & -\sin\psi\cos\phi+\cos\psi\sin\phi\sin\vartheta & \cos\psi\cos\vartheta\end{bmatrix}$$

$$A_{121}=\begin{bmatrix}\cos\vartheta & \sin\phi\sin\vartheta & -\cos\phi\sin\vartheta\\ \sin\psi\sin\vartheta & \cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta & \cos\psi\sin\phi+\sin\psi\cos\phi\cos\vartheta\\ \cos\psi\sin\vartheta & -\sin\psi\cos\phi-\cos\psi\sin\phi\cos\vartheta & -\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta\end{bmatrix}$$

$$A_{131}=\begin{bmatrix}\cos\vartheta & \cos\phi\sin\vartheta & \sin\phi\sin\vartheta\\ -\cos\psi\sin\vartheta & -\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta & \sin\psi\cos\phi+\cos\psi\sin\phi\cos\vartheta\\ \sin\psi\sin\vartheta & -\cos\psi\sin\phi-\sin\psi\cos\phi\cos\vartheta & \cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta\end{bmatrix}$$

$$A_{212}=\begin{bmatrix}\cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta & \sin\psi\sin\vartheta & -\cos\psi\sin\phi-\sin\psi\cos\phi\cos\vartheta\\ \sin\phi\sin\vartheta & \cos\vartheta & \cos\phi\sin\vartheta\\ \sin\psi\cos\phi+\cos\psi\sin\phi\cos\vartheta & -\cos\psi\sin\vartheta & -\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta\end{bmatrix}$$

$$A_{232}=\begin{bmatrix}-\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta & \cos\psi\sin\vartheta & -\sin\psi\cos\phi-\cos\psi\sin\phi\cos\vartheta\\ -\cos\phi\sin\vartheta & \cos\vartheta & \sin\phi\sin\vartheta\\ \cos\psi\sin\phi+\sin\psi\cos\phi\cos\vartheta & \sin\psi\sin\vartheta & \cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta\end{bmatrix}$$

$$A_{313}=\begin{bmatrix}\cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta & \cos\psi\sin\phi+\sin\psi\cos\phi\cos\vartheta & \sin\psi\sin\vartheta\\ -\sin\psi\cos\phi-\cos\psi\sin\phi\cos\vartheta & -\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta & \cos\psi\sin\vartheta\\ \sin\phi\sin\vartheta & -\cos\phi\sin\vartheta & \cos\vartheta\end{bmatrix}$$

$$A_{323}=\begin{bmatrix}-\sin\psi\sin\phi+\cos\psi\cos\phi\cos\vartheta & \sin\psi\cos\phi+\cos\psi\sin\phi\cos\vartheta & -\cos\psi\sin\vartheta\\ -\cos\psi\sin\phi-\sin\psi\cos\phi\cos\vartheta & \cos\psi\cos\phi-\sin\psi\sin\phi\cos\vartheta & \sin\psi\sin\vartheta\\ \cos\phi\sin\vartheta & \sin\phi\sin\vartheta & \cos\vartheta\end{bmatrix}$$

If angles are small, i.e., up to 10°- 15°, we can assume cos x =1, sin x = x, x*x =0 (with x in radians), therefore the dependence on the order of rotations is lost and we can write:

$$A_{312}(\phi,\vartheta,\psi)=\begin{bmatrix}1 & \phi & -\psi\\ -\phi & 1 & \vartheta\\ \psi & -\vartheta & 1\end{bmatrix}=A_{321}(\phi,\psi,\vartheta)=A_{213}(\psi,\vartheta,\phi)=\ldots.$$

It is reminded that in any case the correspondence between axis and angle must be kept. Matrix A can also be written as:

$$A=I-[angles\wedge]$$

If the first and last indexes are coincident, due to loss of importance of the order of rotations, we are always in the condition that the first and last rotations are almost around the same physical direction in space, so it is like having only two rotations:

$$A_{313}(\phi,\vartheta,\psi)=\begin{bmatrix}1 & \phi+\psi & 0\\ -\phi-\psi & 1 & \vartheta\\ 0 & -\vartheta & 1\end{bmatrix}$$

For small angles, quaternion can be simplified and directly connected to Euler angles when the three indexes are different. For the sequence 312 we can write:

$$\begin{cases} q_1 = \frac{1}{2}\vartheta \\ q_2 = \frac{1}{2}\psi \\ q_3 = \frac{1}{2}\phi \\ q_4 = 1 \end{cases}$$

Attitude parameterization adopting Euler angles has a clear physical meaning but is numerically time consuming. Moreover, the rule of consecutive rotations, although existing, is really hard to implement due to its complexity, therefore it will not be illustrated.

Attitude kinematics

Direction cosines

We now want to define how attitude parameters change with time, and how they are related to the angular velocity expressed in body frame.
We begin by considering the direction cosines matrix:

$$\frac{dA}{dt} = f(\omega_u, \omega_v, \omega_w)$$

If matrix A is known at time t, then it is possible to evaluate the matrix at time t+Δt adopting the rule of consecutive rotations:

$$A(t + \Delta t) = A'A(t)$$

where A' represents the rotation in the interval Δt. Express matrix A' as a function of Euler axis and angle in the interval Δt:

$$A' = I\cos\phi + (1 - \cos\phi)\underline{e}\underline{e}^T - \sin\phi[\underline{e}\wedge]$$

To define the derivative we must consider short time intervals, therefore φ will be small:

$$A' = I - \phi[\underline{e}\wedge]$$

Euler axis for small rotations will be coincident with the angular velocity vector direction, so that:

$$\phi = \omega\Delta t \qquad \underline{\omega} = \omega\underline{e}$$

$$\phi[\underline{e}\wedge] = \omega\Delta t \begin{bmatrix} 0 & -e_w & e_v \\ e_w & 0 & -e_u \\ -e_v & e_u & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\omega_w & \omega_v \\ \omega_w & 0 & -\omega_u \\ -\omega_v & \omega_u & 0 \end{bmatrix} \Delta t = [\underline{\omega}\wedge]\Delta t$$

Therefore we can write:

$$A' = I - [\underline{\omega}\wedge]\Delta t$$
$$A(t + \Delta t) = A(t) - \Delta t[\underline{\omega}\wedge]A(t)$$

Applying the definition of derivative, we have:

$$\frac{dA}{dt} = \lim_{\Delta t\to 0} \frac{A(t + \Delta t) - A(t)}{\Delta t} = \lim_{\Delta t\to 0} -\frac{\Delta t[\underline{\omega}\wedge]A(t)}{\Delta t} = -[\underline{\omega}\wedge]A(t)$$

Then, knowing ω we can integrate dA/dt in order to update matrix A. Since integration is done numerically, after some steps matrix A will no longer be orthogonal. It is therefore necessary to orthogonalize matrix A after a certain number of integration steps.

This operation is long and complex, if done exactly, so that approximate solutions to the orthogonalization are normally used. In this way, approximate solutions can be evaluated with a higher frequency (at every integration step) since the numerical process is fast enough. It can be demonstrated that, starting from an initial guess $A_0(t)$ of the direction cosines matrix, not orthogonal due to numerical errors, the recursive formula

$$A_{k+1}(t)= A_k(t)*3/2- A_k(t)*A_k^T(t)*A_k(t)/2$$

converges rapidly, with increasing k, to the exact value of A. In a first order approximation it is possible to adopt a single step iteration

$$A(t)= A_0(t)*3/2- A_0(t)*A_0^T(t)*A_0(t)/2$$

The kinematic equation of the form $\dot{A} = -[\underline{\omega}\wedge]A$ has the following properties

$$A[\underline{\omega}\wedge]A^T = [A\underline{\omega}\wedge]$$
$$\underline{\omega}\wedge\underline{x} = [\underline{\omega}\wedge]\underline{x}$$
$$tr([\underline{\omega}\wedge]A) = -\underline{\omega}^T(A - A^T)^V$$

where ^{V} denotes the inverse cross-product map, i.e., it extracts from a matrix the off-diagonal elements.

Quaternions

The procedure adopted to derive the kinematic relation for direction cosines can now be repeated to analyze the quaternion kinematics. We can assume that quaternion at time t + Δt is given by the composition of quaternion at time t and a quaternion representing the rotation in the interval Δt:

$$q(t+\Delta t) = \begin{bmatrix} q_4' & q_3' & -q_2' & q_1' \\ -q_3' & q_4' & q_1' & q_2' \\ q_2' & -q_1' & q_4' & q_3' \\ -q_1' & -q_2' & -q_3' & q_4' \end{bmatrix} q(t) \qquad \text{with} \qquad \begin{cases} q_1' = e_u \sin\dfrac{\phi}{2} \\ q_2' = e_v \sin\dfrac{\phi}{2} \\ q_3' = e_w \sin\dfrac{\phi}{2} \\ q_4' = \cos\dfrac{\phi}{2} \end{cases}$$

Then:

$$q(t+\Delta t) = \left\{ I\cos\frac{\phi}{2} + \begin{bmatrix} 0 & e_w & -e_v & e_u \\ -e_w & 0 & e_u & e_v \\ e_v & -e_u & 0 & e_w \\ -e_u & -e_v & -e_w & 0 \end{bmatrix} \sin\frac{\phi}{2} \right\} q(t)$$

For short intervals Δt we can approximate:

$$\phi = \omega\Delta t \qquad \cos\frac{\phi}{2} = 1 \qquad \sin\frac{\phi}{2} = \frac{\phi}{2} = \frac{\omega\Delta t}{2}$$

and evaluate e_u, e_v, e_w, as a function of $\underline{\omega} = \omega\ \underline{e}$ and therefore write:

$$q(t+\Delta t) = \left[I + \frac{1}{2}\Omega\Delta t\right] q(t)$$

where:

$$\Omega = \begin{bmatrix} 0 & \omega_w & -\omega_v & \omega_u \\ -\omega_w & 0 & \omega_u & \omega_v \\ \omega_v & -\omega_u & 0 & \omega_w \\ -\omega_u & -\omega_v & -\omega_w & 0 \end{bmatrix}$$

Taking the limit for t→0:

$$\frac{dq}{dt} = \lim_{\Delta t \to 0} \frac{q(t+\Delta t) - q(t)}{\Delta t} = \frac{1}{2}\Omega q(t)$$

An alternative way to derive the quaternion kinematics is to substitute

$$A = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_2 - q_3q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix}$$

into

$$\frac{dA}{dt}A^T = -\left[\underline{\omega}\wedge\right]$$

Which yields

$$\begin{aligned} \omega_1 &= 2(\dot{q}_1q_4 + \dot{q}_2q_3 - \dot{q}_3q_2 - \dot{q}_4q_1) \\ \omega_2 &= 2(\dot{q}_2q_4 + \dot{q}_3q_1 - \dot{q}_1q_3 - \dot{q}_4q_2) \\ \omega_3 &= 2(\dot{q}_3q_4 + \dot{q}_1q_2 - \dot{q}_2q_1 - \dot{q}_4q_3) \end{aligned}$$

Also, differentiating the constraint, we have

$$0 = 2(\dot{q}_1q_1 + \dot{q}_2q_2 + \dot{q}_3q_3 + \dot{q}_4q_4)$$

And rearranging gives

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2}\begin{pmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{pmatrix}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Therefore, also in this case, knowing the angular velocity we can integrate the derivative of the quaternion to propagate attitude. As for the direction cosines, there is a problem of normalization, but since q is a vector its normalization is trivial.

Gibbs vector

It is not possible to evaluate the derivative of the Euler axis/angle parameterization, due to the lack of rule for consecutive rotations. For the Gibbs vector this can instead be evaluated, with the usual procedure already adopted for quaternion and direction cosines:

$$g(t+\Delta t) = \frac{\underline{g}(t) + \underline{g}' - \underline{g}' \wedge \underline{g}(t)}{1 - \underline{g}(t) \cdot \underline{g}'}$$

where:

$$\underline{g}' = \underline{e}\tan\frac{\phi}{2} = \quad \xrightarrow{\text{small } \Delta t} \quad = \frac{1}{2}\underline{\omega}\Delta t$$

The derivative is then:

$$\frac{dg}{dt} = \frac{1}{2}\left[\underline{\omega} - \underline{\omega} \wedge \underline{g}(t) + \left(\underline{g}(t) \cdot \underline{\omega}\right)\underline{g}(t)\right]$$

Euler angles

Euler angles have no convenient rule for combining two consecutive rotations, however it is possible to calculate the derivatives of the angles for each sequence of rotations. Let consider the sequence 313 (φ,ϑ,ψ):

If we have variation of angle φ alone, then rotation will be about axis 3, so that we can write:

$$\underline{\omega} = \dot{\phi}\underline{3}$$

If we have variation of ϑ alone, then rotation will be about axis 1', so that:

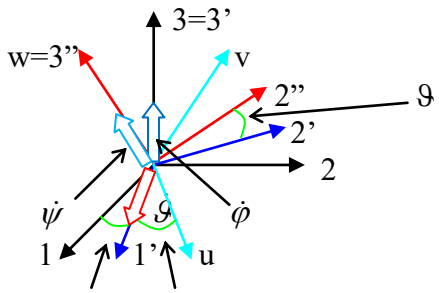
$$\underline{\omega} = \dot{\vartheta}\underline{1}'$$

Again, if we have variation of angle ψ alone, rotation will be about axis w, so that:

$$\underline{\omega} = \dot{\psi}\underline{w}$$

The three contributions are independent one from the other, therefore by simple combination of the three contributions we obtain:

$$\underline{\omega} = \dot{\phi}\underline{3} + \dot{\vartheta}\underline{1}' + \dot{\psi}\underline{w}$$



Taking the projections of $\underline{\omega}$ on axes u,v,w :

$$\begin{cases} \omega_u = \underline{\omega} \cdot \underline{u} = \dot{\phi}\underline{3} \cdot \underline{u} + \dot{\vartheta}\underline{1}' \cdot \underline{u} + \dot{\psi}\underline{w} \cdot \underline{u} = \dot{\phi}\underline{3} \cdot \underline{u} + \dot{\vartheta}\underline{1}' \cdot \underline{u} \\ \omega_v = \underline{\omega} \cdot \underline{v} = \dot{\phi}\underline{3} \cdot \underline{v} + \dot{\vartheta}\underline{1}' \cdot \underline{v} + \dot{\psi}\underline{w} \cdot \underline{v} = \dot{\phi}\underline{3} \cdot \underline{v} + \dot{\vartheta}\underline{1}' \cdot \underline{v} \\ \omega_w = \underline{\omega} \cdot \underline{w} = \dot{\phi}\underline{3} \cdot \underline{w} + \dot{\vartheta}\underline{1}' \cdot \underline{w} + \dot{\psi}\underline{w} \cdot \underline{w} \end{cases}$$

Inner products $\underline{3}\ \underline{u}$, $\underline{3}\ \underline{v}$, $\underline{3}\ \underline{w}$ represent the third column of matrix A_{313}, while inner products $\underline{1}'\underline{u}$, $\underline{1}'\underline{v}$, $\underline{1}'\underline{w}$ represent the first column of matrix A_{313} if we consider $\varphi = 0$, in fact the first column represent axis 1, that is coincident with axis 1' in case $\varphi = 0$. Finally, $\underline{w}\ \underline{w} = 1$ since $\underline{w}$ is a unit vector.

We then have:

$$\begin{cases} \omega_u = \dot{\phi}\, sin\,\vartheta\, sin\,\psi + \dot{\vartheta}\, cos\,\psi \\ \omega_v = \dot{\phi}\, sin\,\vartheta\, cos\,\psi - \dot{\vartheta}\, sin\,\psi \\ \omega_w = \dot{\phi}\, cos\,\vartheta + \dot{\psi} \end{cases}$$

from which the kinematic relations are obtained as:

$$\begin{cases} \dot{\phi} = \dfrac{(\omega_u\, sin\,\psi + \omega_v\, cos\,\psi)}{sin\,\vartheta} \\ \dot{\vartheta} = \omega_u\, cos\,\psi - \omega_v\, sin\,\psi \\ \dot{\psi} = \omega_w - (\omega_u\, sin\,\psi + \omega_v\, cos\,\psi)\dfrac{cos\,\vartheta}{sin\,\vartheta} \end{cases}$$

where angles φ,ϑ,ψ are taken at time t.
Note that an alternative derivation of the kinematics can be performed by substituting the parametrized matrix into the kinematic equations in DCM form, for example substituting

$$A_{313} = \begin{bmatrix} cos\,\psi\, cos\,\phi - sin\psi sin\phi\, cos\,\vartheta & cos\,\psi\, sin\phi + sin\psi\, cos\,\phi\, cos\,\vartheta & sin\psi sin\vartheta \\ -sin\psi\, cos\,\phi - cos\,\psi\, sin\phi\, cos\,\vartheta & -sin\psi sin\phi + cos\,\psi\, cos\,\phi\, cos\,\vartheta & cos\,\psi\, sin\vartheta \\ sin\phi sin\vartheta & -cos\,\phi\, sin\vartheta & cos\,\vartheta \end{bmatrix}$$

Into

$$\dot{A}_{B/N} A_{B/N}^T = -[\omega\,\wedge]$$

yields the same result.

For the other sequences of rotations, we obtain different sets of kinematic equations, adopting the same procedure and considering the composition of appropriate angles and axis of rotation. It is important to notice that in any case we have some singular conditions, for which the derivatives are

infinite, and these singular conditions occur for the same angle values that make singular the inverse transformation of the parameters.
Here follow the sets of kinematic equations for the derivatives of Euler angles for the 12 possible sequences of Euler angles, assuming that the first rotation corresponds to angle φ, the second rotation to angle ϑ and the third rotation to angle ψ. It is assumed that angular velocities ω_u,ω_v, ω_w are those of an arbitrary moving frame **M**=[**u:v:w**] with respect to an inertial frame

$$\text{sequence } 123 \begin{cases} \dot{\phi} = \dfrac{(\omega_u \cos\psi - \omega_v \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_v \cos\psi + \omega_u \sin\psi \\ \dot{\psi} = \omega_w - (\omega_u \cos\psi - \omega_v \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 132 \begin{cases} \dot{\phi} = \dfrac{(\omega_u \cos\psi + \omega_w \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_w \cos\psi - \omega_u \sin\psi \\ \dot{\psi} = \omega_v + (\omega_u \cos\psi + \omega_w \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 231 \begin{cases} \dot{\phi} = \dfrac{(\omega_v \cos\psi - \omega_w \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_w \cos\psi + \omega_v \sin\psi \\ \dot{\psi} = \omega_u - (\omega_v \cos\psi - \omega_w \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 213 \begin{cases} \dot{\phi} = \dfrac{(\omega_v \cos\psi + \omega_u \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_u \cos\psi - \omega_v \sin\psi \\ \dot{\psi} = \omega_w + (\omega_v \cos\psi + \omega_u \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 312 \begin{cases} \dot{\phi} = \dfrac{(\omega_w \cos\psi - \omega_u \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_u \cos\psi + \omega_w \sin\psi \\ \dot{\psi} = \omega_v - (\omega_w \cos\psi - \omega_u \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 321 \begin{cases} \dot{\phi} = \dfrac{(\omega_w \cos\psi + \omega_v \sin\psi)}{\cos\vartheta} \\ \dot{\vartheta} = \omega_v \cos\psi - \omega_w \sin\psi \\ \dot{\psi} = \omega_u + (\omega_w \cos\psi + \omega_v \sin\psi)\dfrac{\sin\vartheta}{\cos\vartheta} \end{cases}$$

$$\text{sequence } 121 \begin{cases} \dot{\phi} = \dfrac{(\omega_v \sin\psi + \omega_w \cos\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_v \cos\psi - \omega_w \sin\psi \\ \dot{\psi} = \omega_u - (\omega_v \sin\psi + \omega_w \cos\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

$$\text{sequence } 131 \begin{cases} \dot{\phi} = \dfrac{-(\omega_v \cos\psi - \omega_w \sin\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_w \cos\psi + \omega_v \sin\psi \\ \dot{\psi} = \omega_u + (\omega_v \cos\psi - \omega_w \sin\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

$$\text{sequence } 212 \begin{cases} \dot{\phi} = \dfrac{-(\omega_w \cos\psi - \omega_u \sin\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_u \cos\psi + \omega_w \sin\psi \\ \dot{\psi} = \omega_v + (\omega_w \cos\psi - \omega_u \sin\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

$$\text{sequence } 232 \begin{cases} \dot{\phi} = \dfrac{(\omega_w \sin\psi + \omega_u \cos\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_w \cos\psi - \omega_u \sin\psi \\ \dot{\psi} = \omega_v - (\omega_w \sin\psi + \omega_u \cos\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

$$\text{sequence } 313 \begin{cases} \dot{\phi} = \dfrac{(\omega_u \sin\psi + \omega_v \cos\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_u \cos\psi - \omega_v \sin\psi \\ \dot{\psi} = \omega_w - (\omega_u \sin\psi + \omega_v \cos\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

$$\text{sequence } 323 \begin{cases} \dot{\phi} = \dfrac{-(\omega_u \cos\psi - \omega_v \sin\psi)}{\sin\vartheta} \\ \dot{\vartheta} = \omega_v \cos\psi + \omega_u \sin\psi \\ \dot{\psi} = \omega_w + (\omega_u \cos\psi - \omega_v \sin\psi)\dfrac{\cos\vartheta}{\sin\vartheta} \end{cases}$$

Euler angles are integrable, this meaning that:

$$\phi = \phi_0 + \int_0^t \dot{\phi} dt$$
$$\vartheta = \vartheta_0 + \int_0^t \dot{\vartheta} dt$$
$$\psi = \psi_0 + \int_0^t \dot{\psi} dt$$

This is not true for angular velocities, that when integrated do not provide angles, since angular velocity is not a fixed vector. To verify this statement, we recall the integrability condition:

$$dz = Adx + Bdy \quad \rightarrow \quad \frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}$$

$$\omega_u = \dot{\phi}\, sin\,\vartheta\, sin\,\psi + \dot{\vartheta}\, cos\,\psi \quad \rightarrow \quad \omega_u dt = sin\,\vartheta\, sin\,\psi\, d\phi + cos\,\psi\, d\vartheta = Ad\phi + Bd\vartheta$$

$$\left.\begin{array}{l} \frac{\partial A}{\partial \vartheta} = sin\,\psi\, cos\,\vartheta \\ \frac{\partial B}{\partial \phi} = 0 \end{array}\right\} \neq \qquad \text{not integrable}$$

The same integrability condition verified for Euler angles:

$$\dot{\phi} = \omega_u \frac{sin\,\psi}{sin\,\vartheta} + \omega_v \frac{cos\,\psi}{sin\,\vartheta} \quad \rightarrow \quad d\phi = \frac{sin\,\psi}{sin\,\vartheta}\omega_u dt + \frac{cos\,\psi}{sin\,\vartheta}\omega_v dt = A\omega_u dt + B\omega_v dt$$

$$\frac{\partial A}{\partial(\omega_v \mathrm{dt})} = \frac{\partial\left(\frac{\sin\psi}{\sin\vartheta}\right)}{\partial(\omega_v \mathrm{dt})} = \frac{1}{\sin^2\vartheta}\left(\cos\psi \frac{\partial\psi}{\partial(\omega_v \mathrm{dt})}\sin\vartheta - \sin\psi\cos\vartheta \frac{\partial\vartheta}{\partial(\omega_v \mathrm{dt})}\right)$$

$$\frac{\partial\psi}{\partial(\omega_v \mathrm{dt})} = \frac{\partial\dot{\psi}}{\partial\omega_v} = -\frac{\cos\psi\cos\vartheta}{\sin\vartheta}$$

$$\frac{\partial\vartheta}{\partial(\omega_v \mathrm{dt})} = \frac{\partial\dot{\vartheta}}{\partial\omega_v} = -\sin\psi$$

therefore:

$$\frac{\partial A}{\partial(\omega_v \mathrm{dt})} = \frac{1}{\sin^2\vartheta}(\sin^2\psi\cos\vartheta - \cos^2\psi\sin\vartheta)$$

In similar way we can evaluate $\frac{\partial B}{\partial(\omega_u dt)}$ and check that it has the same expression, this demonstrates integrability of Euler angles.

Euler equations

Euler equations are the analytic form adopted to describe the attitude motion of a rigid body. We start by writing the fundamental equation:

$$\frac{d\underline{h}}{dt} = \underline{M}$$

In body axes the angular momentum vector is:

$$\underline{h} = \begin{Bmatrix} I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z \\ I_{yx}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z \\ I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z \end{Bmatrix}$$

Writing the fundamental equation in body axes we have:

$$\dot{\underline{h}} + \underline{\omega} \wedge \underline{h} = \underline{M}$$

and expanding the expression of the angular momentum we obtain a system of first order differential equations, nonlinear and strongly coupled:

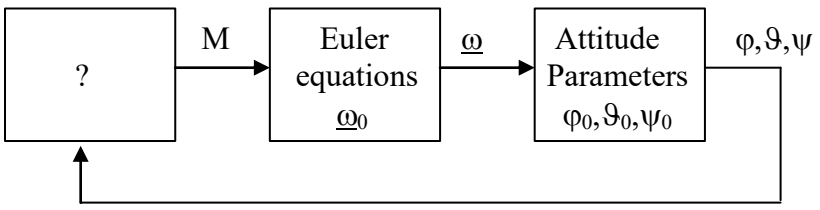
$$\begin{cases} I_{xx}\dot{\omega}_x + I_{xy}\dot{\omega}_y + I_{xz}\dot{\omega}_z - I_{xy}\omega_x\omega_z - I_{yy}\omega_y\omega_z - I_{yz}\omega_z^2 + I_{xz}\omega_x\omega_y + I_{zz}\omega_z\omega_y + I_{yz}\omega_y^2 = M_x \\ I_{yx}\dot{\omega}_x + I_{yy}\dot{\omega}_y + I_{yz}\dot{\omega}_z - I_{yz}\omega_x\omega_y - I_{zz}\omega_x\omega_z - I_{xz}\omega_x^2 + I_{xx}\omega_x\omega_z + I_{xy}\omega_z\omega_y + I_{xz}\omega_z^2 = M_y \\ I_{zx}\dot{\omega}_x + I_{zy}\dot{\omega}_y + I_{zz}\dot{\omega}_z - I_{xx}\omega_x\omega_y - I_{xz}\omega_y\omega_z - I_{xy}\omega_y^2 + I_{yy}\omega_x\omega_y + I_{yz}\omega_z\omega_x + I_{xy}\omega_x^2 = M_z \end{cases}$$

If we refer the dynamics to the principal inertia axes, we can simplify the system dynamics since inertia products will vanish:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = M_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = M_y \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = M_z \end{cases}$$

In these equations, if the coupling term vanishes then the rotation around one axis will be driven only by the torque around the same axis.

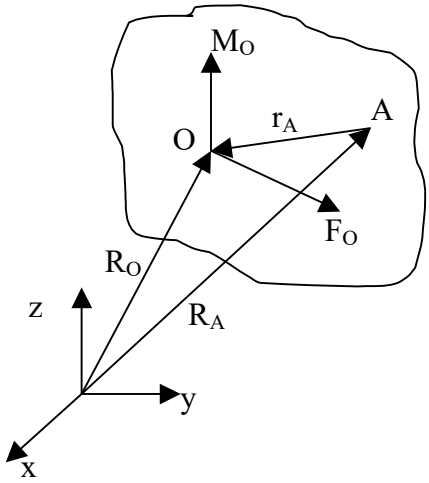
Rotational dynamics of a satellite is then described by joining Euler equations and an attitude parameterization.



On a satellite we have torques influencing directly the dynamics (Euler equations) and induce changes in the components of angular velocity. Adopting one attitude parameterization we can, knowing the

angular velocity, calculate the attitude parameters that indicate the orientation of the satellite in space and that allow to evaluate the position-dependent torques. In Euler equations it is therefore required to assign initial conditions for angular velocity, while in attitude parameterization we must assign the initial satellite attitude.

If the reference system is not centered in the system center of mass, in deriving the rotational dynamics equations we must also consider the linear motion of the origin of the reference frame. With reference to the following figure, assuming O is the center of mass and A the origin of the reference system, assuming that we define with M_o and F_o the torque and force referred to the system center of mass, we have



$$\frac{d\underline{h_O}}{\mathrm{dt}} = \underline{M_O} \quad ; \quad \underline{F_O} = \mathrm{m}\,\underline{a_O}$$

$$\underline{h_A} = \underline{h_O} + \mathrm{m}\,\underline{r_A} \wedge \underline{\dot{r}_A} \quad ; \quad \underline{M_A} = \underline{M_O} + \underline{r_A} \wedge \underline{F_O} \quad ; \quad \underline{a_A} = \underline{a_O} - \underline{\ddot{r}_A}$$

that is

$$\underline{h_O} = \underline{h_A} - \mathrm{m}\,\underline{r_A} \wedge \underline{\dot{r}_A} \quad ; \quad \underline{M_O} = \underline{M_A} - \underline{r_A} \wedge \underline{F_O} \quad ; \quad \underline{a_O} = \underline{a_A} + \underline{\ddot{r}_A}$$

Substituting these expressions in the equation of the angular momentum derivative, we have

$$\underline{M_O} = \frac{d\underline{h_O}}{\mathrm{dt}} \quad \rightarrow \quad \underline{M_A} - \underline{r_A} \wedge \underline{F_O} = \frac{d}{\mathrm{dt}}\left(\underline{h_A} - \mathrm{m}\,\underline{r_A} \wedge \underline{\dot{r}_A}\right)$$

and, since $\underline{F_O} = \mathrm{m}\,\underline{a_O}$

$$\underline{M_A} - \underline{r_A} \wedge \mathrm{m}\,\left(\underline{a_A} + \underline{\ddot{r}_A}\right) = \frac{d}{\mathrm{dt}}\left(\underline{h_A} - \mathrm{m}\,\underline{r_A} \wedge \underline{\dot{r}_A}\right)$$

For a rigid body we have $\underline{\dot{r}_A} = 0$, $\underline{\ddot{r}_A} = 0$, therefore $\underline{h_O} = \underline{h_A}$, $\underline{a_O} = \underline{a_A}$ and

$$\underline{M_A} = \frac{d\underline{h_A}}{\mathrm{dt}} + \underline{r_A} \wedge \mathrm{m}\,\underline{a_A} = \frac{d\underline{h_O}}{\mathrm{dt}} + \underline{S_A} \wedge \underline{a_A}$$

where $\underline{S_A}$ stands for the static moment of the rigid body with respect to point A. If the reference axes are parallel to the principal inertia axes, then we can write

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y + (S_y a_z - S_z a_y) = M_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z + (S_z a_x - S_x a_z) = M_y \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x + (S_x a_y - S_y a_x) = M_z \end{cases}$$

It is pointed out that the same result would have been obtained by considering the equation of the derivative of the angular momentum vector, expressed in the general form $\underline{\dot{h}}_o = -\underline{v}_o \wedge S_o + I\underline{\dot{\omega}}$, with no additional assumptions on the reference frame (fixed or with origin in the system center of mass).

Even in the simplest case, assuming as reference system the principal inertia system, Euler equations are difficult to integrate and therefore numerical integrations is adopted. In some simplified cases we can obtain an analytical solution of the equations, to be used to verify the numerical integration results.

Torque-free motion of a simple spin satellite

Assume the satellite is axial-symmetric. Having two equal principal inertia moments, one equation is decoupled from the other two.

Assume that $I_x = I_y$:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = M_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_z\omega_x = M_y \\ I_z\dot{\omega}_z = M_z \end{cases}$$

Assume that no external torque is applied, then we can study the torque-free motion knowing the initial conditions:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_z\omega_x = 0 \\ I_z\dot{\omega}_z = 0 \end{cases}$$

The last equation leads immediately to:

$$\omega_z = \text{const} = \omega_{z0}$$

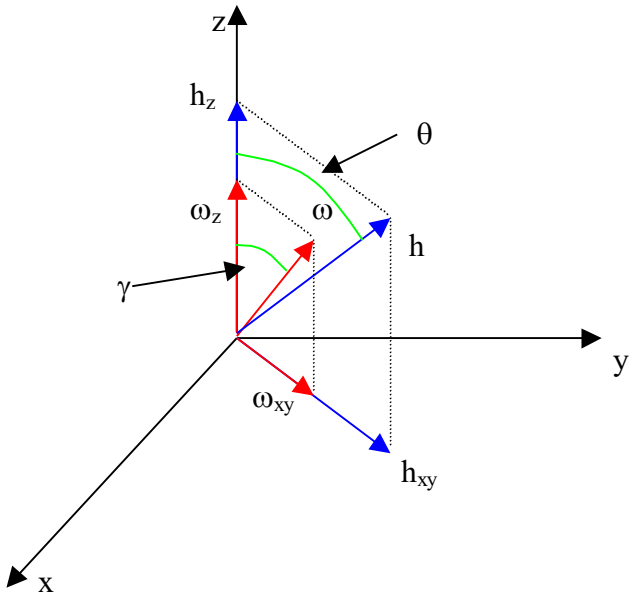
Furthermore, since no torque is applied, angular momentum and kinetic energy are constant. Combining the first two equations, the first multiplied by ω_x and the second multiplied by ω_y, we obtain:

$$\begin{aligned} &I_x\dot{\omega}_x\omega_x + I_x\dot{\omega}_y\omega_y = 0 \\ &\dot{\omega}_x\omega_x + \dot{\omega}_y\omega_y = 0 \\ &\frac{d}{dt}\left(\omega_x^2 + \omega_y^2\right) = 0 \\ &\left(\omega_x^2 + \omega_y^2\right) = \text{const} \end{aligned}$$

The magnitude of $\underline{\omega}$ is then constant. The same result would have been obtained by considering the constant kinetic energy. To have the magnitude constant, both the x and y components of the angular velocity must be constant or the projection of the angular velocity on the x-y plane is on a circle. In fact we know that the angular momentum is constant, therefore:

$$\begin{aligned} &\underline{h} = \text{const} \\ &\underline{h} = I_x\left(\omega_x\underline{i} + \omega_y\underline{j}\right) + I_z\omega_z\underline{k} = h_{xy} + h_z \end{aligned}$$

h_z is constant, while h_{xy} is parallel to ω_{xy} (component on x-y plane) but not constant. Therefore vectors $\underline{h}$ and $\underline{\omega}$ describe cones with axis coincident with axis z if we look at the motion from a body axes reference. If we look at the motion from an inertial reference, $\underline{h}$ will be fixed and $\underline{\omega}$ and the unit vector z describe cones with axes coincident with vector $\underline{h}$. Furthermore, $\underline{\omega}$ and $\underline{h}$ are constantly on a plane intersecting axis z. In case the only non-zero component of the angular velocity is the z component, then motion is referred to as simple spin satellite motion.



We can now solve the first two equations. Define:

$$\lambda = \frac{(I_z - I_x)\omega_z}{I_x}$$

as a characteristic constant of the rigid body, the equations become:

$$\begin{cases} \dot{\omega}_x + \lambda\omega_y = 0 \\ \dot{\omega}_y - \lambda\omega_x = 0 \end{cases}$$

Deriving the first and substituting the value of $\dot{\omega}_y$ from the second equation, we have:

$$\ddot{\omega}_x + \lambda^2\omega_x = 0$$

This is the typical equation of an harmonic system that has the following solution:

$$\omega_x = a\cos(\lambda t) + b\sin(\lambda t)$$
$$\dot{\omega}_x = -a\lambda\sin(\lambda t) + b\lambda\cos(\lambda t)$$

with initial conditions:

$$\omega_x(0) = \omega_{x0}$$
$$\dot{\omega}_x(0) = \dot{\omega}_{x0}$$
$$\omega_y(0) = \omega_{y0}$$
$$\dot{\omega}_y(0) = \dot{\omega}_{y0}$$

We then have:

$$\omega_x = \omega_{x0}\cos(\lambda t) + \frac{\dot{\omega}_{x0}}{\lambda}\sin(\lambda t)$$

From the second equation we obtain:

$$\omega_y = -\frac{\dot{\omega}_x}{\lambda}$$

so that:

$$\omega_x = \omega_{x0}\cos(\lambda t) - \omega_{y0}\sin(\lambda t)$$
$$\omega_y = \omega_{x0}\sin(\lambda t) + \omega_{y0}\cos(\lambda t)$$

In order to demonstrate that ω_{xy} describes a circle, we can write it as a complex number:

$$\omega_{xy} = \omega_x + i\omega_y = \omega_{x0}[\cos(\lambda t) + i\sin(\lambda t)] - \omega_{y0}[\sin(\lambda t) - i\cos(\lambda t)]$$

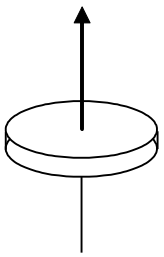
or:

$$\omega_{xy} = \omega_{x0}[cos(\lambda t) + i\, sin(\lambda t)] + i\omega_{y0}[cos(\lambda t) + i\, sin(\lambda t)]$$
$$\omega_{xy} = (\omega_{x0} + i\omega_{y0})[cos(\lambda t) + i\, sin(\lambda t)]$$

that is a harmonic motion on a plane, with magnitude given by the initial values of $\underline{\omega}$ and constant angular velocity equal to λ.

At this point we must analyze the case in which I_z is the maximum inertia moment and the case in which I_z is the minimum inertia moment:

I_z maximum $\lambda > 0$ I_z minimum $\lambda < 0$





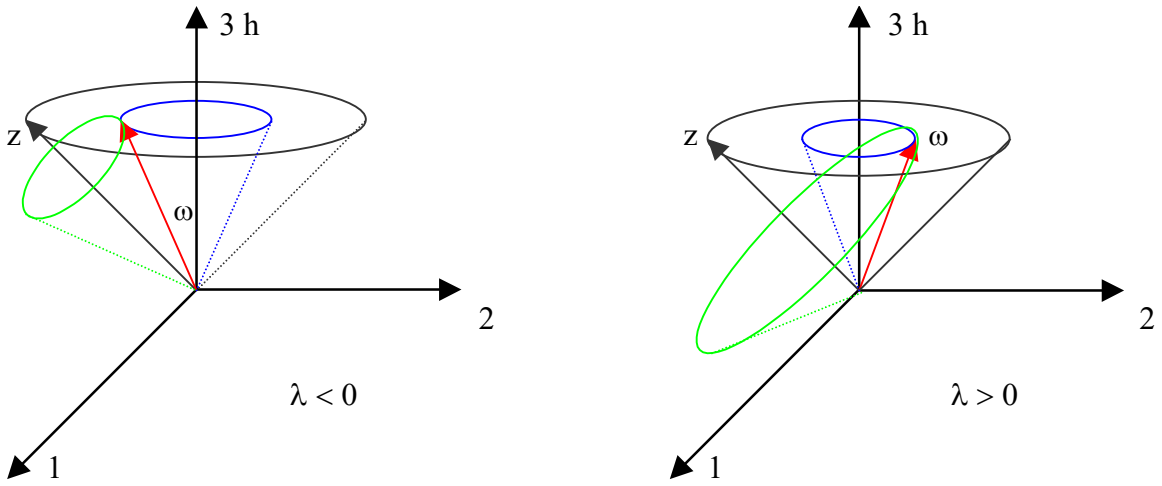
Now we can look at the angles that $\underline{h}$ and $\underline{\omega}$ form with the z axis:

$$tan\,\gamma = \frac{\omega_{xy}}{\omega_z}$$
$$tan\,\vartheta = \frac{h_{xy}}{h_z} = \frac{I_x\omega_{xy}}{I_z\omega_z} = \frac{I_x}{I_z}tan\,\gamma$$

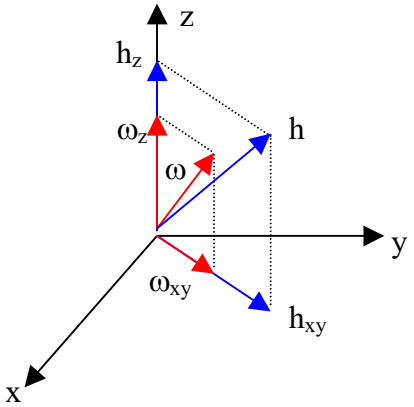
If I_z is the maximum inertia moment, then $\lambda > 0$ and $\vartheta < \gamma$, while if I_z is the minimum inertia moment, $\lambda < 0$ and $\vartheta > \gamma$.

So far we have seen the motion in the body axes frame, and analyzed the motion of the angular momentum vector around the z body axis.
In reality vector $\underline{h}$ is fixed in an inertial frame, so if we allow one of the inertial frame axis to be coincident with $\underline{h}$ we will see the body axis rotating around it, along with the angular velocity vector.

The z axis draws a cone with axis coincident with h, with aperture ϑ, while the angular velocity draws a cone with axis coincident with z, with aperture γ. Moreover we recall that ω draws also a cone with axis coincident with h. We then have two cones, one fixed (ω around h) called "space cone", and one that revolves on it (ω around z), called "body cone". The two cones revolve one outside the other if λ is negative, while they revolve one inside the other if λ is positive.

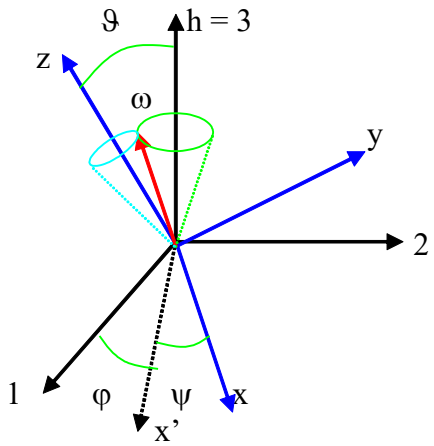


We can now apply one attitude representation to the axial symmetric satellite:



$$\lambda = \frac{(I_z - I_x)\omega_z}{I_x}$$

Adopting the Euler angles with sequence 313, we can represent the inertial reference with the third axis coincident with h.



$$\begin{cases} \omega_x = \dot{\phi}\, sin\,\vartheta\, sin\,\psi \\ \omega_y = \dot{\phi}\, sin\,\vartheta\, cos\,\psi \\ \omega_z = \dot{\phi}\, cos\,\vartheta + \dot{\psi} \end{cases}$$

We notice that ϑ is constant due to the satellite symmetry, and axis z draws a cone of amplitude ϑ around axis h.

Taking time derivatives, we have:

$$\begin{cases} \dot{\omega}_x = \ddot{\phi}\, sin\,\vartheta\, sin\,\psi + \dot{\phi}\dot{\psi}\, cos\,\psi\, sin\,\vartheta \\ \dot{\omega}_y = \ddot{\phi}\, sin\,\vartheta\, cos\,\psi - \dot{\phi}\dot{\psi}\, sin\,\psi\, sin\,\vartheta \\ \dot{\omega}_z = \ddot{\phi}\, cos\,\vartheta + \ddot{\psi} \end{cases}$$

Replacing in the Euler equations we represent the dynamics as a function of φ,ϑ,ψ and recalling that:

$$\dot{\omega}_z = 0$$
$$\omega_x\dot{\omega}_x + \omega_y\dot{\omega}_y = 0$$

from the second equation we obtain:

$$\ddot{\phi}\dot{\phi}\, sin^2\,\vartheta\, sin^2\,\psi + \dot{\phi}^2\dot{\psi}\, sin^2\,\vartheta\, sin\,\psi\, cos\,\psi + \ddot{\phi}\dot{\phi}\, sin^2\,\vartheta\, cos^2\,\psi - \dot{\phi}^2\dot{\psi}\, sin^2\,\vartheta\, sin\,\psi\, cos\,\psi =$$
$$\ddot{\phi}\dot{\phi}\, sin^2\,\vartheta\, (sin^2\,\psi + cos^2\,\psi) = \ddot{\phi}\dot{\phi}\, sin^2\,\vartheta = 0$$

Since ϑ must be constant the following must hold:

$$\ddot{\phi}\dot{\phi} = 0 \quad \Rightarrow \quad \dot{\phi} = cos\,t$$

From the first equation we now have:

$$\ddot{\psi} = 0 \quad \Rightarrow \quad \dot{\psi} = cos\,t$$

We now have the following expressions:

$$\begin{cases} \psi = \psi_0 + \dot{\psi}_0 t \\ \phi = \phi_0 + \dot{\phi}_0 t \\ \vartheta = \vartheta_0 \end{cases}$$

We must now calculate the initial values of the derivatives of φ e ψ, and to do so we make use of the first two Euler equations:

$$\ddot{\phi}\sin\vartheta\sin\psi + \dot{\phi}\dot{\psi}\cos\psi\sin\vartheta + \lambda\dot{\phi}\sin\vartheta\cos\psi = 0$$
$$\ddot{\phi}\sin\vartheta\cos\psi - \dot{\phi}\dot{\psi}\sin\psi\sin\vartheta - \lambda\dot{\phi}\sin\vartheta\sin\psi = 0$$

Since the second time derivative of φ is zero we have:

$$\dot{\phi} = \frac{\omega_z + \lambda}{\cos\vartheta} \qquad \dot{\psi} = -\lambda \Rightarrow \begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = 0 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = 0 \end{cases}$$

In our case $I_z < I_x$ and therefore the time derivative of ψ is positive.

Solution of Euler equations in the phase plane $(\dot{\omega}, \omega)$

Reconsider the Euler equations:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = 0 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = 0 \end{cases}$$

If we assume that the external torques are zero, then angular momentum and kinetic energy will be constant:

$$2\mathrm{T} = \mathrm{const}$$
$$h^2 = \mathrm{const}$$

We now look for a method to decouple and solve the three equations, making use of the conservation laws just mentioned. Starting from the first equation, taking its time derivative, we have:

$$I_x\ddot{\omega}_x + (I_z - I_y)\dot{\omega}_z\omega_y + (I_z - I_y)\omega_z\dot{\omega}_y = 0$$

From the second and third equations we can compute the time derivative of the angular velocity components, and substitute it in the above equation:

$$I_x\ddot{\omega}_x + (I_z - I_y)\left(\frac{I_x - I_y}{I_z}\right)\omega_y^2\omega_x + (I_z - I_y)\left(\frac{I_z - I_x}{I_y}\right)\omega_z^2\omega_x = 0$$

Now we evaluate:

$$h^2 - 2TI_z = I_x^2\omega_x^2 + I_y^2\omega_y^2 - I_xI_z\omega_x^2 - I_yI_z\omega_y^2$$

so that:

$$\omega_y^2(I_y - I_z)I_y = h^2 - 2TI_z - \omega_x^2(I_x - I_z)I_x$$
$$\omega_y^2(I_y - I_z) = \frac{h^2 - 2TI_z - \omega_x^2(I_x - I_z)I_x}{I_y}$$

Substituting in the previous equation, we get to:

$$I_x\ddot{\omega}_x + \left(\frac{I_x - I_y}{I_z}\right)\frac{\left(2TI_z - h^2\right)}{I_y}\omega_x + \frac{\left(I_x - I_y\right)}{I_z}\frac{(I_x - I_z)}{I_y}\omega_x^3 + \left(I_z - I_y\right)\left(\frac{I_z - I_x}{I_y}\right)\omega_z^2\omega_x = 0$$

Now evaluate:

$$h^2 - 2TI_y = I_z^2\omega_z^2 + I_x^2\omega_x^2 - I_yI_z\omega_z^2 - I_xI_y\omega_x^2$$

so that:

$$\omega_z^2\left(I_z - I_y\right)I_z = h^2 - 2TI_y + \omega_x^2\left(I_y - I_x\right)I_x$$

$$\omega_z^2\left(I_z - I_y\right) = \frac{h^2 - 2TI_y + \omega_x^2\left(I_y - I_x\right)I_x}{I_z}$$

Substituting in the previous equation, we get to:

$$I_x\ddot{\omega}_x + \left(\frac{I_x - I_y}{I_z}\right)\frac{\left(2TI_z - h^2\right)}{I_y}\omega_x + \frac{\left(I_x - I_y\right)}{I_z}\frac{(I_x - I_z)}{I_y}\omega_x^3 + \left(\frac{I_z - I_x}{I_y}\right)\frac{\left(h^2 - 2TI_y\right)}{I_z}\omega_x + \frac{(I_z - I_x)}{I_y}\frac{\left(I_y - I_x\right)}{I_z}\omega_x^3 = 0$$

or:

$$\ddot{\omega}_x + \left[\frac{\left(I_y - I_x\right)\left(h^2 - 2TI_z\right) + (I_z - I_x)\left(h^2 - 2TI_y\right)}{I_xI_yI_z}\right]\omega_x + \left[\frac{2(I_z - I_x)\left(I_y - I_x\right)I_x}{I_xI_yI_z}\right]\omega_x^3 = 0$$

The other two equations are obtained by following the same procedure, simply with a permutation of all the operation indexes:

$$\ddot{\omega}_y + \left[\frac{\left(I_z - I_y\right)\left(h^2 - 2TI_x\right) + \left(I_x - I_y\right)\left(h^2 - 2TI_z\right)}{I_xI_yI_z}\right]\omega_y + \left[\frac{2\left(I_z - I_y\right)\left(I_x - I_y\right)I_y}{I_xI_yI_z}\right]\omega_y^3 = 0$$

$$\ddot{\omega}_z + \left[\frac{(I_x - I_z)\left(h^2 - 2TI_y\right) + \left(I_y - I_z\right)\left(h^2 - 2TI_x\right)}{I_xI_yI_z}\right]\omega_z + \left[\frac{2(I_x - I_z)\left(I_y - I_z\right)I_z}{I_xI_yI_z}\right]\omega_z^3 = 0$$

The three equations all have the same structure:

$$\ddot{\omega} + P\omega + Q\omega^3 = 0$$

Integrating the equation we obtain:

$$\dot{\omega}^2 + \mathrm{P}\omega^2 + \frac{1}{2}\mathrm{Q}\omega^4 = K$$

that, for the x equation, becomes:

$$\dot{\omega}_x^2 + P_x\omega_x^2 + \frac{1}{2}Q_x\omega_x^4 = K_x$$

$$\dot{\omega}_x^2 + \omega_x^2\left(P_x + \frac{1}{2}Q_x\omega_x^2\right) = K_x$$

The last expression can be considered as a conic section in the phase plane $(\dot{\omega}, \omega)$. We can evaluate the sign of each coefficient.

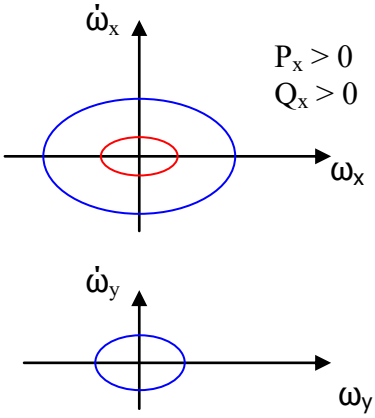
Assuming that $I_z > I_y > I_x$, we have:

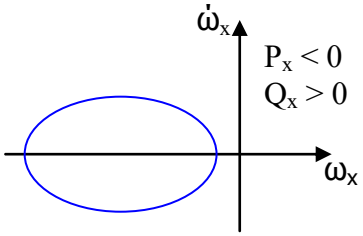
$$I_x < \frac{h^2}{2T} < I_z$$

The coefficients have therefore the following signs:

P_x ?	$P_y > 0$	P_z ?
$Q_x > 0$	$Q_y < 0$	$Q_z > 0$

Due to the structure of the coefficients, if the angular velocity component is large enough the Q term will predominate over the P term, while if the angular velocity component is small enough the P term will predominate over the Q term. Then, in the y phase plane if the angular velocity is large the conic section will be a hyperbola, if the angular velocity is small the conic section will be an ellipse. In the x and z phase planes the conic sections are ellipses for large velocities, for small velocities the type of conic section is undefined. This is not really important, since in any case, even for divergent phase plane trace (hyperbola) it is clear that as the angular velocity grows the trace will change into an ellipse. Notice that in the y phase plane (intermediate inertia phase plane), should the velocity grow, we would have a diverging motion. This is not possible due to the conservation of kinetic energy, therefore the conclusion is that in the y phase plane the solution must be such that the angular velocity is small enough to prevent the trace from being a hyperbola.





Attitude stability

We want to understand the stability of the equilibrium points of the Euler equations, as naturally stable equilibria can be exploited for passive stability. Firstly, we need to define the notions of stability we will used in this course. Note that in the following stability definitions and analysis we determine the stability in the vicinity of equilibrium points. For linear systems there is only ever one equilibrium point. Whereas for nonlinear systems there are multiple equilibria. For an autonomous nonlinear dynamical system.

$$\dot{\underline{x}} = f(\underline{x}), \;\; \underline{x}(0) = \underline{x}_0$$

An equilibrium point $\underline{x}_e$is defined as $0 = f(\underline{x}_e)$.

Stability definitions

Consider an autonomous nonlinear dynamical system

$$\dot{\underline{x}} = f(\underline{x}), \underline{x}(0) = \underline{x}_0$$

defined on an open set containing the origin, and *f* is continuous on this open set. Then an equilibrium point $x_e$is said to be:

1. **Lyapunov stable**, if, for every $\varepsilon > 0$, there exists a $\partial > 0$such that, if $\|x(0) - x_e\| < \partial$, then for every $t > 0$ we have $\|x(t) - x_e\| < \varepsilon$.
2. The equilibrium of the above system is said to be **asymptotically stable** if it is Lyapunov stable and if $\|x(t) - x_e\| \to 0$as $t \to \infty$

These definitions relate to local stability since $\|x(0) - x_e\| < \partial$ means that it is only stable in a region of the equilibrium point. If ∂ is unbounded then the definitions describe global stability.

First recall how we can deduce the stability for a simple1-DOF linear systems of the form:

$m\ddot{x} + c\dot{x} + kx = 0$

It is possible to use the ansatz solution $x = x_0 e^{\lambda t}$ to determine the stability. Then the characteristic equation is

$$m\lambda^2 + c\lambda + k = 0$$

Which is solved as

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

It is possible to use the ansatz solution $x = x_0 e^{\lambda t}$ to determine the stability. One immediately sees that if $Re(\lambda) < 0$ then the system is asymptotically stable. If $Re(\lambda) = 0$ then it is marginally stable and for $Re(\lambda) > 0$ it is unstable. $\lambda$is known as the eigenvalue which completely determines the stability of the system. In general, for a time-invariant linear system of the form $\dot{\underline{x}} = A\underline{x}$ where *A* is a *n* x *n* matrix, the stability of the equilibrium point $\underline{x} = \underline{0}$ can be determined by substituting in the Ansazt solution $\underline{x} = \underline{x}_0 e^{\lambda t}$. This yields:

$$e^{\lambda t}\lambda \underline{x}_0 = e^{\lambda t} A \underline{x}_0$$

This can be re-written as:

$$(A - \lambda I)\underline{x}_0 = 0$$

then multiplying both sides by the inverse of $(A - \lambda I)$ then we have

$$(A - \lambda I)^{-1}(A - \lambda I)\underline{x}_0 = \underline{x}_0 = 0$$

Since $\underline{x}_0 \neq 0$ in general then the above cannot be true. Therefore, the solution to $(A - \lambda I)\underline{x}_0 = 0$ only exists when this inverse does not exist. Equivalently, this condition can be expressed as:

$$det(A - \lambda I) = 0$$

Solving the above expression yields the eigenvalues of the matrix A which in turn indicates the stability of the Linear system.

However, the Euler equations are nonlinear so in order to get information about stability of the equilibrium points we must first linearize about its equilibrium configurations.

To first linearize a nonlinear system of the form $\dot{\underline{x}} = f(\underline{x}), \underline{x}(0) = \underline{x}_0$ we define $\underline{x} = \underline{x}_e + \partial\underline{x}$ i.e. $\underline{x}$ is displaced infinitesimally from the equilibrium point and we observe how $\partial\underline{x}$ behaves. Since the displacement is infinitesimal, we can assume $\partial x_i \partial x_j \approx 0$. In this way our resulting system is linear and the stability can be deduced by considering the eigenvalues. For example, to linearize the Euler equations we simply substitute in $\omega_x = \overline{\omega}_x + \partial\omega_x,\ \omega_y = \overline{\omega}_y + \partial\omega_y,\ \omega_z = \overline{\omega}_z + \partial\omega_z$and cancel all the higher order terms e.g. $\partial\omega_x \partial\omega_y \approx 0$. The following theorem the preservation of the stability properties of equilibrium points after linearization.

Lyapunov's First Stability Theorem

1) If the equilibrium point $\partial\underline{x} = 0$ of the linearized system $\partial\dot{\underline{x}} = A\partial\underline{x}$ is asymptotically stable, then the equilibrium point $\underline{x}_e$ of the original nonlinear system, is also asymptotically stable.
2) If the equilibrium point $\partial\underline{x} = 0$ of the linearized system $\partial\dot{\underline{x}} = A\partial\underline{x}$ is unstable, then the equilibrium point $\underline{x}_e$ of the original nonlinear system is also unstable.

The main implication of this theorem is that marginal stability of the linearized system does not guarantee marginal stability of the equilibrium point of the original nonlinear system. However, in the case of conserved systems i.e. with no dissipative forces or torques then marginal stability of the linearized system implies marginal stability of the nonlinear system.

At this stage we can discuss attitude stability with time, starting by the analysis of Euler equations in case of torque-free motion:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = 0 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = 0 \end{cases}$$

There is a distinction between torque-free stability and forced-motion stability. The latter depends on the type of torque applied, rarely known in a suitable analytical form, therefore we will mostly concentrate on torque-free stability. To analyze stability the first step consists in evaluating an equilibrium configuration. This is obtained by imposing zero time derivatives of the system states and then solving the corresponding system of equations:

$$\begin{cases}(I_z - I_y)\overline{\omega}_z\overline{\omega}_y = 0\\ (I_x - I_z)\overline{\omega}_x\overline{\omega}_z = 0\\ (I_y - I_x)\overline{\omega}_y\overline{\omega}_x = 0\end{cases}$$

The solution is found for one single angular velocity component different from zero. This condition is known as simple spin satellite. Assume, without loss of generality, that the non-zero component is the z component:

$$\overline{\omega}_z \neq 0$$
$$\overline{\omega}_x = \overline{\omega}_y = 0$$

Attitude stability of simple spin satellites

To analyze the stability of this condition, we must first write the linearized equations of motion, considering small perturbations around the equilibrium position:

$$\omega_z^{'} = (\overline{\omega}_z + \omega_z)$$
$$\omega_x^{'} = (\overline{\omega}_x + \omega_x) = \omega_x$$
$$\omega_y^{'} = (\overline{\omega}_y + \omega_y) = \omega_y$$

Angular velocity components are now expressed as the sum of the nominal value plus the small perturbation. In the rest of the analysis, should the assumption of small perturbations hold this will mean that the condition is stable. Rewriting the equations, the product of two perturbations become negligible, so that the linearized equations are:

$$\begin{cases}I_x\dot{\omega}_x + (I_z - I_y)\overline{\omega}_z\omega_y = 0\\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\overline{\omega}_z = 0\\ I_z\dot{\omega}_z = 0\end{cases}$$

The derivative are taken only for the perturbation component of the angular velocity, since the nominal components are constant by definition. The system can now be integrated:

$$\omega_z = \text{const}$$

The perturbation of the z component is therefore constant. For the two remaining equations we can write:

$$\ddot{\omega}_x + \lambda^2\omega_x = 0$$
$$\lambda^2 = \frac{(I_z - I_y)(I_z - I_x)}{I_xI_y}\overline{\omega}_z^2$$

The solution is then of the general form:

$$\omega_x = ae^{i\lambda t}$$

The condition that guarantees a stable motion, harmonic, is $\lambda^2 > 0$, while for $\lambda^2 < 0$ the solution is exponentially diverging, so unstable.
The condition for stability is verified when:

$$\lambda^2 > 0 \quad \Rightarrow \quad I_z > I_y \text{ and } I_z > I_x \quad \text{or} \quad I_z < I_y \text{ and } I_z < I_x$$

that means that I_z must be either the maximum or the minimum inertia moment.

Stability of simple spin satellites with energy dissipation

Now we look at how the two previous stability conditions are changed due to presence of energy dissipation, that is:

$$\dot{T} < 0$$
$$|h| = cost$$

Calling with ω_z the angular velocity, in case the rotation occurs around the maximum inertia axis or the minimum inertia axis we can write:

$$2T_{I_{z\max}} = I_{z_{\max}} \omega^2_{z_{\max}}$$
$$2T_{I_{z\min}} = I_{z_{\min}} \omega^2_{z_{\min}}$$
$$|h| = I_{z_{\max}} \omega_{z_{\max}} = I_{z_{\min}} \omega_{z_{\min}}$$

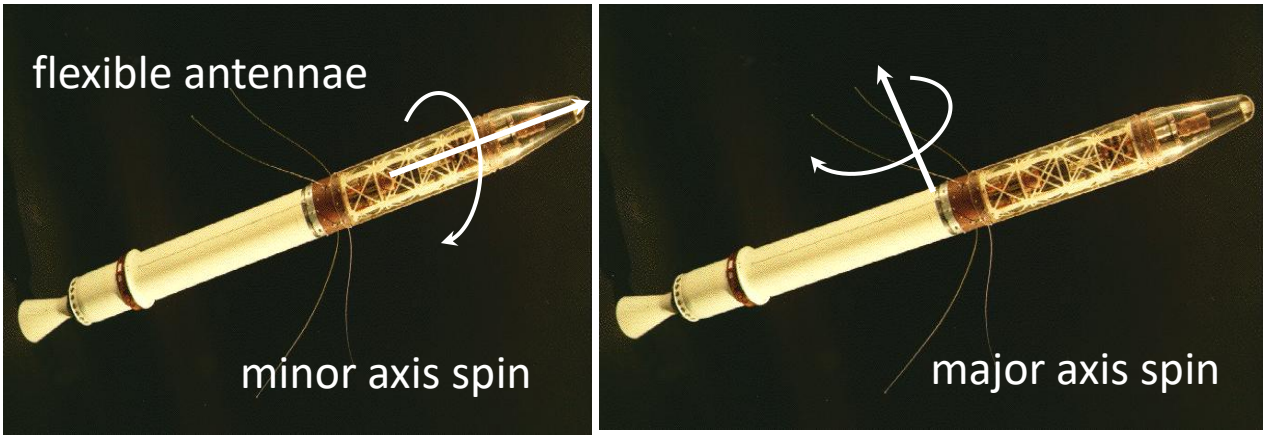
therefore, for conservation of angular momentum, $\omega_{zmax} < \omega_{zmin}$, and substituting this inequality into the expression of kinetic energy we have:

$$T_{I_{zmax}} < T_{I_{zmin}}$$

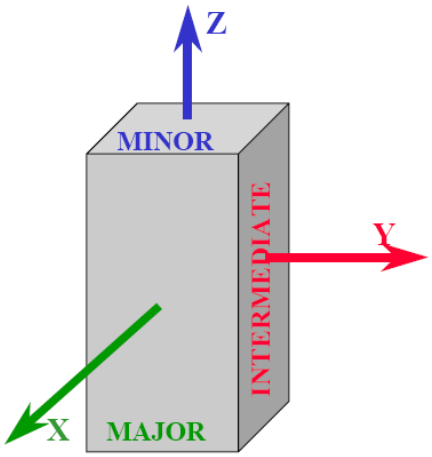
Therefore, in case of energy loss, the satellite will tend to the motion condition that has the minimum energy, rotating around the maximum inertia axis.
The only stable condition, with the usual assumptions on the inertia moments, is then:

$$I_z > I_y > I_x$$

Explorer 1 in the Figure below (first US satellite, 1958) was designed as a minor axis spinner, unexpectedly at the time energy dissipation caused by flexing of the wire antennae on the spaceraft body caused the spacecraft rotation to transition to major axis spin.

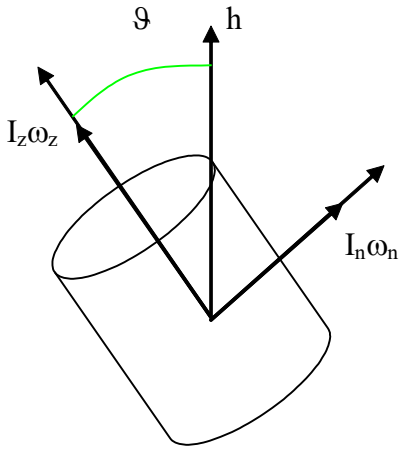


Therefore, we have the Major axis spin rule.



- $I_{xx} > I_{yy} > I_{zz}$
- Major axis spin is stable
- Minor axis spin is stable
- Intermediate axis spin is unstable
- Energy dissipation changes these results
  → Minor axis spin becomes unstable
- This is called the Major-Axis Rule

We can now analyze how the satellite reorients itself from a generic motion condition to the stable one. For simplicity we take the case of axial symmetric satellite.



We take as reference a rotating system, whose plane n-z includes the three vectors ω, z and h. This reference is also principal inertia, and vector h has only 2 components, along the two axes n and z:

$$h^2 = I_z^2\omega_z^2 + I_n^2\omega_n^2$$
$$2T = I_z\omega_z^2 + I_n\omega_n^2$$

The energy loss can be represented as:

$$2\dot{T} = 2I_z\omega_z\dot{\omega}_z + 2I_n\omega_n\dot{\omega}_n < 0$$

In addition, the following holds:

$$2h\dot{h} = 2I_z^2\omega_z\dot{\omega}_z + 2I_n^2\omega_n\dot{\omega}_n = 0 \quad \text{in fact} \quad \dot{h} = 0$$

so that we have:

$$\omega_n\dot{\omega}_n = -\frac{I_z^2}{I_n^2}\omega_z\dot{\omega}_z$$

and the energy loss is expressed as:

$$\dot{T} = \omega_z \dot{\omega}_z \left(I_z - \frac{I_z^2}{I_n} \right) = \omega_z \dot{\omega}_z \left(\frac{I_z}{I_n} (I_n - I_z) \right) < 0$$

If I_z is the minimum inertia axis, then:

$$\omega_z \dot{\omega}_z < 0$$

$$\omega_z > 0 \quad \dot{\omega}_z < 0$$
$$\omega_z < 0 \quad \dot{\omega}_z > 0$$

In both cases, ω_z decreases in amplitude, and ω_n increases, therefore the angular velocity will tend to align itself with the maximum inertia axis. If instead I_z is the maximum inertia moment, then it can be seen that ω_z increases in amplitude, and ω_n decreases, so again the angular velocity will tend to align itself with the maximum inertia axis. We finally have:

$$\tan\vartheta = \frac{I_n \omega_n}{I_z \omega_z}$$

It is possible to repeat the analysis taking the time derivative of angle ϑ. Kinetic energy will vary only until the angular velocity is aligned with the maximum inertia axis, that in this case is coincident with $\underline{h}$. In this final condition all internal dissipation sources must vanish, since loss of energy will no longer be possible without reduction of angular momentum, and this may vary only if external torques are applied to the satellite.

Attitude stability relative to a rotating frame

Previously we have considered the attitude of a spacecraft $A_{B/N}$ defined as the relative orientation between the body fixed frame **B** and an inertially fixed frame **N** such that:

$$\mathbf{B}=A_{B/N}\mathbf{N}$$

However, we may want to describe the attitude of the spacecraft body frame with respect to a moving frame such as the local-vertical-local-horizontal frame.

We can now consider the stability of the satellite taking as parameters the relative rotation of the principal inertia axis relative to a rotating frame. This is the typical case of a simple spin satellite spinning at one rotation per orbit along a circular orbit, for which the stability implies that the principal inertia axis remain aligned with the LVLH frame (local-vertical local-horizontal). For this analysis we must rewrite the equations of motion and then introduce the relative attitude parameters. Calling the LVLH frame **L** we have the relative attitude matrix defined by:

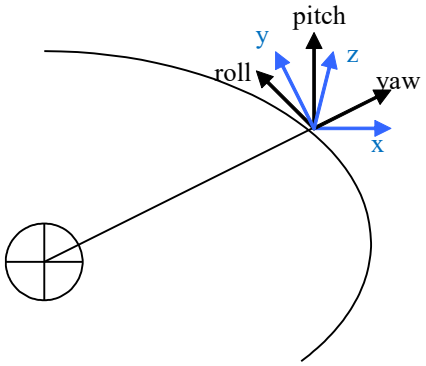
$$\mathbf{B} = A_{B/L}\ \mathbf{L}$$

with the relative angular velocity $\underline{\omega}_{B/L} = \underline{\omega}_B - A_{B/L}\underline{\omega}_L$. Note the main difference with the previous case is that $\underline{\omega}_L \neq \underline{0}$. Since $\underline{\omega}_B = \underline{\omega}_{B/N}$ we can simply substitute $\underline{\omega}_B = \underline{\omega}_{B/L} + A_{B/L}\underline{\omega}_L$ into the Euler equations derived earlier.

The equations are:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = 0 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = 0 \end{cases}$$

We consider a rotation around the z axis of the LVLH frame (the pitch axis), with an angular velocity of magnitude n such that $\underline{\omega}_L = [0 \quad 0 \quad n]$, small rotations between the body fixed frame **B** and the LVLH frame **L** represented by a set of Euler angles with three different indexes, to avoid singularity in the nominal condition.



Having considered small rotations, the order in which they are defined is not relevant. Indicating the three small angles as α_x, α_y, α_z, the rotation matrix representing the relative attitude of the satellite in the LVLH frame is:

$$A_{B/L} = \begin{bmatrix} 1 & \alpha_z & -\alpha_y \\ -\alpha_z & 1 & \alpha_x \\ \alpha_y & -\alpha_x & 1 \end{bmatrix}$$

and $\underline{\omega}_L = [0 \quad 0 \quad n]$ where n is the angular velocity of the moving frame about the z-axis (typically, one rotation per orbit). Angular velocity becomes:

$$\underline{\omega}_B = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} = \begin{bmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z \end{bmatrix} + \begin{bmatrix} 1 & \alpha_z & -\alpha_y \\ -\alpha_z & 1 & \alpha_x \\ \alpha_y & -\alpha_x & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}$$

Therefore, the time derivatives are:

$$\begin{cases} \dot{\omega}_x = \ddot{\alpha}_x - \dot{\alpha}_y n \\ \dot{\omega}_y = \ddot{\alpha}_y + \dot{\alpha}_x n \\ \dot{\omega}_z = \ddot{\alpha}_z \end{cases}$$

Now we can substitute in the Euler equations:

$$\begin{cases} I_x(\ddot{\alpha}_x - \dot{\alpha}_y n) + (I_z - I_y)(\dot{\alpha}_z + n)(\dot{\alpha}_y + \alpha_x n) = 0 \\ I_y(\ddot{\alpha}_y + \dot{\alpha}_x n) + (I_x - I_z)(\dot{\alpha}_z + n)(\dot{\alpha}_x - \alpha_y n) = 0 \\ I_z\ddot{\alpha}_z + (I_y - I_x)(\dot{\alpha}_x - \alpha_y n)(\dot{\alpha}_y + \alpha_x n) = 0 \end{cases}$$

and expand the products, still neglecting the products of infinitesimal terms:

$$\begin{cases} I_x\ddot{\alpha}_x + n(I_z - I_y - I_x)\dot{\alpha}_y + n^2(I_z - I_y)\alpha_x = 0 \\ I_y\ddot{\alpha}_y + n(I_x + I_y - I_z)\dot{\alpha}_x + n^2(I_z - I_x)\alpha_y = 0 \\ I_z\ddot{\alpha}_z = 0 \end{cases}$$

We then have:

$$\ddot{\alpha}_z = 0 \quad \Rightarrow \quad \dot{\alpha}_z = \text{const} \quad \Rightarrow \quad \alpha_z = \text{const} \cdot t + \text{const}$$

Dividing the first equation by I_x and the second by I_y we get:

$$\begin{cases} \ddot{\alpha}_x + n(K_x - 1)\dot{\alpha}_y + n^2 K_x \alpha_x = 0 \\ \ddot{\alpha}_y + n(1 - K_y)\dot{\alpha}_x + n^2 K_y \alpha_y = 0 \end{cases}$$

where the coefficients K_x and K_y are non-dimensional and always in the range between –1 and +1:

$$K_x = \frac{I_z - I_y}{I_x}$$
$$K_y = \frac{I_z - I_x}{I_y}$$

To evaluate the stability, we can take the Laplace transform of the system, and analyze the characteristic polynomial:

$$\begin{bmatrix} (s^2 + K_x n^2) & sn(K_x - 1) \\ sn(1 - K_y) & (s^2 + K_y n^2) \end{bmatrix} \begin{Bmatrix} \alpha_x(s) \\ \alpha_y(s) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

The roots of the characteristic polynomial must have non positive real part in order to have a stable system:

$$\begin{aligned} &det[A(s)] = 0 \Rightarrow \\ &(s^2 + n^2 K_x)(s^2 + n^2 K_y) - s^2 n^2 (1 - K_y)(K_x - 1) = 0 \\ &s^4 + s^2 \left(n^2 K_x + n^2 K_y + n^2 (1 - K_y)(1 - K_x)\right) + n^4 K_x K_y = 0 \\ &s^4 + n^2 s^2 (K_x + K_y + 1 - K_x - K_y + K_x K_y) + n^4 K_x K_y = 0 \\ &s^4 + n^2 s^2 (1 + K_x K_y) + n^4 K_x K_y = 0 \end{aligned}$$

For stability, s^2 cannot be real or complex, since in this case its square root will correspond to two complex numbers with phase difference equal to π, so one will have positive real part. The only stable condition will then be $s^2<0$, that will give two purely imaginary solutions, with zero real part. The stability conditions are then:

$$\begin{aligned} &b^2 > 4c \qquad a = 1 \\ &b > \sqrt{b^2 - 4c} \end{aligned}$$

Since n^2 is positive, it is possible to solve the characteristic equation for the variable $n^2 s^2$, so the stability conditions are written as:

$$(1 + K_x K_y)^2 - 4K_x K_y > 0 \quad \rightarrow \quad (1 - K_x K_y)^2 > 0$$

$$(1 + K_x K_y) > \sqrt{(1 - K_x K_y)^2} \quad \rightarrow \quad K_x K_y > 0$$

The first condition is trivial, so the second is the real stability condition, that is transformed into:

$$I_z > I_x \quad , \quad I_z > I_y$$
$$\text{or} \qquad I_z < I_x \quad , \quad I_z < I_y$$

We have therefore found the same stability condition that must hold for stability in terms of angular velocity. This because angular velocity components ω_x and ω_y are harmonic, so also angles show a harmonic oscillation. Rotation around axis z is unstable and tends to misalign axis z and the pitch axis.

To analyze the actual motion of the satellite we must evaluate the system characteristic roots:

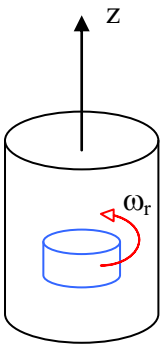
$$s^2 = \frac{-n^2(1 + K_x K_y) \pm n^2\sqrt{(1 - K_x K_y)^2}}{2}$$

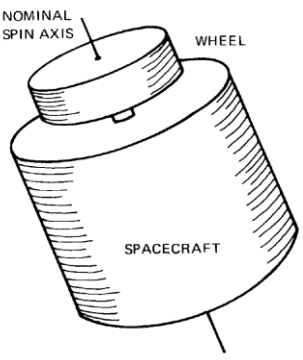
The solution depends upon the nominal angular velocity n, therefore if nominal motion is slow also the oscillations around axes x and y will be slow. The oscillation is not asymptotically stable since the characteristic roots have zero real part, while for asymptotic stability a negative real part is required.

Stability of dual spin satellites

Following the previous analysis, we know that if I_z is the maximum inertia moment then the rotation is stable, and if the nominal angular velocity is high the frequency of oscillation is also high and the amplitude of the resulting oscillations will be small. We would like however to have a satellite whose stability and frequency of oscillation are independent from the axis of rotation and from the nominal value of the angular velocity, so that we could have stable motion even if rotation is around an axis that is not the maximum inertia axis.

In the simple spin satellite the stability is dependent upon $I_z\omega_z$, that is the nominal angular momentum. In order to increase the nominal angular momentum, without varying ω_z, we can introduce an additional element that can rotate relatively to the z axis. This is the dual spin satellite.





Left: Momentum bias design. Right: Dual-Spin spacecraft

The angular momentum becomes:

$$\overline{h} = I_x\omega_x\underline{i} + I_y\omega_y\underline{j} + (I_z\omega_z + I_r\omega_r)\underline{k}$$

where I_x, I_y and I_z include the presence of the rotor and ω_r is the relative velocity of the rotor around the z axis. We can write the Euler equations for this system, simply replacing the term $I_z\omega_z$ with the term $I_z\omega_z+I_r\omega_r$:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y + I_r\omega_r\omega_y = M_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z - I_r\omega_r\omega_x = M_y \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x + I_r\dot{\omega}_r = M_z \\ I_r\dot{\omega}_r = M_r \end{cases}$$

The fourth equation is now required since we added one degree of freedom to the system. M_r is the relative torque between rotor and satellite.
We can now analyze the stability of the torque-free motion of the dual spin satellite. The equilibrium condition is obtained with the nominal angular velocity of the satellite aligned with axis z, and arbitrary relative velocity of the rotor. Linearizing the equations in this condition, we have:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\overline{\omega}_z\omega_y + I_r\overline{\omega}_r\omega_y = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\overline{\omega}_z - I_r\overline{\omega}_r\omega_x = 0 \\ I_z\dot{\omega}_z + I_r\dot{\omega}_r = 0 \\ I_r\dot{\omega}_r = 0 \end{cases}$$

The last two equations are decoupled, that is we can decouple the equations representing rotations around the nominal direction of $\underline{h}$. From these two equations we have ω_z and ω_r constant.
The first two equations are again solved by taking the time derivative of the first and substituting the derivative obtained by the second equation:

$$\dot{\omega}_y = \lambda_y\omega_x$$
$$\ddot{\omega}_x + \lambda_x\lambda_y\omega_x = 0$$

having introduced:

$$\lambda_x = \frac{(I_z - I_y)\overline{\omega}_z + I_r\overline{\omega}_r}{I_x} = K_x\overline{\omega}_z + \frac{I_r}{I_x}\overline{\omega}_r$$
$$\lambda_y = -\frac{(I_x - I_z)\overline{\omega}_z - I_r\overline{\omega}_r}{I_y} = K_y\overline{\omega}_z + \frac{I_r}{I_y}\overline{\omega}_r$$

Stability condition now becomes:

$$\lambda_x\lambda_y > 0$$

Taking the case:

$$\lambda_x > 0$$
$$\lambda_y > 0$$

we have two inequalities whose solution depends on I_r and $\overline{\omega}_r$:

$$\begin{cases}(I_z - I_y)\overline{\omega}_z + I_r\overline{\omega}_r > 0\\(I_z - I_x)\overline{\omega}_z + I_r\overline{\omega}_r > 0\end{cases}$$

For given $\overline{\omega}_z$, $\overline{\omega}_r$, I_r we can evaluate the conditions on I_x, I_y, I_z that guarantee stability. This is normally not the best way to look at the problem, since in general we want to analyze stability for given $\overline{\omega}_z$, I_x, I_y, I_z, evaluating the conditions on I_r and $\overline{\omega}_r$. So, normally we evaluate the characteristics of the rotor that guarantee the stability of the overall system. If $\overline{\omega}_z = 0$then the system is stable for any $\overline{\omega}_r$, while if $\overline{\omega}_r = 0$ we have the same result of the simple spin satellite.

The most interesting case is the general case, for which stability holds if:

$$\begin{cases}(I_y - I_z)\overline{\omega}_z < I_r\overline{\omega}_r\\(I_x - I_z)\overline{\omega}_z < I_r\overline{\omega}_r\end{cases}$$

This system of inequalities has always a set of solutions, since it is enough to solve the inequality with the greatest left side term.

Moreover, there is the condition:

$$\lambda_x < 0$$
$$\lambda_y < 0$$

that would lead to:

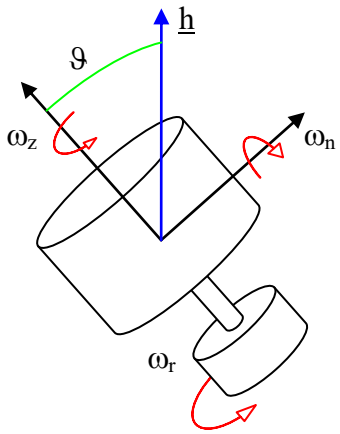
$$\begin{cases}(I_y - I_z)\overline{\omega}_z > I_r\overline{\omega}_r\\(I_x - I_z)\overline{\omega}_z > I_r\overline{\omega}_r\end{cases}$$

This second set of inequalities are valid only for the case with no energy loss.
The dual spin satellite can then be stable even if rotating around an axis that is not the maximum inertia axis.

Effects of energy loss on axial symmetric dual spin satellite

The system angular momentum is:

$$\underline{h} = (I_z\omega_z + I_r\omega_r)\underline{k} + I_n\omega_n\underline{i}$$
$$h^2 = (I_z\omega_z + I_r\omega_r)^2 + (I_n\omega_n)^2$$
$$2h\dot{h} = 2(I_z\omega_z + I_r\omega_r)(I_z\dot{\omega}_z + I_r\dot{\omega}_r) + 2I_n^2\omega_n\dot{\omega}_n = 0$$



The kinetic energy is:

$$2T = I_z\omega_z^2 + I_r\omega_r^2 + I_n\omega_n^2$$
$$2\dot{T} = 2I_z\omega_z\dot{\omega}_z + 2I_r\omega_r\dot{\omega}_r + 2I_n\omega_n\dot{\omega}_n$$
$$\dot{T} = I_z\omega_z\dot{\omega}_z + I_r\omega_r\dot{\omega}_r + I_n\omega_n\dot{\omega}_n = \dot{T}_s + \dot{T}_r$$

We can evaluate an equivalent angular velocity as:

$$\omega_e = \frac{(I_z\omega_z + I_r\omega_r)}{I_n}$$
$$\Rightarrow I_n\omega_n\dot{\omega}_n = -\omega_e(I_z\dot{\omega}_z + I_r\dot{\omega}_r) \quad \text{since } 2\mathrm{h}\dot{\mathrm{h}} = 0$$
$$\dot{T}_s + \dot{T}_r = I_z\dot{\omega}_z(\omega_z - \omega_e) + I_r\dot{\omega}_r(\omega_r - \omega_e)$$

It is an arbitrary distribution of the energy loss, with the first term relative to the satellite and the second relative to the rotor:

$\dot{T}_s = I_z\dot{\omega}_z\lambda_z$ misalignment of $\underline{\mathrm{h}}$
$\dot{T}_r = I_r\dot{\omega}_r\lambda_r$ viscous energy loss due to the relative velocity

with:

$$\lambda_z = (\omega_z - \omega_e)$$
$$\lambda_r = (\omega_r - \omega_e)$$

We therefore have:

$$I_n\omega_n\dot{\omega}_n = -\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)$$

Now, evaluate angle ϑ:

$$\sin\vartheta = \frac{h_n}{h}$$

Taking the time derivative:

$$\dot{\vartheta}\cos\vartheta = \frac{I_n\dot{\omega}_n}{h}$$

$$\dot{\vartheta} = \frac{I_n\dot{\omega}_n}{h\cos\vartheta} = \frac{I_n\omega_n\dot{\omega}_n}{h\cos\vartheta\omega_n} = \frac{-\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)}{h\cos\vartheta\omega_n} =$$
$$= -\frac{I_n\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)}{h\cos\vartheta I_n\omega_n} = -\frac{I_n\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)}{h\cos\vartheta h_n} =$$
$$= -\frac{I_n\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)}{h\cos\vartheta h\sin\vartheta} = -\frac{2\mathrm{I}_n\omega_e\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)}{h^2\sin2\vartheta} =$$
$$= -\frac{2\mathrm{I}_n\omega_e}{h^2\sin2\vartheta}\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right) = -A\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right)$$

since:

$$0 < \vartheta < \pi/2$$

therefore $\sin(\vartheta) > 0$ and where A > 0.
For stability we should have $\dot{\vartheta} < 0$, that is:

$$\left(\frac{\dot{T}_s}{\lambda_z} + \frac{\dot{T}_r}{\lambda_r}\right) > 0$$

Assuming the contribution of the nominal value of h is given mostly by the presence of the rotor, that is:

$$I_r\omega_r >> I_z\omega_z$$

we then have:

$$\omega_e = \frac{I_r\omega_r}{I_n}$$

$$\lambda_z = \omega_z - \frac{I_r\omega_r}{I_n} \cong -\frac{I_r\omega_r}{I_n}$$

$$\lambda_r = \omega_r - \frac{I_r\omega_r}{I_n}$$

In the case $I_r >> I_n$ (case only theoretical, since this would imply a rotor bigger than the satellite):

$$\begin{cases}\lambda_z < 0 \\ \lambda_r < 0\end{cases} \quad \rightarrow \quad \sum \frac{\dot{T}}{\lambda} > 0 \quad \rightarrow \quad \text{stable}$$

The more general case is then $I_r < I_n$, for which:

$$\frac{I_n\dot{T}_s}{(I_n\omega_z - I_r\omega_r)} + \frac{I_n\dot{T}_r}{(I_n\omega_r - I_r\omega_r)} > 0$$

This stability condition is transformed into a condition on the energy loss terms:

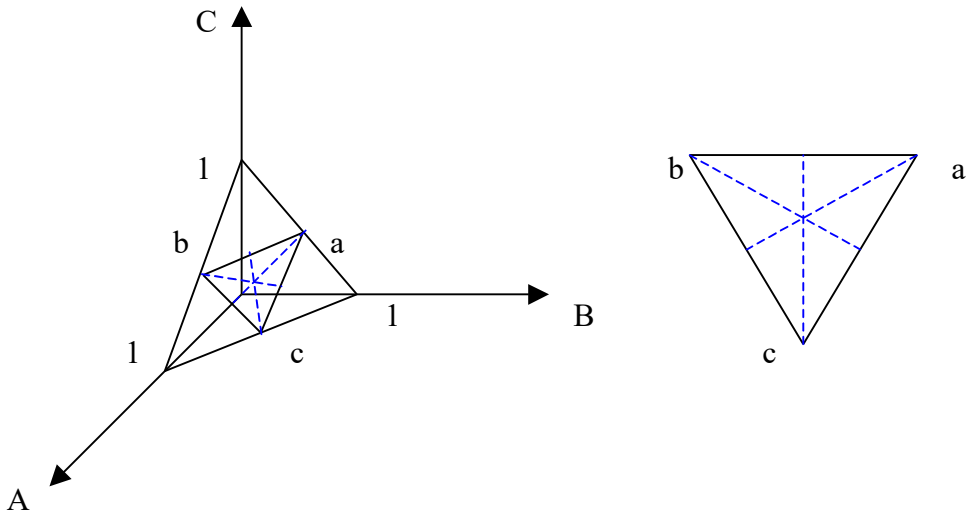
$$\dot{T}_r > \frac{\dot{T}_s(I_n\omega_r - I_r\omega_r)}{(I_r\omega_r - I_n\omega_z)}$$

In general, the rotor energy loss can be tuned with some viscous damping mechanism. Notice that now we have a condition that is no longer purely geometrical. In fact, in the case $\dot{\omega}_r = 0$, then the rotor will show no energy loss and $\dot{T}_r$ would be zero, so that even with energy loss in the satellite the system would be unstable. Therefore the rotor should generate some energy loss to ensure stability.

Stability diagrams

We can now take a look at a few diagrams, known as canonical representations, used to evaluate attitude stability of satellites.

Let consider a plane including the three points on the axes at unit distance from the origin, represented by the equation:



$$A + B + C = 1$$

or, defining:

$$A = \frac{I_x}{I_x + I_y + I_z} \quad ; \quad \mathrm{B} = \frac{I_y}{I_x + I_y + I_z} \quad ; \quad \mathrm{C} = \frac{I_z}{I_x + I_y + I_z}$$

$$\frac{I_x}{\sum I} + \frac{I_y}{\sum I} + \frac{I_z}{\sum I} = 1$$

Connecting the three mid points on the sides of the triangle, we form a second triangle (a,b,c) that includes all the possible combinations of inertia moments.

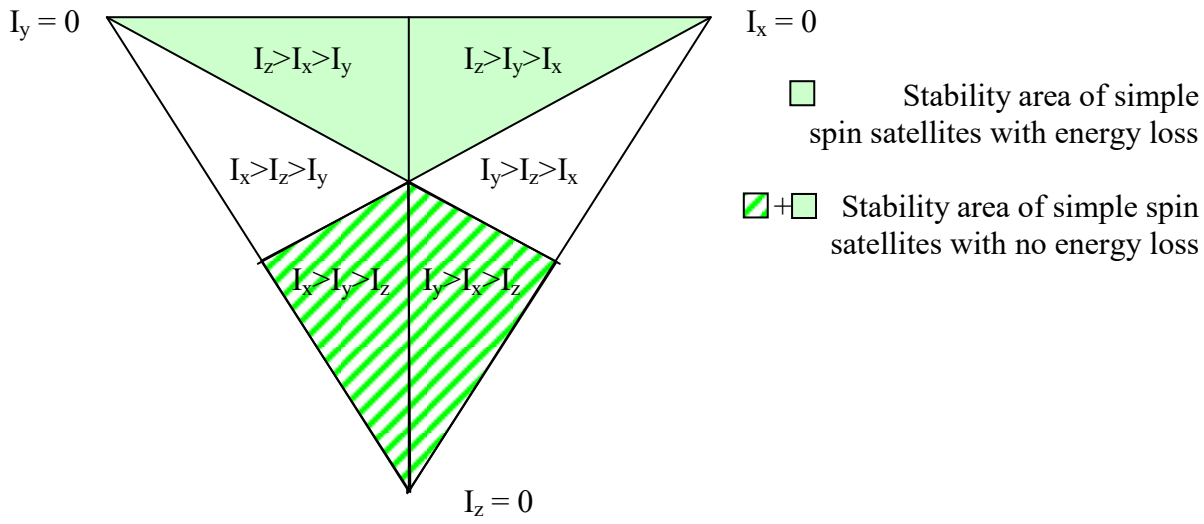
In fact, the following holds:

- In point *a*: $I_x = 0$ and $I_y = I_z$
- In point *b*: $I_y = 0$ and $I_x = I_z$
- In point *c*: $I_z = 0$ and $I_y = I_x$

Along the bisectors two inertia moments are equal:

- Along the bisector from *a*: $I_z = I_y$
- Along the bisector from *b*: $I_z = I_x$
- Along the bisector from *c*: $I_x = I_y$

At the intersection of the three bisectors the three inertia moments are all equal.

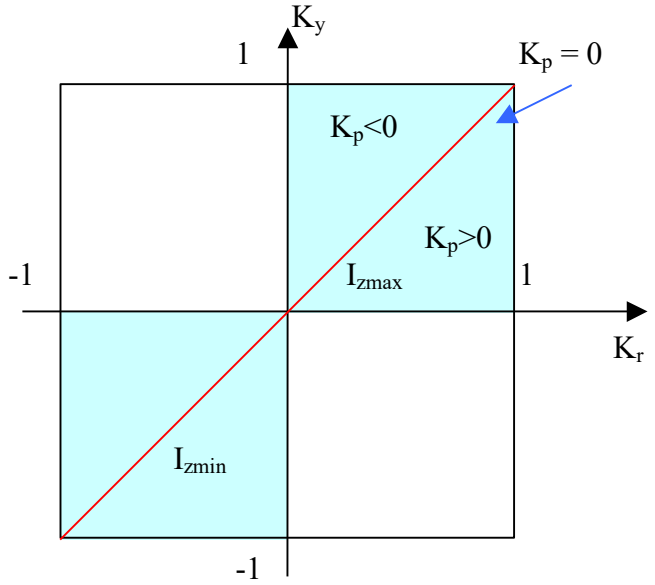


Alternatively, through a simple change of variables, it is possible to represent in the same way the plane as a function of K_x, K_y, K_z, in fact:

$$K_x = \frac{I_z - I_y}{I_x}$$
$$K_y = \frac{I_z - I_x}{I_y}$$
$$K_z = \frac{I_y - I_x}{I_z}$$

It is then possible to reduce the representation to the K_x – K_y plane. In this case, typically the coefficients are renamed as K_p, K_r, and K_y, where the correspondence with the previously defined K_x, K_y, and K_z coefficients is:

$K_x = K_y$	Yaw
$K_y = K_r$	Roll
$K_z = K_p$	Pitch



It is possible to see that:

$$\begin{cases} I_x K_y = I_z - I_y \Rightarrow I_z = I_x K_y + I_y \\ I_y K_r = I_z - I_x \Rightarrow I_z = I_y K_r + I_x \end{cases}$$
$$\Rightarrow I_x K_y + I_y = I_y K_r + I_x$$

Therefore:

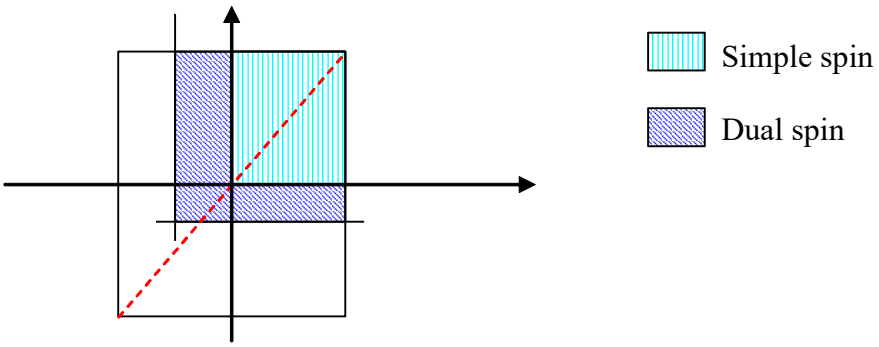
$$(K_y - 1) = \frac{I_y}{I_x}(K_r - 1)$$

If $I_y > I_x$ we then have $K_p > 0$, $(K_r - 1) < 0$, $(K_y - 1) < 0$, so that $(K_y - 1) < (K_r - 1) \quad \rightarrow \quad K_r > K_y$.

The simple spin satellite is stable in the dashed regions of the following figure, while for the dual spin satellite the stability holds if:

$$\lambda_x = K_y \overline{\omega}_z + \frac{I_r}{I_x}\overline{\omega}_r > 0$$
$$\lambda_y = K_r \overline{\omega}_z + \frac{I_r}{I_y}\overline{\omega}_r > 0$$
$$K_y > -\frac{I_r}{I_x}\frac{\overline{\omega}_r}{\overline{\omega}_z}$$
$$K_r > -\frac{I_r}{I_y}\frac{\overline{\omega}_r}{\overline{\omega}_z}$$

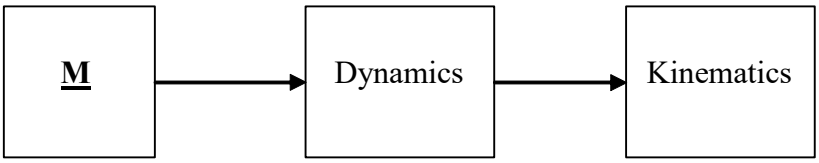
If the satellite angular velocity and the rotor angular velocity have the same sign, the stability region is increased as shown in the figure.



If the angular velocity of satellite and rotor have opposite signs, then the stability region is reduced.

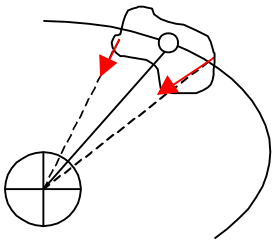
Disturbing torques

Now we can analyze the disturbing torques that act on the satellite:

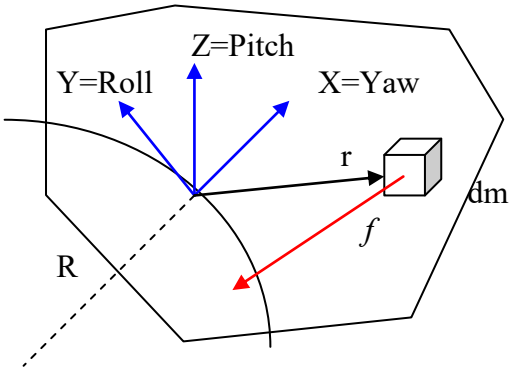


Start with the torque due to gravity gradient:

Gravity gradient torque



The gravity field is not uniform, therefore there could be a torque acting on the satellite. This is mainly true for large satellites, and even if the resulting torque is small the effect can be considerable due to the long time of action.



We can evaluate the torque generated by the elementary force *f*, due to the gravity acting on the elementary mass dm:

$$dM = -\underline{r} \wedge \frac{Gm_t dm}{|R+r|^3}(\underline{R}+\underline{r})$$

$$M = -\int_B \underline{r} \wedge \frac{Gm_t}{|R+r|^3}(\underline{R}+\underline{r})dm$$

r is much smaller than R, therefore it can be considered as a perturbation:

$$|R+r|^{-3} = \left(R^2 + r^2 + 2\underline{R}\cdot\underline{r}\right)^{-\frac{3}{2}} = R^{-3}\left(1 + \frac{r^2}{R^2} + 2\frac{\underline{R}\cdot\underline{r}}{R^2}\right)^{-\frac{3}{2}} \cong R^{-3}\left(1 + 2\frac{\underline{R}\cdot\underline{r}}{R^2}\right)^{-\frac{3}{2}}$$

$$\text{taking the series expansion at} r = 0 \Rightarrow \cong R^{-3}\left(1 - \frac{3}{2}(1)^{-\frac{5}{2}}2\frac{\underline{R}\cdot\hat{\underline{r}}}{R^2}r\right) = R^{-3}\left(1 - 3\frac{\underline{R}\cdot\underline{r}}{R^2}\right)$$

Then:

$$M = -\frac{Gm_t}{R^3}\int_B \underline{r}\wedge\left(1 - 3\frac{\underline{R}\cdot\underline{r}}{R^2}\right)(\underline{R}+\underline{r})dm$$

If the reference system has origin in the center of mass:

$$\int \underline{r}\wedge(\underline{r}+\underline{R})dm = \int(\underline{r}\wedge\underline{r}) + (\underline{r}\wedge\underline{R})dm = -\underline{R}\wedge S_0 = 0$$

so that:

$$M = -\frac{Gm_t}{R^3}\int_B \underline{r}\wedge\left(-3\frac{\underline{R}\cdot\underline{r}}{R^2}\right)(\underline{R}+\underline{r})dm = -\frac{Gm_t}{R^5}\int_B \underline{r}\wedge(\underline{R}+\underline{r})(-3\underline{R}\cdot\underline{r})dm$$

Finally, we get to:

$$M = \frac{3Gm_t}{R^5}\int_B (\underline{r}\cdot\underline{R})(\underline{r}\wedge\underline{R})dm$$

Taking as reference system the LVLH frame (x-yaw, y-roll, z-pitch), we have:

$$\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$$
$$\underline{R} = R\underline{i}$$

So that:

$$M = \frac{3Gm_t}{R^5}\int_B Rx\left(z\underline{j} - y\underline{k}\right)Rdm = \frac{3Gm_t}{R^3}\int_B x\left(z\underline{j} - y\underline{k}\right)dm = \frac{3Gm_t}{R^3}\int_B \left(xz\underline{j} - xy\underline{k}\right)dm$$
$$= \frac{3Gm_t}{R^3}\left(-I_{xz}\underline{j} + I_{xy}\underline{k}\right)$$

We therefore have a torque with components in roll and pitch, and in case one principal axis is aligned with the yaw axis then the torque will vanish, since in this case I_{xz} e I_{xy} are zero. Furthermore the torque is inversely proportional to the third power of the distance from the attracting body (Earth).
In order to evaluate the effect of the torque on the satellite dynamics, this has to be evaluated in the principal axis frame and inserted in the Euler equations.
Taking as reference the principal inertia axes, the expression of $\underline{r}$ does not change but:

$$\underline{R} = R\left(c_1\underline{i} + c_2\underline{j} + c_3\underline{k}\right)$$

where c_1, c_2, c_3 are the direction cosines of the radial direction in the principal axes. Then:

$$M = \frac{3Gm_t}{R^3}\int_B (xc_1 + yc_2 + zc_3)\begin{pmatrix} yc_3 - zc_2 \\ zc_1 - xc_3 \\ xc_2 - yc_1 \end{pmatrix}dm$$

Evaluating all the terms under the integral sign, considering that the reference is principal inertia, we have:

$$M = \frac{3Gm_t}{R^3}\int_B \begin{Bmatrix}(y^2-z^2)c_2c_3\\(z^2-x^2)c_1c_3\\(x^2-y^2)c_1c_2\end{Bmatrix} dm = \frac{3Gm_t}{R^3}\begin{Bmatrix}(I_z-I_y)c_2c_3\\(I_x-I_z)c_1c_3\\(I_y-I_x)c_1c_2\end{Bmatrix}$$

Therefore if one of the principal axes is aligned with the radial direction the torque is zero because only one of the direction cosines is non-zero.

Stability of simple spin satellites subject to gravity gradient torque

In this case the Euler equations become:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = \dfrac{3Gm_t}{R^3}(I_z - I_y)c_3c_2 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = \dfrac{3Gm_t}{R^3}(I_x - I_z)c_1c_3 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = \dfrac{3Gm_t}{R^3}(I_y - I_x)c_2c_1 \end{cases}$$

This torque is generated by the same force field that causes the orbital motion, so it is acting continuously, and it depends on the attitude of the satellite (the orientation of the radial direction in the principal inertia frame). We can introduce a set of three small Euler angles to define the relative orientation of the principal axes and the LVLH frame:

$$\begin{Bmatrix}\omega_x\\\omega_y\\\omega_z\end{Bmatrix} = \begin{bmatrix}\dot{\alpha}_x\\\dot{\alpha}_y\\\dot{\alpha}_z\end{bmatrix} + \begin{bmatrix}1 & \alpha_z & -\alpha_y\\-\alpha_z & 1 & \alpha_x\\\alpha_y & -\alpha_x & 1\end{bmatrix}\begin{bmatrix}0\\0\\n\end{bmatrix}$$

$$\begin{Bmatrix}c_1\\c_2\\c_3\end{Bmatrix} = \begin{bmatrix}1 & \alpha_z & -\alpha_y\\-\alpha_z & 1 & \alpha_x\\\alpha_y & -\alpha_x & 1\end{bmatrix}\begin{Bmatrix}1\\0\\0\end{Bmatrix}$$

In this way we can write the system dynamics as a function of the Euler angles and their derivatives, including the gravity gradient torque that now becomes an integral part of the system dynamics. Assuming that n is equal to one rotation per orbit, then:

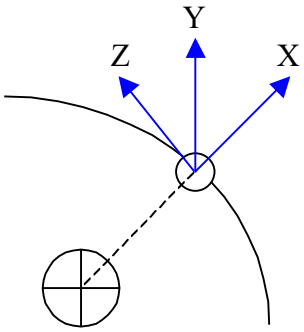
$$n^2 = \frac{V^2}{R^2} = \frac{1}{R^2}\left(\sqrt{\frac{K}{R}}\right)^2 = \frac{K}{R^3} = \frac{Gm_t}{R^3}$$

Therefore, in case of circular orbit, the attitude dynamics including gravity gradient becomes:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y = 3n^2(I_z - I_y)c_3c_2 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = 3n^2(I_x - I_z)c_1c_3 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x = 3n^2(I_y - I_x)c_2c_1 \end{cases}$$

where n is the nominal angular velocity along the orbit.

In this set of equations we have a twofold coupling between angular velocities and angles. Attitude is represented by the direction cosines c_1, c_2, c_3.
Assuming the principal inertia axes are sufficiently close to the corresponding axes of the LVLH frame, we can write:



$$\begin{Bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{Bmatrix} = \begin{bmatrix}1 & \alpha_z & -\alpha_y\\ -\alpha_z & 1 & \alpha_x\\ \alpha_y & -\alpha_x & 1\end{bmatrix}\begin{Bmatrix}\dot{\alpha}_x\\ \dot{\alpha}_y\\ \dot{\alpha}_z + n\end{Bmatrix}$$

$$\begin{Bmatrix}c_1\\ c_2\\ c_3\end{Bmatrix} = \begin{bmatrix}1 & \alpha_z & -\alpha_y\\ -\alpha_z & 1 & \alpha_x\\ \alpha_y & -\alpha_x & 1\end{bmatrix}\begin{Bmatrix}1\\ 0\\ 0\end{Bmatrix}$$

We can now substitute angular velocities and direction cosines with the corresponding terms depending on angles and their derivatives. Neglecting the terms that include two angles or derivatives (small terms) we get to the linearized form:

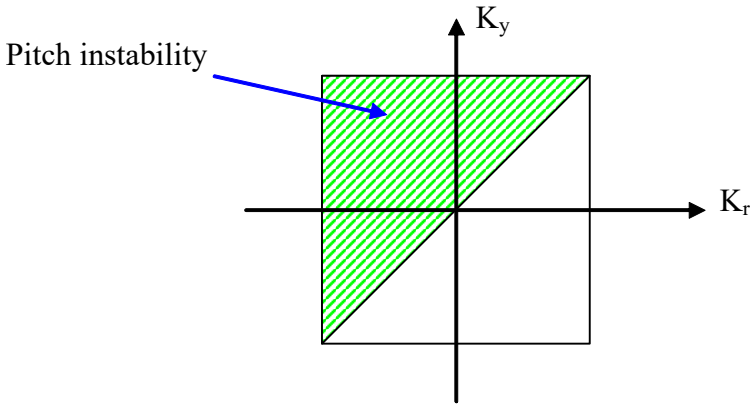
$$\begin{cases} I_x\ddot{\alpha}_x + (I_z - I_y - I_x)n\dot{\alpha}_y + (I_z - I_y)n^2\alpha_x = 0\\ I_y\ddot{\alpha}_y + (I_x + I_y - I_z)n\dot{\alpha}_x + (I_z - I_x)n^2\alpha_y = 3n^2(I_x - I_z)\alpha_y\\ I_z\ddot{\alpha}_z = -3n^2(I_y - I_x)\alpha_z \end{cases}$$

That can be simplified into:

$$\begin{cases} I_x\ddot{\alpha}_x + (I_z - I_y - I_x)n\dot{\alpha}_y + (I_z - I_y)n^2\alpha_x = 0\\ I_y\ddot{\alpha}_y + (I_x + I_y - I_z)n\dot{\alpha}_x + 4n^2(I_z - I_x)\alpha_y = 0\\ I_z\ddot{\alpha}_z + 3n^2(I_y - I_x)\alpha_z = 0 \end{cases}$$

The third equation is decoupled from the first two. We can now look at the stability of the system. Starting from the z axis, the stability condition is:

$I_y > I_x$ therefore $K_p > 0 \Rightarrow K_r > K_y$



Consider now the first two equations. Divide the first by I_x and the second by I_y:

$$\begin{cases} \ddot{\alpha}_x + (K_y - 1)n\dot{\alpha}_y + K_y n^2 \alpha_x = 0 \\ \ddot{\alpha}_y + (1 - K_r)n\dot{\alpha}_x + 4K_r n^2 \alpha_y = 0 \end{cases}$$

The characteristic equation is:

$$\begin{aligned} &(s^2 + n^2 K_y)(s^2 + 4n^2 K_r) - [ns(K_y - 1)][ns(1 - K_r)] = 0 \\ &s^4 + n^2 s^2 (K_y + 4K_r + 1 - K_r - K_y + K_r K_y) + 4n^2 K_r K_y = 0 \\ &s^4 + n^2 s^2 (1 + 3K_r + K_r K_y) + 4n^2 K_r K_y = 0 \end{aligned}$$

The conditions for stability are:

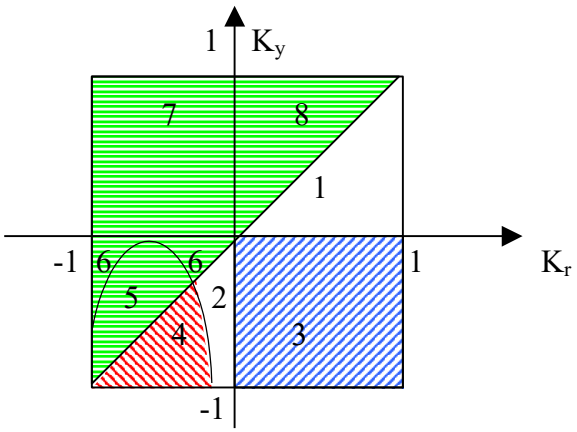
$$\begin{aligned} &x^2 + \mathrm{b}x + c = 0 \\ &\begin{cases} c > 0 \\ \dfrac{-b \pm \sqrt{b^2 - 4\mathrm{c}}}{2} < 0 \\ b^2 - 4\mathrm{c} > 0 \end{cases} \\ &(1 + 3\mathrm{K}_r + K_r K_y)^2 > 16\mathrm{K}_r K_y \\ &K_r K_y > 0 \end{aligned}$$

That is:

$$\begin{cases} 1 + 3K_r + K_r K_y > 4\sqrt{K_r K_y} \\ K_r K_y > 0 \end{cases}$$

The presence of gravity gradient therefore reduces the stability region. In detail, the first condition can be analyzed by evaluating a few solutions for particular values of coefficient K_y:

$$\begin{aligned} &K_y = 0 \quad K_r = -\frac{1}{3} \\ &K_y = -1 \quad 1 + 2K_r = 4\sqrt{-K_r} \\ &K_y = K_r \quad 1 + 3K + K^2 = -4K \end{aligned}$$



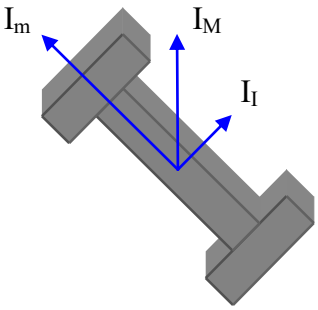
The unmarked regions represent the stable regions. In fact, in region (1) we have:

$$I_z > I_y > I_x \qquad I_p > I_r > I_y \qquad \text{always}$$

In region (2) we have:

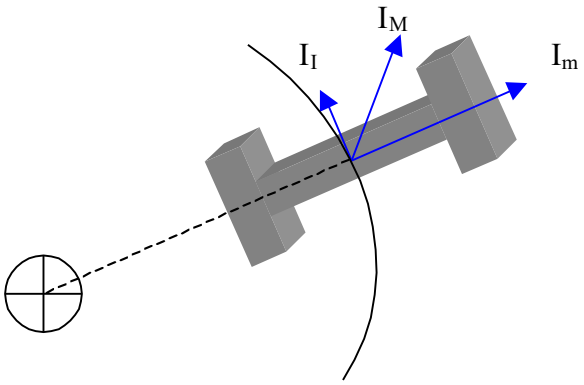
$$I_y > I_x > I_z \qquad I_r > I_y > I_p \qquad \text{limited by the additional curve}$$

In the remaining regions we have some kind of instability. In regions (3) and (4) we have yaw and roll instability, in regions (5) and (7) yaw, roll and pitch instability while in regions (6) and (8) only pitch instability. Furthermore in region (8) we have $I_p > I_y > I_r$ while in region (7) $I_y > I_r > I_p$.
Now assume we have an orbiting satellite with three different inertia moments as in the following figure:

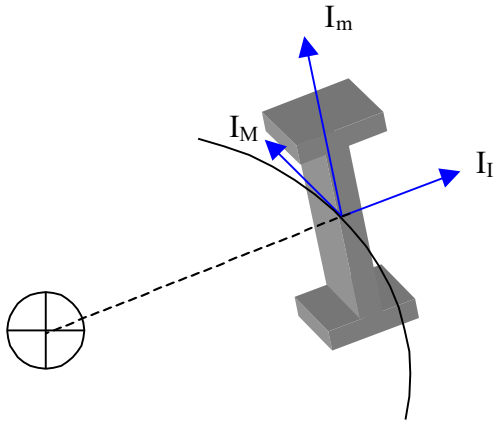


I_m = I minimum
I_M = I maximum
I_I = I intermediate

To have stability identified by the region (1), the satellite must be oriented in orbit as follows:



To have stability identified by the region (2) instead orientation must be:



In order to stabilize a generic configuration, we can have one mass positioned at an appropriate distance from the main body of the satellite, so to modify the inertia moments, but with the limitation due to proper orientation of the satellite, on the contrary we could generate instability.
In this case we could even solve the characteristic equation to evaluate the frequency of natural oscillation of the system. For the pitch equation we have:

$$s^2 + 3K_p n^2 = 0$$
$$\omega_p = n\sqrt{3K_p}$$

For the roll/yaw equations the solution is not trivial, in fact the characteristic equation is:

$$s^4 + n^2 s^2\left(1 + 3\mathrm{K}_r + K_r K_y\right) + 4\mathrm{n}^4 K_r K_y = 0$$
$$\frac{2}{n^2}\omega_{ry}^2 = -\left(1 + 3K_r + K_r K_y\right) \pm \sqrt{\left(1 + 3K_r + K_r K_y\right)^2 - 16K_r K_y}$$

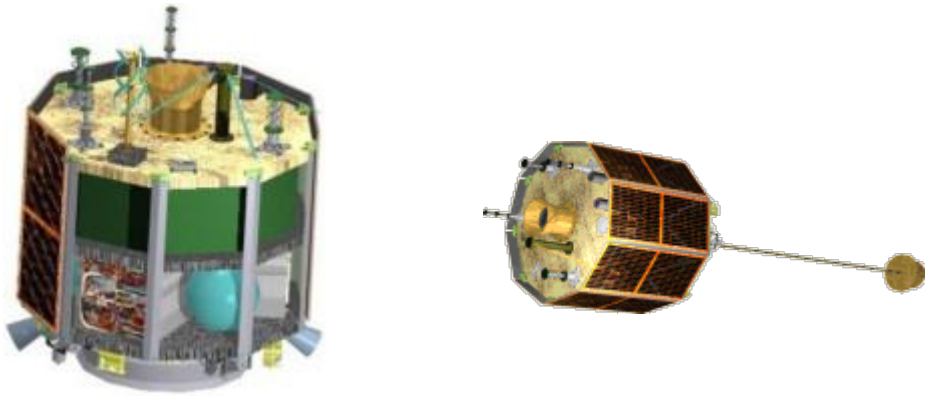
In the special case for which the satellite mass tends to be aligned with the yaw axis, we have I_p=I_r and:

$$I_y \to 0 \Rightarrow K_r \to 1$$
$$K_p \to 1$$
$$K_y = 0$$

The solution of the characteristic equation is then:

$$\frac{2}{n^2}\omega_{ry}^2 = -8$$
$$\omega_{ry} = 2n$$
$$\omega_p = n\sqrt{3}$$

Disturbing torques acting with frequency close to 2n can exist for some geocentric inclined orbits. As example, in two orbit positions symmetric with respect to Earth it is possible to face the same atmospheric conditions and magnetic field conditions, so that resonant motion can be generated.



SSTL UOSAT-12

Gravity gradient torque is exploited for gravity gradient stabilization whereby a boom can be deployed to increase the relative principal moments of inertia. We note that the spacecraft is always stable with configuration (1) where $I_z > I_y > I_x$. In addition, we can observe that with a much larger I_z than the other inertias the gravity gradient torque is much larger for a particular relative attitude. This larger this relative value the more robustness the gravity gradient torque can give relative to other disturbance torques. A gravity gradient boom essentially increases I_z with respect to the other moments of inertia.

Dynamics of dual spin satellites subject to gravity gradient

A dual spin satellite can be subject to gravity gradient. In such case, its dynamics becomes:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y + I_r\omega_r\omega_y = \dfrac{3Gm_t}{R^3}(I_z - I_y)c_3c_2 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z - I_r\omega_r\omega_x = \dfrac{3Gm_t}{R^3}(I_x - I_z)c_1c_3 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x + I_r\dot{\omega}_r = \dfrac{3Gm_t}{R^3}(I_y - I_x)c_2c_1 \end{cases}$$

The rotor could be aligned with an axis not coincident with any of the principal inertia axes, and in this case the equations of rotational motion are written starting from the expression of the angular momentum:

$$\underline{h} = I\underline{\omega} + I_r\omega_r\underline{r} = \begin{Bmatrix} I_x\omega_x + I_r\omega_r r_x \\ I_y\omega_y + I_r\omega_r r_y \\ I_z\omega_z + I_r\omega_r r_z \end{Bmatrix}$$

where $\underline{r}$ is the direction of the rotor axis.

Adopting this procedure, it is straightforward to extend the case to a general configuration with n rotors having distinct rotation axis.
Assuming the direction $\underline{r}$ is fixed (design parameter), by taking time derivatives we have, for the case of single rotor:

$$\frac{d\underline{h}}{dt} = \underline{M}$$
$$\frac{d\underline{h}}{dt} = \dot{\underline{h}} + \underline{\omega} \wedge \underline{h}$$

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y + I_r\dot{\omega}_r r_x + I_r\omega_r(\omega_y r_z - \omega_z r_y) = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z + I_r\dot{\omega}_r r_y + I_r\omega_r(\omega_z r_x - \omega_x r_z) = 0 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x + I_r\dot{\omega}_r r_z + I_r\omega_r(\omega_x r_y - \omega_y r_x) = 0 \end{cases}$$

We now seek equilibrium configurations characterized by ω_x and ω_y different from zero.

$$\begin{cases} I_r\omega_r\omega_y r_z = 0 \\ -I_r\omega_r\omega_x r_z = 0 \\ (I_y - I_x)\omega_y\omega_x + I_r\omega_r(\omega_x r_y - \omega_y r_x) = 0 \end{cases}$$

Solution of the first two equations leads to $r_z = 0$ (I_r, ω_r, ω_x, ω_y, are all non-zero).
From the last equation we see that the rotor axis must lie on the plane in which we want a non-zero angular velocity. Adding now the gravity gradient:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_z\omega_y + I_r\dot{\omega}_r r_x + I_r\omega_r(\omega_y r_z - \omega_z r_y) = \dfrac{3Gm_t}{R^3}(I_z - I_y)c_2c_3 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z + I_r\dot{\omega}_r r_y + I_r\omega_r(\omega_z r_x - \omega_x r_z) = \dfrac{3Gm_t}{R^3}(I_x - I_z)c_1c_3 \\ I_z\dot{\omega}_z + (I_y - I_x)\omega_y\omega_x + I_r\dot{\omega}_r r_z + I_r\omega_r(\omega_x r_y - \omega_y r_x) = \dfrac{3Gm_t}{R^3}(I_y - I_x)c_2c_1 \end{cases}$$

Consider all the angular velocity components as non-zero, and analyze equilibrium:

$$\begin{cases} (I_z - I_y)\omega_z\omega_y + I_r\omega_r(\omega_y r_z - \omega_z r_y) = \dfrac{3Gm_t}{R^3}(I_z - I_y)c_2c_3 \\ (I_x - I_z)\omega_x\omega_z + I_r\omega_r(\omega_z r_x - \omega_x r_z) = \dfrac{3Gm_t}{R^3}(I_x - I_z)c_1c_3 \\ (I_y - I_x)\omega_y\omega_x + I_r\omega_r(\omega_x r_y - \omega_y r_x) = \dfrac{3Gm_t}{R^3}(I_y - I_x)c_2c_1 \end{cases}$$

This system represents equilibrium in the LVLH frame. Adopting a generic attitude parameterization, non-necessarily linearized since in principle the relative attitude can be a large rotation, we get to three equations that allow evaluating the equilibrium configurations in the LVLH frame.

$$\begin{cases} f_1(\phi, \vartheta, \psi, r_x, r_y, r_z) = 0 \\ f_2(\phi, \vartheta, \psi, r_x, r_y, r_z) = 0 \\ f_3(\phi, \vartheta, \psi, r_x, r_y, r_z) = 0 \end{cases}$$

The solution of the system of equations can lead to equilibrium configurations with the principal axes not aligned with the LVLH frame, with appropriate orientation of the rotor. This allows having Earth pointing satellites whose principal axes are not aligned with the radial direction.

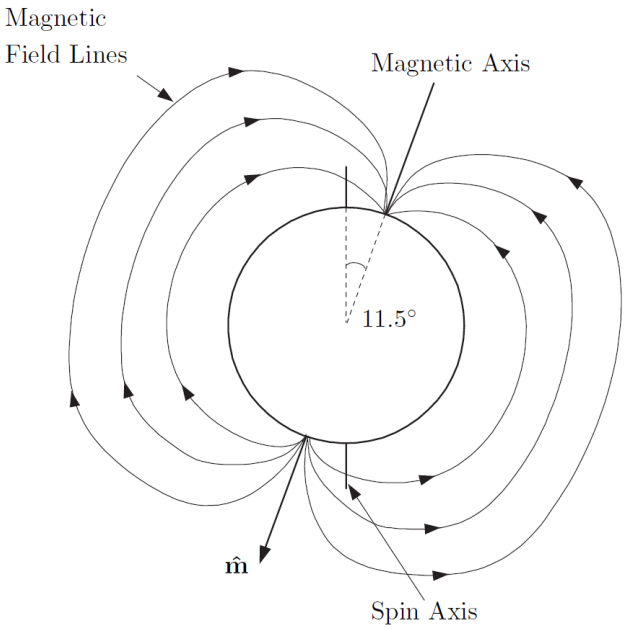
Magnetic torques

The magnetic torque is generated according to the general law $\underline{M} = \underline{m} \wedge \underline{B}$ where $\underline{m}$ is the residual magnetic induction due to parasitic currents in the satellite and $\underline{B}$ is the Earth's magnetic field.

Normally, m is an undesired effect, while B is always somehow present. There are models that, given the satellite position, allow evaluating its components. Magnetic torques on a satellite therefore do not depend on its inertia properties but rather on its position and attitude.

Magnetic field model

The Earth's magnetic field is conceptually similar to that generated by a dipole inclined with respect to the Earth's axis by about 11.5 degrees. It is not a fixed field, it slowly rotates and decreases its amplitude in time. The field is more intense at the poles, along the equator its intensity is about half of the polar value, and decreases with the inverse cubic power of the distance to the ground.
A uniform magnetic field would generate sinusoidal iso-intensity curves on the ground, but this is not the case due to perturbations. The magnetic field can be considered similar to a dipole field only above 7000 km from ground. For distances from ground beyond 10 Earth radii (more than the Geosynchronous distance) the magnetic field is deformed by the effect of the solar radiation.



The magnetic field B can be modeled in precise way as the gradient of a scalar potential V, that is B=-∇V, where V is the potential function, normally modeled as a series expansion of spherical harmonics

$$V(\mathrm{r},\theta,\varphi) = R\sum_{n=1}^{k}\left(\frac{R}{r}\right)^{n+1}\sum_{m=0}^{n}\left(g_n^m\cos(\mathrm{m}\varphi) + h_n^m\sin(\mathrm{m}\varphi)\right)P_n^m(\theta)$$

R is the Earth's equatorial radius (6371.2 km, IGRF standard), g_n^m and h_n^m are called Gaussian (available in tabular form), r, θ and φ are the spherical coordinates of the position of the satellite (distance from the center of the Earth, colatitude and East longitude from Greenwich). Coefficients g and h are evaluated from experimental data and are known up to order 13. These coefficients are subject to time variation. The following table reports as example the coefficients for years 1995 and 2000, expressed in nT.

		IGRF 1995		IGRF 2000	
n	m	g_n^m	h_n^m	g_n^m	h_n^m
1	0	-29682	-	-29615	-
1	1	-1789	5318	-1728	5186
2	0	-2197	-	-2267	-
2	1	3074	-2356	3072	-2478
2	2	1685	-425	1672	-458
3	0	1329	-	1341	-
3	1	-2268	-263	-2290	-227
3	2	1249	302	1253	296
3	3	769	-406	715	-492
4	0	941	-	935	-
4	1	782	262	787	272
4	2	291	-232	251	-232
4	3	-421	98	-405	119
4	4	116	-301	110	-304

		IGRF 1995		IGRF 2000	
n	m	g_n^m	h_n^m	g_n^m	h_n^m
5	0	-210	-	-217	-
5	1	352	44	351	44
5	2	237	157	222	172
5	3	-122	-152	-131	-134
5	4	-167	-64	-169	-40
5	5	-26	99	-12	107
6	0	66	-	72	-
6	1	64	-16	68	-17
6	2	65	77	74	64
6	3	-172	67	-161	65
6	4	2	-57	-5	-61
6	5	17	4	17	1
6	6	-94	28	-91	44

		IGRF 1995		IGRF 2000	
n	m	g_n^m	h_n^m	g_n^m	h_n^m
7	0	78	-	79	-
7	1	-67	-77	-74	-65
7	2	1	-25	0	-24
7	3	29	3	33	6
7	4	4	22	9	24
7	5	8	16	7	15
7	6	10	-23	8	-25
7	7	-2	-3	-2	-6
8	0	24	-	25	-
8	1	4	12	6	12
8	2	-1	-20	-9	-22
8	3	-9	7	-8	8
8	4	-14	-21	-17	-21
8	5	4	12	9	15

		IGRF 1995		IGRF 2000	
n	m	g_n^m	h_n^m	g_n^m	h_n^m
8	6	5	10	7	9
8	7	0	-17	-8	-16
8	8	-7	-10	-7	-3
9	0	4	-	5	-
9	1	9	-19	9	-20
9	2	1	15	3	13
9	3	-12	11	-8	12
9	4	9	-7	6	-6
9	5	-4	-7	-9	-8
9	6	-2	9	-2	9
9	7	7	7	9	4
9	8	0	-8	-4	-8
9	9	-6	1	-8	5

The components of $\underline{B}$ are then

$$B_r = \frac{-\partial V}{\partial r}$$

$$B_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta}$$

$$B_\varphi = \frac{-1}{r\sin\theta}\frac{\partial V}{\partial \varphi}$$

where B_r represents the radial component, positive outward, B_θ the coelevation component, positive if directed towards south, and B_φ the azimuth component, positive towards east. Coefficients g_n^m and h_n^m are evaluated in the assumption that polynomials P_n^m are normalized according to (Schmidt)

$$\int_0^{\pi} \left[P_n^m(\theta)\right]^2 \sin\theta \, d\theta = \frac{2\left(2-\delta_m^0\right)}{2n+1}$$

where $\delta_m{}^0$ is the Kronecker delta, $\delta_i{}^j$=1 if i=j, =0 if i≠j. In alternative, a normalization due to Gauss is possible, according to which

$$P_n^m = S_{n,m} P^{n,m}$$

with

$$S_{n,m} = \left[\frac{(2-\delta_m^0)(n-m)!}{(n+m)!}\right]^{\frac{1}{2}} \frac{(2n-1)!!}{(n-m)!}$$

Coefficients $S_{n,m}$ are independent from r, θ and φ and can be evaluated from the Gaussian coefficients $g_n{}^m$ and $h_n{}^m$, following the rule:

$$g^{n,m} = S_{n,m} g_n^m$$
$$h^{n,m} = S_{n,m} h_n^m$$
$$S_{0,0} = 1$$
$$S_{n,0} = S_{n-1,0} \frac{(2n-1)}{n} \quad \text{for } n \geq 1$$
$$S_{n,m} = S_{n,m-1} \left[\frac{(\delta_m^1+1)(n-m+1)}{(n+m)}\right]^{\frac{1}{2}} \quad \text{for } m \geq 1$$

In a similar way it is possible to get recursive formulas for polynomials $P^{n,m}$

$$P^{0,0} = 1$$
$$P^{n,n} = \sin\theta \, P^{n-1,n-1}$$
$$P^{n,m} = \cos\theta \, P^{n-1,m} - K^{n,m} P^{n-2,m}$$

with

$$K^{n,m} = 0 \quad \text{for } n = 1$$
$$K^{n,m} = \frac{(n-1)^2 - m^2}{(2n-1)(2n-3)} \quad \text{for } n > 1$$

With these models, the components of magnetic field <u>B</u> are

$$B_r = \frac{-\partial V}{\partial r} = \sum_{n=1}^{k} \left(\frac{R}{r}\right)^{n+2} (n+1) \sum_{m=0}^{n} \left(g^{n,m}\cos(m\varphi) + h^{n,m}\sin(m\varphi)\right) P^{n,m}(\theta)$$

$$B_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta} = -\sum_{n=1}^{k} \left(\frac{R}{r}\right)^{n+2} \sum_{m=0}^{n} \left(g^{n,m}\cos(m\varphi) + h^{n,m}\sin(m\varphi)\right) \frac{\partial P^{n,m}(\theta)}{\partial \theta}$$

$$B_\varphi = \frac{-1}{r\sin\theta}\frac{\partial V}{\partial \varphi} = \frac{-1}{\sin\theta}\sum_{n=1}^{k} \left(\frac{R}{r}\right)^{n+2} \sum_{m=0}^{n} m\left(-g^{n,m}\sin(m\varphi) + h^{n,m}\cos(m\varphi)\right) P^{n,m}(\theta)$$

In a geocentric inertial reference, the components of <u>B</u> are then

$$B_1 = \left(B_r\cos(\delta) + B_\theta\sin(\delta)\right)\cos(\alpha) - B_\varphi\sin(\alpha)$$

$$B_2 = \big(B_r\cos(\delta) + B_\theta\sin(\delta)\big)\sin(\alpha) + B_\varphi\cos(\alpha)$$
$$B_3 = \big(B_r\sin(\delta) - B_\theta\cos(\delta)\big)$$

where δ is the satellite declination (δ=π/2-θ) and α is the right ascension, linked to the longitude φ by the relation $\alpha=\varphi+\alpha_G$, where α_G is the Greenwich meridian right ascension.

Dipole model

As the distance from ground increases, the magnetic field can be more and more approximated to a dipole, obtained by taking only the first term of the series expansion.

$$V(\mathrm{r},\theta,\varphi) = \frac{R^3}{r^2}\left[g_1^0 P_1^0(\theta) + (g_1^1\cos(\varphi) + h_1^1\sin(\varphi)P_1^1(\theta)\right] =$$
$$= \frac{1}{r^2}\left[g_1^0 R^3\cos(\theta) + g_1^1 R^3\cos(\varphi)\sin(\theta) + h_1^1 R^3\sin(\varphi)\sin(\theta)\right]$$

We then have:

$$B_r = 2\left(\frac{R}{r}\right)^3\left[g_1^0\cos(\theta) + g_1^1\cos(\varphi)\sin(\theta) + h_1^1\sin(\varphi)\sin(\theta)\right]$$
$$B_\theta = \left(\frac{R}{r}\right)^3\left[g_1^0\sin(\theta) - g_1^1\cos(\varphi)\cos(\theta) - h_1^1\sin(\varphi)\cos(\theta)\right]$$
$$B_\phi = \left(\frac{R}{r}\right)^3\left[g_1^1\sin(\varphi) - h_1^1\cos(\varphi)\right]$$

An alternative form considers the vector $\underline{B}$ as a dipole of given amplitude and orientation, both evaluated from the previous expression, so that

$$H_0 = \left((g_1^0)^2 + (g_1^1)^2 + \left(h_1^1\right)^2\right)^{1/2}$$
$$\underline{B}(\underline{r}) = \frac{R^3 H_0}{r^3}\left[3(\underline{\hat{m}}\cdot\underline{\hat{r}})\underline{\hat{r}} - \underline{\hat{m}}\right] \quad (\text{^indicates unit vector})$$

where $\underline{r}$ is the position vector at the point where vector $\underline{B}$ is evaluated. In the geocentric inertial frame, the components of B can be calculated starting from a unit dipole vector, as

$$\hat{m} = \begin{Bmatrix}\sin\theta'_m\cos\alpha_m \\ \sin\theta'_m\sin\alpha_m \\ \cos\theta'_m\end{Bmatrix} \quad \text{with } \alpha_m = \alpha_{G0} + \frac{d\alpha_G}{dt}(t - t_0) + \phi'_m$$

where α_{G0} is the right ascension of the Greenwich meridian at the reference time t_0, $d\alpha_G/dt$ the average rotation speed of the Earth, θ'_m and φ'_m give the orientation of the magnetic dipole. Then we have

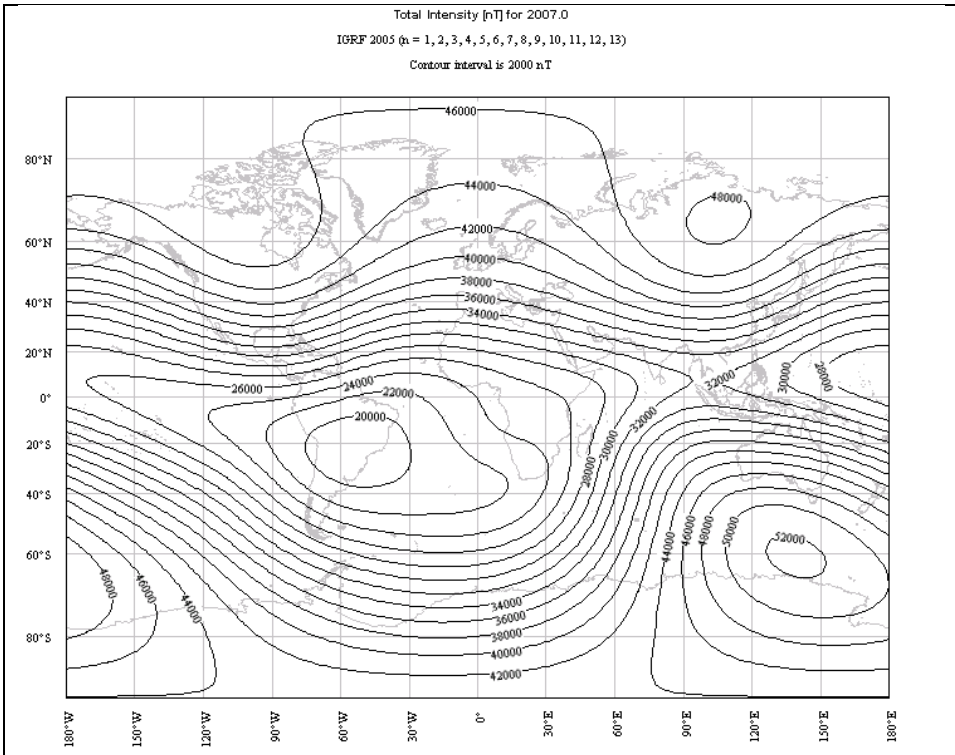
$$\theta'_m = \mathrm{acos}\left(\frac{g_1^0}{H_0}\right) \quad ; \quad \phi'_m = \mathrm{atan}\left(\frac{h_1^1}{g_1^1}\right)$$
$$\underline{\hat{m}}\cdot\underline{\hat{r}} = \hat{r}_x\sin\theta'_m\cos\alpha_m + \hat{r}_y\sin\theta'_m\sin\alpha_m + \hat{r}_z\cos\theta'_m$$

As a consequence, the projection into the geocentric inertial frame gives:

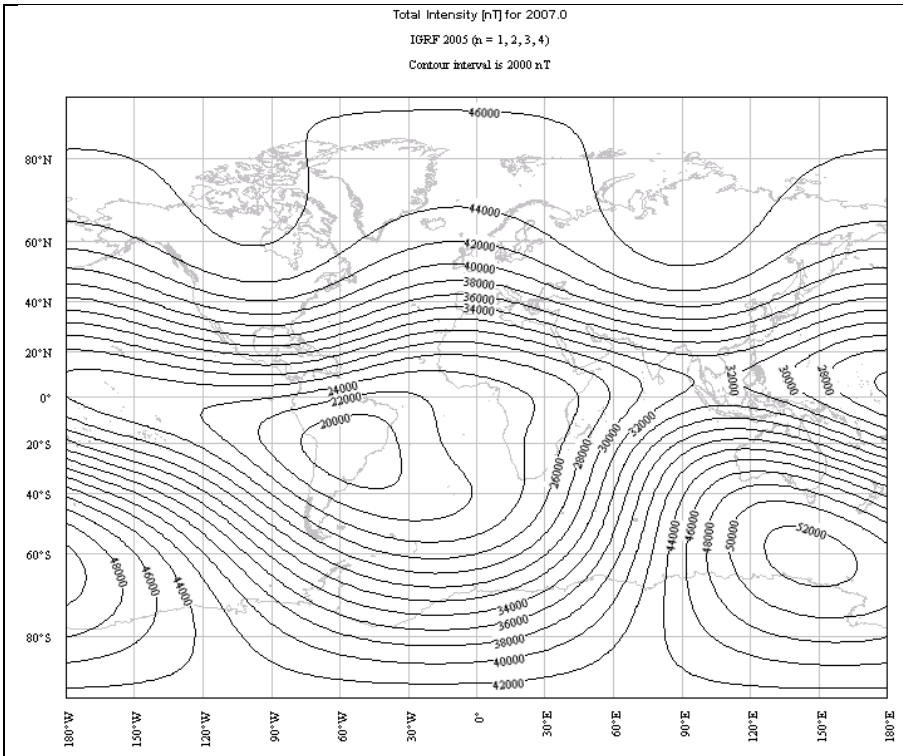
$$B_x = \frac{R^3 H_0}{r^3}\left[3(\underline{\hat{m}} \cdot \underline{\hat{r}})\hat{r}_x - \sin\theta'_m \cos\alpha_m\right]$$
$$B_y = \frac{R^3 H_0}{r^3}\left[3(\underline{\hat{m}} \cdot \underline{\hat{r}})\hat{r}_y - \sin\theta'_m \sin\alpha_m\right]$$
$$B_z = \frac{R^3 H_0}{r^3}\left[3(\underline{\hat{m}} \cdot \underline{\hat{r}})\hat{r}_z - \cos\theta'_m\right]$$

Then to obtain the magnetic field as measured in the body frame we use $\underline{B}_B = A_{B/N}\underline{B}_N$.

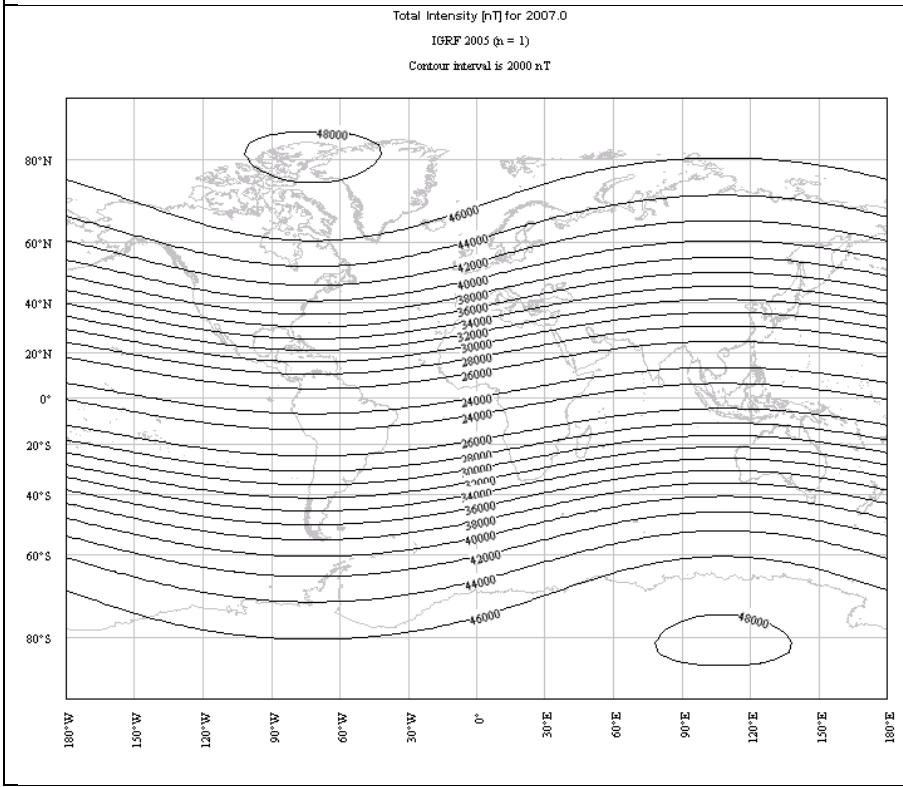
Next follow some figures reporting the contour plots of the magnitude of the magnetic field for different altitudes from ground and for different orders of expansion of the magnetic field model. The images have been created using a software developed by the Finnish Meteorological Institute (http://www.ava.fmi.fi/MAGN/igrf/applet.html). It can be noticed that, as height from ground increases, the minimum expansion order that well approximates the true field decreases. At low altitude a fourth order expansion might not be sufficient, while above 20000 km even the dipole model can be acceptable.



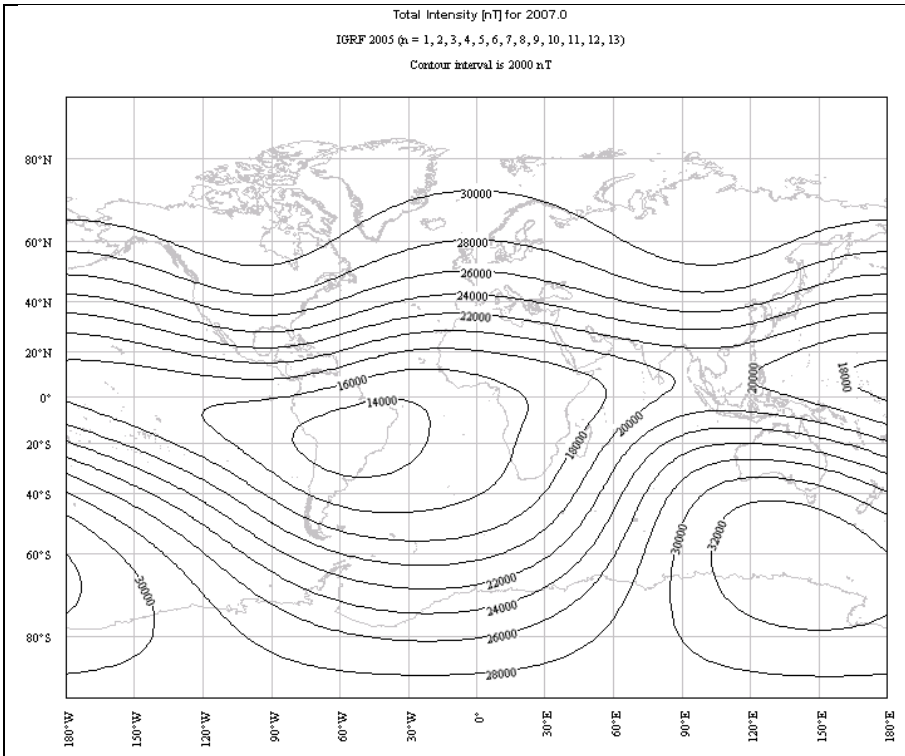
Height = 500 km
Order 13



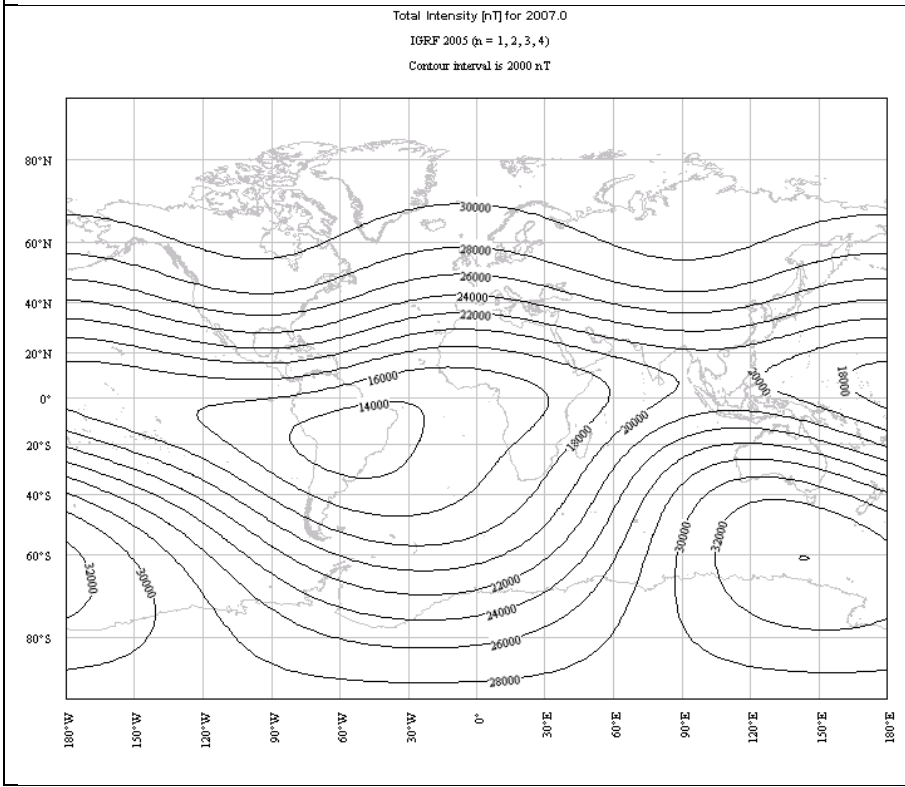
Height = 500 km
Order 4



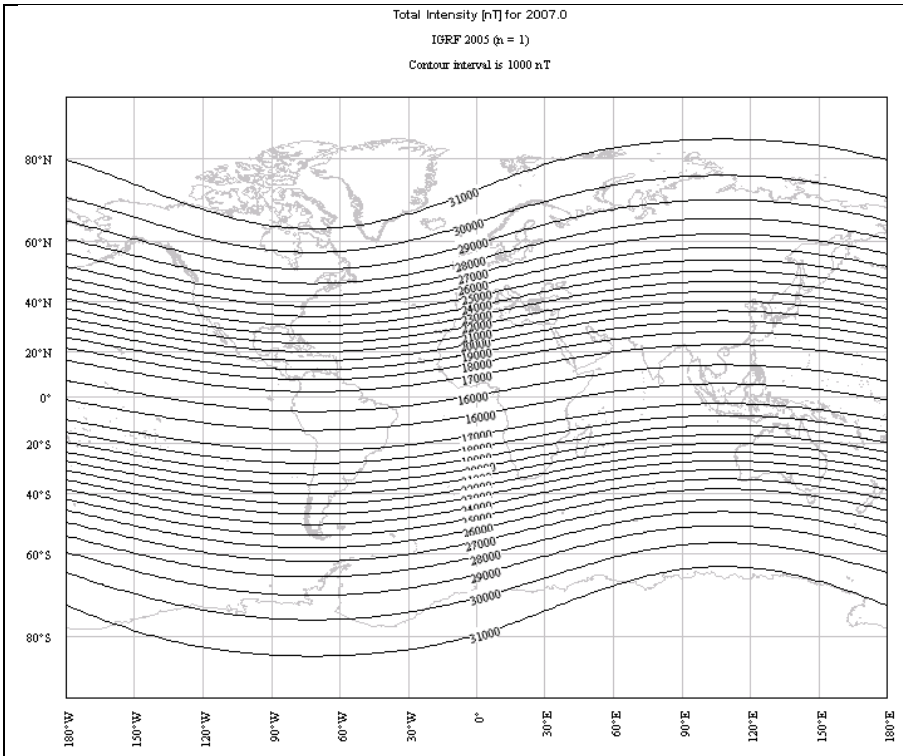
Height = 500 km
Dipole model



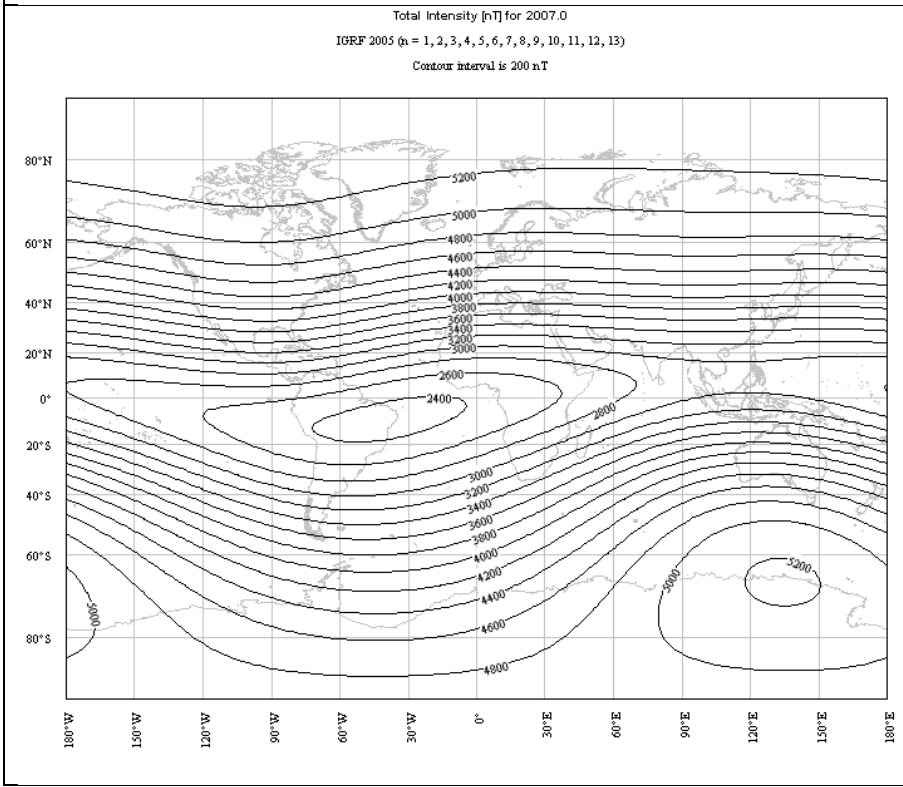
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Order 13



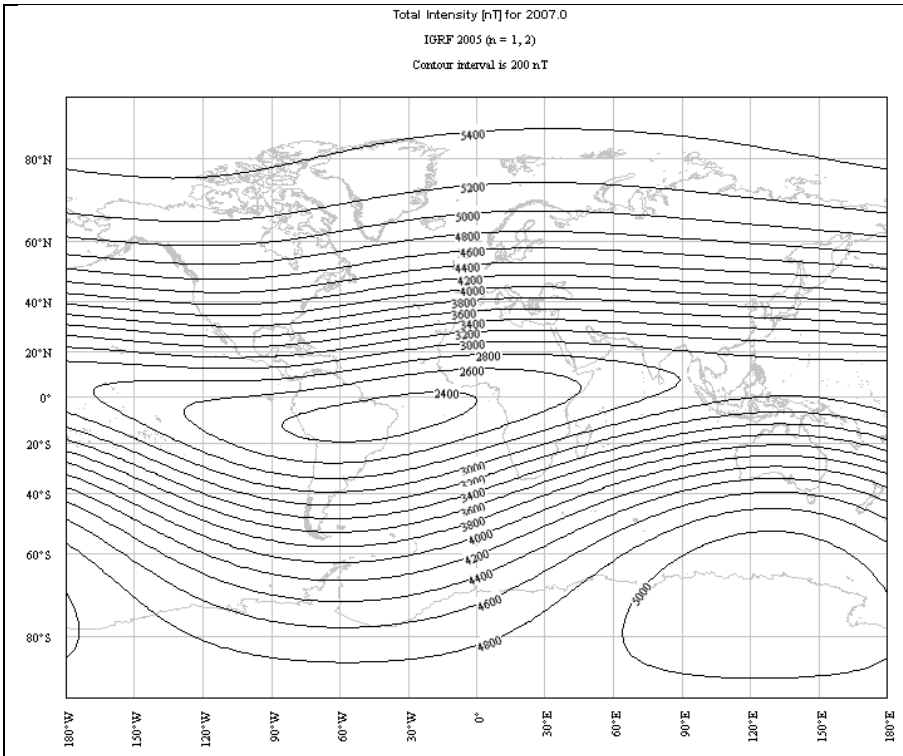
Height = 1500 km
Order 4



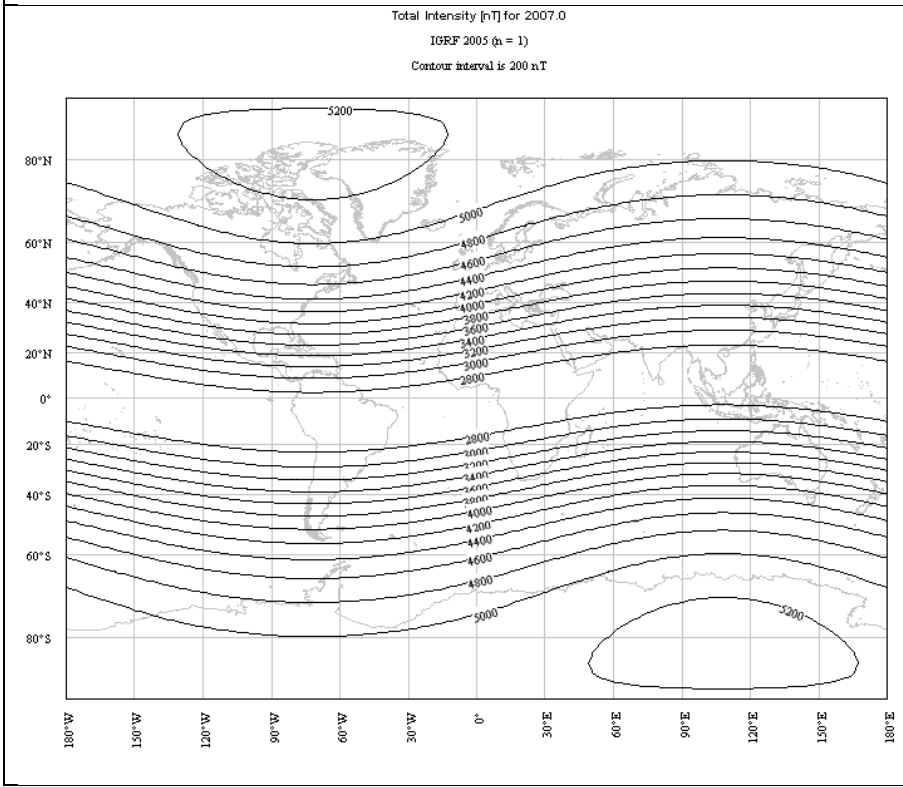
Height = 1500 km
Dipole model



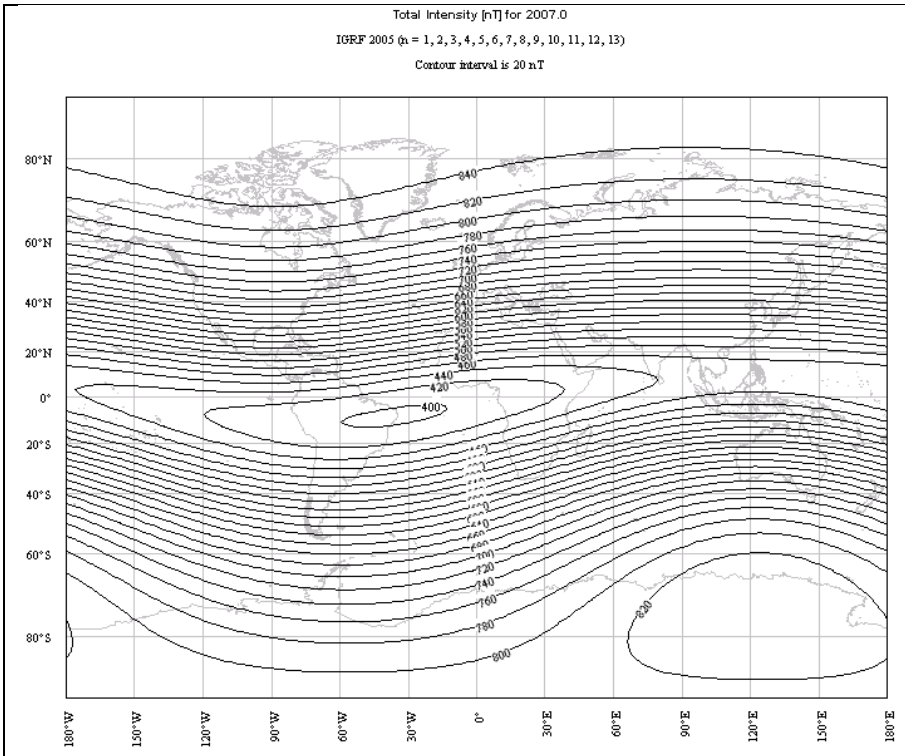
Height = 8000 km
Order 13



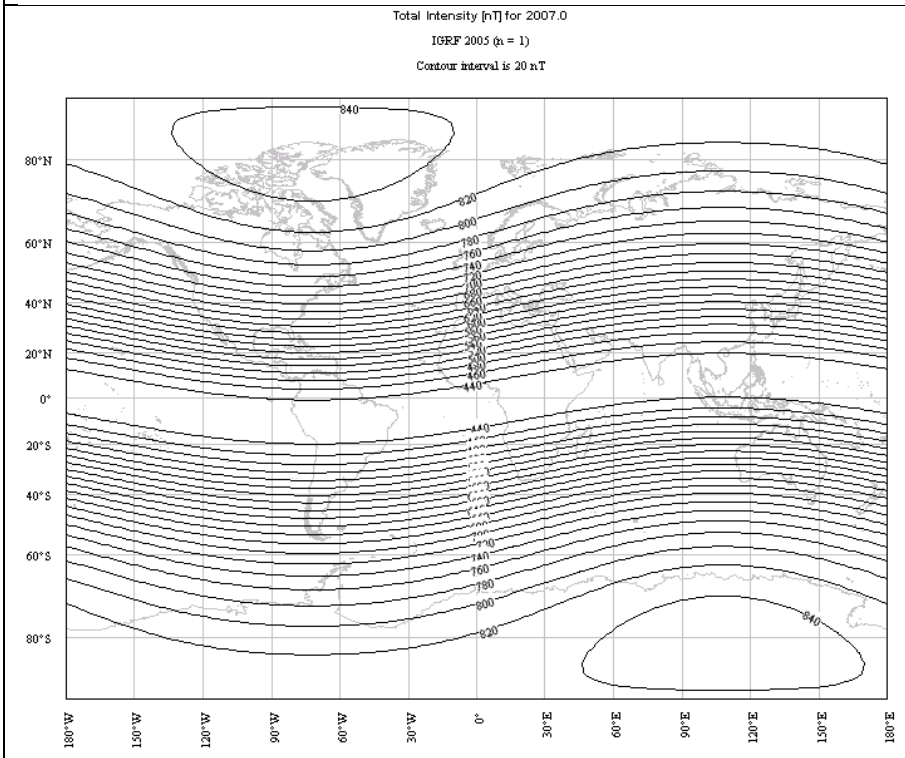
Height = 8000 km
Order 2



Height = 8000 km
Dipole model



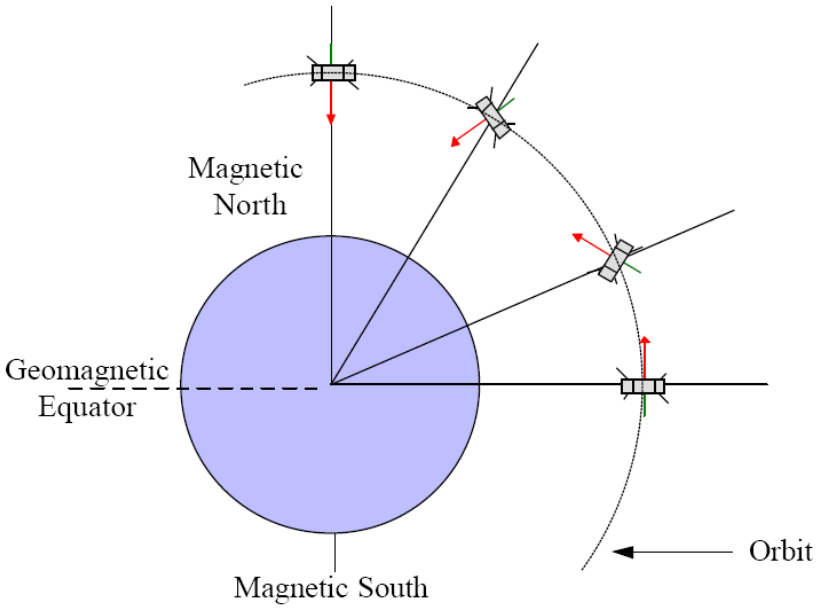
Height = 20000 km
Order 13



Height = 20000 km
Dipole model

Uses and implications of Earth's magnetic field for attitude dynamics and control

A simple low cost passive stabilization uses a permanent magnet for coarse Earth pointing over the polar regions.



The Earth's magnetic field has several implications for attitude determination and control. Firstly, the interaction between local magnetic fields on the spacecraft induced by the flow of current induces a torque on the spacecraft given by the formula:

$$\underline{M} = \underline{m} \wedge \underline{b}_B$$

$\underline{m}$ is the residual magnetic induction due to currents in the satellite. Units of $\underline{m}$ can be stated as $Nm/T = Am^2$. The magnetic torque induced on a spacecraft can be an undesired effect due to parasitic magnetic induction. The parasitic induction is difficult to model in simulation, but we can use an average constant value of $\underline{m} = [0.1 \quad 0.1 \quad 0.1]^T$ which represents quite an extreme case. However, when designing our controls, it ensures that we can test them against a worst case scenario. A more precise estimate of the spacecraft dipole moment $\underline{m}$ can be evaluated adopting the guidelines indicated by NASA in its report NASA SP-8018, summarized in the following two tables.

Table I.–*Criteria for Magnetic Properties Control*

	Class I	Class II	Class III
Design	Formal specification on magnetic properties control; approved materials and parts lists; cancellation of moments by preferred mounting arrangements and control of current loops.	Advisory specifications and guidelines for material and parts selection. Avoidance of "soft" magnetic materials or current loops and awareness of good design practices.	Nominal control over current loops; guidelines for avoidance of "soft" magnetic materials.
Quality control	Complete magnetic inspection of parts and testing of subassemblies.	Inspection or test of suspect parts.	Test of subassemblies that are potentially major sources of dipole moment.
Test and compensation	Deperming either at subassembly or spacecraft level; test of final spacecraft assembly and compensation if required.	Deperming and compensation frequently used.	Test and compensation optional.

Note.–Class I–Magnetic torques dominant when compared with other torques.
II–Magnetic torques comparable to other torques.
III–Magnetic torques insignificant when compared with other torques.

Table IV.–*Factors for Estimating Spacecraft Dipole Moment (M).*

Category of magnetic properties control (see table I)	Estimate of dipole moment per unit mass for nonspinning spacecraft		Estimate of dipole moment per unit mass for spinning spacecraft	
	$A\text{-}m^2/kg$	(pole-cm/lb)	$A\text{-}m^2/kg$	(pole-cm/lb)
Class I	1×10^{-3}	(0.45)	0.4×10^{-3}	(0.18)
Class II	3.5×10^{-3}	(1.6)	1.4×10^{-3}	(0.63)
Class III	10×10^{-3} and higher	(4.5)	4×10^{-3} and higher	(1.8)

Torque due to atmospheric drag

The interaction between satellite and upper atmosphere generates aerodynamic forces, that in turn can generate a torque about the center of mass. Aerodynamic torques represent the major disturbance for satellites orbiting at altitudes lower than 400 Km.
To have an approximated mathematical model of the forces, we assume that air particles hit the external surface of the satellite and their kinetic energy is totally transferred to the satellite. The aerodynamic force acting on the elementary area dA defined by its perpendicular direction $\hat{N}$ is:

$$d\bar{F}_i = \frac{1}{2} C_D \cdot \rho \cdot v^2 \left(\hat{N} \bullet \bar{v}\right) \bar{u}_v \cdot dA$$

where ρ is the air density, $\bar{v}$ is the relative velocity and C_D is the drag coefficient that is evaluated experimentally. It is a constant parameter that depends on the type of surface and on the local angle of attack.

The aerodynamic torque is then:

$$\bar{M}_{aero} = \int_S \bar{r}_i \Lambda d\bar{F}_i.$$

The integral is evaluated for all the surfaces of the satellites for which $\hat{N} \bullet \bar{v} > 0$: surfaces defined by a negative scalar product are not exposed to aerodynamic flow. The relative velocity of the elementary area dA is:

$$\bar{v} = \bar{v}_0 + \omega \Lambda \bar{r}$$

where $\bar{v}_0$ is the relative velocity of the center of mass and ω is the angular velocity of the satellite. After substitution we obtain:

$$\bar{M}_{aero} = \frac{1}{2} C_D \cdot \rho \left(v^2 \int_S \left(\hat{N} \bullet \hat{v}_0\right)(\hat{v}_0 \Lambda \bar{r}_i) dA + \bar{v}_0 \int_S \left\{ \hat{N}(\omega \Lambda \bar{r}_i)(\hat{v}_0 \Lambda \bar{r}_i) + \left(\hat{N} \bullet \hat{v}_0\right)[(\omega \Lambda \bar{r}_i) \Lambda \bar{r}_i] \right\} dA \right)$$

where the first term represents the torque due to the misalignment between satellite center of mass and center of pressure, while the second term represents the torque produced by the rotation of the satellite.
It is possible to decompose the outer surface of the satellite into a larger series of simple geometric shape, in order to evaluate the aerodynamic forces and torque as a sum of elementary terms. The force due to atmospheric drag is given by

$$\bar{F}_{a_i} = -\frac{1}{2} C_D \rho(h,t) {v_r}^2 \frac{\underline{v}_r}{\|\underline{v}_r\|} A_{cross}$$

where $\rho(h,t)$ is the air density which depends on the altitude of the orbit, $\underline{v}_r$ is the relative velocity between the satellite and the atmosphere, C_D is the drag coefficient that is evaluated experimentally ($C_D \approx 2.2, 1.5 < C_D < 2.6$ assuming a flat plate) and A_{cross} is the cross sectional surface area perpendicular to $\frac{\underline{v}_r}{\|\underline{v}_r\|}$ which can be expressed as $A_{cross} = (\underline{n}_s . \frac{\underline{v}_r}{\|\underline{v}_r\|}) A$ where $\underline{n}_s$ is the unit vector normal to the surface and A is the area of the surface. Then it is possible to write for a given surface:

$$\bar{F}_{a_i} = -\frac{1}{2} C_D \rho(h,t) {v_r}^2 \frac{\underline{v}_r}{\|\underline{v}_r\|} (\bar{n}_s . \frac{\underline{v}_r}{\|\underline{v}_r\|}) A$$

Each surface evaluated for $\bar{n}_{si} . \frac{\underline{v}_r}{\|\underline{v}_r\|} > 0$
Then the total torque is given by

$$M = \sum \bar{r}_i \wedge \bar{F}_{a_i}$$

where $\bar{r}_i$ is the position vector from the centre of mass of the spacecraft to the centre of pressure of the surface A_i. Recall, to model the torque in the body frame the above equations must be evaluated in body coordinates. For example, evaluating the relative velcoity $\underline{v}_r$ with respect to the inertial frame we have:

$$\underline{v}_r = \underline{v}_o - \underline{\omega}_\oplus \times \underline{R}$$

Where $\underline{v}_o$is the velocity of the spacecraft in the inertial frame, $\underline{R}$is the position of the spacecraft and $\underline{\omega}_\oplus$ is the angular rotation of the Earth (assuming the atmosphere rotates with the Earth). In Cartesian coordinates and assuming an equatorial orbit we could write:

$$\underline{v}_r = \begin{bmatrix} \dot{x} + \omega_{\oplus} y \\ \dot{y} - \omega_{\oplus} x \\ \dot{z} \end{bmatrix}$$

where $\omega_{\oplus} = 0.000072921 rad/sec$. However, this must be expressed in the body frame such that

$$\underline{F}_i^b = -\frac{1}{2} C_D \rho(h,t) {v_r}^2 \frac{\underline{v}_r^b}{\left\|\underline{v}_r^b\right\|} (\hat{\underline{n}}_{si}^b . \frac{\underline{v}_r^b}{\left\|\underline{v}_r^b\right\|}) A_i$$

where $\underline{v}_r^b = A_{B/N} \underline{v}_r$. Furthermore, the normal to the surfaces $\hat{\underline{n}}_{si}^b$ in the body frame. However, these are fixed with respect to the body frame and so are constant values. For example, a simple cuboid spacecraft, with the principal axis aligned with the normal to the 6 surfaces would be:

$$\hat{\underline{n}}_{s1}^b = [1 \quad 0 \quad 0]^T, \hat{\underline{n}}_{s2}^b = [0 \quad 1 \quad 0]^T, \hat{\underline{n}}_{s3}^b = [0 \quad 0 \quad 1]^T$$
$$\hat{\underline{n}}_{s4}^b = [-1 \quad 0 \quad 0]^T, \hat{\underline{n}}_{s5}^b = [0 \quad -1 \quad 0]^T, \hat{\underline{n}}_{s6}^b = [0 \quad 0 \quad -1]^T$$

To perform all the calculations it is necessary to evaluate correctly the aerodynamic coefficient and the air density as a function of the distance from Earth. To make the simplified expression compatible with the experimental data, the air density can be made dependent upon the altitude as well as other parameters, such as temperature, solar activity (F10.7 constant), surface and material properties, and for ease of use the drag coefficient is taken as the flat plate drag coefficient. One example of tabular data of atmospheric density as a function of altitude is reported here below.

altit. (km)	10	20	30	40	50	60	70	80	90	100
density (g/cm^3)	4,02e-04	8,34e-05	1,57e-05	3,18e-06	8,37e-07	2,33e-07	5,86e-08	1,40e-08	2,99e-09	5,17e-10
altit. (km)	110	120	130	140	150	160	170	180	190	200
density (g/cm^3)	8,42e-11	1,84e-11	7,36e-12	3,78e-12	2,19e-12	1,37e-12	9,00e-13	6,15e-13	4,32e-13	3,10e-13
altit. (km)	210	220	230	240	250	260	270	280	290	300
density (g/cm^3)	2,27e-13	1,68e-13	1,26e-13	9,58e-14	7,35e-14	5,68e-14	4,43e-14	3,48e-14	2,75e-14	2,18e-14
altit. (km)	310	320	330	340	350	360	370	380	390	400
density (g/cm^3)	1,74e-14	1,40e-14	1,13e-14	9,10e-15	7,39e-15	6,02e-15	4,92e-15	4,03e-15	3,31e-15	2,72e-15
altit. (km)	410	420	430	440	450	460	470	480	490	500
density (g/cm^3)	2,25e-15	1,86e-15	1,54e-15	1,28e-15	1,07e-15	8,89e-16	7,43e-16	6,22e-16	5,22e-16	4,39e-16
altit. (km)	510	520	530	540	550	560	570	580	590	600
density (g/cm^3)	3,71e-16	3,13e-16	2,66e-16	2,26e-16	1,93e-16	1,65e-16	1,41e-16	1,22e-16	1,05e-16	9,14e-17
altit. (km)	610	620	630	640	650	660	670	680	690	700
density (g/cm^3)	7,96e-17	6,97e-17	6,12e-17	5,41e-17	4,79e-17	4,27e-17	3,82e-17	3,44e-17	3,10e-17	2,82e-17
altit. (km)	710	720	730	740	750	760	770	780	790	800
density (g/cm^3)	2,57e-17	2,35e-17	2,16e-17	1,99e-17	1,84e-17	1,71e-17	1,59e-17	1,49e-17	1,39e-17	1,31e-17
altit. (km)	810	820	830	840	850	860	870	880	890	900
density (g/cm^3)	1,23e-17	1,16e-17	1,10e-17	1,04e-17	9,86e-18	9,37e-18	8,91e-18	8,48e-18	8,09e-18	7,72e-18
altit. (km)	910	920	930	940	950	960	970	980	990	1000
density (g/cm^3)	7,37e-18	7,04e-18	6,74e-18	6,45e-18	6,18e-18	5,92e-18	5,68e-18	5,44e-18	5,22e-18	5,02e-18

Aerodynamic torques are never used to control the satellite rotation, and their effect is totally negligible above 700 km altitude. Aerodynamic torques represent the major disturbance for satellites orbiting at altitudes lower than 400 Km.

Solar radiation torque

Solar radiation illuminating the outer surface of a satellite generates a pressure, that in turn generates a force and a torque around the center of mass of the satellite.
The main sources of electromagnetic radiation are the direct solar radiation, the solar radiation reflected by the Earth (or by any other planet) and the radiation directly emitted by the Earth. Solar radiation intensity varies with the inverse square distance from the source, so that direct solar radiation is almost constant for geocentric orbits, independent from the orbit radius since the distance to the Sun does not change significantly. On the contrary, the reflected radiation and the Earth radiation intensity are strongly dependent on the orbit radius.

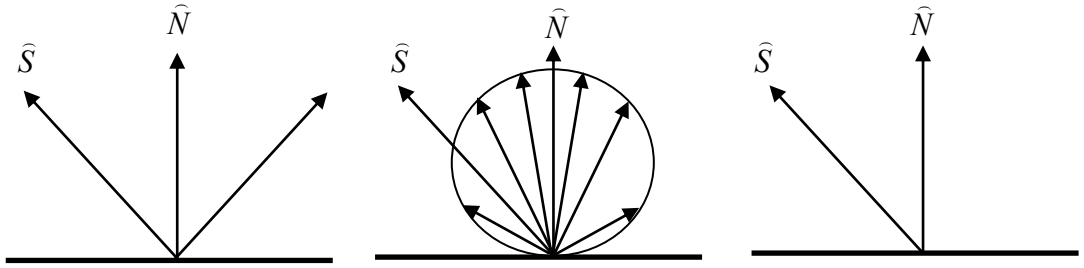
Altitude (Km)	**Direct solar radiation** (W/m^2)	**Radiation reflected by the Earth** (W/m^2)	**Earth radiation** (W/m^2)
500	1358	600	150
1000	1358	500	117
2000	1358	300	89
4000	1358	180	62
8000	1358	75	38
15000	1358	30	14
30000	1358	12	3
60000	1358	7	2

The average pressure due to radiation can be evaluated as

$$P = \frac{F_e}{c}$$

where c is the speed of light and F_e is the power per unit surface.

Radiation forces can be modeled assuming that part of the incident radiation is absorbed, part is reflected specularly and part reflected with diffusion.



Specular reflection Diffuse reflection Absorption

Therefore the three components of the resulting force acting on a surface element dA having orthogonal direction $\hat{N}$ will be:

$$\begin{cases} d\bar{F}_{assorbita} = -Pc_a \cos\vartheta \cdot \hat{S} dA & (0 \le \vartheta \le 90°) \\ d\bar{F}_{diffusa} = Pc_d \left(-\frac{2}{3}\cos\vartheta \cdot \hat{N} - \cos\vartheta \cdot \hat{S}\right) dA & (0 \le \vartheta \le 90°) \\ d\bar{F}_{speculare} = -2Pc_s \cos^2\vartheta \cdot \hat{N} dA & (0 \le \vartheta \le 90°) \end{cases}$$

with $\hat{S}$ denoting the unit direction of the satellite-Sun vector and ϑ is the angle between the surface orthogonal direction and direction $\hat{S}$.
When ϑ is greater than 90°, so that $\cos\vartheta$ is negative, the surface is not illuminated and not subject to radiation forces.
Coefficients c_a, c_d, c_s are respectively the coefficient of absorption, diffuse reflection and specular reflection, and are constrained by the following equality:

$$c_a + c_d + c_s = 1$$

Merging the three components due to the three possible effects, the total force due to radiation is evaluated as:

$$d\bar{F}_i = -P \int_S \left[(1 - c_s)\hat{S} + 2\left(c_s \cos\vartheta + \frac{1}{3}c_d\right)\hat{N}\right] \cos\vartheta \cdot dA$$

The torque due to radiation will then be:

$$\bar{M}_{sol} = \int_S \bar{r}_i \Lambda d\bar{F}_i$$

where the integral has to be evaluated for all surfaces illuminated by radiation. In many cases it is advisable to decompose the external surface into several simple planar surfaces, so that the resulting torque will be the sum of the contributions of each surface element:

$$\bar{M}_{sol} = \sum_{i=1}^{N} \bar{r}_i \Lambda \bar{F}_i.$$

Taking the case of a flat panel of area A, the resulting force is

$$\bar{F}_i = -A \cdot P \left[(1 - c_s)\hat{S} + 2\left(c_s \cos\vartheta + \tfrac{1}{3}c_d\right)\hat{N}\right] \cos\vartheta \qquad \vartheta = \cos^{-1}(\hat{S} \cdot \hat{N}).$$

In the case of a cylindrical surface, the effects can be divided into forces acting on the base of the cylinder and forces acting on the lateral surface.

$$F_{lateral} = PA \sin\vartheta \left\{\left[\left(1 + \frac{1}{3}c_s\right) + \frac{\pi}{6}c_d\right] \cdot \hat{S} + \left[-\frac{4}{3}c_s - \frac{\pi}{6}c_d\right] \cdot \cos\vartheta \cdot \hat{N}\right\}$$

$$F_{base} = PA \cos\vartheta \left[(1 - c_s) \cdot \hat{S} + 2\left(c_s \cos\vartheta + \frac{1}{3}c_d\right) \cdot \hat{N}\right]$$

The relative position of the Sun with respect to the spacecraft measured in the inertial frame is given as $\hat{S}_N$ and in the body frame is given as:

$$\hat{S}_b = A_{B/N}\hat{S}_N$$

Then the torque is given as

$$\underline{T}_{SRP} = \begin{cases} \sum_{i=1}^{n} \underline{r}_i \Lambda \underline{F}_i & \hat{\underline{S}}_b \cdot \hat{\underline{n}}_s^b > 0 \\ 0 & \hat{\underline{S}}_b \cdot \hat{\underline{n}}_s^b < 0 \end{cases}$$

where $\underline{r}_i$is the position of the centre of pressure of the flat surface with respect to the centre of mass. As before a simple cuboid geometry with the principal axis aligned with the normal to the surfaces can be evaluated using:

$$\hat{\underline{n}}_{s1}^b = [1 \quad 0 \quad 0]^T, \hat{\underline{n}}_{s2}^b = [0 \quad 1 \quad 0]^T, \hat{\underline{n}}_{s3}^b = [0 \quad 0 \quad 1]^T$$
$$\hat{\underline{n}}_{s4}^b = [-1 \quad 0 \quad 0]^T, \hat{\underline{n}}_{s5}^b = [0 \quad -1 \quad 0]^T, \hat{\underline{n}}_{s6}^b = [0 \quad 0 \quad -1]^T$$

Suggested readings

- M.H. Kaplan: *Modern Spacecraft Dynamics and Control*, Ed. Wiley & Sons.
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- K.J. Ball, G.F. Osborne: *Space Vehicle Dynamics*, Clarendon Press.
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- F.P.J. Rimrott: *Introductory Attitude Dynamics*, Springer-Verlag.
- M.J. Sidi: *Spacecraft dynamics and control: a practical engineering approach*, Cambridge University Press.
- B. Wie, *Space Vehicle Dynamics and Control*, AIAA Education Series.
- F. Landis Markley, J.L. Crassidis, *Fundamentals of Spacecraft attitude determination and control*, Space Technology Library, Springer.

Course Notes

Spacecraft Attitude Dynamics and Control

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Part 3:
Introduction to state-space control and attitude control

Note for the reader

These short notes are in support of the course “Spacecraft attitude dynamics”, they are not intended to replace any textbook. Interested readers are encouraged to consult also printed textbooks and archival papers.

Franco Bernelli Zazzera

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Introduction

The main tasks of a control system are:
- to stabilise an unstable system;
- to increase the stability of a stable system but with a rather slow transient.

It is always very important, when studying any system, to be able to identify the quantities (states or measurements) that are of interest. Especially since such quantities are often not directly measurable but must be deduced from a smaller number of parameters. We must therefore identify the system's inputs and outputs. It can be seen that in general we need to deal with Multi-Input-Multi-Output (MIMO) systems. The main difficulty encountered in moving from the analysis of Single-Input-Single-Output (SISO) systems to MIMO systems is not in the number of inputs and outputs but in moving from 1-input/output to multiple inputs/outputs. Extending the analysis from a 2-input/output system to an n-m-input/output system is then relatively simple.

We have therefore somewhat defined
- the variables to be controlled: i.e. what we have called the state of the system;
- the control variables: what we have called the input of the system.

It is now necessary to have a mathematical model of the physical system. In the most general case you will have non-linear differential equations. For a non-linear system, it is not possible to calculate the transfer function. We will try in many cases to go back to a linear type of analysis so, in the case of non-linearities, it will be necessary to perform two fundamental operations:
1) analysis of the equilibrium states;
2) linearisation of the system around an equilibrium state to obtain a linear system that can be treated analytically. It should be noted that linearisation around one of the "few" equilibrium states that a system can assume is not a reduction of control capacity because in practice one will want to control a system to keep it in a very precise equilibrium state and not to make it wander between different equilibrium states.

After having, if necessary, linearised the equations that mathematically describe the behaviour of the physical system (and bear in mind that these equations, however detailed they may be, will always be a simplification of what happens in reality), it will be possible to design the controller that deals with stabilising (or increasing the stability of) the physical system. At this point it is necessary to carry out checks on the controller designed. The steps to be followed are roughly as follows:
1) verification of operation on the linear mathematical model of the physical system;
2) verification of operation on the non-linear mathematical model of the physical system (only if it has been necessary to linearise the equations of motion);
3) verification of operation on the real system, which will also involve differences with the non-linear model, if any, that is certainly more accurate than the linear model adopted for the project.

It should be noted that designing the controller after having carried out these operations implies limits on the initial conditions that the system may have when the control actively intervenes. If the initial conditions deviate much from the considered equilibrium, the linearised model of the equations of motion will no longer be valid (i.e. it will deviate from the linear model) and, consequently, the controller we have designed will probably be ineffective in its action. It will also be our task to understand up to what values of initial conditions (for the system we are considering) the controller can still work. At this point we assume that we have designed and verified (at least analytically) the desired controller. The problem of its practical implementation arises. There are basically two methods of introducing the controller into the physical system:

1) Analogue controller;
2) Digital controller.

In the first case, it will be necessary to construct an electric circuit with resistors, capacitors, op-amps, etc., which has, as a whole, the transfer function of the controller we have analytically designed. One problem with this type of controller is that in order to modify its characteristics (this must always be taken into account), there are two different ways of doing so
- replacing certain elements from the circuit;
- introducing variable elements from the outset (such as variable resistors) in order to be able to intervene without damaging the circuit.

An alternative to analogue systems are digital control systems. These systems are much more adaptable to changing operating conditions for the simple reason that the control calculations are typically performed by software. The main problem with this type of control systems is that, due to their binary nature, they are inherently discrete and therefore require a suitable interface with the system to be controlled. In short, this means that the measurements read by the physical system and the actions exercised on the physical system by the digital controller are, of necessity, timed according to a precise schedule. In order for this to work correctly (the physical system is continuous), two converters will be needed (in reality, other elements will be needed, but for now, this is fine), one to convert the analogue quantities into digital form upstream of the controller (A/D) and one to do the reverse operation downstream of the controller (D/A), before intervening on the physical system.
Because of this different way of interacting with the system, it may happen that a controller designed according to "continuous logic" does not have the same effect when implemented in digital form. In these cases, therefore, it is more convenient to design the controller directly with methods that take into account the discretisation linked to digital control.

The important thing is to bear in mind when designing the control system that it must also be physically feasible. The controller we have designed tells us the extent of the actions that must take place according to the measurements of the outputs of the physical system. At this point we need to go into a little more detail. How are these forces applied? In general, we will have a device (actuator) which is distinct from the physical system to be controlled. This device will, in turn, be a system described analytically by a mathematical model of some kind. For now, it is sufficient to note that there may be situations in which the force required by the controller (i.e. the action which will be the input of the system to be controlled) is not available (saturation of the actuators) or arrives at the physical system with a certain delay. To be aware of the possibility of such phenomena, it is necessary to know in sufficient detail the dynamic behaviour of the device used, which we have so far assumed to be ideal (i.e. with a transient = 0 s and steady state equal to that required). It should also be noted that the dynamic behaviour of an actuator, by partly influencing the dynamics of the system to be controlled (since it will necessarily have to come into contact with the system to intervene), could also render ineffective the controller we have designed based solely on the dynamics of the physical system. The same applies to the devices (transducers) used to measure the quantities required by the controller (i.e. the system output).

Finally, one thing we must not forget is disturbances, understood as noise in the measurements taken or disturbances acting directly on the physical system. This means that the controller we have designed, apart from being strictly valid only for the linearised mathematical model of the physical system, does not consider the effects just listed. Typically, the influence of actuators and transducers and noise is introduced in the controller verification phase. As it happens, although they can all be modelled, introducing them at the controller design stage would be an unnecessary complication.

Therefore, if it is not necessary for other reasons, these elements will only be considered after the controller has been designed.

Modeling physical systems

Linearization

All (or almost all) dynamical systems can be described, in the most general case, by non-linear, time-varying equations which, symbolically, look as follows.

$$\begin{cases} \dot{x} = f(x, u, t) \\ y = g(x, u, t) \end{cases}$$

If a system has one or more equilibrium configuration, then it is possible to linearize the dynamics in the vicinity of the equilibrium state and express the system dynamics in state-space form. First the equilibrium configuration (state and control input) needs to be evaluated by solving

$$0 = f(\bar{x}, \bar{u})$$

Then, the state and control matrices of the state-space model can be evaluated as

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\bar{x} \\ u=\bar{u}}}; \quad B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=\bar{x} \\ u=\bar{u}}}$$

Typically, we will act on dynamic systems described by equations of the following type.

$$\begin{cases} \dot{x} = A \cdot x + B \cdot u \\ y = C \cdot x + D \cdot u \end{cases}$$

These equations are a major simplification of the reality of things. The two most obvious simplifications are:
- linearity of the equations;
- time invariant coefficients.

However, these equations are used very often for the following reasons:
- they are easy to solve;
- they still give good results.

In practice, it is a matter of performing a linearisation of the equations around their equilibrium point. This operation allows in fact to:
- study the stability of the equilibrium points of the system;
- evaluate the response of the system (in equilibrium) to small perturbations.

One might think of using linear time-varying equations, but this type of system is not well suited to control synthesis (i.e. designing a system to control them) and, in general, their simulation is quite complex.
To get around this problem (if it is possible), it is assumed that the variations in time (of the mathematical model of the physical system) are sufficiently slow with respect to the phenomenon being studied.
Then, as already said, one studies linear time-invariant systems that are obtained from the linearisation of a system.

It is, however, necessary to remember that, behind the equations we are using, there are much more complex equations which, in turn, cannot describe the real behavior of the system.

The importance of the effect of sensors

Let us consider again the system described in state-space. The quantities within the matrices are often measured by means of appropriate sensors. It is therefore important to remember that these sensors have their own dynamics which may or may not influence that of the physical system in which they are "introduced". Typically, they will introduce a time delay.
For example, a sensor can be described by an expression such as:

$$\begin{cases} \dot{x}_a = A_a \cdot x_a + B_a \cdot y_a \\ a = c_a \cdot x_a \end{cases}$$

There are some cases in which sensor dynamics can be neglected. It should also be borne in mind that a design that is too refined (and possibly expensive in computing resources) may not make much sense. For example, if our calculations foresee the use of an electrical resistor of 45.2279 , we will never be able to adopt these results in precise form, for the simple reason that such a resistor cannot be found on the market and it is not at all convenient to have it custom-made (also because probably already in the order of 10^{-2} the precision in the production of a resistor can be very poor).

The time response of the system

Let us suppose for now that we have the linearised equations describing the physical system we are studying, in the well-known form

$$\begin{cases} \dot{x} = A \cdot x + B \cdot u \\ y = C \cdot x + D \cdot u \end{cases}$$

What should we do now? Generally, two steps are taken:
- stability study according to Liapunov;
- study of the response.

Stability is studied with respect to initial conditions of the system. That is, we are interested in seeing whether the perturbed motion is stable or not. Remember that there are both concepts of simple stability and asymptotic stability. The general integral of a differential system, is usually composed of two parts:
- general integral of the associated homogeneous system;
- particular integral.

It is interesting to note that stability is only associated with the first part of this solution since the particular integral describes the system's transient from the perturbed initial conditions. The system is stable when the real part of the eigenvalues is negative or null. In the case where it is null, it is possible for the system to be stable or unstable depending on the multiplicity of these null eigenvalues. The system can be stable only in the case in which the multiplicity of the eigenvectors is equal to the multiplicity of the eigenvalues. The solution in time domain of the state-space system is given, in general, by an expression of the type:

$$x = \Phi \cdot x_0 + \int_0^t \Phi(t-\tau) \cdot B \cdot u(\tau) d\tau$$

where Φ (known as the transition matrix) is the solution of:

$$\dot{\Phi} = A \cdot \Phi$$
$$\rightarrow \Phi = e^{A \cdot t}$$

Substituting this solution into the state-space model, we observe that we obtain an identity, that confirms this is a solution of the initial system.

The system output is then

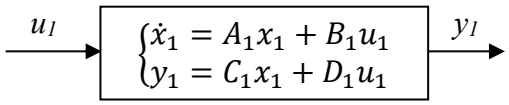
$$y = C \cdot \Phi(t) \cdot x_0 + C \int_0^t \Phi(t-\tau) \cdot B \cdot u(\tau) d\tau + D \cdot u(t)$$

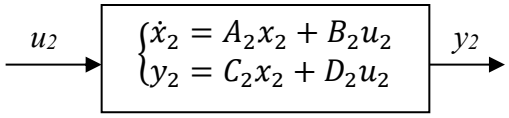
The integral present in the solution is called convolution integral. Its generic form is

$$\int f(t-\tau) g(\tau) d\tau$$

Systems connected in series, parallel, feedback

Let us now suppose that we have two systems described in the state space form:

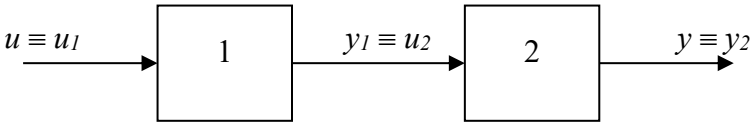




We want to see what the equations of a system made up of these two systems look like, connecting systems in three different ways:
- series;
- parallel;
- feedback.

Series

The series of the two systems is represented in the following block diagram.

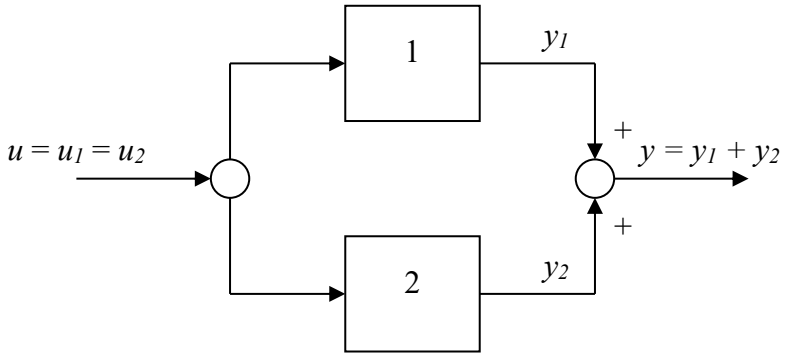


The global system obtained is represented by the following equations that describe the state-space model of two connected linear systems.

$$\begin{Bmatrix}\dot{x}_1\\ \dot{x}_2\end{Bmatrix} = \begin{bmatrix}A_1 & 0\\ B_2C_1 & A_2\end{bmatrix}\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + \begin{bmatrix}B_1\\ B_2D_1\end{bmatrix}\{u_1\}$$
$$y = [D_2C_1 \quad C_2]\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + [D_2D_1]\{u_1\}$$

Parallel

Let us now see what is achieved by linking them in parallel. The block diagram shows that they have the input in common and the output of the overall system is the sum of the two outputs.

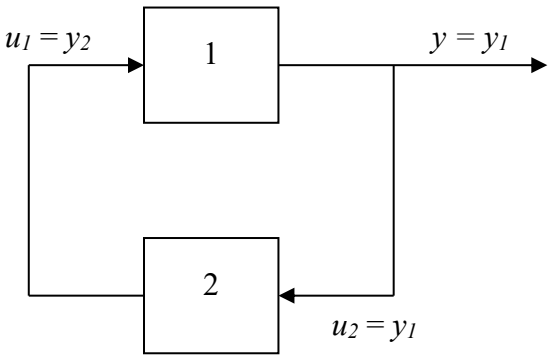


In this case the state-space model is as follows.

$$\begin{Bmatrix}\dot{x}_1\\ \dot{x}_2\end{Bmatrix} = \begin{bmatrix}A_1 & 0\\ 0 & A_2\end{bmatrix}\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + \begin{bmatrix}B_1\\ B_2\end{bmatrix}\{u\}$$
$$y = [C_1 \quad C_2]\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + [D_1 + D_2]\{u\}$$

Feedback loop

This is the typical case of control in which the output of the system to be controlled is fed back to process it and eventually intervene on the system. The block diagram shows that in the analysis that will be carried out, the output that is of most interest is the one of the system that is controlled, while the system on the feedback loop will process the output of the other system.



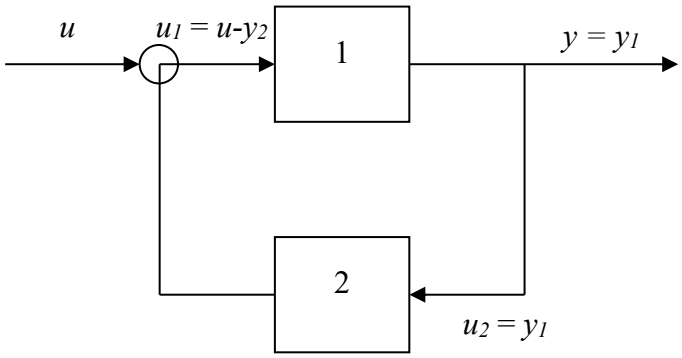
The definition of the equations of the system is more complex because there is a loop which complicates the explication of the equations. The substitutions that must be made in order to continue with the calculation are:

$$u_1 = y_2$$
$$u_2 = y_1$$
$$y = y_1$$
$$u = 0$$

The final state-space model is as follows.

$$\begin{Bmatrix}\dot{x}_1\\ \dot{x}_2\end{Bmatrix} = \begin{bmatrix} A_1 + B_1(I - D_2D_1)^{-1}D_2C_1 & B_1(I - D_2D_1)^{-1}C_2 \\ B_2C_1 + B_2D_1(I - D_2D_1)^{-1}D_2C_1 & A_2 + B_2D_1(I - D_2D_1)^{-1}C_2 \end{bmatrix}\begin{Bmatrix}x_1\\ x_2\end{Bmatrix}$$
$$y = [C_1 + D_1(I - D_2D_1)^{-1}D_2C_1 \quad D_1(I - D_2D_1)^{-1}C_2]\begin{Bmatrix}x_1\\ x_2\end{Bmatrix}$$

It is observed that there is no input *u* because the output of one system becomes the input of the other. In a more general case, the system can be subject to feedback control in addition to a reference input.



In such configuration, the state-space model becomes.

$$\begin{Bmatrix}\dot{x}_1\\ \dot{x}_2\end{Bmatrix} = \begin{bmatrix} A_1 - B_1D_2(I + D_1D_2)^{-1}C_1 & -B_1C_2 - B_1D_2(I + D_1D_2)^{-1}D_1C_2 \\ B_2(I + D_1D_2)^{-1}C_1 & A_2 - B_2(I + D_1D_2)^{-1}D_1C_2 \end{bmatrix}\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + \begin{bmatrix} B_1 - B_1D_2(I + D_1D_2)^{-1}D_1 \\ B_2(I + D_1D_2)^{-1}D_1 \end{bmatrix}\{u\}$$
$$y = [(I + D_1D_2)^{-1}C_1 \quad -(I + D_1D_2)^{-1}D_1C_2]\begin{Bmatrix}x_1\\ x_2\end{Bmatrix} + [(I + D_1D_2)^{-1}D_1]\{u\}$$

Global vs. individual system stability

It is interesting to see how the individual stability of the two systems is "transported" once they are connected in the ways just seen. Looking at matrix A of the different interconnected systems we can deduce that:

- *Systems in series*: we note that, although there is an extra-diagonal term, if we evaluate the eigenvalues of the matrix we obtain the set of eigenvalues of A_1 and A_2 since the matrix is triangular. Therefore, if two stable systems are connected in series, the stability of the overall system is guaranteed.
- *Systems in parallel*: in this case even the dynamics of the two systems are completely decoupled and the global eigenvalues will be nothing but the set of eigenvalues of system 1 and system 2. Even for systems connected in parallel, individual stability is maintained in the global system.
- *Feedback systems*: In this case, a more complex relationship between the two systems can be observed, the dynamics of which are strongly linked. It can therefore be deduced that the global eigenvalues of the loop system will generally not be equal to the eigenvalues of the two individual systems. This is a disadvantage (stability analysis is also required for the global system), but it also makes it possible to have a stable global system even when one of the two systems is unstable or even both unstable. This consequence is an advantage for those who are about to design a control system, especially if the system to be controlled is unstable.

Control proportional to the system state

We want to see control theory developed in the time domain using models in the state space. The system model we will refer to is the linear one

$$\begin{cases}\dot{x} = Ax + Bu \\ y = Cx + Du\end{cases}$$

Let assume that the generic control u is formulated as

$$u = K \cdot x$$

such that the dynamics of the controlled system become, symbolically:

$$\dot{x} = (A + BK)x$$

This is the state matrix of a feedback system on which a state-proportional control is applied.
The characteristic equation (the one that makes it possible to calculate the eigenvalues, which are essential for stability analysis) of this, in symbolic form, is

$$|sI - (A + BK)| = 0$$

from which we obtain a polynomial of degree equal to the number of states of the system, which can be used to calculate the gains k to correct the stability of the system at will.

Pole-placement control

Controllability canonical form (first companion form, FCF)

Let us now suppose that we have the transfer function of a generic system, given by the expression:

$$\frac{y(s)}{u(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

From this transfer function (which links an input to an output) we want to derive the model of the system in the state space to design its control proportional to the system states. We will have to proceed by processing the transfer function to obtain the matrices A, B, C and D that describe the system in state space. Note that the denominator of the transfer function is the characteristic polynomial of the matrix A.
The transfer function can also be written as follows:

$$\frac{z(s)}{u(s)}\frac{y(s)}{z(s)} = \frac{1}{s^n + a_1 s^{n-1} + \cdots + a_n}(b_0 s^n + b_1 s^{n-1} + \cdots + b_n)$$

where *z(s)* is a generic function of *s* that we do not define for now. From this expression, we can write

$$\frac{z(s)}{u(s)} = \frac{1}{s^n + a_1 s^{n-1} + \cdots + a_n}$$
$$u(s) = z(s)(s^n + a_1 s^{n-1} + \cdots + a_n)$$

Similarly

$$\frac{y(s)}{z(s)} = (b_0 s^n + b_1 s^{n-1} + \cdots + b_n)$$

from which, performing the anti-transform (L^{-1}), we obtain an expression of *u(t)* as a linear combination of the function *z* and all its derivatives up to the *n*-th order

$$u(t) = D^n z(t) + a_1 D^{n-1} z(t) + \cdots + a_n z(t)$$

If we now define, in a completely arbitrary way (but with a precise intention), the state of the system as the set of *z* and its time derivatives up to the order *n*−1

$$x = \begin{Bmatrix} D^{n-1} z(t) \\ D^{n-2} z(t) \\ \vdots \\ z(t) \end{Bmatrix}; \quad \dot{x} = \begin{Bmatrix} D^n z(t) \\ D^{n-1} z(t) \\ \vdots \\ D^1 z(t) \end{Bmatrix}$$

we can write the model of the system states in the usual form $x = Ax + Bu$. In fact, by expressing $D^n z(t)$ the following is obtained

$$D^n z(t) = u(t) - a_1 D^{n-1} z(t) - a_2 D^{n-2} z(t) - \cdots - a_n z(t)$$

From which the structure of matrices *A* and *B* is obtained

$$A = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}; \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

We must now find a way to identify, in association with matrices *A* and *B*, the matrices *C* and *D* which describe the output of the system. To this end, we extract the two remaining terms, thanks to which we can write:

$$y(s) = z(s)(b_0 s^n + b_1 s^{n-1} + \cdots + b_n)$$

In analogy to what was done to find *A* and *B*, by doing the anti-transform, we obtain

$$y(t) = b_0 D^n z(t) + b_1 D^{n-1} z(t) + \cdots + b_n z(t)$$

From which matrices *C* and *D* are otained

$$C = [b_1 - a_1 b_0 \quad b_2 - a_2 b_0 \quad \ldots \quad b_n - a_n b_0]; \quad D = [b_0]$$

We have therefore managed to obtain a model of the system's states from its generic transfer function. The characteristic aspect of this model (in which matrix *A* contains the coefficients of the characteristic polynomial of the system in the first row) makes it particularly interesting and we give it the name First Companion Form (FCF) or Controllability Canonical Form. The convenience of this form lies in the simplicity of its deduction from the transfer function. If we then introduce a control proportional to the state, in this case there will be one single input and therefore the gain matrix *K*

will be a row vector, denoted as k. We can note that the matrix $A+Bk$ is particularly simple and becomes

$$A + Bk = \begin{bmatrix} k_1 - a_1 & k_2 - a_2 & \cdots & k_{n-1} - a_{n-1} & k_n - a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

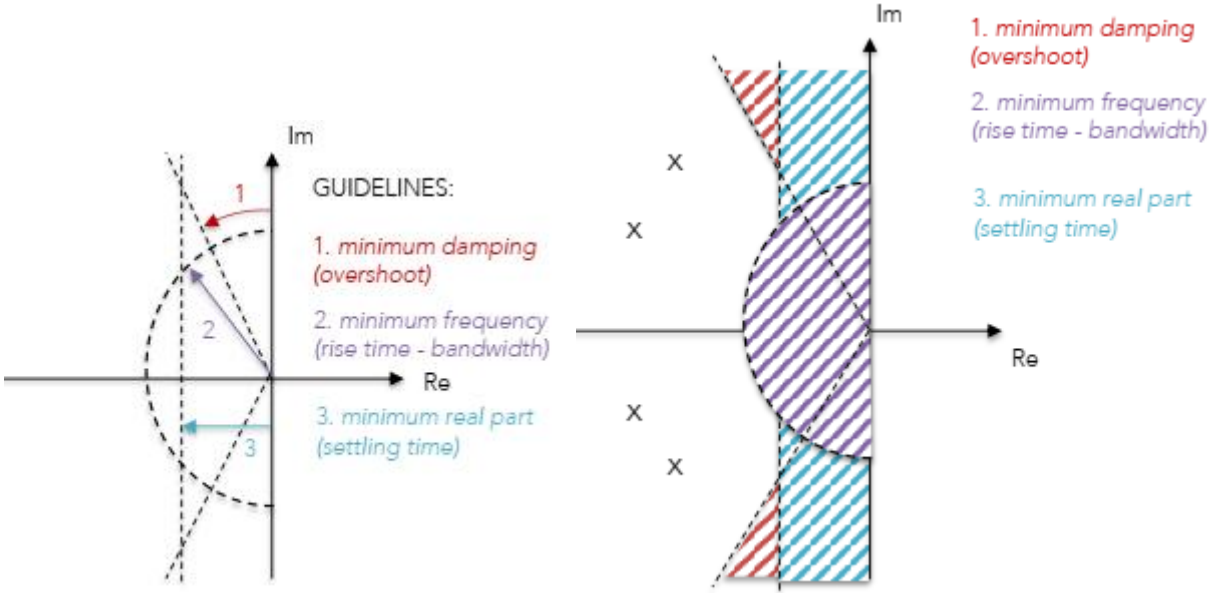
It can be seen that, because of the particular form, each gain modifies only one of the coefficients of the characteristic polynomial, guaranteeing complete freedom in positioning the poles of the system. If we identify the closed loop coefficients as $a_{i_{CL}}$, then the control gains are immediately computed as

$$a_{i_{CL}} = k_i - a_i \quad \rightarrow \quad k_i = a_{i_{CL}} + a_i$$

We have thus obtained a very interesting result of a rather general nature (at least for linear time-invariant systems): if it is possible to write the system in the FCF, through a feedback proportional to the state one can place the poles of the system (in a closed loop) where one wishes in order to have the best possible dynamic response. The problem with this form of the state model is that the state is unlikely to have any physical meaning and is therefore difficult to interpret in practice. The state will usually have to be transformed through a state transformation to obtain the system as a function of a more meaningful state. We shall see, however, that it is also possible to overcome this problem.

Where to place the closed-loop poles?

Recalling the general characteristics of the time response as a function of the poles of the system, preliminary relationship between characteristics of the step response (rise time, settling time, overshoot) and the location of closed-loop poles can be deduced. Note that this is rather straightforward for typical second-order response, while for higher order systems it is not so obvious. Also, it must be remarked that the response is also affected by the zeros.



The concept of controllability

We have seen that a control proportional to the states of a system allows us to position all the poles of the system through the relation:

$$\dot{x} = Ax + Bu \equiv (A + BK)x$$
$$u = Kx$$

We have also seen that the positioning of the poles is particularly simple if the system is represented by the model expressed in the FCF, although we will often have to find a state transformation that allows us to transform the matrix A into the form of the FCF.
This operation is not always possible. For this reason, we define the concept of controllability.

Controllability of a system: a system is controllable if it is possible to bring its state from an initial (finite) value to a final (finite) value in a limited time, using only the control variable.

Pole placement is possible if and only if the system is controllable. Tools are therefore needed to establish the controllability of a system in a rigorous manner. There is an algebraic condition that can also be applied to non-linear systems. Having defined the matrix

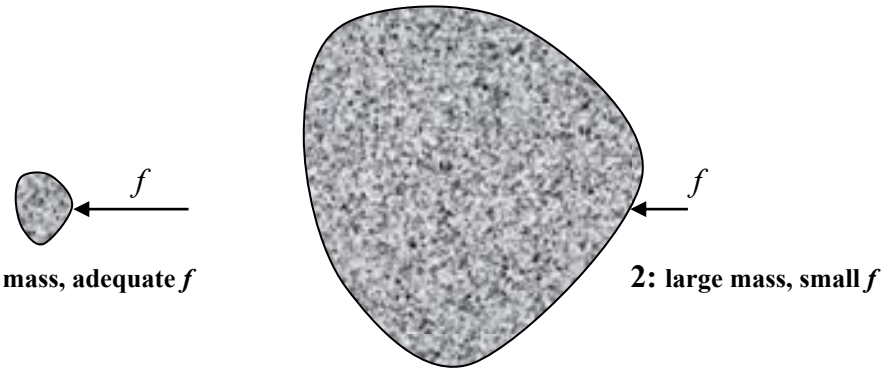
$$P = \int_0^t \Phi(\lambda)B(\lambda)B^T(\lambda)\Phi(\lambda)d\lambda$$

if P is full rank, the system is controllable. Considering linear systems, the analysis can be simplified by looking at the rank of the following matrix:

$$Q = [B \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B]$$

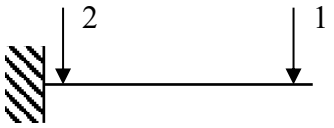
If Q is full rank (equal to the number of states in the system), the linear system (characterised by the matrices A and B) is controllable. Note that the dimensions of Q are given by:
- rows: number of states of the system;
- columns: product of the number of states and inputs.

It is interesting to observe that as the number of inputs increases (i.e. the actions that can be exercised on the system to control it), the columns of B and, consequently, of Q increase, so that the latter is more likely to have maximum rank. In other words, it means that a system is easier to control if one acts on it with a greater number of actions. It is also interesting to see that Q depends not only on the matrix that determines the free dynamics of the system (A), but also on the matrix that introduces the effect of the control actions into the system (B). One thing missing from the analytical verification of controllability just introduced is the quantitative aspect. In fact, it is possible to verify whether a system is controllable or not, but it is not possible to assess how controllable a system can be. An example that can make the importance of the quantitative aspect clear is the following comparison.



1: small mass, adequate *f* **2: large mass, small *f***

System 1 (a small, isolated mass) is certainly controllable. System 2 is also controllable (it is a system with the same dynamics as system 1 but with different physical characteristics), but it has a problem: the large mass will move slowly under the effect of the small force. This means that, although both are controllable, system 2 is "less" controllable than system 1. One of the first resources we call upon to distinguish these cases is common sense. There is also the possibility of identifying parameters (of little physical significance) that indicate in some way how controllable a system is. However, these cannot be deduced from a theorem but are defined case by case by intuition. If, for example, the matrix Q is square and the system is controllable (i.e. $\det(Q) \neq 0$), one can try to evaluate the actual value of the determinant of Q. An example may be clearer. Let us consider a clamped beam.



If a force in the vertical direction allows me to control, for example, the position of the tip of the beam (and if Q is square), it is to be expected that the determinant of Q will be closer to zero in the case of force 2 because this is the case in which more action is required (compared to the case with force 1) to obtain the same result.

The concept of observability

In addition to controllability characteristics, a system must also fulfil other requirements. We define the concept of observability.

Observability of a system: a system is observable if a record of the outputs y can be used to determine the value of the state x.

This characteristic is fundamental because to apply proportional control to the states, it is necessary to know the value of state x of the system, and this can only be known from the system output y. Similarly to what has been done for controllability, it is possible to define a matrix

$$N = [C^T \quad A^T C^T \quad \cdots \quad A^{T^{n-1}} C^T]$$

If N is full rank (also equal to the number of states of the system), the system is observable. The dimensions of N are given by:

- - rows: number of states;
- - columns: product of the number of states and the number of outputs.

Here again, the quantitative aspect can be deduced more on the basis of intuition than mathematical rigour. One can certainly observe, in the case of a square N-matrix, the value of the determinant. It is clear that this method makes more sense if several measurement possibilities are considered for the same system in order to compare the different determinants of the different N obtained.
As can be seen, N depends not only on the matrix describing the dynamics of the free system (A), but also on the matrix establishing the state combination provided by the system's outputs (C).

Note that both the matrix Q and the matrix N are calculated in practice by a recursive method of the type B, $A \cdot B$, $A \cdot AB$, $A \cdot A^2B$, … and C^T, $A^T \cdot C^T$, $A^T \cdot A^T C^T$, ecc.

Partially controllable systems

There are systems that are only partially controllable in the sense that only part of their state can be controlled by the inputs under consideration. Alternatively, some rather complex systems may require

a subdivision of the control into subsystems. In this case the controllability analysis can be done on the subsystems and not on the complete system. But the controllability analysis can also be used to identify, once a control system is known, which states are not controllable. Suppose we know a state transformation *T* that allows us to write the system in the following form:

$$\begin{Bmatrix}\dot{\bar{x}}_C\\ \dot{\bar{x}}_N\end{Bmatrix} = \begin{bmatrix}A_{CC} & 0\\ A_{NC} & A_{NN}\end{bmatrix}\begin{Bmatrix}\bar{x}_C\\ \bar{x}_N\end{Bmatrix} + \begin{bmatrix}B_C\\ 0\end{bmatrix}\{u\}$$
$$x = T\bar{x}$$

where controllable and non-controllable states are separated. It has been said that to control a state means bringing it from one condition to another (both limited) in a finite time. Suppose then that we want to bring the state of the system up to its steady state ($x = 0$). Then, by setting the derivative of the state equal to zero and solving for x_C and x_N, we obtain:

$$\bar{x}_C = -{A_{CC}}^{-1}B_C u$$
$$\bar{x}_N = -{A_{NN}}^{-1}A_{NC}\bar{x}_C$$

These show that, once the steady state behaviour of the state controlled by u (i.e. x_C) has been decided, the steady state behaviour of the non-controllable state (x_N) cannot be decided and will depend on the value assumed by x_C. It is clear that in order to obtain the system in the form just seen it is necessary to find the appropriate matrix *T*.

Pole placement without the FCF

In this paragraph we will consider again the case of a single input evaluated obviously through a linear combination of the state (proportional control to states). The problem with pole placement in the FCF is, as we have already seen, the poor physical interpretation that the state takes on in that particular form. Since the model of a physical system is usually written by choosing a state that has some meaning, it would be nice to be able to perform pole placement (i.e. determine the *k* matrix of gains that gives $u = kx$) without having to go through the representation of the system in the FCF. Suppose we have a generic system whose state has some comprehensible physical meaning:

$$\dot{x} = Ax + Bu$$

Consider the linear transformation:

$$x = T\bar{x}$$

That transforms the original system into its FCF:

$$\dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u$$
$$\bar{A} = T^{-1}AT$$
$$\bar{B} = T^{-1}B$$

Let us now say that we do the pole placement by introducing the matrix *k* that multiplies the transformed state $\bar{x}$. Then

$$u = \bar{k}\bar{x}$$
$$\dot{\bar{x}} = \left(\bar{A} + \bar{B}\bar{k}\right)\bar{x}$$

The fact that the system is transformed does not affect the correction of the stability of the system since we know that the poles of the system are always the same (the matrix A, even if transformed, always has the same eigenvalues). The problem arises due to the fact that it is the state of the FCF that is multiplied by $\bar{k}$. At this point we can take two paths

1. transform the $\bar{k}$ to adapt it to the state of the initial system (but this requires knowledge of the matrix T);
2. calculate the k suitable for the initial system directly, without going through the FCF.

Evaluation of the transfromation matrix T

If we write the controllability matrix $\bar{Q}$ and then substitute the expressions of $\bar{A}$ and $\bar{B}$ as a function of A, B and T, we can see that:

$$\bar{Q} = T^{-1}Q$$

At this point it should be noted that it is possible to calculate both Q and $\bar{Q}$ because the starting system is certainly known (i.e. A and B) and, once the characteristic polynomial of A is known, $\bar{A}$ and $\bar{B}$ are also known, from which we can deduce $\bar{Q}$. Then it is possible to calculate the transformation matrix T and the gain matrix k

$$T^{-1} = \bar{Q}Q^{-1}$$
$$k = \bar{k}T^{-1}$$

The operations to be performed to find k by this method are:
- place the poles in the FCF (which involves evaluating the characteristic polynomial of A) and find $\bar{k}$;
- calculate the matrix Q, invert it and multiply it with $\bar{Q}$ (also calculated) to obtain T^{-1};
- multiply $\bar{k}$ by T^{-1} to finally obtain the desired k.

This method is not very convenient because it involves the inversion of the matrix Q, which can present numerical problems due to the possible ill-conditioning of Q.

Avoid using the FCF

Let us consider the coefficients of the characteristic polynomial of the open-loop system (a_i) and those desired for the closed-loop system ($\breve{a}_i$). The gain k for the FCF is computed as

$$\bar{k}_i = \breve{a}_i - a_i$$

Let us now consider the characteristic polynomials in an open loop ($\alpha(s)$) and closed loop ($\alpha_C(s)$). Since a property of the characteristic polynomials of square matrices is that "every matrix is a solution of its characteristic polynomial" it can be written that:

$$\alpha(\bar{A}) \equiv \bar{A}^n + a_1\bar{A}^{n-1} + \cdots + a_{n-1}\bar{A} + a_nI = 0$$

The same holds if we replace $\bar{A}$ with A (note that $A^0 = I$). If instead we substitute $\bar{A}$ in the polynomial α_C (that is not its characteristic polynomial) we have

$$\alpha_C(\bar{A}) = \bar{A}^n + \breve{a}_1\bar{A}^{n-1} + \cdots \breve{a}_{n-1}\bar{A} + \breve{a}_nI \neq 0$$

From $\alpha(\bar{A})$ we make explicit $\bar{A}^n$ and substitute in $\alpha_C(\bar{A})$, obtaining:

$$\alpha_C(\bar{A}) = (\breve{a}_1 - a_1)\bar{A}^{n-1} + (\breve{a}_2 - a_2)\bar{A}^{n-2} + \cdots + (\breve{a}_n - a_n)I \neq 0$$

in which the coefficients at the second member are the k_i. Unfortunately, we cannot extract the gain vector k since its elements each multiply a matrix.

Then, introduce a vector ${e_n}^T$ defined as:
- it is a row vector of order n;
- it is composed by zeros with the exception of position n (indicated as subscript) where it is valued at 1.

Then, ${e_n}^T$ is

$${e_n}^T = [0 \quad 0 \quad \cdots \quad 0 \quad 1]$$

Recalling the structure of $\bar{A}$, it is observed that its product with ${e_n}^T$ is

$${e_n}^T \cdot \bar{A} = [0 \quad 0 \quad \cdots \quad 1 \quad 0] \equiv {e_{n-1}}^T$$

Implementing this property recursively, it is then

$$\underbrace{{e_n}^T \bar{A} \cdot \bar{A} \cdot \ldots \cdot \bar{A}}_{m \text{ times}} = \underbrace{\underbrace{\underbrace{{e_n}^T \bar{A}}_{{e_{n-1}}^T} \cdot \bar{A}}_{{e_{n-2}}^T} \cdot \ldots \cdot \bar{A}}_{{e_{n-m}}^T} = {e_{n-m}}^T$$

Therefore, pre-multiplying the polynomial α_C by ${e_n}^T$ the following is obtained:

$${e_n}^T \cdot \alpha_C(\bar{A}) = {e_1}^T(\breve{a}_1 - a_1) + {e_2}^T(\breve{a}_2 - a_2) + \cdots + {e_n}^T(\breve{a}_n - a_n)$$

$$[\breve{a}_1 - a_1 \quad 0 \quad \cdots \quad 0 \quad 0]$$

$$[0 \quad \breve{a}_2 - a_2 \quad \cdots \quad 0 \quad 0]$$

$$\vdots$$

$$[0 \quad 0 \quad \cdots \quad 0 \quad \breve{a}_n - a_n]$$

In synthesis

$${e_n}^T \cdot \alpha_C(\bar{A}) = \bar{k}$$

Now introduce the state transformation T, i.e. substitute $\bar{A}$ with its expression function of A ($T^{-1}AT$) and substitute $\bar{k}$ with its expression function of k (kT):

$$({e_n}^T \alpha_C(T^{-1}AT) \equiv) \quad {e_n}^T T^{-1} \alpha_C(A)T = kT \quad (\equiv \bar{k})$$

From which the expression for k is obtained as

$$k = e_n{}^T T^{-1} \alpha_C(A)$$

Recalling now the expression of T^{-1} and considering that, for the particular structure of Q the relation $e_n{}^T \cdot Q = e_n{}^T$ holds, it is finally possible to write

$$k = e_n{}^T Q^{-1} \alpha_C(A)$$

which we note is independent of FCF. Since we do not want to invert the matrix Q, we make the following substitution, to use the well-known numerical methods for solving linear systems that allow us to avoid the inversion of Q, with all its advantages

$$e_n{}^T Q^{-1} = b^T$$

and b is computed by solving the linear system

$$Q \cdot b = e_n$$

Thus, the gain matrix k is obtained, after solving for b, as

$$k = b^T \cdot \alpha_C(A)$$

We were therefore able to see that the positioning of the poles of a system (as long as it is controllable!) can also be done without resorting to FCF, which requires rather complex and numerically critical calculations.

The advantage of using more than one actuator (input)

Until now, we have dealt with systems with only one input. If we move on to control with more than one actuator, we can immediately emphasise that the control possibilities increase (which is intuitive but is confirmed by the controllability analysis introduced previously). In fact, matrix B has more than one column and there is a greater probability of obtaining a Q with maximum rank. In practice, the advantage, if the system is controllable, is certainly in the possibility of positioning the poles. Moreover, it may be possible to intervene in other aspects of the system (such as minimising the effort of controlling).

The state observer

To apply a control proportional to the state ($u = Kx$) it is necessary to know the entire state, otherwise such control is impossible. We know that the only information directly available regarding the physical system is the one obtained from the y-outputs. It will therefore be necessary to pass from the measurements of the outputs y (which may or may not be equal to the state or a part of it) to the value of the state x. We have already said that for this to be possible, the system must be observable. Once this property has been verified, it is necessary to find a way to compute x once the outputs have been measured. A first trivial case that is extremely rare to encounter (but is best illustrated) is when the linearized system is proper and has non-singular square matrix C. In this case the state can be obtained directly from

$$x = C^{-1} y$$

The fact that this case never occurs is related to its physical meaning. A matrix C that is square and invertible but different from a diagonal matrix implies that the outputs y measure linear combinations

of the state variables. If it were possible to measure a linear combination of states then it is presumably easier to measure the states directly (i.e. C is diagonal and y coincides with x). Therefore, the case of non-diagonal and non-singular square C is, for these practical considerations, very rare to find. However, the case of a diagonal C-matrix is also very rare because the systems have, typically:
- many states;
- few possible measures.

This should not discourage because we have seen that, for the system to be observable, it is sufficient that the observability matrix N is of maximum rank. If the available y are few, we will have to find a way to compute all the states from this few information. This is the purpose of the systems known as observers. An observer is a fictitious system which, by processing the outputs of the real system y, allows us to compute its state x. The observer will be characterised by its own state ($\hat{x}$) which will follow a certain dynamics. To describe the evolution of this dynamics, we consider the following mathematical model:

$$\begin{cases} \dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}u + Ly \\ \hat{y} = \hat{x} \end{cases}$$

The observer receives the measurements y and introduces them into his own dynamics through the matrix L. We want the observer's state ($\hat{x}$) to be representative of the state of the real system (i.e. as close to it as possible). Thus, the size of $\hat{x}$ and x will be the same. We will now see how the model of the observer is derived from the model of the real system. We recall that the equations of the real system are

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

The observation error is defined as

$$e = \hat{x} - x$$

Deriving the error with respect to time and substituting the dynamics of the real system and the observer, we obtain the dynamics of the observation error which, reworked properly, becomes

$$\dot{e} = \hat{A}e + \left(\hat{A} - A + LC\right)x + \left(\hat{B} - B + LD\right)u$$

The observation error must be (according to our requirements) zero after a limited (observation) transient. From this we deduce two conditions that allow us to express the observer matrices as a function of the system matrices and L:

1. For the error to be null, it is necessary that the two terms in x and u in the system are null, which leads to the following two necessary conditions:
$$\begin{cases} \hat{A} = A - LC \\ \hat{B} = B - LD \end{cases}$$

2. If the system is not asymptotically stable, the steady-state error can never be zero. Then a new condition is that the matrix $\hat{A}$ must be characteristic of an asymptotically stable system.

What needs to be done is then:
- find an L that makes the matrix $\hat{A}$ asymptotically stable;
- also calculate, after finding the appropriate L, the matrix $\hat{B}$.

We must therefore see how the matrix L can be calculated so as to guarantee the asymptotic stability of $\hat{A}$. Recall that from $A+Bk$ (i.e. by introducing a control proportional to the system state) it is possible

to place the poles of the system as desired. In other words, this means that we can make the matrix $A+BK$ asymptotically stable. Note that $\hat{A}$ is in a similar form:

$$\hat{A} = A - LC$$

The difference is that the matrix L premultiplies C instead of multiplying it to the right as in $A+BK$. However, the problem is only apparent because we can transpose $\hat{A}$ and obtain:

$$\hat{A}^T = A^T - C^T L^T$$

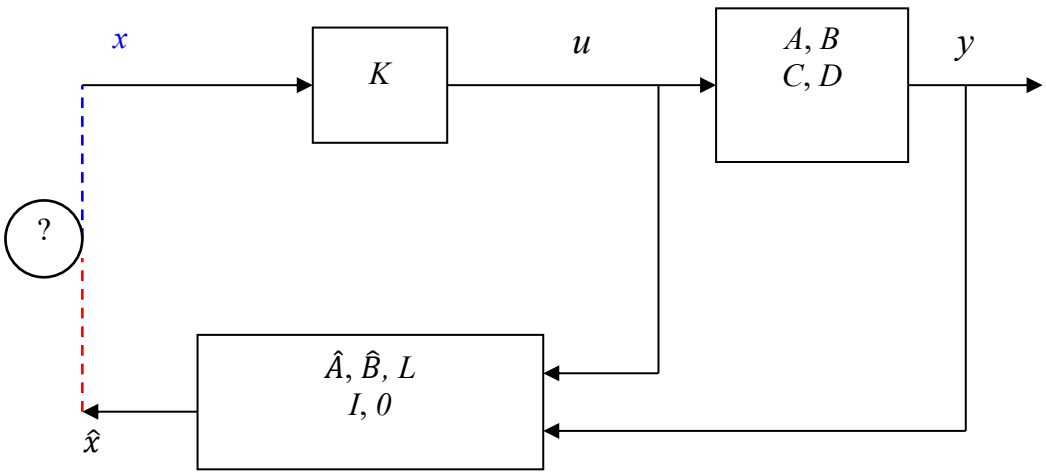
which is therefore correct and allows us to follow the methods seen above for positioning the A^T poles. Note that transposing a matrix does not change its eigenvalues. We can therefore deduce the L^T through the positioning of the poles of A^T. The operations to be done to identify the mathematical model of the observer are therefore

1. the positioning of the poles of the matrix A^T is done and the matrix L^T is computed;
2. L^T is transposed to obtain the matrix L;
3. Calculate (obviously, the characteristics of the true system A, B, C and D are known) the matrices $\hat{A}$ e $\hat{B}$.

We have then found the mathematical model of a fictitious system (the observer) that allows us to compute the state of the real system from its outputs. Note that if the real system is a proper system, matrix $\hat{B}$ coincides with matrix B due to the absence of matrix D.

Controlled system and observer

Observe the following block diagram.



We have already shown (by analysing the dynamics of the observation error) that, after a certain observation transient, the observer's state must coincide with the true state of the system. However, it is inevitable that the input to the control block (k) is the observed state and not the true state. How can we be sure that the observer has completed his transient (or, in other words, that it can accurately track the variations of the true state)? If in fact this were not true, the pole placement that is being performed through block k, is not the one that was previously designed because it processes a state that is not that of the true system. Consequently, it is not certain that the system behaves in a stable way (and therefore it is likely that the observer will never "see" the true state).

Let us consider the two systems (true and observer):

$$\begin{cases}\dot{x} = Ax + Bu \\ y = Cx + Du\end{cases}$$
$$\begin{cases}\dot{\hat{x}} = (A - LC)\hat{x} + (B - LD)u + Ly \\ \hat{y} = \hat{x}\end{cases}$$

In which the control u is:

$$u = K\hat{x}$$

Rewrite in matrix form:

$$\begin{Bmatrix}\dot{x} \\ \dot{\hat{x}}\end{Bmatrix} = \begin{bmatrix}A & BK \\ LC & A - LC + BK\end{bmatrix}\begin{Bmatrix}x \\ \hat{x}\end{Bmatrix}$$

Having already substituted Ly with its equivalent function of the real system
The system describes the dynamics of both systems, true and observer. If we assume that we have computed the matrices K and L by the methods described in the previous paragraphs, it should be noted that:
- we have guaranteed the stability of the true system through the control of the states, neglecting the possible influence of the observer;
- we have designed the observer of the true state without worrying about the possible influence of the control system.

Recalling the properties of loop systems, we want to see if the eigenvalues of the overall system are the union of the eigenvalues of the two systems (true system in a closed loop with the controller and observer). This would in fact guarantee the validity of the design method adopted (designs of K and L "decoupled").

In order to see that in reality the dynamics of the two systems do not influence each other (i.e. the eigenvalues are just the union of the eigenvalues of the two separate systems!) we make a state transformation

$$\hat{x} = x + e$$

we have substituted for the observed state the observation error whose dynamics is known. The system thus obtained is

$$\begin{Bmatrix}\dot{x} \\ \dot{e}\end{Bmatrix} = \begin{bmatrix}A + Bk & BK \\ 0 & A - LC\end{bmatrix}\begin{Bmatrix}x \\ e\end{Bmatrix}$$

It can be seen that the state matrix
- is a triangular matrix for which the contribution to the eigenvalues comes only from the main diagonal;
- the two sub-matrices on the main diagonal are precisely those that we have made asymptotically stable, the first through the positioning of the poles of A with the proportional control K and the second through the positioning of the poles of A^T by means of L^T.

It therefore means that the eigenvalues of the global system are precisely the union of
- eigenvalues of the system subjected to proportional control (i.e. the eigenvalues of $A+BK$);
- eigenvalues of the observer ($\hat{A} = A–LC$).

The above shows that it is possible to position the poles of the real system (i.e. design the gain matrix K) without worrying about the number of known states. If states are missing from the measurements

on the real system, it is possible to design (independently from the controller) an observer that provides the whole system state. When the two systems are coupled (as shown in the previous diagram) we are sure that the stability of the global system is maintained since its eigenvalues are the union of the eigenvalues of the two designed systems.

Some considerations on the observer

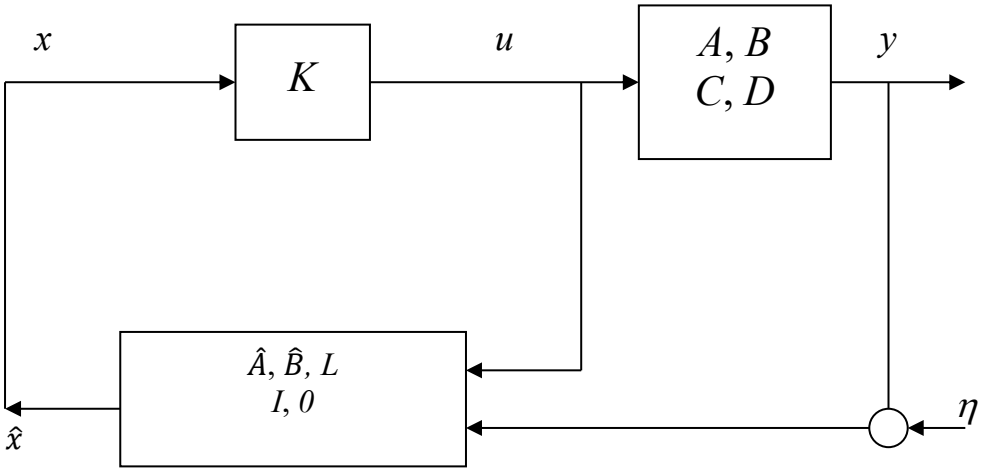
Let us consider a real system and the design of a proportional control system. We have already shown that it is very important to bear in mind that the availability of control action is not infinite (i.e. the actuators saturate). Therefore, when designing the k, care should be taken because the larger it is, the sooner actuator saturation can occur. This is why it is often important to keep the gain low in order to avoid such saturation.
If we now take into account that k is the means by which we reposition the poles of the open-loop system (because the new poles are those of the $A+BK$ matrix), this limitation translates into a limited freedom of repositioning. This in turn limits the speed of the system's response. This is a physical limitation that shows that it is impossible to design control systems that are too fast (of course, it depends on the actuators, but they are always characteristic of mechanical systems).
Let us now turn to the observer. Its design (i.e. the determination of the L-matrix) also involves the positioning of the poles (of the A^T-matrix), but there is a notable difference with respect to control systems: the observer is a fictitious system calculated by a computer, so the displacement of the poles can be as large as one likes (as far as possible!). This means that the speed of the observer can be "chosen" in such a way that the observation transients are negligible with respect to the variations of the state of the system, i.e. in such a way that the observer follows the state of the system without problems and almost as if it had no dynamics of its own (i.e. as if it were always at steady state).

Let us now consider the moment when the control system is turned on. The conditions prior to start-up will be:
- the physical system will have its state x;
- the observer will have a state equal to zero.

It therefore means that when the control/observation system is switched on, the observation error will be $e = -x$. If the observer is too fast (which usually implies low "damping" values), overshoots of the observed state may occur before the observed state reaches steady state (and properly "locks on" to the true state).
The problem is that these overshoots also enter the control system (which processes the observed state and not the true state) and could damage the true system.
The trick used to get around this problem is to first switch on the observer (to make it "lock" the true state) and then, once the observation transient is over, switch on the control system as well. Unfortunately, this trick can only be used if the true system is already stable.

Note that the speed of the observer is not established by a precise rule. It will depend on the physical system and the actuators considered. A somewhat empirical rule is that the response speed of the observer should be about 10 times that of the system "under observation".

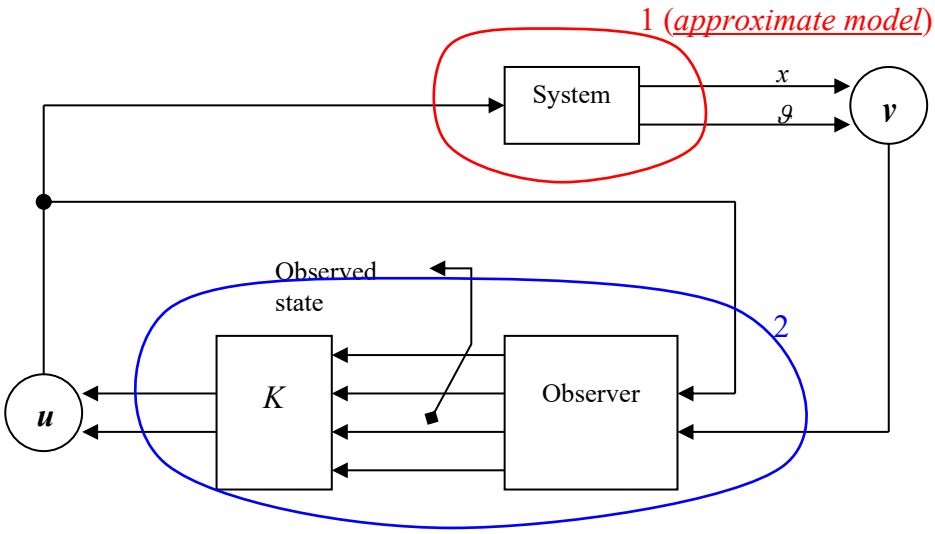
Another aspect to be considered on the speed of the observer is the presence of disturbances (which are usually high frequency, i.e. they have very fast variations) in the acquisition branch of the outputs y :



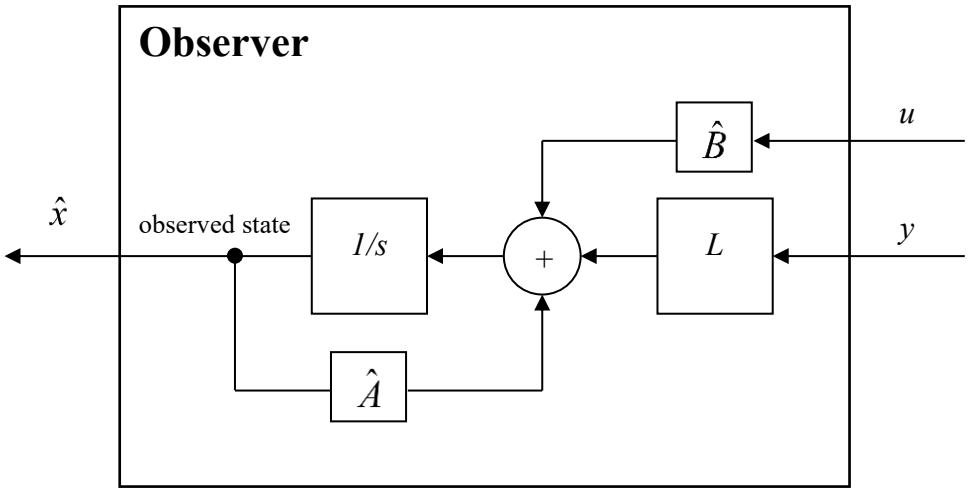
Let us assume that the system's output y coincides with its state x (as far as we are concerned, this is fine). If the observer is very fast, the observed state will be equal to $(x + \eta)$ because it will also be able to follow disturbance oscillations. Since this observed state then enters the controller k, you may find yourself having a control action even when you don't need it. ($x = 0$ but $\hat{x} = \eta$ and thus $u = K \cdot \eta$!). One might think of introducing a low-pass filter so that disturbances are eliminated before they enter the observer. In this case, however, it may be convenient to make the observer slower so that the disturbances are naturally filtered out. Furthermore, the speed of the observer is determined not only by the disturbances, but also by the physical characteristics of the real system, which leads us to think that filtering by means of an observer is more efficient than any other filter, whose cut-off frequency is certainly less related to the characteristics of the system. Obviously, in the absence of noise, it is preferable to have a fast observer, but this is a difficult case to find in practice because noise is everywhere.

Details for the observer design

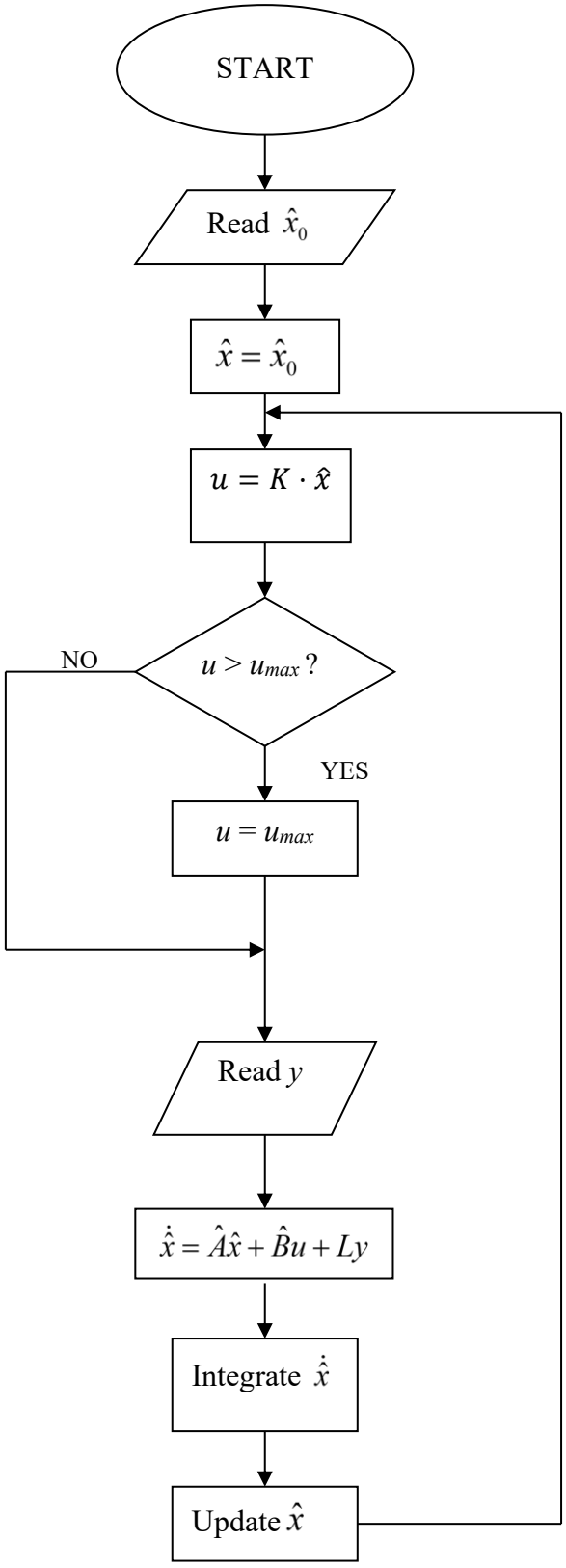
It should be noted that the mathematical model of the real system, which is used to evaluate the system's dynamics, differs from the real behaviour of the system the more simplifying the hypotheses considered are. Its "uncertain" outputs will enter the observer which, on the contrary, is described exactly by the model because it is a fictitious system whose dynamics were chosen by us in the design phase. In turn, the outputs of the observer will enter the gain block K (control proportional to the observed state) which will allow the stabilising intervention on the real system (assuming that the observed state deviates not much from the real one). The whole can be schematised in the following block diagram (assuming a system with 4 states, 2 outputs and 2 inputs).



Block 1 is the actual physical system while block 2 (the compensating block) is often implemented on the computer. So let's suppose we do the observer on a computer with appropriate software. The block diagram of the observer would become as follows.



The flow chart describing the operations (cyclical and performed at each output measurement, at the appropriately chosen sampling rate) carried out from the moment the compensator block is switched on is shown here below. Note that it also presents a logical control on the intensity of the output in case the limits of the actuators are already known.



As far as actuators and sensors are concerned, we assume that they are contained in the representative block of the physical system. If these are not ideal, it will be necessary to introduce their dynamics to be able to understand their influence on the dynamics of the system "under control". However, we have already underlined the fact that if the bandwidth of a sensor is considerably wider than that of

the real system, it is possible to consider it as an ideal sensor (i.e. without dynamics or, in other words, with zero-order dynamics).

The observer and the real system

So far, we have assumed that we have a perfect knowledge of the model of the system to be compensated. There are a number of experimental methods that allow us to refine our knowledge of poorly known systems.

1. Transfer function. It is possible to deduce the transfer function experimentally by relating the system's outputs to suitable known inputs. However, this method is more related to the inputs and outputs and it is difficult to infer the physical characteristics of the system (the system looks like a black box of which we only know the ratio between the input and the outputs; the state of the system is totally unknown).
2. State space model. The state space model contains (if the states are appropriately chosen) the physical parameters of the system (e.g. masses, elastic or viscous friction constants, etc.). If through a (well-designed) observer we "look" at the state of the real system, we can see how close the analytical model of the system (obviously approximated) we used for the design is to that of the real system. In this way it is possible to "calibrate" the analytical model in order to have a better performance of the compensator.

This means that, if the model of a physical system is known (at least approximately), it is possible to do more than just predict its output as a function of a certain input.

Reduced state observer

Let us now suppose that we have a system in which only a portion of the state can be observed. We then partition the system into two blocks

$$\begin{cases} \dot{x}_1 = A_{11}x_1 + A_{12}x_2 + B_1u \\ \dot{x}_2 = A_{21}x_1 + A_{22}x_2 + B_2u \\ y = C_1x_1 \end{cases}$$

Let us assume that x_1 is completely observable while the x_2 portion of the state is not. We have seen that the observer requires a certain amount of computing power. In order to reduce the computational effort of the observer (and thus increase its speed), one could think (for such a system) of designing a "partial" observer that observes only the "invisible" state (x_2). Let's try to see if, proceeding as for the global observer, we reach a good solution. Considering the second block of the model, we write the observer's dynamics for the part x_2 of the system

$$\begin{aligned} &\dot{\hat{x}}_2 = (A_{21} - L_2C_1)\hat{x}_1 + A_{21}\hat{x}_2 + B_2u + L_2y \\ &\rightarrow \hat{x}_1 = {C_1}^{-1}y \\ &\dot{\hat{x}}_2 = A_{21}\hat{x}_2 + B_2u + A_{21}{C_1}^{-1}y \end{aligned}$$

This is achieved by:

- designing a full-state observer;
- writing the reduced-state observer as a portion of the full-state observer.

It has already been shown that the stability of the observer is guaranteed by making the matrix *A-LC* (representative of the dynamics of the observer) asymptotically stable through an appropriate choice of the coefficients of the matrix *L*. Unfortunately, in this case we cannot decide the stability of the reduced order observer because the matrix describing its dynamics is A_{22} and there is no way to modify its poles. It follows that the dynamics of the reduced order observer will inevitably be comparable to that of the true system (A_{22} being a portion of *A* of the true system). We do not want

the observer's velocity to be fixed. It therefore means that constructing the reduced order observer as a sub-block of the global observer is not the best approach for what we expect to achieve.
In order to shape the dynamics of the reduced order observer we introduce an auxiliary dynamics (dynamics added to the dynamics of the global observer) described by the system

$$\dot{z} = Fz + Gu + Hy$$

Also, assume the following dynamics of the reduced-order observer

$$\dot{\hat{x}}_2 = Ly + z$$

At this point we identify (as done for the full state observer) the dynamics of the observation error

$$\begin{aligned} \dot{e} &= \dot{\hat{x}}_2 - \dot{x}_2 \\ &= L\dot{y} + \dot{z} - \dot{x}_2 \end{aligned}$$

we are interested in writing the error in terms of *e, x1* and *u*. We then substitute

$$\begin{aligned} x_2 &= \hat{x}_2 - e \\ z &= \hat{x}_2 - Ly \end{aligned}$$

After collecting the common coefficients for the different variables present, we obtain

$$\begin{aligned} \dot{e} = (A_{22} - LC_1A_{12})e + (LC_1A_{11} - FLC_1 + HC_1 - A_{21})x_1 + \\ +(LC_1A_{12} - A_{22} + F)\hat{x}_2 + (LC_1B_1 + G - B_2)u \end{aligned}$$

Now, in analogy to what has been done for the full state observer, we impose the coefficients of x_1, $\hat{x}_2$ and *u* equal to zero in order to make the error independent from them. This allows us to obtain the three matrices *F, G* and *H* of the auxiliary dynamics and, imposing the zero error at steady state (i.e. the asymptotic stability on the matrix of the coefficients of *e*), the matrix *L* of the observer.

The expressions sought are:

$$\begin{aligned} F &= A_{22} - LC_1A_{12} \\ G &= B_2 - LC_1B_1 \\ H &= FL + (A_{21} - LC_1A_{11}){C_1}^{-1} \end{aligned}$$

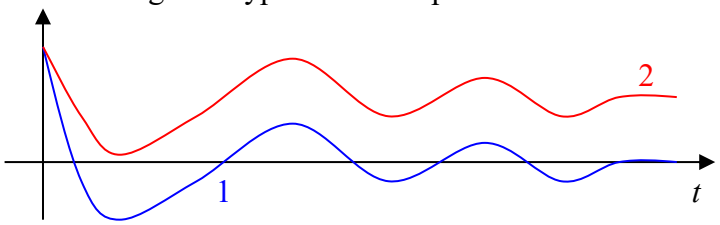
As a result of what has been said, we note that the asymptotic stability is imposed on the matrix *F* (equivalent to the matrix of the coefficients of *e*) or on the dynamics of the auxiliary system (just as before it was imposed on the dynamics of the observer). A summary of the operations to do to deduce the reduced order observer of a system is the following:
1. place the poles of the matrix A_{22} through the matrix *L* (obviously one must always resort to the trick of transposition to adjust things). In this way the matrix *F* characteristic of the dynamics of the introduced auxiliary system is obtained (it must be asymptotically stable).
2. determine the matrices *G* and *H*.

In this way the following can be decided
- the best possible dynamics (by means of *F*) of the system being integrated for observation (*z*);
- the best combination of *z* and *y* (through *G* and *H*) that allows me to obtain the portion of the state that cannot be observed directly (x_2).

The block scheme of the operations to be carried out is analogous to the one of the full state observer with the difference that the dynamics to be integrated is the one of the state z from which then the evolution of $\hat{x}_2$ is deduced.

Optimal control

Let us now look at a different philosophy of control system design. The optimal control, instead of aiming at obtaining only an asymptotically stable system (in closed loop), tries to guarantee stability together with the optimisation of some quantity linked to the system to be controlled. It is therefore clear that "optimal" is not to be understood as "best" but as the possibility of optimising something. The quality of the control will depend on what you decide to optimise. In classical control theory, the problem is studied through the system's transfer functions (defined in the frequency domain) which allow only one output to be fed back to one input. In the state-space domain, we can bind multiple outputs to multiple inputs by means of state-proportional control. Again, however, when it came to establishing the stability of the system (positioning of the poles), we returned to the frequency domain (where the poles of the system are defined).
Let us now consider the following two hypothetical responses.



From the point of view of oscillations and damping they are indistinguishable. What changes is the value they take on at steady state. Assuming that the correct value of the response is zero, we can also say that response 1 is better than response 2. In the frequency domain, one method of noticing this difference is the final value theorem. However, this only helps us if we are designing a controller in the frequency domain, whereas it cannot help us in the case of state space control. The step forward we now propose is to solve the control problem by reasoning exclusively in the time domain. In this case we could, for example, "adjust" the problems of response 2 by evaluating the integral in a certain time interval and minimising it, i.e. evaluating the gain matrix k of the proportional state control in such a way that this integral is minimised.

$$\int_0^{t_f} x \cdot dt$$

Since we have always considered the state x to be a vector, it is more correct if written as

$$\int_0^{t_f} x^T x \cdot dt$$

where in practice the norm of x is optimised. Minimising this without taking into account the amount of control (u) doesn't make much sense because it would result in an exaggerated control request (through K) which would certainly guarantee an excellent response but would result in a control expense/performance ratio which would be too high. So, let's rewrite it introducing also the inputs (indices of the control expenditure)

$$\int_0^{t_f} (x^T x + u^T u) \cdot dt$$

This is more appropriate. However, it would be nicer to be able to decide which quantity to give more importance to (either x or u). We then arrive at the classical form found for the optimal control:

$$\int_0^{t_f} (x^T Q x + u^T R u) \cdot dt$$

in which Q and R allow us to shift the focus of the minimisation between x and u. If Q is larger than R we favour the response to control cost (high expenditure/performance ratio) and vice versa.

The cost function to minimise

Finally, we need to take into account the end of the control transient (which we assume occurs at time t_f) with a term outside the integral:

$$J_0 = \frac{1}{2} {x_f}^T S x_f + \frac{1}{2} \int_0^{t_f} (x^T Q x + u^T R u) \cdot dt$$

This is the cost function we want to minimise. Matrix S has roughly the same meaning as Q and R and, together with these two, it guides us in the process of minimisation. Let us try to deduce some properties of the above matrices:
- they are certainly square;
- they have no reason not to be symmetric.

There are two reasons for them to be symmetrical: one mathematical and one, more important, of a physical nature. We write $u^T R u$ by splitting u down into two elements:

$$\begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} R_1 & R_2 \\ R_3 & R_4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = R_1 {u_1}^2 + (R_2 + R_3) u_1 u_2 + R_4 {u_2}^2$$

As can be seen, what ultimately matters in the mixed product $u_1 u_2$ is the sum of the two extra-diagonal terms. Therefore, from a mathematical point of view, there is no reason for R_2 and R_3 to be different. From a physical point of view, optimising by considering R_2 different from R_3 would mean imposing, on the same contribution ($u_1 u_2$), two different weights. We must therefore minimise J_0. Note that the minimisation cannot be done freely because there is a link between x and u (the equations of dynamics). We take this link into account through appropriate multipliers. Then the complete expression to be minimised is

$$\min_u J = \frac{1}{2} {x_f}^T S x_f + \frac{1}{2} \int_0^{t_f} [x^T Q x + u^T R u + 2\lambda^T (Ax + Bu - \dot{x})] \cdot dt$$

Note that we minimise in terms of u since it is the quantity we can use to intervene in the system. We must then impose grad(J) = 0. In other words, we will have to derive J with respect to the different variables (x, u and λ) and set all these derivatives equal to zero. The different terms of grad(J) are as follows:

$$\frac{\partial J}{\partial \lambda} = \frac{1}{2} \int_0^{t_f} 2 \frac{\partial}{\partial \lambda} [\lambda^T (Ax + Bu - \dot{x})] dt = 0$$

$$\rightarrow \frac{\partial}{\partial \lambda} [\lambda^T (Ax + Bu - \dot{x})] = 0$$

we derive for λ^T. We then obtain the constraint imposed by the equations of dynamics

$$\dot{x} = Ax + Bu$$

So the first condition we have found is that the minimum of J can exist as long as the constraint given by the equations of dynamics is respected.

$$\frac{\partial J}{\partial u} = \frac{1}{2}\int_0^{t_f} \frac{\partial}{\partial u}[u^T R u + 2\lambda^T B u]dt = 0$$
$$\rightarrow \frac{\partial}{\partial u}[u^T R u + 2\lambda^T B u] = 0$$

Now, derive with respect to u^T, to obtain

$$2B^T\lambda + 2Ru = 0$$
$$\rightarrow u = -R^{-1}B^T\lambda$$

Clarification is now necessary with regard to the R-matrix. For it to be invertible, it must be positive definite (we have already seen that it will also be symmetric). To say that R is positive definite means to say that in the cost functional J the contribution of u (i.e. of the control cost) is always present and is only null when u is null (i.e. when there is no control). This mathematical requirement has a very precise physical meaning: if one wants to impose a control on the system (through a certain u) it is indispensable to put a limit to this control (which is done for sure because, once u is there, having to be R positive definite, the control action is surely minimised). We note that anyway u is not known because it depends on λ (which we do not know). Note also that since R and B are constant (linear time-invariant systems), it will inevitably be variable (otherwise the control u would be constant and would not make much sense).
The last term of the gradient of J is given by:

$$\frac{\partial J}{\partial x} = \int_0^{t_f} \frac{\partial}{\partial x}\left[\frac{1}{2}x^T Q x + \lambda^T A x + \dot{\lambda}^T x\right] dt = 0$$
$$\rightarrow \frac{\partial}{\partial x}\left[\frac{1}{2}x^T Q x + \lambda^T A x + \dot{\lambda}^T x\right] = 0$$

where we have made an integration by parts to eliminate the term in x

$$\int_0^{t_f} \lambda^T(-\dot{x})dt = [-\lambda x]_0^{t_f} + \int_0^{t_f} \dot{\lambda}^T x dt$$

And the term outside the integral sign ($-\lambda x_f + \lambda x_0$) is constant and is not derivated (as done for the term x_f in J).
To obtain a meaningful condition, we transpose the term in brackets and derive for x^T instead of x. We obtain the following condition:

$$\dot{\lambda} + A^T\lambda + Qx = 0$$
$$\rightarrow \dot{\lambda} = -A^T\lambda - Qx$$

In summary:

$$\dot{x} = Ax + Bu$$
$$u = -R^{-1}B^T\lambda$$
$$\dot{\lambda} = -A^T\lambda - Qx$$

It is remarked that the third equation describes the dynamics of the multipliers which are directly linked to the state. The second one instead allows us to obtain, indirectly through λ, the control action

u to be exercised on the system. If we manage to integrate the third equation, we have solved the control problem.
We have seen that R must be positive definite. As far as Q is concerned, however, it can also be the case that it is positive semi-definite. The meaning is that one can accept large values of the state (because if Q is positive semi-definite, it will have null terms that will prevent the whole state from being minimised). The important point is to have a limited control action (and this is guaranteed by R).

The Riccati equation

Continuing with the optimal control, the system we have to solve is

$$\begin{Bmatrix} \dot{x} \\ \dot{\lambda} \end{Bmatrix} = \begin{bmatrix} A & -BR^{-1}B^T \\ -Q & -A^T \end{bmatrix} \begin{Bmatrix} x \\ \lambda \end{Bmatrix}$$

The problem with what is obtained is that λ is not defined before completing the optimisation. To circumvent the problem, we can impose a constraint on the multipliers λ

$$\lambda = Px$$

In this way, an approximation is made by saying that the multipliers λ are directly proportional to the state of the system. It is also possible to write a direct link between the state x and the control action u

$$u = -R^{-1}B^T Px$$

Rewrite J_0 substituting the expression for u.

$$J_0 = \frac{1}{2}{x_f}^T S x_f + \frac{1}{2}\int_0^{t_f} x^T (Q + P^T B R^{-1} B^T P) x dt$$

We must now determine P. Substituting then $\lambda=Px$ in the equation describing the dynamics of λ, we obtain an equation describing the dynamics of P which contains inside it also the constraint given by the dynamics of x

$$\dot{x} = Ax + Bu$$
$$\dot{P}x + P\dot{x} = -A^T Px - Qx$$
$$u = -R^{-1}B^T Px = Kx$$
$$\dot{P} = -A^T P - Q - PA + PBR^{-1}B^T P$$

The differential equation for matrix P is known as Riccati equation. It is the solution of this equation that allows us to solve the control problem. The solution P is a symmetric and positive definite matrix. Note that if P is a solution of the Riccati equation, it is guaranteed that the controlled system with the resulting K is stable.

Boundary conditions for the integration of the Riccati equation

The Riccati equation, being differential, requires boundary conditions. It also suggests to us that P may vary in time and so we may have, at least in principle, a variable gain. We are interested in

relating the initial and final values of P to the other parameters involved in our equations. We start from the following identity:

$$x_f{}^T P_f x_f - x_0{}^T P_0 x_0 = \int_0^{t_f} \frac{d}{dt}(x^T P x)dt$$

Let's take the time derivative of the term in brackets

$$\frac{d}{dt}(x^T P x) = \left(\dot{x}^T P x + x^T \dot{P} x + x^T P \dot{x}\right)$$

The three terms can be expressed in terms of A, B, P, Q and R leading to

$$\frac{d}{dt}(x^T P x) = -x^T (Q + PBR^{-1}B^T P)x$$

Recalling the expression of J_0, the following holds:

$$J_0 = \frac{1}{2}x_f{}^T S x_f - \frac{1}{2}x_f{}^T P_f x_f + \frac{1}{2}x_0{}^T P_0 x_0 = \frac{1}{2}x_f{}^T (S - P_f)x_f + \frac{1}{2}x_0{}^T P_0 x_0$$

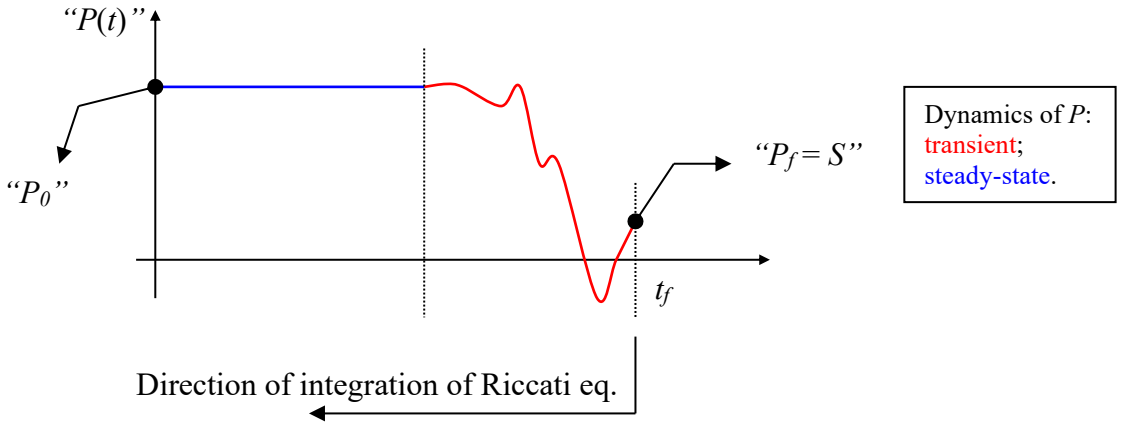
Since our aim is to minimise J_0 and since we know that P is symmetric and positive definite (so the term P_0 can never be negative) we deduce that the most favourable condition is that:

$$P_f = S$$

So these are the boundary conditions we were looking for the solution of the Riccati equation.

Integration of the Riccati equation

In the decision of the different weights of the optimisation, we can also decide the duration of the transient by imposing the time t_f (indeed, it is necessary that we assume for it a value). To obtain the $P(t)$ we integrate backwards in time the Riccati equation from the time t_f to the time t. This means that P, evolving backwards in time, will have a transient between t_f and a slightly earlier time (as in the following figure) and then it will be constant at its value P_0.



In practice it is often assumed that t_f is larger than the characteristic time of the transient of the system under control. The reason is that in this way the transient of P can be neglected (because by now the

system to be controlled will already be at steady state) and it can be assumed that P is constant. This hypothesis allows us to simplify considerably the calculations since it's no more necessary to integrate the Riccati equation but it's enough to solve it algebraically (in this case it's called Algebraic Riccati Equation). If we assume a sufficiently large t_f, the cost function can be also written:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt$$

where steady state values of x and u are zero, so the term x_f vanishes.

Alternatives to optimising the system state (standard problem)

For some systems it may not be the best thing to optimise the behaviour of state x. For example, in those cases where the physical meaning of the state is not very clear and therefore it is not possible to identify well what to optimise. An alternative may be the following:

$$J = \int_0^\infty (y^T Q y + u^T R u) dt$$

where the decision was made to optimise the output instead of the state of the system (obviously the limit on the control action is always there). This is a transformation of the standard problem that we have already seen and solved. The cases we can run into in solving the new problem depend on the system which may or may not be proper (i.e. D may or may not be null). If $D = 0$ the problem is easily solved because:

$$y = Cx$$
$$\rightarrow y^T Q y = x^T (C^T Q C) x$$

so that matrix Q of the standard problem is substituted by $C^T Q C$, then the optimal control problem is solved to obtain k. If $D \neq 0$ the problem is slightly more complex. In fact, the cost function becomes

$$J = \int_0^\infty [x^T (C^T Q C) x + u^T (D^T Q D + R) u + x^T C^T Q D u + u^T D^T Q C x] dt$$

Note that the mixed terms are each the transpose of the other and, since they give a scalar (doing all the products), they will be the same number. Therefore, we also collect the mixed contribution and rewrite the cost function as follows:

$$J = \int_0^\infty [x^T \bar{Q} x + u^T \bar{R} u + 2 x^T \bar{W} u] dt$$
$$\bar{Q} = C^T Q C$$
$$\bar{R} = D^T Q D + R$$
$$\bar{W} = C^T Q D$$

We observe that the aspect of this functional is quite general (dealing with linear systems) because it contains two "pure quadratic" terms and one "mixed quadratic" term. Through this functional (and through the weight matrices) we can decide to optimise not only the state and the control action, but also the interaction between the two (which could also have, for some systems, an interesting physical meaning). We can anyway go back to an expression analogous to the standard one even if the mixed term is present. Let's suppose that the control u is composed of two terms (i.e. we make a change of variables):

$$u = \bar{u} - \bar{R}^{-1}\bar{W}^T x$$

If we replace this expression in the cost function, we obtain

$$J = \int_0^\infty \left(x^T \bar{\bar{Q}} x + \bar{u}^T \bar{\bar{R}} \bar{u}\right) dt$$
$$\bar{\bar{Q}} = \bar{Q} - \bar{W}\bar{R}^{-1}\bar{W}^T$$
$$\bar{\bar{R}} = \bar{R}$$

So, we managed to show that, in the seen cases, the problem of optimal control can be always recast as the standard problem. The solution of the problem, however, is not yet finished because the matrix of the gains that we obtain is now relative to a system different from the initial one (because of the change of variables). In fact, the solution reads

$$\bar{u} = -\bar{\bar{R}}^{-1} B^T \bar{\bar{P}} x$$

which provides the control $\bar{u}$ and not the one of the initial system. In other words, we obtain the following system

$$\dot{x} = Ax + B\bar{u} - B\bar{R}^{-1}\bar{W}^T x = \bar{\bar{A}}x + \bar{\bar{B}}\bar{u}$$
$$\bar{\bar{A}} = A - B\bar{R}^{-1}\bar{W}^T$$
$$\bar{\bar{B}} = B$$

The gain matrix needs to be transformed to fit the original system. For this it's sufficient to do the inverse change of variables, leading to

$$u = \cdots = \left(-\bar{\bar{R}}^{-1}\bar{\bar{B}}^T\bar{\bar{P}} - \bar{R}^{-1}\bar{W}^T\right)x$$

This shows the value of *K* to be applied to the state to obtain the required optimum control of the initial system.

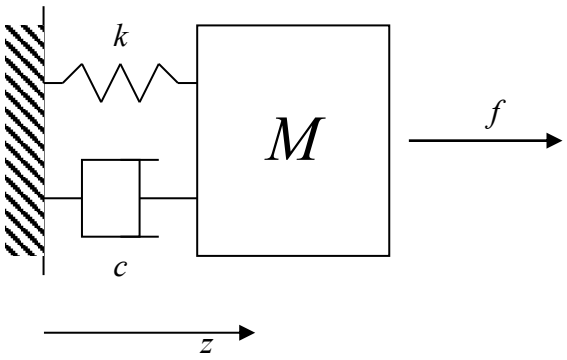
The important thing is to have seen that it is also possible to introduce in the functional *J* a mixed term (allowing to optimise in some way the contribution of *x* and *u* together) because it is always possible to go back to the standard form.

Criteria for the definition of weights Q and R

Having seen how to solve the optimal control problem from an analytical point of view (by solving the Riccati equation), we need to see how to define the optimisation weight matrices.

System energy

Let us look at an example. Consider the spring-mass-damper system in the figure.



Its dynamics are described by the equations

$$\begin{Bmatrix} \ddot{z} \\ \dot{z} \end{Bmatrix} = \begin{bmatrix} -\frac{c}{M} & -\frac{k}{M} \\ 1 & 0 \end{bmatrix} \begin{Bmatrix} \dot{z} \\ z \end{Bmatrix} + \begin{bmatrix} \frac{1}{M} \\ 0 \end{bmatrix} \{f\}$$

The total energy of the system is composed of the kinetic energy of the mass plus the elastic energy of the spring

$$E = \frac{1}{2} M \dot{z}^2 + \frac{1}{2} k z^2$$

This expression can also be written in matrix form

$$E = [\dot{z} \quad z] \begin{bmatrix} \frac{M}{2} & 0 \\ 0 & \frac{k}{2} \end{bmatrix} \begin{bmatrix} \dot{z} \\ z \end{bmatrix}$$

which looks just like the term in x of the cost function J ($x^T Q x$). Selecting then the matrix Q equal to the matrix multiplying the state vector in the expression of the energy, the optimal control guarantees the minimization of the total energy of the system. Setting then $R = 1$ it's possible to optimise (minimise) the energy spent for the control of the system. Wanting to be more general, one can consider a $R = \rho$ in order to move more or less weight on the minimization of the control expenditure.

Unfortunately, it is not always so easy to find a possible expression of Q and R. Remaining in fact in the case of the total energy of the system, it could be possible to have a non-linear expression of E so that, in order to write it in matrix form, it will be necessary to linearise it and then write a Q that will be close to the matrix of E only if we are close to the position of linearisation.

Limiting the maximum values of states and control

We may wish to optimise control according to a different criterion. For example, we may wish to establish the "freedom of movement of states" and/or controls by imposing the maximum value they can assume. This can be achieved by defining the matrices Q and R as follows.

$$Q = diag\left(\frac{1}{{x_{iMax}}^2}\right); \quad R = diag\left(\frac{1}{{u_{iMax}}^2}\right)$$

In this way, for the states that have more freedom of movement (and for the controls that can intervene "more") the weight will be lower than that of the states that are more "constrained" and for this reason

the optimal control will guarantee a more energetic action on those more limited states. Weights of this kind can be used also try to compare quantities with different dimensions (or when different units of measure are used for quantities of the same nature). With this method it's then possible to control a system without knowing anything about its energy but giving exclusively more or less importance to its states. It should be noted that a disadvantage of this type of weight is the impossibility of knowing the behaviour of the system during the transient. For example, there may be too many oscillations. To solve this problem, the system's response must be analysed once a first solution has been computed (and therefore in closed loop) and, in the event of a poor response, Q and R must be adjusted and the response checked again.

Emulating the behavior of a target system

Another interesting method of determining Q and R depends on a good knowledge of the mathematical model of the system to be controlled. Assuming that the open loop system is described by the generic model:

$$\dot{x} = Ax + Bu$$

It could be interesting to make this system behave like another similar system but described by equations with different coefficients (i.e. behaving better)

$$\dot{x}_m = A_m x + B_m u$$

where obviously matrices A_m and B_m will be different from A and B. What you want to optimise (minimise) then is the difference between these two systems, i.e.

$$e = \dot{x} - \dot{x}_m = (A - A_m)x + (B - B_m)u$$

To obtain the minimisation of e, it is necessary to define the cost function as follows

$$J = \int_0^\infty (e^T Q e + u^T R u) dt$$

The new cost function (replacing the expression of e as a function of x and u) is given by

$$\begin{aligned} J = \int_0^\infty (x^T [(A - A_m)^T Q (A - A_m)] x + \\ + u^T [(B - B_m)^T Q (B - B_m) + R] u + \\ + 2x^T [(A - A_m)^T Q (B - B_m)] u) dt \end{aligned}$$

With this we can minimize the "distance" between the behaviour of the real system and the one of the system we chose as model. Note that in this case, we could also consider the initial R as null because the term in (B - B_m) appears, which guarantees anyway a constraint on the control action. The meaning of $R = 0$ in this case is that, since there is a constraint on the action u anyway, one wants to bring the two systems closer "whatever it takes". In this type of optimal control, it is noted that the weight matrices Q and R play a secondary role (because the main purpose is to "bring" the two systems as close as possible) and one could take both equal to the identity matrix (i.e. giving the same weight to all errors on the state). In this way the weight matrices will depend exclusively on the difference between the model of the true system and the model that we would like the system to follow.

Minimum guaranteed damping

To guarantee a certain degree of damping the closed loop system, it is possible to modify the cost function *J*. The modification to introduce is:

$$J = \int_0^\infty e^{2\alpha t}(x^T Q x + u^T R u)dt$$

Minimizing this cost function means being able to subtract from the bracketed term a contribution opposite to $e^{2\alpha t}$ which results in a damping of the state (and of the control). We notice however that this cost function has a different form from the one of the standard problem and, as we have seen up to now, it is not even equal to the form containing the mixed term. To bring it back to the standard problem, we make the following substitution:

$$x = e^{-\alpha t}\bar{x}$$
$$u = e^{-\alpha t}\bar{u}$$

This transformation must also be taken into account in the equations of the system's dynamics. The equations that describe the dynamics of the new state *x* are

$$\dot{\bar{x}} = \alpha e^{\alpha t} x + e^{\alpha t}\dot{x}$$

That can be also written as

$$\dot{\bar{x}} = (\alpha I + A)\bar{x} + B\bar{u}$$

With the appropriate change of state and control variable, the cost function assumes the well-known standard form. The solution (matrix *k*) is then obtained considering the new state

$$\bar{u} = K\bar{x}$$

In this case, however, returning to the original state is very simple because both *x* and *u* undergo the same transformation and, in applying the inverse transformation, the contribution of the exponentials cancels, and we obtain therefore the same matrix *K* obtained for the "auxiliary" state

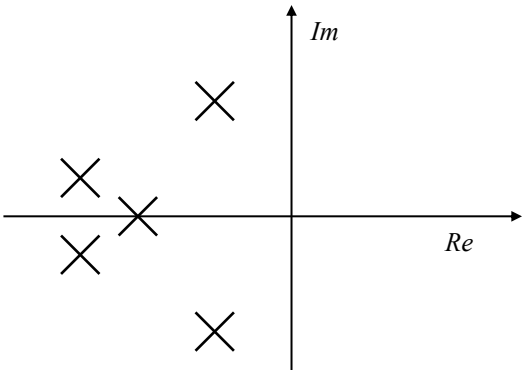
$$u = Kx$$

The gain matrix *K* obtained applying the optimal control to the modified system is then the same that we have to consider for the true system. The solution guarantees the stability of the state $\bar{x}$ because it is obtained by solving a Riccati equation. It follows that, if this state is stable, so will be the state *x* (given by $e^{-\alpha t}\bar{x}$) which will have, in addition, an exponential decay of order α (in the worst case in which $\bar{x}$ is constant). This method therefore allows us to impose a minimum damping on the state of the system by simply modifying (in practice) matrix *A* by adding to it the contribution αI where α is the desired degree of exponential damping.

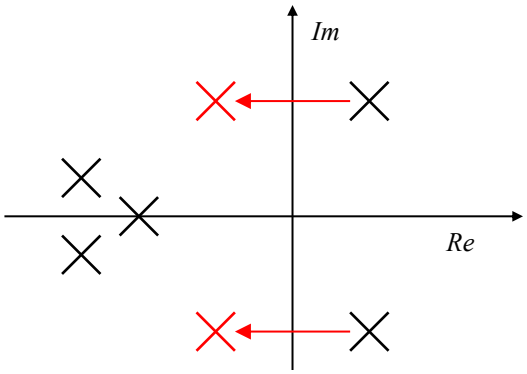
Closed-loop stability

We have already said that, through optimal control, it is not possible to know where the poles of the closed-loop system will be placed. Let's see what happens (qualitatively speaking) to the poles of a

system in the case of $Q = 0$ and $R > 0$. In particular, we want to compare a system with stable poles with one with unstable poles. The following figure summarises the consequences in the two cases.



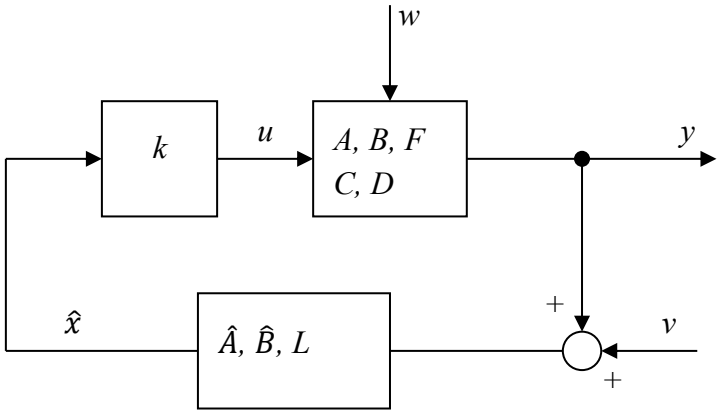
Stable open loop system: since there are no constraints on the movement of the state, free movement is already fine and therefore the smallest control that can be applied is $u = 0$!



Unstable open loop system: since the solution of the control problem guarantees a stable system, the fact of accepting any movement of the state, as long as it is stable, implies a control u that places the closed-loop poles in symmetrical position of the unstable poles with respect to the *Im* axis.

Optimal observer

We have seen how to design (by placing the poles) any linear observer that allows us to estimate the whole state of a physical system. If we are in the presence of disturbances, the concept of an optimal observer may arise. The optimisation of the observer can be done instead in the time domain, following a method similar to the one seen for optimal control. Let's consider the following scheme, representative of the situation we want to study:



Assume the following model for the system, where both states and measurements are corrupted by noise

$$\begin{cases} \dot{x} = Ax + Bu + Fw \\ y = Cx + v \end{cases}$$

which is particularly simple when w and v are white noise. The matrix F is an analogue of B for the disturbances w. The observer, assuming it is represented by a linear system, will be described by

$$\begin{cases} \dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}u + Ly \\ \hat{y} = \hat{x} \end{cases}$$

where $\hat{A}$ and $\hat{B}$ are of appropriate dimensions. Write the dynamics of the error between the true state and the observed state

$$\dot{x} - \dot{\hat{x}} \equiv \dot{e} = Ax + Bu + Fw - \hat{A}\hat{x} - \hat{B}u - Ly$$

and, noticing that

$$\begin{aligned} e &= x - \hat{x} \quad \rightarrow \quad \hat{x} = x - e \\ y &= Cx + v \end{aligned}$$

reformulate the error dynamics as

$$\dot{e} = \left(A - LC - \hat{A}\right)x + \left(B - \hat{B}\right)u + \hat{A}e + Fw - Lv$$

By analogy with the development of the linear observer, in order to make the dynamics of e independent of x and u we will impose

$$\begin{aligned} \hat{A} &= A - LC; \\ \hat{B} &= B \end{aligned}$$

Then

$$\dot{e} = (A - LC)e + Fw - Lv$$

Assuming now that *w* and *v* are white noises, we can say that their algebraic sum will be another white noise and so we'll pose:

$$\eta = Fw - Lv$$

such that

$$\dot{e} = (A - LC)e + \eta$$

Note that noise η is continuously forcing the observation error. One can then think of applying an optimisation (i.e. a minimisation) on the variance of *e*. In this way, the error, although forced by the noise *w* and *v* (and thus η) will be distributed around its mean value (hopefully equal to zero) with a minimum variance (i.e. all the values it takes on can be considered very close to the mean value = 0). For this, it is necessary to calculate the statistical indicators of *e*. The optimisation can then be applied. We will then look for the matrix *L* that allows us to minimise the variance of *e*, which, in general, will also be a matrix. Let us call the variance

$$\sigma_{ee}{}^2 = P$$

The dynamics of the variance is represented by the Lyapunov equation

$$\dot{P} = (A - LC)P + P(A^T - C^T L^T) + Q_\eta$$

where Q_η is the variance of η ($\sigma_{\eta\eta}{}^2$) that is given by

$$Q_\eta = \int \big(Fw(t) - Lv(t)\big)\big(Fw(t) - Lv(t)\big)^T dt$$

Knowing the variance matrices *W*, *V* associated to the noise signals *w* and *v* , and the cross-variance *X*, Q_η is computed as:

$$Q_\eta = FWF^T + LVL^T - FXL^T - LX^T F^T$$

In most cases *v* and *w* are considered uncorrelated, that means that the cross-variance matrix *X* is zero. The Lyapunov equation then becomes

$$\dot{P} = (A - LC)P + P(A^T - C^T L^T) + FWF^T + LVL^T - FXL^T - LX^T F^T$$

This equation has the same structure as the Riccati equation used to solve the optimal control problem. We then look for a solution by carrying out an analogy with the results obtained for optimal control. We observe in particular that the solution of the optimal problem in the case of the control was provided by:

$$K = -R^{-1}B^T P$$

P is the solution of the Riccati equation. Let us then try to relate this to the solution we are looking for (i.e. the matrix *L* that optimises the observer by minimising the variance of the error). Let us suppose that the solution for *L* is given by an expression analogous to the one of the optimal control

$$L^T = V^{-1}CP$$

in which the following analogy is implemented

$$-R \leftrightarrow V$$
$$B^T \leftrightarrow C$$

Note that L is transposed because, while k multiplies the state of the system (and so has a number of columns equal to the number of states), L will have the number of rows equal to the number of states of the system (because $y = Cx + v$). If we also assume that X is null, we obtain, after a simplification:

$$\dot{P} = AP + PA^T + FWF^T - PC^T V^{-1} CP$$

which is perfectly equivalent to Riccati's equation if we consider the following associations

$$-A^T \leftrightarrow A$$
$$-Q \leftrightarrow FWF^T$$

To find the L that optimises the observation, I can then (assuming $X = 0$) solve the Riccati equation and then substitute P in the $L^T = V^{-1}CP$.

The term X can be seen as the cross term in the optimal control problem. Another way to take it into account is to see it as a perturbation of the optimal condition. Suppose, for example, that we decompose P and L into the optimal solution and perturbation:

$$P = \hat{P} + U$$
$$L = \hat{L} + Z$$

where U and Z are the perturbations of the solutions, respectively, $\hat{P}$ ed $\hat{L}$. The equation obtained by substituting these into the Lyapunov equation is made up of two groups of separable terms and, putting these two groups in order, looks as follows:

$$\begin{aligned}\dot{\hat{P}} + \dot{U} = &\left(A - \hat{L}C\right)\hat{P} + \hat{P}\left(A^T - C^T\hat{L}^T\right) + FWF^T + \hat{L}V\hat{L}^T - FX\hat{L}^T - \hat{L}X^TF^T + \\ &+\left(A - \hat{L}C - ZC\right)U + U\left(A^T - C^T\hat{L}^T - C^TZ^T\right) + ZVZ^T + \\ &-Z\left(C\hat{P} - V\hat{L}^T + X^TF^T\right) - \left(\hat{P}C^T - \hat{L}V + FX\right)Z^T\end{aligned}$$

Since $\hat{P}$ is, by definition, solution of the equation

$$\dot{\hat{P}} = \left(A - \hat{L}C\right)\hat{P} + \hat{P}\left(A^T - C^T\hat{L}^T\right) + FWF^T + \hat{L}V\hat{L}^T - FX\hat{L}^T - \hat{L}X^TF^T$$

they cancel out from the above equation. Furthermore, the terms $Z\left(C\hat{P} - V\hat{L}^T + X^TF^T\right)$ and $\left(\hat{P}C^T - \hat{L}V + FX\right)Z^T$ need to be zero since they are not symmetric and would not be compatible with the definition of Riccati equation. Therefore, the following must hold

$$\hat{P}C^T - \hat{L}V + FX = 0$$
$$\rightarrow \hat{L} = \left(\hat{P}C^T + FX\right) \cdot V^{-1}$$

This provides the solution for $\hat{L}$ as function of $\hat{P}$. Note that this solution reverts to the previous one in case FX is missing. This guarantees that U is a solution of a Riccati equation (represented by the remaining terms of the equation). Consequently, U will surely be positive definite and this implies that $\hat{P}$ is really a minimum (because $\hat{P} + U$ will surely be greater) and therefore an optimal solution (as we had originally assumed). What we have done so far shows that if we are able to solve the

optimal control problem, we can, by applying the appropriate substitutions, solve in the same way the optimal observer problem.

Notes on the optimal observer

Let us see the differences in the behaviour of this observer with respect to the classical designed observer (i.e. L evaluated by positioning the poles of A). Let us summarise the expressions obtained that allow us to design the optimal observer (in the case where $X = 0$).

$$\begin{cases}\dot{\hat{x}} = \hat{A}x + \hat{B}u + Ly \\ \hat{y} = \hat{x}\end{cases}$$
$$\hat{A} = A - LC$$
$$\hat{B} = B$$
$$L = V^{-1}CP \quad \leftarrow \quad \dot{P} = AP + PA^T + FWF^T - PC^TV^{-1}CP$$

The good performances of the optimal observer in the presence of disturbances is due to the dependence of L on V^{-1}. This is seen from the following considerations.

Very noisy measurements (V with large elements)	Low noise measurements (V with small elements)
In this case, L will be small because it derives from the product of CP with V^{-1} which has small values. It follows that the observer will have a dynamics similar to that of the model considered for the observed system ($\hat{A} \to A$): "…the observer "trusts poorly" the measurements (because the contribution of y to his dynamics is damped by the small L) and relies more on the model we have assumed for the real system …"	In this case, L will be quite large. Thus, the dynamics of the observer will be quite different from that of the model of the observed system. ($\hat{A} \neq A$): "…the observer "trusts greatly" the y-measurements and because of this it can be faster than the physical system providing a better observation …"

If we look at what has been said now, it seems that the contribution of noise w is not considered. This is not true because matrix W (of the variances of the noises on the physical system) is contained within the Lypunov equation (which we then solve in P as if it were a Riccati equation). Therefore, we can say that their contribution enters, indirectly, through the matrix P of the variances of the observation error. Bearing in mind that it is necessary to invert matrix V, it will be appropriate to have sensors with similar "noise". In this way V will be well conditioned and will not present problems in the inversion.

Example of control system based on inertia and reaction wheels

We can now consider one example of control system with one IW along the pitch axis (z) and two RW along the roll and yaw axes (x and y). The Euler equations are in this case:

$$\begin{cases} I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega} + \underline{\dot{h}}_r + \underline{\omega} \wedge \underline{h}_r = \underline{M}_d \\ \underline{\dot{h}}_r = \underline{M}_c \end{cases}$$

We have 6 equations and each rotor is along one of the principal axes, so A is an identity matrix. Assuming we want to control the attitude and not the angular velocity, we must add equations for the attitude kinematics:

$$f\left(\underline{\alpha}, \underline{\dot{\alpha}}, \underline{\omega}\right) = 0$$

We can consider the case for which we refer the attitude to the LVLH frame. The nominal condition is a constant angular velocity around the pitch axis, one rotation per orbit, and a nominal angular velocity for the IW. The remaining nominal parameters are zero. The kinematic equations are then:

$$\begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} = \begin{bmatrix} 1 & \alpha_z & -\alpha_y \\ -\alpha_z & 1 & \alpha_x \\ \alpha_y & -\alpha_x & 1 \end{bmatrix} \begin{Bmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z + n \end{Bmatrix}$$

Since along x and y we have RW, h_{rx}=h_{ry}=0 in nominal conditions. The linearized equations are:

$$\begin{cases} I_x\left(\ddot{\alpha}_x - \dot{\alpha}_y n\right) + \left(I_z - I_y\right)\left(\mathrm{n}\dot{\alpha}_y + n^2\alpha_x\right) + \dot{h}_{\mathrm{rx}} + h_{\mathrm{rz}}\left(\dot{\alpha}_y + \mathrm{n}\alpha_x\right) = M_{\mathrm{dx}} \\ I_y\left(\ddot{\alpha}_y + \dot{\alpha}_x n\right) + \left(I_x - I_z\right)\left(\mathrm{n}\dot{\alpha}_x - n^2\alpha_y\right) + \dot{h}_{\mathrm{ry}} - h_{\mathrm{rz}}\left(\dot{\alpha}_x - \mathrm{n}\alpha_y\right) = M_{\mathrm{dy}} \\ I_z\ddot{\alpha}_z + \dot{h}_{\mathrm{rz}} = M_{\mathrm{dz}} \end{cases}$$

$$\begin{cases} \dot{h}_{rx} = -M_{cx} \\ \dot{h}_{ry} = -M_{cy} \\ \dot{h}_{rz} = -M_{cz} \end{cases}$$

Now we reformulate the problem inserting the control, getting to the equations:

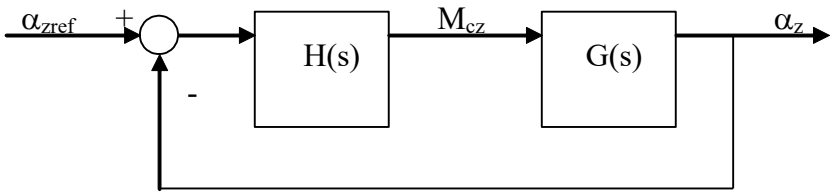
$$\begin{cases} I_x\left(\ddot{\alpha}_x - \dot{\alpha}_y n\right) + \left(I_z - I_y\right)\left(\mathrm{n}\dot{\alpha}_y + n^2\alpha_x\right) + h_{\mathrm{rz}}\left(\dot{\alpha}_y + \mathrm{n}\alpha_x\right) = M_{\mathrm{cx}} \\ I_y\left(\ddot{\alpha}_y + \dot{\alpha}_x n\right) + \left(I_x - I_z\right)\left(\mathrm{n}\dot{\alpha}_x - n^2\alpha_y\right) - h_{\mathrm{rz}}\left(\dot{\alpha}_x - \mathrm{n}\alpha_y\right) = M_{\mathrm{cy}} \\ I_z\ddot{\alpha}_z = M_{\mathrm{cz}} \\ \underline{\dot{h}}_r = -\underline{M}_c \end{cases}$$

The design of the controller can be based on classical transfer function techniques. For the pitch axis (z) the transfer function is

$$s^2\alpha_z(s) = \frac{M_{cz}}{I_z}$$

$$G_z(s) = \frac{\alpha_z(s)}{M_{cz}} = \frac{1}{I_z s^2}$$

The block diagram of the controller is:



The closed loop transfer function becomes:

$$F(s) = \frac{H(s)G(s)}{1 + H(s)G(s)} = \frac{\alpha_z(s)}{\alpha_{\text{zref}}}$$

In order to define H(s), we should first define the performances required by F(s). In addition to the above scheme, we introduce the transfer function between a disturbance torque and the attitude angle, that is:

$$F_d(s) = \frac{G(s)}{1 + H(s)G(s)} = \frac{\alpha_z(s)}{T}$$

To evaluate H(s), we can assume a PD-like control, since the system dynamics is of second order:

$$M_{cz} = K_d s\alpha + K_p \alpha$$
$$H(s) = \frac{M_{cz}}{\alpha_z(s)} = \left(K_d s + K_p\right)$$

We then have:

$$F(s) = \frac{\left(K_d s + K_p\right)\frac{1}{I_z s^2}}{1 + \frac{\left(K_d s + K_p\right)}{I_z s^2}} = \frac{\frac{\left(K_d s + K_p\right)}{I_z s^2}}{1 + \frac{\left(K_d s + K_p\right)}{I_z s^2}} = \frac{K_d s + K_p}{I_z s^2 + K_d s + K_p}$$

This is actually a second order system characterized by:

$$\omega = \sqrt{\frac{K_p}{I_z}}$$
$$2\xi\omega = \frac{K_d}{I_z}$$

ω is the closed loop natural frequency of the pitch oscillations and ξ is the damping. The ideal damping would be around 0.7, but in order to reduce the control action a damping in the order of 0.3 can be accepted. The natural frequency can be related to the orbital frequency n, imposing that steady-state is reached within a specific fraction of orbit period. A choice of ω = 20n is commonly accepted as a good solution. With these assumptions, the steady state error angle can be evaluated by using the final value theorem:

$$\alpha_{z\infty} = \lim_{s \to 0} \mathrm{sF}_d\,(s) d(s)$$

where d(s) is the Laplace transform of the disturbing torque. Maximizing the disturbance, considering it as constant, we have:

$$d(s) = \frac{1}{s}$$

$$\alpha_{z\infty} = \lim_{s\to 0} F_d(s) = \frac{\frac{1}{I_z s^2}}{1 + \frac{(K_d s + K_p)}{I_z s^2}} = \frac{1}{I_z s^2 + K_d s + K_p} = \frac{1}{K_p}$$

This relation can also be used to tune the proportional gain in order to reach a specified steady state error. The performance of the IW can then be analyzed:

$$\underline{\dot{h}}_r = \underline{M}_c$$

Taking the Laplace transform of the IW with the control:

$$I_r s\omega_r = (K_d s + K_p)\alpha(s)$$

The above equation can be used to check if the IW reaches saturation.

To study the control for x and y axes we can make some simplifying assumptions in order to design the controller with Laplace transform techniques. Consider M_{cx}=0, and assume only α_y is measured, as could be done with a fixed head Earth sensor that does not provide α_x:

$$\begin{cases} I_x(s^2\alpha_x - s\alpha_y n) + (I_z - I_y)(ns\alpha_y + n^2\alpha_x) + h_{rz}(s\alpha_y + n\alpha_x) = 0 \\ I_y(s^2\alpha_y + s\alpha_x n) + (I_x - I_z)(ns\alpha_x - n^2\alpha_y) - h_{rz}(s\alpha_x - n\alpha_y) = M_{cy} \end{cases}$$

The transfer function is then:

$$G_y(s) = \frac{As^2 + B}{Cs^4 + Ds^2 + E}$$
$$A = I_x$$
$$B = (I_z - I_y)n^2 + h_{rz}n$$
$$C = I_x I_y$$
$$D = \left\{n^2 I_y(I_z - I_y) + I_y h_{rz} n - I_x(I_x - I_z)n^2 - I_x h_{rz} n + \left[(I_x + I_y - I_z)^2 - h_{rz}^2\right]\right\}$$
$$E = \left[(I_z - I_x)(I_z - I_y)n^4 - h_{rz}(I_x - I_z)n^3 + h_{rz}(I_z - I_y)n^3 + h_{rz}^2 n^2\right]$$

The transfer function can be manipulated so that the terms depending on h_{rz} are explicitly taken into account. In this context, these should be considered constant, in the nominal condition.

$$\alpha_y[as^4 + bs^3 + cs^2 + ds + e + h_{rz}f(s)] = sh_{ry}g(s)$$
$$\alpha_y H(s) = M_{cy}(s)$$

For this system a PD control might not be appropriate to impose specified performances, but a fourth order compensator can be designed and target performances reached.

Active Control of Spacecraft in the LVLH frame

$$\begin{cases} I_x\ddot{\alpha}_x + (I_z - I_y - I_x)n\dot{\alpha}_y + (I_z - I_y)n^2\alpha_x = 0 \\ I_y\ddot{\alpha}_y + (I_x + I_y - I_z)n\dot{\alpha}_x + (I_z - I_x)n^2\alpha_y = 3n^2(I_x - I_z)\alpha_y \\ I_z\ddot{\alpha}_z = -3n^2(I_y - I_x)\alpha_z \end{cases}$$

Simple proportional controllers can be used to control the pitch axis in the LVLH frame

$$I_z\ddot{\alpha}_z = T_3$$

The torque T_3can then be chosen to provide asymptotic stability for example:

$$T_3 = -k_1\alpha_z - k_2\dot{\alpha}_z$$

The closed-loop system can then be expressed as:

$$I_z\ddot{\alpha}_z + k_2\dot{\alpha}_z + k_3\alpha_z = 0$$

where $k_3 = k_1 + 3n^2(I_y - I_x)$

The eigenvalues related to this closed-loop system are:

$$\lambda = \frac{-k_2 \pm \sqrt{k_2^2 - 4k_3}}{2}$$

So, we can see that (linear) asymptotic stability is achieved. It is also possible to use this information to reduce the tuning requirement since setting $k_2^2 = 4k_3$ provides the fastest system response (analogous to critical damping).

In the other two axis (yaw and roll) the equations are coupled. To design and tune a control law in this case it is useful to recast the equations into first order

$$\begin{cases} \dot{\alpha}_x = v_x \\ \dot{\alpha}_y = v_y \\ \dot{v}_x = -n(K_x - 1)v_y - n^2K_x\alpha_x + u_1 \\ \dot{v}_y = n(1 - K_y)v_x - n^2K_y\alpha_y + u_2 \end{cases}$$

With $x = [\alpha_x \quad \alpha_y \quad v_x \quad v_y]^T$ this can be expressed as $\dot{x} = Ax + Bu$ with $u = [u_1 \quad u_2]^T$and

$$B = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^T$$

Firstly, controllability of the system muct be checked since the first order differnetial equation is 4 dimensional and we only have to control inputs. This is achieved through the computation:

$$C = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

If the matrix C is full rank, then the system is controllable.

If the system is controllable a simple continuous control law of the form:

$$\underline{u} = -K\underline{x}$$

Yields the following closed-loop system

$$\dot{\underline{x}} = (A - BK)\underline{x}$$

Then *K* is chosen such that the eigenvalues of the matrix $(A - BK)$ lie in the left-hand side of the complex plane.

The gain matrix *K* can be determined by minimizing thr Linear quadratic performance index:

$$J = \frac{1}{2}\int_0^\infty (\underline{x}^T Q\underline{x} + \underline{u}^T R\underline{u})\, dt$$

Where Q is a symmetric and positive semidefinite weighting matrix and R a symmetric positive definite are weighting matrices then the optimal gain matrix is:

$$K = R^{-1}B^T P$$

With

$$0 = A^T P + PA - PBR^{-1}B^T P + Q$$

General control problem

We have seen that when the actuators are IW or RW we can write:

$$M = A\dot{h}_r$$

The question now is how to evaluate M in a general case. Assuming the satellite is inertial pointing, Euler equations assume a particularly simple form, since no nominal value of angular velocity is present. Equations are decoupled:

$$M_i = I\dot{\omega}_i i = 1,2,3$$

Linearizing the system, we have three second order decoupled equations:

$$M_i = I\ddot{\alpha}_i$$

Lack of a nominal angular momentum forces us to have three independent actuators for the three axes; otherwise the system would not be controllable. Each equation represents a linear second order system, so that we can assume a simple PID control would allow obtaining the desired system performances:

$$M = f(\alpha) = PID(\alpha)$$

In some cases the integral term might not be required, in case a steady state error is allowed provided it is small or in case no constant component of disturbance torque is present.
When the satellite is far from equilibrium, its dynamics should include also the coupling terms due to angular velocities, so that we must write:

$$M = I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega}$$

To consider the coupling terms $\underline{\omega} \wedge I\underline{\omega}$ the control torque should be evaluated as:

$$M_c = I\underline{\dot{\omega}}$$

and then the nonlinear terms can be considered as a correction to the control torque as:

$$M = M_c + \underline{\omega} \wedge I\underline{\omega}$$

In case of a dual spin satellite the problem is formulated as:

$$M = I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega} + A\underline{\dot{h}_r} + \underline{\omega} \wedge \mathrm{Ah}_r \quad \rightarrow \quad \text{nonlinear dynamics}$$
$$0 = I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega} + A\underline{\dot{h}_r} + \underline{\omega} \wedge \mathrm{Ah}_r \quad \rightarrow \quad \text{control equation}$$
$$M_c = I\underline{\dot{\omega}} = \mathrm{PID}(\omega, \alpha) \quad \rightarrow \quad \text{pseudocontrol function}$$
$$M_c = -\underline{\omega} \wedge I\underline{\omega} - A\underline{\dot{h}_r} - \underline{\omega} \wedge \mathrm{Ah}_r \quad \rightarrow \quad \text{actuator equation}$$
$$\underline{\dot{h}_r} = A^*\left[-M_c - \underline{\omega} \wedge I\underline{\omega} - \underline{\omega} \wedge \mathrm{Ah}_r\right] \quad \rightarrow \quad \text{actuator command}$$

If, besides nonzero angular velocities, we consider also large angular rotations, the error angles definition depends on the sequence of rotations considered. A general solution of the control problem must then be sought for, starting from the definition of direction cosine matrix.

Call A_S the satellite attitude matrix in an inertial frame, and assume the target attitude is given by matrix A_T:

$$A_S = \begin{bmatrix} a_{11S} & a_{12S} & a_{13S} \\ a_{21S} & a_{22S} & a_{23S} \\ a_{31S} & a_{32S} & a_{33S} \end{bmatrix} = \begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix}$$

$$A_T = \begin{bmatrix} a_{11T} & a_{12T} & a_{13T} \\ a_{21T} & a_{22T} & a_{23T} \\ a_{31T} & a_{32T} & a_{33T} \end{bmatrix} = \begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix}$$

Each row of A_S and A_T represents one reference axis, either satellite or target. Our goal is to have:

$$A_S A_T^T = I$$

In actual conditions, the error in the attitude is expressed as:

$$A_S A_T^T = A_e$$

where A_e is the attitude error. We want now to relate the attitude error A_e with the control torque, in order to make A_e=I.

$$A_S A_T^T = \begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix} [X_T^T \quad Y_T^T \quad Z_T^T] = \begin{bmatrix} X_S X_T^T & X_S Y_T^T & X_S Z_T^T \\ Y_S X_T^T & Y_S Y_T^T & Y_S Z_T^T \\ Z_S X_T^T & Z_S Y_T^T & Z_S Z_T^T \end{bmatrix}$$

To reach the zero error, the extra diagonal terms must vanish to zero. $X_S Y_T^T = 0$ means that X_S and Y_T must become orthogonal, since this is the representation of the dot vector product. This condition can be obtained if the satellite rotates around its z body axis; so that we can say that the torque acting around the satellite body axis Z_S must take the form:

$$M_{zS} = f_z(X_S Y_T^T)$$

The actual control law is still undefined. The same must apply to the other extra diagonal terms of the attitude error matrix. If we want $X_S Z_T^T = 0$ the torque must be around the satellite body axis Y_S, in the general form:

$$M_{yS} = f_y(X_S Z_T^T)$$

Finally, to have $Y_S X_T^T = 0$ we need a torque around the satellite body axis X_S, so that:

$$M_{xS} = f_x(Y_S Z_T^T)$$

It is remarked that a similar result would be obtained by considering the terms below the diagonal, so that $M_{zS} = f_z(Y_S X_T^T)$ and the same for M_{yS} and M_{xS}.
To understand how the control functions can be designed, consider the case of small errors. In this case the error can be expressed in linearized form as:

$$A_S A_T^T = \begin{bmatrix} 1 & \alpha_Z & -\alpha_y \\ -\alpha_z & 1 & \alpha_x \\ \alpha_y & -\alpha_x & 1 \end{bmatrix}$$

In the case of decoupled motion or small rotations, we can write:

$$M_z = PD(\alpha) = K_{pz}\alpha_z + K_{dz}\dot{\alpha}_z$$

It is noticed that α_z corresponds to $X_S Y_T^T$ in the case of large errors, or to a_{12e}. We can then generalize the control function as:

$$M_{zs} = K_{pz}a_{12e} + K_{dz}\omega_z$$

The derivative should rigorously depend on the derivative of a_{12e}, or$X_S Y_T^T$, but since it is mostly used to introduce viscous damping any term with zero final value and sign opposite to the actual velocity can be used, so that ω_z can be a further candidate, in some cases easier to use for control. We can then evaluate K_p and K_d for the linear approximation of the dynamics and simply extend the validity of the control law as follows.

$$\begin{cases} M_x = K_{\mathrm{px}}\alpha_x + K_{\mathrm{dx}}\dot{\alpha}_x \\ M_y = K_{\mathrm{py}}\alpha_y + K_{\mathrm{dy}}\dot{\alpha}_y \\ M_z = K_{\mathrm{pz}}\alpha_z + K_{\mathrm{dz}}\dot{\alpha}_z \end{cases} \rightarrow \begin{cases} M_{\mathrm{xs}} = K_{\mathrm{px}}a_{23\mathrm{e}} + K_{\mathrm{dx}}\omega_x \\ M_{\mathrm{ys}} = K_{\mathrm{py}}a_{31\mathrm{e}} + K_{\mathrm{dy}}\omega_y \\ M_{\mathrm{zs}} = -K_{\mathrm{pz}}a_{21\mathrm{e}} + K_{\mathrm{dz}}\omega_z \end{cases}$$

The same can be designed adopting the terms below the diagonal, obtaining for each different case a different transient response in case of large initial errors:

$$M_{zs} = -K_{pz}a_{21e} + K_{dz}\omega_z$$

We can finally try to have an intermediate situation, for which:

$$\begin{cases} M_{xS} = f_x(Y_S Z_T^T - Z_S Y_T^T) \\ M_{yS} = f_y(X_S Z_T^T - Z_S X_T^T) \\ M_{zS} = f_z(X_S Y_T^T - Y_S X_T^T) \end{cases}$$

In this case, since

$$a_{12e} - a_{21e} = 2\alpha_z$$

we must divide by 2 the proportional gain if it was evaluated on the basis of a control proportional to α:

$$\begin{cases} M_{\mathrm{xs}} = \dfrac{K_{\mathrm{px}}}{2}(a_{23\mathrm{e}} - a_{32\mathrm{e}}) + K_{\mathrm{dx}}\omega_x \\ M_{\mathrm{ys}} = \dfrac{K_{\mathrm{py}}}{2}(a_{31\mathrm{e}} - a_{13\mathrm{e}}) + K_{\mathrm{dy}}\omega_y \\ M_{\mathrm{zs}} = \dfrac{K_{\mathrm{pz}}}{2}(a_{12\mathrm{e}} - a_{21\mathrm{e}}) + K_{\mathrm{dz}}\omega_z \end{cases}$$

It is also remarked that a_{12e}-a_{21e} is proportional to the z component of the Euler axis of the attitude error matrix. It is likely that this final form of control will try to get the satellite rotating around the Euler axis of the attitude error, so that the transient response might be faster that in the previous formulations. One further option requires writing the terms a_{12} and a_{21} as a function of the quaternions, respectively $2(q_1q_2+q_3q_4)$ and $2(q_1q_2-q_3q_4)$, so that the control functions become:

$$\begin{cases} M_{xs} = \dfrac{K_{px}}{2}(4q_{1e}q_{4e}) + K_{dx}\omega_x = 2K_{px}q_{1e}q_{4e} + K_{dx}\omega_x \\ M_{ys} = \dfrac{K_{py}}{2}(4q_{2e}q_{4e}) + K_{dy}\omega_y = 2K_{py}q_{2e}q_{4e} + K_{dy}\omega_y \\ M_{zs} = \dfrac{K_{pz}}{2}(4q_{3e}q_{4e}) + K_{dz}\omega_z = 2K_{pz}q_{3e}q_{4e} + K_{dz}\omega_z \end{cases}$$

Notice that the quaternions are relative to the attitude error.
As the satellite approaches the target condition, the dynamics become automatically linear. If at steady state we want a nonzero value for one angular velocity component, around that specific axis the control function must be modified as:

$$M_i = K_{\text{pi}}\alpha_i + K_{\text{di}}(\dot{\alpha}_i - \bar{\dot{\alpha}}_i)$$
$$\Rightarrow M_{\text{is}} = 2\text{K}_{\text{pi}}q_{\text{ie}}q_{4\text{e}} + K_{\text{di}}(\omega_i - \bar{\omega}_i)$$

Angular velocity ω can be expressed as a function of quaternions. From the quaternion kinematics we know that:

$$\dot{q} = \frac{1}{2}\Omega q$$

This can be reformulated as:

$$\Omega q = \begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix} \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} = Q\omega$$

and the angular velocity is:

$$\omega = 2Q^*\dot{q}$$

where Q^* is the pseudo inverse of Q, that is also equal to Q^T due to its structure. It is easy to verify that:

$$Q^T Q = I$$

Nonlinear Control Theory

In this section we describe the basics of nonlinear control theory. The theory is based on Lyapunov's second stability theorem and can be used to prove global stability of a system, as opposed to linear stability which can only prove global stability for linear systems. Lyapunov's second theorem can be applied to prove stability without the need for linearization of the nonlinear dynamics about an equilibrium point.

Lyapunov's Second Stability Theorem

Consider an autonomous nonlinear dynamic system described by:

$$\dot{\underline{x}} = f(\underline{x}), f(\underline{x}^*) = 0$$

Where $\underline{x}^*$ is an isolated equilibrium point. If there exists in some finite neighbourhood D , of the equilibrium point $\underline{x}^*$a scalar function $V(\underline{x})$ with continuous first partial derivative with respect to $\underline{x}$ such that the following conditions hold

$$\begin{aligned} &\text{(i)} \quad V(\underline{x}) > 0 \text{for all } \underline{x} \neq \underline{x}^* \text{in } D \text{ and } V(\underline{x}^*) = 0 \\ &\text{(ii)} \quad \dot{V}(\underline{x}) < 0 \text{for all } \underline{x} \neq \underline{x}^* \text{in } D \text{ except for } \dot{V}(\underline{x}^*) = 0 \end{aligned}$$

Then the system is said to be **asymptotically stable.** If D includes all possible states then the system is said to be **globally asymptotically stable.**

If $\dot{V}(\underline{x}) = 0 \text{for all } \underline{x} \neq \underline{x}^* \text{in } D$

then the system is **Lyapunov stable**.

The above theorem can be used to deduce if a feedback-control law is asymptotically stable or not. For example, lets take the case of designing a de-tumbling controller for a spacecraft. We have the equations of motion (ignoring environmental disturbances for now):

$$\begin{aligned} \dot{\omega}_x &= \frac{I_y - I_z}{I_x}\omega_y\omega_z + \frac{u_x}{I_x} \\ \dot{\omega}_y &= \frac{I_z - I_x}{I_y}\omega_x\omega_z + \frac{u_y}{I_y} \\ \dot{\omega}_z &= \frac{I_x - I_y}{I_z}\omega_y\omega_x + \frac{u_z}{I_z} \end{aligned}$$

Where $\underline{u} = [u_x \quad u_y \quad u_z]^T$is the control torque input assumed to be given perfectly by an actuator. We can use the kinetic energy as a candidate Lyapunov function to determine suitable controls:

$$V(\underline{\omega}) = \frac{1}{2}(I_x\omega_x^2 + I_y\omega_y^2 + I_z\omega_z^2)$$

then

$$\dot{V}(\underline{\omega}) = I_x\omega_x\dot{\omega}_x + I_y\dot{\omega}_y + I_z\dot{\omega}_z$$

Then substituting in the dynamic equation (Euler equations plus the control) we get

$$\dot{V}(\underline{\omega}) = \omega_x u_x + \omega_y u_y + \omega_z u_z$$

Then we need to define a control law such that $\dot{V}(\underline{\omega})$ <0 to guarentee aymptotic stability i.e $\omega_x \to 0, \omega_y \to 0, \omega_z \to 0$. We can set $\underline{u} = -k\underline{\omega}$to yield global asymptotic stability where k is a tuning parameter and $\underline{\omega} = [\omega_x \quad \omega_y \quad \omega_z]^T$. However, this choice is not unique and other choices exist such as a bang-bang type control (instantaneous switching)

$$\underline{u} = -T\, \underline{sgn}\, \omega_{max}$$

Where $\underline{sgn}\, \omega = [sgn\, \omega_x \quad sgn\, \omega_y \quad sgn\, \omega_z]^T$where the sgn function returns 1 if the component is positive and -1 if it is negative.

For example, the kinetic energy of a rigid-body gives

$$E(\underline{\omega}) = \frac{1}{2}(I_x\omega_x + I_y\omega_y + I_z\omega_z)$$

and with $\underline{M} = -k\underline{\omega}$ it can be easily shown that

$$\dot{E}(\underline{\omega}) = -k(\omega_x^2 + \omega_y^2 + \omega_z^2)$$

Therefore, the kinetic energy in this case is a Lyapunov function and the equilibrium point (0,0,0) of the closed-loop system is globally asymptotically stable.

Lyapunov Control functions

In classical mechanics Lyapunov's second stability theorem is used to prove nonlinear stability of an autonomous nonlinear system. However, it can also be used to design control algorithms for systems of the form:

$$\dot{\underline{x}} = f(\underline{x}, \underline{u}), f(\underline{x}^*, \underline{u}) = 0$$

In this case we choose a suitable Lyapunov candidate function that satisfies the condition:

(i) $V(\underline{x}) > 0$for all $\underline{x} \neq \underline{x}^*$in *D and* $V(\underline{x}^*) = 0$

Then differentiate this function with respect to time which will yield:

$$\frac{dV(\underline{x})}{dt} = s(\underline{x}, \underline{u})$$

Then the control $\underline{u}$is chosen such that

$$s(\underline{x}, \underline{u}) < 0$$

For all $\underline{x} \neq \underline{x}^*$. This provides a constructive approach for nonlinear control design. On differentiation of this Lyapunov function and for the nonlinear control system

$$J\frac{d\underline{\omega}}{dt} = J\underline{\omega} \times \underline{\omega} + \underline{u}$$

Yields

$$\dot{V}(\underline{\omega}) = \underline{\omega}^T \underline{u}$$

Then we must choose a $\underline{u}$such that $\underline{\omega}^T \underline{u} < 0$. Obviously, we can set $\underline{u} = -k\underline{\omega}$to yield global asymptotic stability. However, other choices exist such as

$$\underline{u} = -T\, sgn\, \underline{\omega}_{max}$$

Which also guarantees global asymptotic stability of the closed-loop system. This type of control is an on-off type controller that would be more suited to a thruster.

Slew maneuvers

$$\dot{\omega}_1 = \frac{I_2 - I_3}{I_1}\omega_2\omega_3 + \frac{u_1}{I_1}$$
$$\dot{\omega}_2 = \frac{I_3 - I_1}{I_2}\omega_1\omega_3 + \frac{u_2}{I_2}$$
$$\dot{\omega}_3 = \frac{I_1 - I_2}{I_3}\omega_2\omega_1 + \frac{u_3}{I_3}$$

$$\begin{bmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\end{bmatrix} = \frac{1}{\cos\theta_2}\begin{pmatrix}\cos\theta_2 & \sin\theta_1\sin\theta_2 & \cos\theta_1\sin\theta_2\\ 0 & \cos\theta_1\cos\theta_2 & -\sin\theta_1\cos\theta_2\\ 0 & \sin\theta_1 & \cos\theta_1\end{pmatrix}\begin{bmatrix}\omega_1\\ \omega_2\\ \omega_3\end{bmatrix}$$

Use a P(I)D controller $\underline{u} = -k_1\underline{e} - k_2\dot{\underline{e}} - k_3\int \underline{e}dt$

Closed-loop system

$$\ddot{\theta}_i + \frac{k_{1i}}{I_i}\dot{\theta}_i + \frac{c_{1i}}{I_i}\theta_i = 0$$

A slew motion is defined here as a controlled motion between two attitudes. The motion is also defined as a rest-to-rest motion. In this case the desired equilibrium point is $\underline{\omega}_d = \underline{0}$ and $I = A_d$. We can try the following Lyapnuov control function which clearly satisfirs $V(\underline{\omega}_d = \underline{0}, I = A_d) = 0$ and $V > 0$ for all other values

$$V = \frac{1}{2}\underline{\omega}^T J\underline{\omega} + k_2 tr(I - A)$$

then

$$\dot{V} = \underline{\omega}^T\underline{u} + k_2 tr(-\dot{A}) = \underline{\omega}^T\underline{u} + k_2 tr([\omega]\hat{}A)$$
$$= \underline{\omega}^T(\underline{u} + k_2(A^T - A))$$

It follows that the control

$$\underline{u} = -k_1\underline{\omega} - k_2(A^T - A)^V$$

asymptotically stabilizes the equilibrium point $\underline{\omega}_d = \underline{0}$ and $I = A_d$. If using quaternions to represent the attitude kinematics we have

$$\frac{d}{dt}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Then the control $\underline{u} = -k_1\underline{\omega} - k_2\underline{q}_e$ with

$$\begin{bmatrix} q_{1e} \\ q_{2e} \\ q_{3e} \\ q_{4e} \end{bmatrix} = \begin{bmatrix} q_{4c} & q_{3c} & -q_{2c} & -q_{1c} \\ -q_{3c} & q_{4c} & q_{1c} & -q_{2c} \\ q_{2c} & -q_{1c} & q_{4c} & -q_{3c} \\ q_{1c} & q_{2c} & q_{3c} & q_{4c} \end{bmatrix}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Asymptotically stabilizes the desired equilibrium $\hat{q}_d = [q_{1c} \quad q_{2c} \quad q_{3c} \quad q_{4c}]^T$. Proof of asymptotic stability can be shown by using the Lyapunov function

$$V = \frac{1}{2}\underline{\omega}^T J\underline{\omega} + 2k_2(1 - q_{4e})$$

Then on differentiating with respect to time we have

$$\dot{V} = \underline{\omega}^T J\dot{\underline{\omega}} + 2k_2(-\dot{q}_{4e})$$

Then substituting in the kinematics and dynamics we get

$$\dot{V} = \underline{\omega}^T\underline{u} + k_2(\underline{\omega}_e{}^T\underline{q}_e)$$

Since in this case $\underline{\omega}_e = \underline{\omega}$then we can write

$$\dot{V} = \underline{\omega}^T(\underline{u} + k_2\underline{q}_e)$$

Then the control $\underline{u} = -k_1\underline{\omega} - k_2\underline{q}_e$gives:

$$\dot{V} = -k_1\underline{\omega}^T J\underline{\omega}$$

Nonlinear control methods and tracking control

A general Lyapunov function for designing slew motions can be given as:

$$V = \frac{1}{2}\underline{\omega} \cdot J\underline{\omega} + 2k_2H(q_{4e})$$

Where $H(q_{4e})$ is a function of the scalar part of the quaternion error q_{4e}. Moreover, to satisfy the conditions of a Lyapunov function $H(\pm 1) = 0$and $H(q_{4e}) > 0$ for $q_{4e} \neq \pm 1$. This function is used to design a class of controls that depend on the control engineer's selection of $H(q_{4e})$

$$\begin{bmatrix} q_{1e} \\ q_{2e} \\ q_{3e} \\ q_{4e} \end{bmatrix} = \begin{bmatrix} q_{4d} & q_{3d} & -q_{2d} & -q_{1d} \\ -q_{3d} & q_{4d} & q_{1d} & -q_{2d} \\ q_{2d} & -q_{1d} & q_{4d} & -q_{3d} \\ q_{1d} & q_{2d} & q_{3d} & q_{4d} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

It follows on differentiation that

$$\dot{V} = \underline{\omega} \cdot J\dot{\underline{\omega}} + 2k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \dot{q}_{4e}$$

And recalling the dynamic equations (assuming negligible disturbance torques for now) and the quaternion kinematic equations we have:

$$J\dot{\underline{\omega}} = J\underline{\omega} \times \underline{\omega} + \underline{u}$$
$$\frac{d\underline{q}_e}{dt} = -\frac{1}{2}[\omega_e\hat{}]\underline{q}_e + q_{4e}\underline{\omega}_e$$
$$\frac{dq_{4e}}{dt} = -\frac{1}{2}\underline{\omega}_e{}^T\underline{q}_e$$

Then substituting these equations into $\dot{V}$yields:

$$\dot{V} = \underline{\omega} \cdot (\underline{u} - k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \underline{q}_e)$$

Then a simple slew controller can be selected

$$\underline{u} = -k_1\underline{\omega} + k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \underline{q}_e$$

Such that

$$\dot{V} = -k_1\underline{\omega} \cdot \underline{\omega}$$

Now it is a matter of selecting the best performing potential $H(q_{4e})$. One example is to select

$$H(q_{4e}) = 1 - q_{4e}$$

Which gives

$$\underline{u} = -k_1\underline{\omega} - k_2\underline{q}_e$$

This ensures that the zero error state $\underline{\omega} = \underline{0}, q_{4e} = 1$ is a stable equilibrium point according to our proof. However, $\underline{\omega} = \underline{0}, q_{4e} = -1$is unstable using this potential function and it can clearly be seen not to satisfy the condition of the Lyapunov function since $H(-1) > 0$. However, $\underline{\omega} = \underline{0}, q_{4e} = -1$also corresponds to the same zero error state. In other words,

$$q_{4e} = [0 \quad 0 \quad 0 \quad \pm 1]^T \rightarrow A_e = I_{3\times 3}$$

Thus, there is an ambiguity due to the non-uniqueness, whereby, the zero error represented by quaternions can be stable or unstable. This can lead to the problem of unwinding. Moreover, if our initial condition is $\underline{\omega} = \underline{0}, q_{4e} = -0.9$and we use the control corresponding to $H(q_{4e}) = 1 - q_{4e}$the the control will drive the system to the stable state $\underline{\omega} = \underline{0}, q_{4e} = 1$ and the system will perform a large rotation. It would be far more efficient to perform a small rotation to the error state $\underline{\omega} = \underline{0}, q_{4e} = -1$, but for this control it is unstable. Another choice is $H(q_{4e}) = 1 + q_{4e}$, but in this case $\underline{\omega} = \underline{0}, q_{4e} = -1$will be a stable equilibrium of the closed loop system and $\underline{\omega} = \underline{0}, q_{4e} = 1$will be

unstable, so potentially problems of unwinding can occur. If we choose any of the following functions:

$$H(q_{4e}) = 1 - q_{4e}^2$$
$$H(q_{4e}) = 1 - sgn(\, q_{4e}(0))q_{4e}$$
$$H(q_{4e}) = arccos^2\, q_{4e}$$

Then $H(\pm 1) = 0$. In this case both $\underline{\omega} = \underline{0}, q_{4e} = 1$and $\underline{\omega} = \underline{0}, q_{4e} = -1$ which are physically the same error are both stable.

Eigenaxis Rotation slew maneuver

A final ideal control that we will use will be a slew that corresponds to an eigenaxis rotation. This control is particularly useful since it can be optimally tuned. We begin with a similar Lyaunov function to one that has been used before, but without the inclusion of the inertia matrix:

$$V = \frac{1}{2}\underline{\omega} \cdot \underline{\omega} + 2k_2 H(q_{4e})$$

Then on differentiation we have

$$\dot{V} = \underline{\omega} \cdot \dot{\underline{\omega}} + 2k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \dot{q}_{4e}$$

And it follows from

$$J\dot{\underline{\omega}} = J\underline{\omega} \times \underline{\omega} + \underline{u}$$

And

$$\frac{dq_{4e}}{dt} = -\frac{1}{2}\underline{\omega}_e{}^T \underline{q}_e$$

That

$$\dot{V} = \underline{\omega} \cdot (J^{-1}(J\underline{\omega} \times \underline{\omega} + \underline{u}) - k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \underline{q}_e)$$

Then to guaranteed that $\dot{V} < 0$ we choose

$$\underline{u} = \underline{\omega} \times J\underline{\omega} - k_1 J\underline{\omega} + k_2 J \frac{\partial H(q_{4e})}{\partial q_{4e}} \underline{q}_e$$

If we compare this to our previously derived slew control:

$$\underline{u} = -k_1\underline{\omega} + k_2 \frac{\partial H(q_{4e})}{\partial q_{4e}} \underline{q}_e$$

The two differences are (i) the term $\underline{\omega} \times J\underline{\omega}$, which essentially cancels the dynamics and makes the system more responsive to the control. (ii) the matrix *J* now appears in each component of the control. In this case the closed loop system is:

$$\dot{\underline{\omega}} = -k_1\underline{\omega} + k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\underline{q}_e$$

Assume the form of an eigenaxis rotation which is a rotation around a fixed axis $\underline{e}$ of magnitude θ. In this case we can write

$$\underline{\omega} = \underline{\omega}_e = \dot{\theta}_e\underline{x}, \underline{q}_e = sin\frac{\theta_e}{2}\underline{x}$$

Substituting into the closed form expression we have:

$$\ddot{\theta} + k_1\dot{\theta} - k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\theta = 0$$

So for example if we choose $H(q_{4e}) = 1 - q_{4e}$ then

$$\frac{\partial H(q_{4e})}{\partial q_{4e}} = -1$$

So

$$\ddot{\theta} + k_1\dot{\theta} + k_2\, sin\frac{\theta}{2} = 0$$

This reduction of the closed-loop dynamics can be used to tune the slew motion i.e. for small angles we have

$$\ddot{\theta} + k_1\dot{\theta} + k_2\frac{\theta}{2} = 0$$

And so the corresponding characteristic equation can be expressed as:

$$\lambda = \frac{-k_1 \pm \sqrt{k_1^2 - 2k_2}}{2}$$

So for optimal tuning we choose

$$k_1^2 = 2k_2$$

And only one parameter is required to be tuned.

Trajectory tracking

Slew motions are rest-to-rest motions, so when we design a slew control $\underline{\omega}_d = \underline{0}$and therefore in the Lyapunov function we set $\underline{\omega} = \underline{\omega}_e$. In the general case that we want to track a time dependent attitude trajectory, that is, a moving frame such as the LVLH frame, we can use the most general form of the Lyapunov function:

$$V = \frac{1}{2}\underline{\omega}_e{}^T J\underline{\omega}_e + 2k_2H(q_{4e})$$

Note that, as proved by differentiating $A_e = AA_d^T$, $\underline{\omega}_e = \underline{\omega} - A_e(q)\underline{\omega}_d$

Note that

$$\dot{\underline{\omega}}_e = \dot{\underline{\omega}} - \frac{d}{dt}(A_e(q)\underline{\omega}_d) = J^{-1}(J\underline{\omega} \times \underline{\omega} + \underline{u}) - \frac{d}{dt}(A_e(q)\underline{\omega}_d)$$

The term $\frac{d}{dt}(A_e(q)\underline{\omega}_d)$ appears since the reference frame that we want to track is time dependent:

$$\dot{V} = \underline{\omega_e}^T(J\underline{\omega} \times \underline{\omega} + \underline{u} - J\frac{d}{dt}(A_e(q)\underline{\omega}_d) - k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\underline{q}_e)$$

Then since $\underline{\omega_e}^T(J\underline{\omega} \times \underline{\omega}) \neq 0$) the control

$$\underline{u} = -k_1\underline{\omega}_e + k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\underline{q}_e + \underline{\omega} \times J\underline{\omega} + J\frac{d}{dt}(A_e(q)\underline{\omega}_d)$$

Leads to

$$\dot{V} = -k_1\underline{\omega_e}^T\underline{\omega}_e$$

And so this control asymptotically tracks the moving frame (time-dependent attitude reference trajectory). Note that

$$\frac{d}{dt}(A_e(q)\underline{\omega}_d) = \dot{A}_e(q)\underline{\omega}_d + A_e(q)\dot{\underline{\omega}}_d$$
$$= -[\omega_e]^{\wedge}A_e(q)\underline{\omega}_d + A_e(q)\dot{\underline{\omega}}_d$$

So we can write the control as:

$$\underline{u} = -k_1\underline{\omega}_e + k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\underline{q}_e + \underline{\omega} \times J\underline{\omega} + J(A_e(q)\dot{\underline{\omega}}_d - [\omega_e]^{\wedge}A_e(q)\underline{\omega}_d)$$

The tracking control is far more complex than the slew motion control. It requires precise knowledge of the inertia matrix, and the angular acceleration $\dot{\underline{\omega}}_d$of the moving frame. Note that for slowly evolving moving frames the term $A_e(q)\dot{\underline{\omega}}_d - [\omega_e]^{\wedge}A_e(q)\underline{\omega}_d$ will be very small so the control can be reduced to

$$\underline{u} = -k_1\underline{\omega}_e + k_2\frac{\partial H(q_{4e})}{\partial q_{4e}}\underline{q}_e + \underline{\omega} \times J\underline{\omega}$$

Note that if you are using DCM the control can be derived in an analogous way with $H(q_{4e})$ replaced with $tr(I - A_e)$ in the Lyapunov function

$$\underline{u} = -k_1\underline{\omega}_e - k_2(A_e^T - A_e)^V + \underline{\omega} \times J\underline{\omega} + J(A_e\dot{\underline{\omega}}_d - [\omega_e]^{\wedge}A_e\underline{\omega}_d)$$

Previously a reference attitude has been designed based on a circular orbit around the Earth. However, a reference attitude can be designed in an arbitrary way. For example, we may be required to point at the Sun or Moon while in an elliptic orbit or even a Libration point orbit. Assuming that the pointing vector is aligned with the unit vector $\vec{r}_{sc}$with respect to the spacecraft and $\vec{v}_{sc}$its relative velocity then we can construct an orthonormal moving frame:

$$A_d = \begin{bmatrix} \vec{x}_1 = \vec{r}_{sc} \\ \vec{x}_2 = \vec{x}_3 \times \vec{x}_1 \\ \vec{x}_3 = \vec{r}_{sc} \times \vec{v}_{sc} \end{bmatrix}$$

$\vec{v}_{sc}$could be replaced with a Sun pointing vector, which may be more optimal for power generation.

Disturbances attenuation

Let us now look at our previous control laws and see if the closed-loop systems are still guarenteed to be asymptotically stable. Using the simple example of a de-tumbing control $\underline{u} = -k\underline{\omega}$

The Euler equations now contain the control torque and the disturbance torques $J\frac{d\underline{\omega}}{dt} = J\underline{\omega} \times \underline{\omega} + \underline{u} + \underline{d}$. Using the Lyapunov function:

$$V = \frac{1}{2}\underline{\omega}^T J \underline{\omega}$$

We have

$$\dot{V}(\underline{\omega}) = \underline{\omega}^T(\underline{u} + \underline{d})$$

It follows that

$$\dot{V}(\underline{\omega}) = -\underline{\omega}^T\underline{\omega} - \underline{\omega}^T\underline{d}$$

Therefore, there is no guarentee that the spacecraft will de-tumble to exactly zero. However, lets augment our control to include a saturated term such that

$$\underline{u} = -k_1\underline{\omega} - k_2\, sgn(\underline{\omega})$$

In this case we would have

$$\dot{V}(\underline{\omega}) = -k_1\underline{\omega}^T\underline{\omega} - k_2\underline{\omega}^T\, sgn(\underline{\omega}) + \underline{\omega}^T\underline{d}$$

Defining $|\underline{x}| = \underline{x}^T\, sgn(\underline{x})$ then we can say that $|\underline{\omega}||\underline{d}| \geq \underline{\omega}^T\underline{d}$then

$$\dot{V}(\underline{\omega}) \leq -k_1\underline{\omega}^T\underline{\omega} - k_2|\underline{\omega}| + |\underline{\omega}||\underline{d}|$$

Then the closed-loop system is asymptotically stable and the modified control is robust to disturbances if $k_2 > |\underline{d}|$ i.e $-k_1\underline{\omega}^T\underline{\omega} - k_2|\underline{\omega}| + |\underline{\omega}||\underline{d}| \leq 0$. So if we know the upperbound on the disturbances then we can design a robust control.

The reduced-attitude control problem

In some applications it may only be required to point in a prescribed direction. We define a desired pointing vector by $\Gamma_d$which is expressedIn such cases full 3-axis stabilization is not required and it may not be necessary to define the entire reference attitude on the pointing direction. Furthermore, in such cases it may not be necessary to determine the entire attitude but only a pointing vector. Define an orthonormal frame **N** such that the desired $\underline{\Gamma}_d = [1 \quad 0 \quad 0]^T_N$. The current pointing vector can be expressed in the body frame as $\underline{\Gamma}$then $\underline{\Gamma} = A_{B/N}\underline{\Gamma}_d$ then:

$$\dot{\underline{\Gamma}} = \dot{A}_{B/N}\underline{\Gamma}_d = -[\omega]^\wedge A_{B/N}\underline{\Gamma}_d = -[\omega]^\wedge\underline{\Gamma} = \underline{\Gamma} \times \underline{\omega}$$

Then define the error angle between the two vectors by:

$$\Gamma_d^T \Gamma = cos\,\theta_e$$

We define the Lyapunov function

$$\boldsymbol{V = \frac{1}{2}\langle \underline{\omega}, J\underline{\omega}\rangle + k_2(1 - \Gamma_d^T \Gamma)}$$

Which satisfies the first two requirements for a Lyapunov function. Then differentiating

$$\boldsymbol{\dot{V} = \langle \underline{\omega}, \underline{u}\rangle - k_2(\underline{\Gamma}_d^T \dot{\underline{\Gamma}}) = \langle \underline{\omega}, \underline{u}\rangle - k_2 \underline{\Gamma}_d^T(\underline{\Gamma} \times \underline{\omega})}$$
$$\boldsymbol{\dot{V} = \langle \underline{\omega}, \underline{u}\rangle - k_2 \underline{\omega}^T(\underline{\Gamma}_d \times \underline{\Gamma}) = \langle \underline{\omega}, \underline{u} + k_2(\underline{\Gamma} \times \underline{\Gamma}_d)\rangle}$$

Then the control

$$\underline{u} = -k_1 \underline{\omega} - k_2(\Gamma \times \Gamma_d)$$

Asymptotically converges to the desired poiting direction Γ_d.

Minimum time maneuvers

The adoption of the classical control techniques is feasible if the system performances are to be evaluated in terms of bandwidth, transient response or steady state error. In some cases, optimal performances are provided in the form of minimum time or minimum fuel maneuvers. In these cases, the controller has to be designed on different basis. To minimize maneuver time, the optimal performance requires minimizing the cost function that includes explicitly time and system dynamics (as a constraint):

$$min J = \int_{t_0}^{t_f} dt + \lambda \int_{t_0}^{t_f} (\dot{x} - Ax - Bu) dt$$

This problem has an exact, analytical solution only in some very special case. The rotation around one principal axis is one of these cases. For small rotations, the dynamics and the cost function become:

$$M = I\ddot{\alpha}$$
$$\dot{\alpha} = \frac{d\alpha}{dt} \Rightarrow dt = \frac{d\alpha}{\dot{\alpha}}$$
$$\Rightarrow J = \int_{\alpha_0}^{\alpha_f} \frac{1}{\dot{\alpha}} d\alpha$$

To minimize maneuver time, we must maximize $\dot{\alpha}$. We assume the available torque is limited to a maximum value:

$$u = \ddot{\alpha} = \frac{M}{I}$$

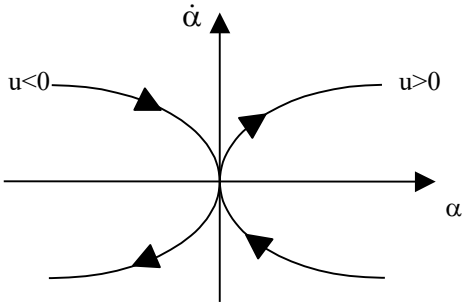
If the torque is sufficiently high the maneuver will be fast, but in cases in which the torque is limited the solution of the minimum time problem becomes of interest.
Integrate the dynamics to get:

$$\dot{\alpha} = \dot{\alpha}_0 + ut$$
$$\alpha = \alpha_0 + \dot{\alpha}_0 t + \frac{1}{2} u t^2$$

The first equation is used to calculate time:

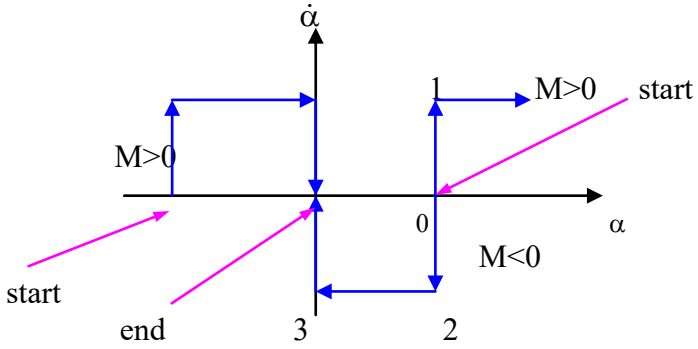
$$t = \frac{\dot{\alpha} - \dot{\alpha}_0}{u}$$
$$\alpha = \alpha_0 + \frac{\dot{\alpha}_0(\dot{\alpha} - \dot{\alpha}_0)}{u} + \frac{1}{2}\frac{(\dot{\alpha} - \dot{\alpha}_0)^2}{u}$$
$$\alpha - \alpha_0 = \frac{1}{u}\dot{\alpha}_0(\dot{\alpha} - \dot{\alpha}_0) + \frac{1}{2u}(\dot{\alpha} - \dot{\alpha}_0)^2$$

The phase plane plot $(\dot{\alpha}, \alpha)$ of the constant torque maneuver is then a family of parabolas, whose concavity depends on the torque limit:



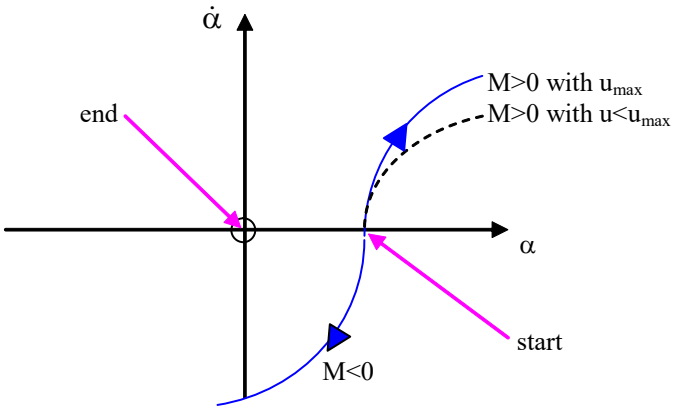
The arrows indicate that each parabola is followed with a specified direction. If for instance motion starts from the origin with M>0, then $\ddot{\alpha} > 0$ and $\dot{\alpha}$ will increase. By increasing the input u the parabolas are more open. Taking the extreme approximation of infinite torque, parabolas would be replaced by vertical lines, so that the maneuver would be impulsive, with a change in $\dot{\alpha}$ associated to a constant α. On the contrary, if u=0 we would have a phase portrait given by horizontal lines, with no change in $\dot{\alpha}$. In general, phase plane maneuvers are designed in order to have both $\dot{\alpha}_0$ and $\dot{\alpha}_f$ equal to zero, that is to say rest-to-rest maneuvers. In addition, since the origin of the phase plane is arbitrary, either α_0 or α_f is set to zero.

We can now design a maneuver in the phase plane assuming impulsive torques. This is the case of maneuvers performed by using high thrust propulsive systems.



Starting from point 0 to end in the origin, if we apply a positive torque we would reach point 1, but here $\dot{\alpha}$ is positive and we would de part from the desired attitude. We must then start with a negative torque, to reach point 2, switch off the controller to keep $\dot{\alpha}$ constant until point 3 is reached and then provide a positive torque to reach the target final attitude. Of course, should the initial attitude be negative all the maneuver has to be performed in the opposite way. Notice also that the vertical arcs of the phase plane are traced in almost zero time, since the torque is assumed infinite, and are equivalent to impulsive maneuvers. The total maneuver time depends then only on the horizontal arcs of the phase plane trace. In theory, we would like to have $\dot{\alpha}$ as high as possible to minimize maneuver time, so that the horizontal arc would be drawn in a short time. The major issue in this case is thruster synchronization, since with high $\dot{\alpha}$ even a small time error would mean reversing the control (point 3 in the example) in a different point on the phase plane, so the target attitude would not be reached.

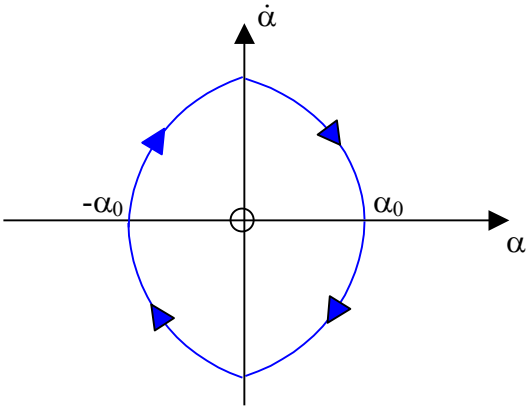
We can now consider a more realistic case, with bounded maximum torque.



We will consider only the parabolas corresponding to u_{max} so that $\dot{\alpha}$ is the maximum possible and time is minimum. The problem is to find the position in which the torque has to be switched in sign. If the maneuver is completed according to the control logic:

$$u = -u(\alpha)_{max}$$

once on the axis $\dot{\alpha}$ the sign of α changes so that the phase plane portrait would look like in the following figure:

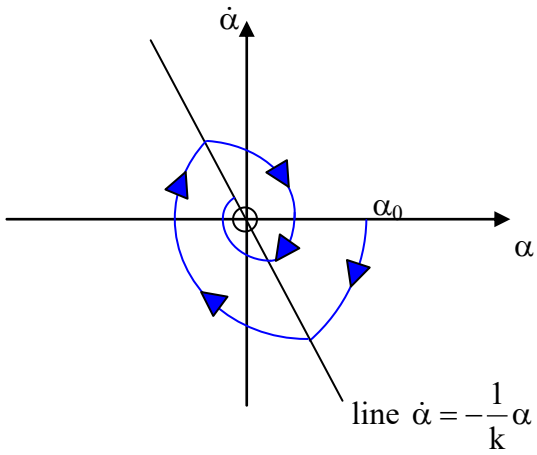


There is evidently a limit cycle, the system would behave like an undamped second order oscillator.

Change the control logic to:

$$u = -u(\alpha + k\dot{\alpha})_{max}$$

so that the switch in the sign of the control torque is along an inclined straight line. We would like k to be positive in order to have a negative inclination of the switching line:



Reaching the switching line, u changes its sign, so that the phase plane trace is switched to a parabola with reversed axis; as the number of torque switches increases, the trace gets closer and closer to the origin. k is then an index of the damping in the oscillations. However, rigorously, an infinite number of switchings are needed to reach exactly the origin in a general case.

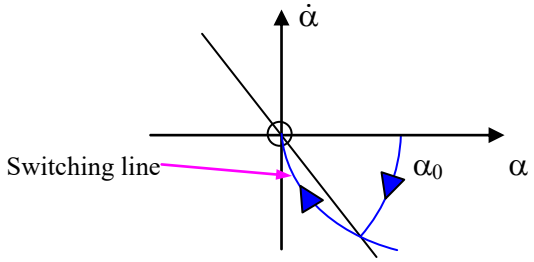
If we draw the two parabolas that pass through the origin, corresponding to positive and negative torque, since for $\dot{\alpha}_0$ equal to zero we have:

$$\alpha = \frac{1}{2u}\dot{\alpha}^2$$

we can consider the following switching curve:

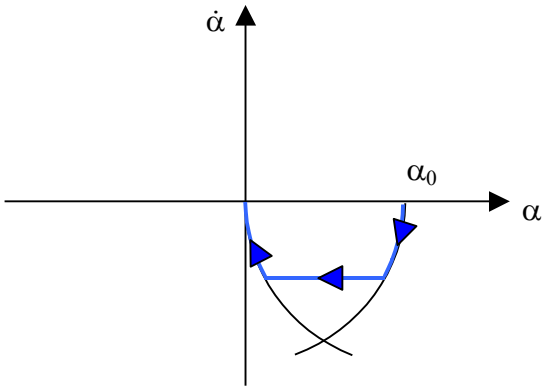
$$u = -u\left(\alpha - \frac{1}{2u}\dot{\alpha}|\dot{\alpha}|\right)_{max}$$

The phase plane portrait of the maneuver will then be:



It can be shown that this is the minimum time maneuver. If the initial and final velocities are zero, the satellite accelerates at the maximum level for half the rotation, then decelerates at same level for the second half of the rotation. The sign of the control torque becomes a function of α and $\dot{\alpha}$.

Finally, if we want to consider a minimum fuel maneuver we should fix a maximum maneuver time. This can be seen as a minimum time maneuver with one intermediate coast arc (at constant $\dot{\alpha}$) if the allowed maneuver time is greater than the minimum maneuver time for the same rotation.

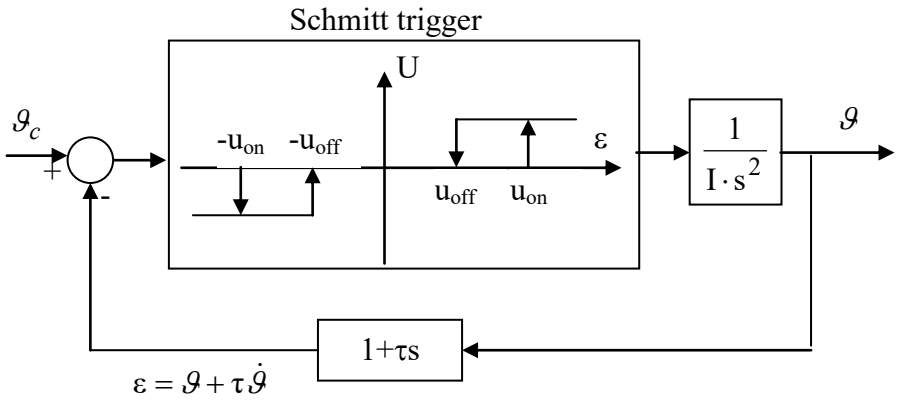


In this case, the switching from one parabola to the other occurs in a finite time. Notice that if t_{max} is equal to the minimum time we would reduce the coast arc to zero and find again the minimum time solution. Fixing t_{max} becomes equivalent to fixing $\dot{\alpha}_{max}$.

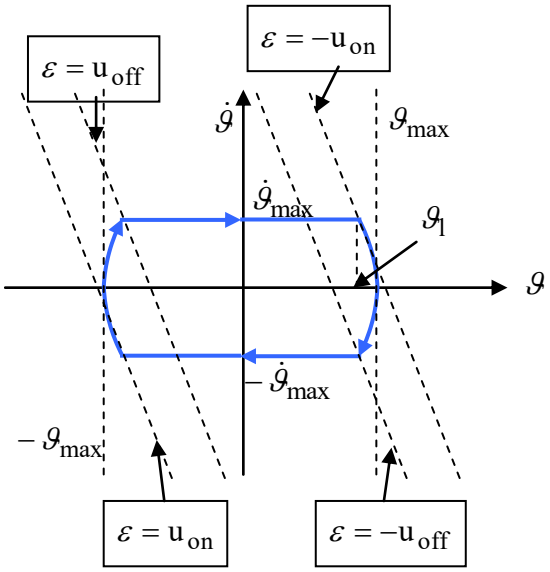
The process just show is valid only if the rotation is around one principal axis; in other cases, the complete set of Euler equations should be used as system dynamics (and optimization dynamic constraint) and a closed form solution can no longer be found.

Nonlinear control with constant thrust actuators

Considering the dynamic behavior of a satellite projected in the phase plane, it is possible to set up a nonlinear controller decoupled for each axis. The control is based on a combination of the angle ϑ and its derivative $d\vartheta$. In particular, a nonlinear switch called "Schmitt trigger" activates the controller on the basis of the value of a variable $\vartheta+\tau d\vartheta$. Assume, for example, that the value of $\vartheta+\tau d\vartheta$ is greater than a given limit u_{on}. In this case the actuators would be switched on until the same variable $\vartheta+\tau d\vartheta$ falls below a second limit u_{off}.



The values of u_{on}, u_{off} can be determined considering the maximum allowable angular error ϑ_{max}, the maximum admissible angular rate $d\vartheta_{max}$ and the time constant τ. With reference to the following figure, we must first of all consider the two parabolas passing from the points $(\pm\vartheta_{max},0)$, corresponding to the controlled dynamics with the maximum torque.



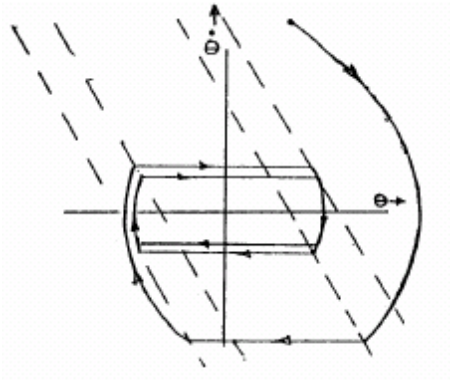
Intersect the two parabolas with the horizontal lines at $d\vartheta_{max}$, to identify ϑ_1, the angle error at which the controller must be switched on in order to prevent the error from getting larger than ϑ_{max}

$$\vartheta_1 = \vartheta_{max} - \frac{d\vartheta_{max}^2}{2u_c}$$

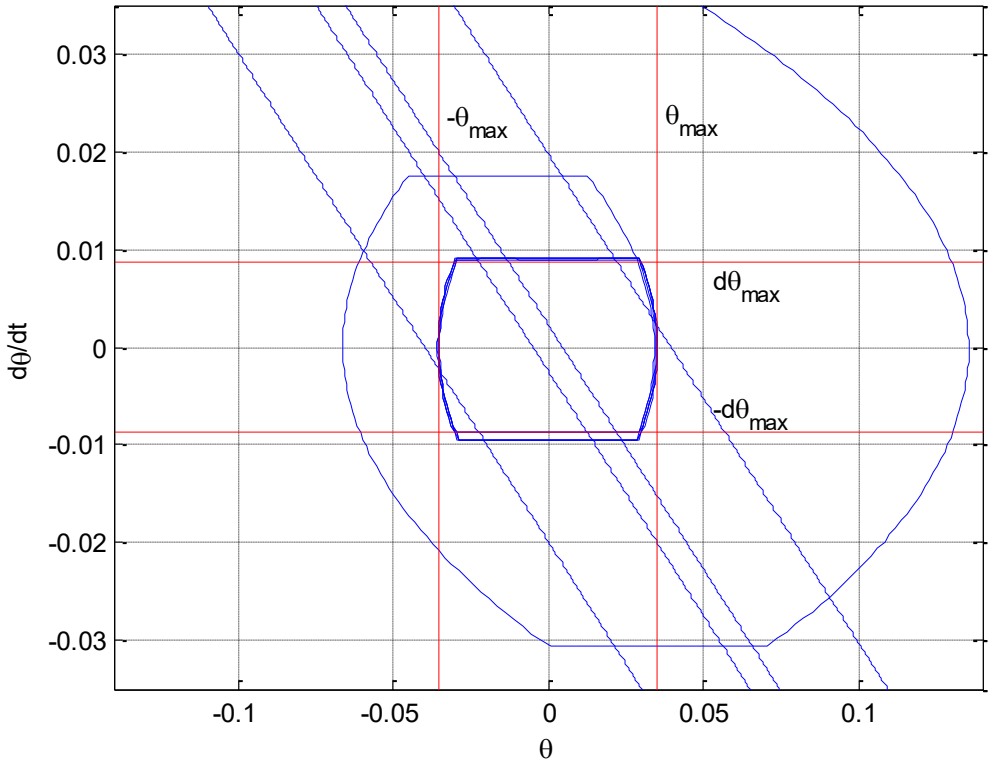
$u_c = M/I$ is still the control command. The switching curve to activate the controller must intercept the point $(\vartheta_1, d\vartheta_{max})$ and have a slope equal to $-1/\tau$. The value of τ can be tuned according to some performance requirements. The values of u_{on} and u_{off} are then evaluated as:

$$u_{on} = \tau d\vartheta + \vartheta_1$$
$$u_{off} = -\tau d\vartheta + \vartheta_1$$

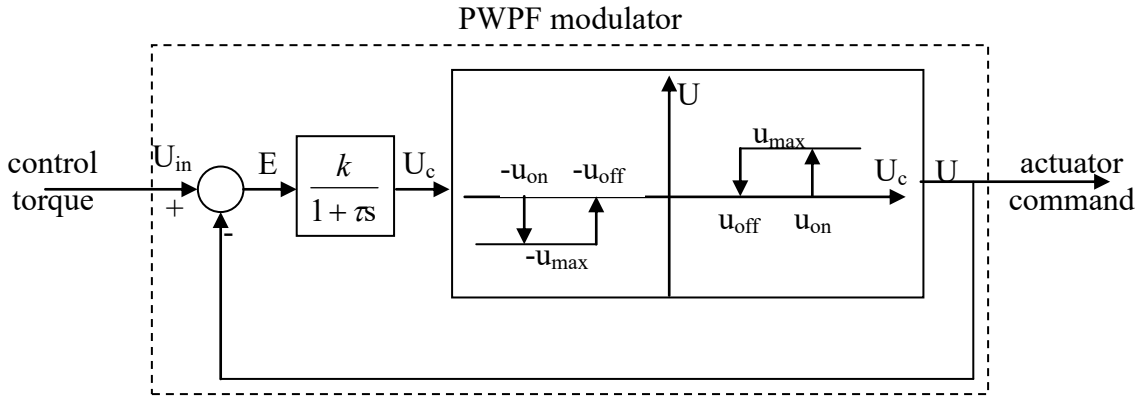
With symmetry considerations, the switching values for negative errors are determined. On the phase plane, in ideal conditions with no disturbance torque and sensor error, the satellite phase portrait must converge to a limit cycle bounded by the values $-\vartheta_{max}/+\vartheta_{max}$ and $-d\vartheta_{max}/+d\vartheta_{max}$.
The transient response for large initial errors will still converge to the same final limit cycle, provided the time constant τ is selected with the correct sign. The parameter τ has an influence on the way the phase portrait converges to the limit cycle.



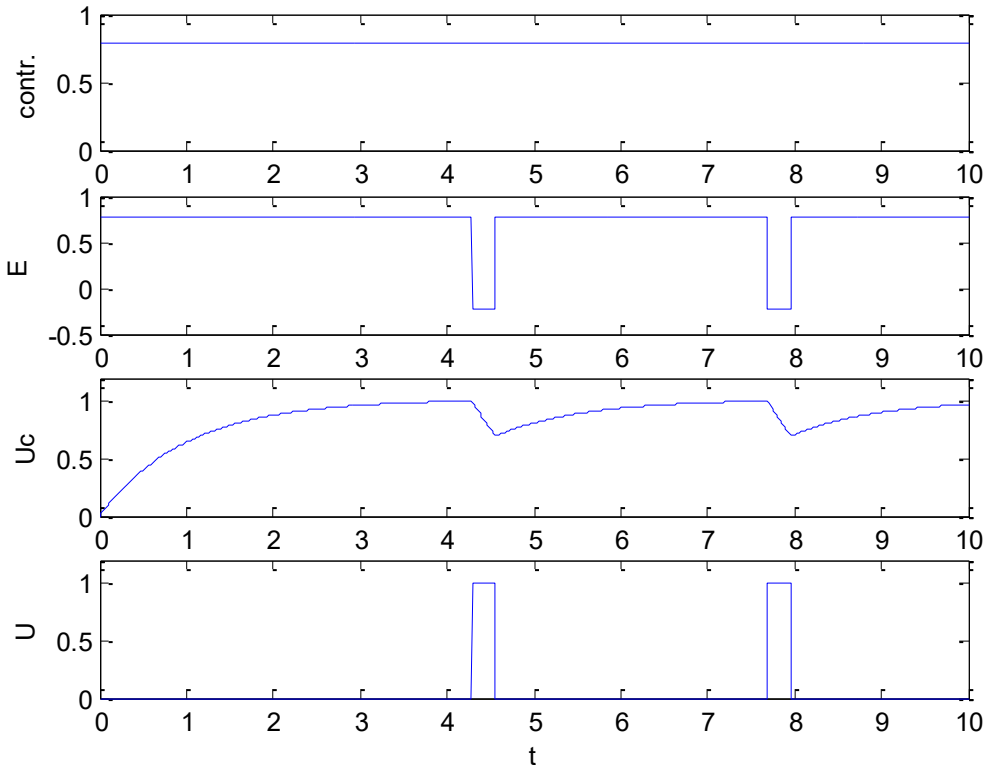
Considering the inevitable presence of sensor errors and delays in the activation of the actuators, the switching of the control will not be exactly on the desired switching lines. This means that the real limit cycle in the phase plane will be slightly different from the ideal one, as shown in the following example.



One alternative to the previous technique is based on the so called "PWPF modulator" (Pulse-Width-Pulse-Frequency Modulator). This varies both the width of the control pulse (the duration of the control action) and the frequency of the switchings. This is obtained as shown in the scheme below, with the integration and filtering of the desired control action until the switching threshold u_{on} is reached.



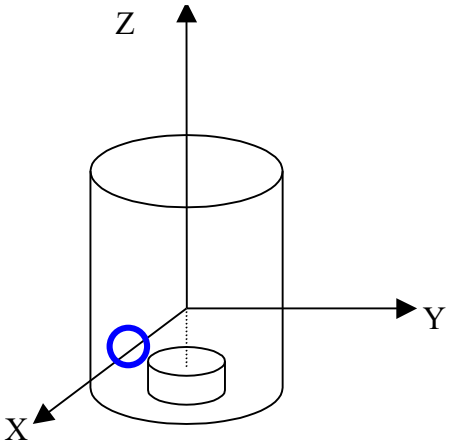
When the PWPF modulator provides a real control torque to the satellite, the torque is subtracted from the request of the controller before being passed to the integrator. Since in principle the supplied torque U is greater than the desired control U_{in}, the integrated signal will reverse its sign so that after a short period of time the PWPF modulator will switch the controller off. In this condition the initial behavior of the PWPF modulator is restored, until the following switch on of the controller. The time separation of two consecutive control pulses depends on the control law designed, while the duration of the control pulses depends on the maximum torque provided by the actuators U_c. To understand the principle of operation, assuming a constant control request U_{in}, the behavior of the PWPF modulator is the following.


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Passive damping systems

It is possible to make a distinction between attitude maneuver control and stability augmentation control. Maneuver control systems in general allow also controlling the stability of the system, while stability augmentation control systems do not allow controlling maneuvers, since they are passive systems. There are in fact some passive systems that can improve the stability of the satellite.
We can see two examples of passive or semi-active systems for the stability augmentation of nutation (conical motion of the spin axis) and libration (pendulum oscillations about the pitch axis) motion. These systems do not allow the execution of rotational maneuvers, in some cases they can even represent obstacles to maneuvers.
In general, nutation damping mechanisms base their effect on viscous damping created by the relative rotation of a viscous fluid within the spacecraft.

To model the damping mechanism, we can consider a dual spin satellite with nominal spin velocity around the z axis. Nutation appears as coupled oscillations around x and y axes, that can be damped with the aid of a fluid ring around x axis. Due to coupling of x and y equations, damping oscillations around x axis also damps oscillations around y axis.



Considering the fluid ring at the same rate of a reaction wheel, we can write the following equations:

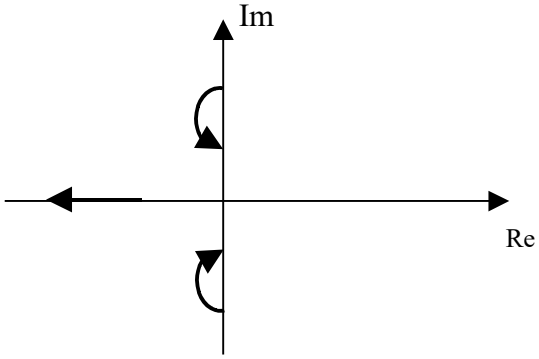
$$\underline{h} = \left(I_x\omega_x + I_f\omega_f\right)\underline{i} + I_y\omega_y\underline{j} + (I_z\omega_z + I_r\omega_r)\underline{k}$$

$$\begin{cases} I_x\dot{\omega}_x + \left(I_z - I_y\right)\omega_z\omega_y + I_r\omega_r\omega_y + I_f\dot{\omega}_f = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z - I_r\omega_r\omega_x + I_f\omega_f\omega_z = 0 \\ I_z\dot{\omega}_z + I_r\dot{\omega}_r + \left(I_y - I_x\right)\omega_x\omega_y - I_f\omega_f\omega_y = 0 \\ I_r\dot{\omega}_r = 0 \\ I_f\dot{\omega}_f + c\left(\omega_x + \omega_f\right) = 0 \end{cases}$$

where c is the viscous damping of the fluid, which creates a relative torque between fluid ring and satellite.
Linearizing the equations we obtain:

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\bar{\omega}_z\omega_y + I_r\bar{\omega}_r\omega_y + I_f\dot{\omega}_f = 0 \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\bar{\omega}_z - I_r\bar{\omega}_r\omega_x + I_f\omega_f\bar{\omega}_z = 0 \\ I_f\dot{\omega}_f + c(\omega_f + \omega_x) = 0 \\ I_z\dot{\omega}_z + I_r\dot{\omega}_r = 0 \\ I_r\dot{\omega}_r = 0 \end{cases}$$

The last two equations represent the motion around axis z, while the first three represent a damped motion in which the damping depends on c.
The root locus, as a function of c, has the following trace on the complex plane:

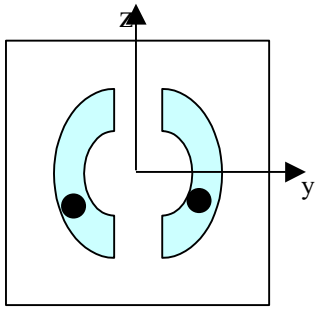


It can be clearly seen that an increase in the coefficient c does not mean a continuous increase in the system damping, there is in any case an optimal value of c that maximizes damping. If $\bar{\omega}_z$ is zero and $I_f << I_x, I_y, I_z$ then the optimal value of c is:

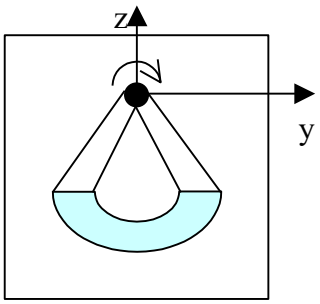
$$c = \frac{I_f I_r \omega_r}{\sqrt{I_x I_y}}$$

It can be seen that each satellite has its optimal value of c, depending on the inertia moments of the spacecraft. This means also that the optimal fluid is spacecraft-dependent. Since it is not always possible to adapt the fluid properties to the satellite, then optimality can be obtained by proper selection of the fluid quantity, through parameter I_f.

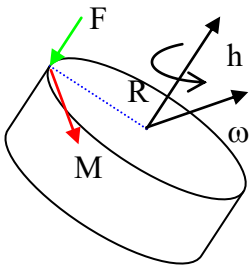
Popular damping configurations are realized through two partial fluid tubes containing a ball free to move inside the tube, increasing the damping:



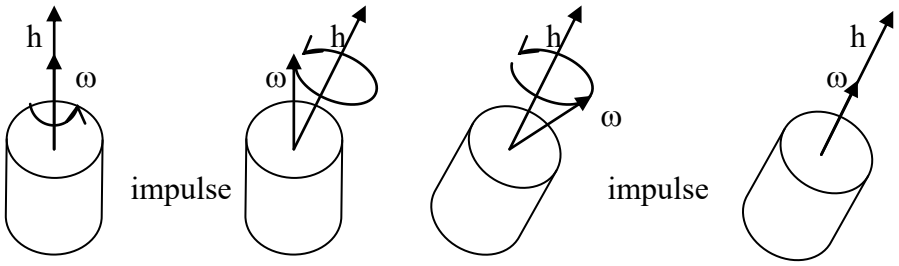
Alternatively, a pendulum configuration can be adopted:



A semi-active system to damp nutation, generated as a consequence of a spin axis reorientation maneuver, can act on ω or on h. To act on h a system based on thrusters can be set up, in order to align ω with h and cancel nutation. This is realized by generating an impulse (activating the thruster) when the phase is correct, so that the torque M aligns ω and h:

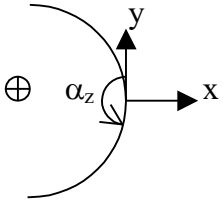


It is also possible to design a two-step maneuver, designed in such a way that at the end of the second step ω and h are aligned:



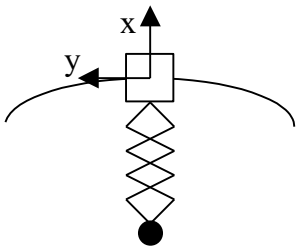
To damp libration oscillations, passive systems must damp α_z:

$$\ddot{\alpha}_z + 3K_p n^2 \alpha_z = 0$$



Angular velocity is $n + \dot{\alpha}_z$. In this case a system based on absolute angular velocity is not useful, since this will never change sign if $n > \dot{\alpha}_z$ for any value of the angular rate $\dot{\alpha}_z$. So, a system based on $\dot{\alpha}_z$ only must be designed. For this, a permanent magnet immersed in a viscous fluid is suitable, if the permanent magnet tends to remain aligned with the radial direction.

Also for the libration damping a semi-active system can be designed:



Moving the tip mass along the radial direction we control the gravity gradient torque and so the libration oscillations.
One interesting semi-active damping mechanism is based on the conservation of angular momentum, in a configuration where the tip mass can be moved closer to the satellite body, starting from a given position. In the initial configuration, angular momentum is:

$$h_0 = I_z(n + \dot{\alpha}_{z0})$$

With the mass closer to the satellite angular momentum is instead:

$$h_1 = I_r(n + \dot{\alpha}_{z1})$$

Since h is constant:

$$I_z(n + \dot{\alpha}_{z0}) = I_r(n + \dot{\alpha}_{z1})$$

Furthermore, since $I_r<I_z$:

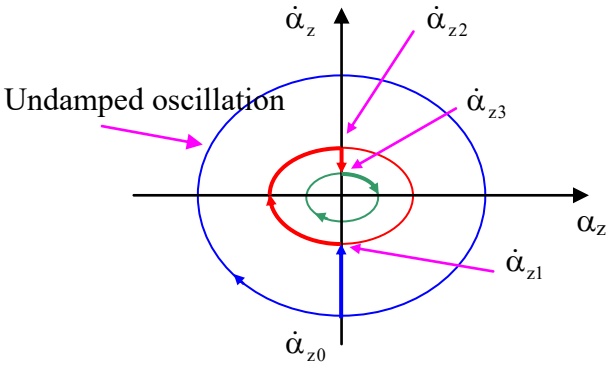
$$n + \dot{\alpha}_{z0} < n + \dot{\alpha}_{z1}$$
$$\dot{\alpha}_{z0} < \dot{\alpha}_{z1}$$

Assuming $\dot{\alpha}_{z0} < 0$ we have:

$$|\dot{\alpha}_{z1}| < |\dot{\alpha}_{z0}|$$

and the amplitude of oscillations, in the rotating orbit frame, is reduced.

Looking at the motion in the phase plane we have:



If the tip mass is repositioned at its original distance when $\dot{\alpha}_z$is positive, we have $\dot{\alpha}_{z3} < \dot{\alpha}_{z2}$. Notice that $\dot{\alpha}_{z2}$ is simply the opposite of $\dot{\alpha}_{z1}$, with same amplitude but changed in sign. So, considering negative $\dot{\alpha}_{z0}$ we have:

$$I_z(n - |\dot{\alpha}_{z0}|) = I_r(n - |\dot{\alpha}_{z1}|)$$

$$I_r(n + \dot{\alpha}_{z2}) = I_z(n + \dot{\alpha}_{z3}) \Rightarrow I_r(n + |\dot{\alpha}_{z1}|) = I_z(n + \dot{\alpha}_{z3})$$

If we impose $\dot{\alpha}_{z2}$ to be zero, then:

$$|\dot{\alpha}_{z1}| = \frac{(I_z - I_r)n}{I_r}$$

from which:

$$I_z(n - |\dot{\alpha}_{z0}|) = I_r\left(n - \frac{(I_z - I_r)n}{I_r}\right) = I_r n\left(\frac{2I_r - I_z}{I_r}\right)$$

$$I_r = \frac{I_z(2n - |\dot{\alpha}_{z0}|)}{2n}$$

This provides the value of I_r, i.e. the closest position of the tip mass, required to damp the libration oscillations in only 2 maneuvers.

This maneuver, in the most favorable case, is completed in half libration oscillation, and in the worst case in 1.5 libration periods, that has a frequency in the order of $\sqrt{3}n\left(K_p \cong 1\right)$. Since there are 1.7 oscillations in one orbit, the overall maneuver is completed in more or less one orbit.

Spin rate damping

Active control with magnetic actuators

Active spin damping can be achieved with feedback of the rate of change of the external magnetic field B_m on a set of magnetic torquers.

$$\underline{m} = -k_b \underline{\dot{B}_m},$$

This control is effective provided the spin rate of the satellite is sufficiently high, so that a change in the magnetic field is really caused by the spin and not by the change in satellite position along the orbit.

$$\underline{M_c} = \underline{m} \Lambda \underline{B} = -k_b \underline{\dot{B}_m} \Lambda \underline{B}.$$

The magnetic field can be represented in the principal axes of the satellite:

$$\underline{B} = A_{\text{rot}} \underline{B_{\text{orb}}}$$

and, taking the derivative of the magnetic field in principal axes, we have:

$$\underline{\dot{B}} = \dot{A}_{\text{rot}} \underline{B_{\text{orb}}} + A_{\text{rot}} \underline{\dot{B}_{\text{orb}}} \approx \dot{A}_{\text{rot}} \underline{B_{\text{orb}}}$$

The rate of change of the magnetic field is the sum of two terms, one due to the rotation of the satellite and one due to its change in position. We can assume that for high spin rates the term due to the change in position is negligible. Recalling that

$$\dot{A} = -[\omega \Lambda] A$$

we also have:

$$\underline{\dot{B}} = -[\omega \Lambda] \underline{B}$$

Evaluating the rate of change of the kinetic energy of the satellite $\dot{E}_k = \omega \underline{I} \dot{\omega}$, considering the satellite dynamics, the control law and the relation between magnetic field and its derivative we get to:

$$\dot{E}_k \approx \underline{\omega}^T \underline{M_c} = \underline{\omega}^T \left(\underline{m} \Lambda \underline{B} \right) = k_b \underline{\dot{B}}^T \left(\underline{\omega}^T \Lambda \underline{B} \right) = -k_b \underline{\dot{B}}^T \underline{\dot{B}},$$

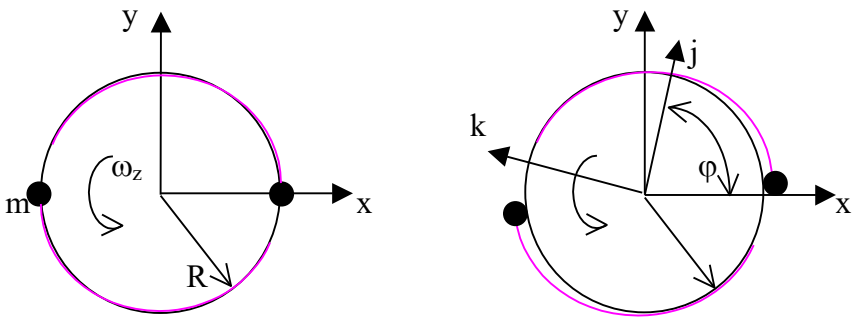
which is negative, showing the spin rate of the satellite is reduced.

Passive control with yo-yo device

Passive control based on the so-called yo-yo device is useful whenever, right after launch, the spacecraft must reset its spin speed due to launch. The mechanism is able to cancel the residual spin velocity, after which the spacecraft can start normal operation and precise control. The mechanism is based on the principle of conservation of angular momentum and kinetic energy, through a symmetric deployment of two small masses.

Assume as example a cylindrical shape and a spin axis aligned with the cylinder axis, labeled with z. If we have two tethers winded around the satellite, with a tip mass, the masses are fixed to the satellite

during launch and released after launch. As the masses are released, the tethers are deployed remaining tangent to the satellite lateral surface.



Consider the rotating frame (j,k) in which j defines the point of contact of the tether with the satellite; call φ the angle between the x axis and the j axis, identifying the deployment angle, and $\dot{\phi}$ the angular velocity of the rotating frame with respect to the body frame. The initial condition is:

$$h_0 = (I_z + 2mR^2)\omega_0$$
$$T_0 = (I_z + 2mR^2)\frac{\omega_0^2}{2}$$

where I_z is the inertia moment of the satellite only and mR^2 is the contribution to inertia due to one tip mass. In a generic time instant, the mass m has a velocity:

$$\underline{r} = R\underline{j} - R\phi\underline{k}$$
$$\underline{v} = \dot{\underline{r}} + \underline{\omega} \wedge \underline{r}$$
$$\omega = \omega_z + \dot{\phi}$$
$$\underline{v} = \begin{cases} R\phi(\omega_z + \dot{\phi}) \\ -R\dot{\phi} + R(\omega_z + \dot{\phi}) = R\omega_z \end{cases}$$

The angular momentum of one mass, aligned with z, will be:

$$h_m = (\underline{r} \wedge \underline{v})m = (R^2\omega_z + R^2(\omega_z + \dot{\phi})\phi^2)m$$
$$T_m = \frac{1}{2}m\underline{v} \cdot \underline{v} = \frac{m}{2}\left(R^2\omega_z^2 + R^2(\omega_z + \dot{\phi})^2\phi^2\right)$$

Due to conservation of h and T with time, we can equate the initial condition to the generic condition to have:

$$h_0 = (I_z + 2mR^2)\omega_0 = I_z\omega_z + 2mR^2\left(\omega_z + \phi^2(\omega_z + \dot{\phi})\right)$$
$$T_0 = (I_z + 2mR^2)\frac{\omega_0^2}{2} = \frac{I_z\omega_z^2}{2} + \frac{2mR^2}{2}\left(\omega_z^2 + \phi^2(\omega_z + \dot{\phi})^2\right)$$

The two equations can be solved for the variables ω_z and $\dot{\phi}$, so that the deployment speed and the satellite angular velocity can be determined:

$$(I_z + 2mR^2)\omega_0 = (I_z + 2mR^2)\omega_z + 2mR^2\phi^2(\omega_z + \dot{\phi})$$
$$(I_z + 2mR^2)\omega_0^2 = (I_z\omega_z^2 + 2mR^2)\omega_z^2 + 2mR^2\phi^2(\omega_z + \dot{\phi})^2$$

To simplify the analysis, we can introduce a nondimensional parameter that indicates the ratio of the inertia of the masses to the overall system inertia:

$$a = \frac{(I_z + 2mR^2)}{2mR^2}$$

The equations become:

$$\begin{cases} a\omega_0 = a\omega_z + \phi^2(\omega_z + \dot{\phi}) \\ a\omega_0^2 = a\omega_z^2 + \phi^2(\omega_z + \dot{\phi})^2 \end{cases}$$
$$\Rightarrow \begin{cases} a(\omega_0 - \omega_z) = \phi^2(\omega_z + \dot{\phi}) \\ a(\omega_0^2 - \omega_z^2) = \phi^2(\omega_z + \dot{\phi})^2 \end{cases}$$
$$\Rightarrow \begin{cases} a(\omega_0 - \omega_z) = \phi^2(\omega_z + \dot{\phi}) \\ a(\omega_0 - \omega_z)(\omega_0 + \omega_z) = \phi^2(\omega_z + \dot{\phi})^2 \end{cases}$$
$$\Rightarrow \phi^2(\omega_z + \dot{\phi})(\omega_0 + \omega_z) = \phi^2(\omega_z + \dot{\phi})^2$$
$$\Rightarrow (\omega_0 + \omega_z) = (\omega_z + \dot{\phi})$$
$$\Rightarrow \dot{\phi} = \omega_0$$

We therefore see that the tether is deployed with constant angular velocity, equal to the initial satellite spin velocity. At this point we can substitute:

$$\phi = \omega_0 t (\phi_0 = 0)$$

so that the following is obtained:

$$a(\omega_0 - \omega_z) = \omega_0^2 t^2(\omega_z + \omega_0)$$
$$\omega_z(\omega_0^2 t^2 + a) = a\omega_0 - \omega_0^3 t^2$$
$$\omega_z = \frac{a - \omega_0^2 t^2}{a + \omega_0^2 t^2}\omega_0$$

Now, since:

$$\phi = \omega_0 t$$
$$L = R\phi$$

with L indicating the length of the deployed tether, we can calculate:

$$\omega_z = \frac{a - R^2/L^2}{a + R^2/L^2}\omega_0$$

We can calculate either the time or the length of tether deployed that lead to ω_z=0:

$$t = \frac{\sqrt{a}}{\omega_0}$$
$$L = R\sqrt{a}$$

It is interesting to notice that the length of the tether does not depend on the initial velocity, so the system works correctly even in presence of large uncertainties in the spin rate. Notice also that, in order to have a non-zero final velocity there is a strong dependence on the initial velocity, so in this case errors are not corrected. This is the main reason why this system has been adopted only to cancel completely the spin.

Course Notes

Spacecraft Attitude Dynamics and Control

Politecnico di Milano
Prof. Franco Bernelli Zazzera

Part 2:
Attitude sensors and actuators, attitude determination

Note for the reader

These short notes are in support of the course "Spacecraft attitude dynamics", they are not intended to replace any textbook. Interested readers are encouraged to consult also printed textbooks and archival papers.

Technological evolution provides continuously upgraded and improved sensors and actuators used for attitude determination and control of spacecraft. These notes, therefore, report some basic concept and operating principle of the most common sensors and actuators, and the reader is encouraged to consult the catalogues of the manufacturers to collect information of the actual functional schemes.

Franco Bernelli Zazzera

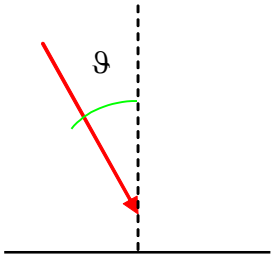
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Spacecraft attitude sensors

Sun sensors

Sun sensors are normally based on materials that produce an electric signal when illuminated by Sun radiation. Measuring the electric output, in general a current, it is possible to evaluate the Sun incidence angle with respect to the surface illuminated.



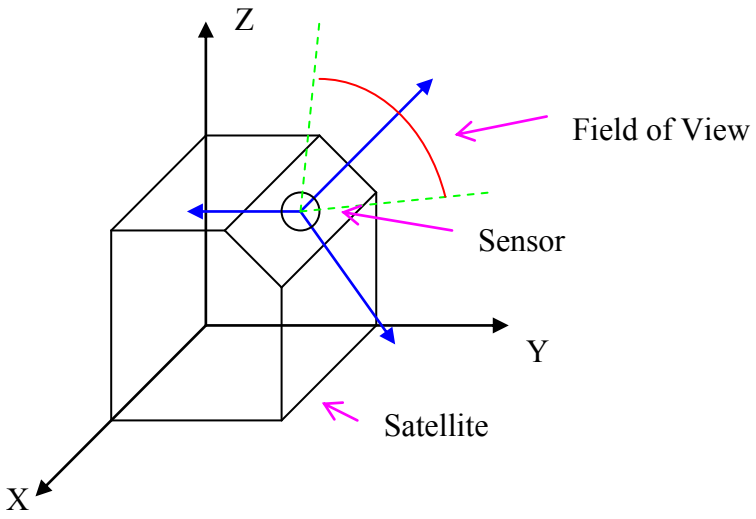
$I = I_0 \cos\vartheta = \alpha SW \cos\vartheta$

where S is the sensor surface, α is a coefficient and W the intensity of the incident radiation.

Sensors are normally built using a series of surfaces with appropriate geometry. Three major categories are identified:

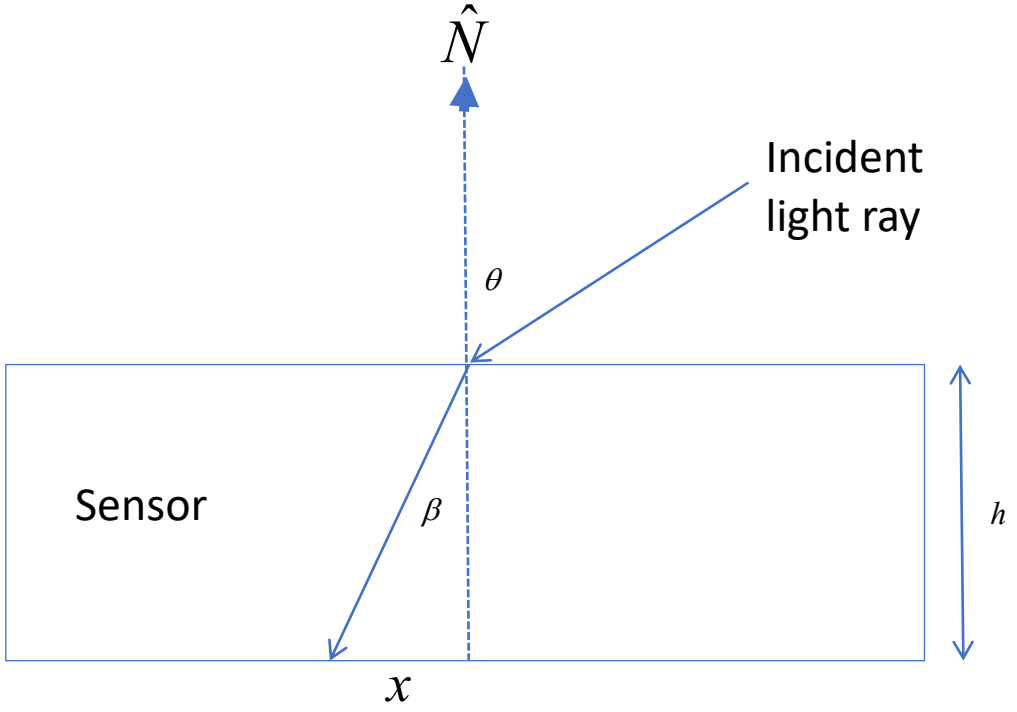
- Sun presence sensors, that essentially detect the presence of the Sun only in a narrow portion of the surrounding space
- Sun sensors, the most used ones, that provide information on the position of the Sun with a wide field of view
- Fine Sun sensors, that provide a precise position of the Sun within a narrower field of view

Every sensor has a well-defined field of view (FOV) and a reference axis that defines the local zero angles. The sensor provides a measure of the relative position of the Sun in the sensor reference. This measurement has to be transformed into information in the satellite reference through a rotation, knowing the orientation of the sensor reference with respect to the principal inertia reference (typically).



The Sun sensor can be calibrated according to the mission of the satellite. For geocentric orbits the sensor will always see the Sun under an apparent angle of 0,53 degrees, that is the apparent Sun diameter at 1 AU, while for interplanetary orbits the angle depends on the position. Normally, Sun sensors are calibrated for use in geocentric orbits.
Sun sensors are widely adopted. They are used not just for attitude determination purposes, but also for instrument pointing, solar panel pointing and/or thermal requirements verification.

One-axis Sun sensor optics can be described using the Figure below:



Snell's law gives $n\ sin\,\beta = sin\,\theta$ where n is the refractive index and, from simple geometry, we have

$$sin\,\beta = \frac{x}{\sqrt{x^2 + h^2}}$$

It follows that

$$sin\,\theta = \frac{nx}{\sqrt{x^2 + h^2}}$$

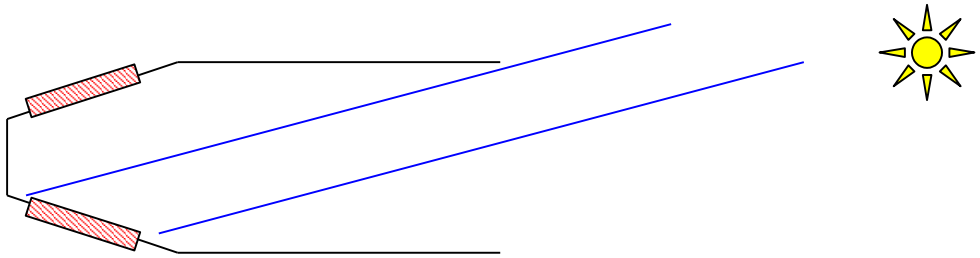
Given that the thickness h is known only the distance x is required to compute the angle between the Sun and the normal to the sensor which can be measured with a digital detector. Three axis Sun sensors are used which yield three independent measurements $\theta_1, \theta_2, \theta_3$

$$\begin{bmatrix} cos\,\theta_1 \\ cos\,\theta_2 \\ cos\,\theta_3 \end{bmatrix} = \begin{bmatrix} \hat{N}_1 \cdot \underline{s} \\ \hat{N}_2 \cdot \underline{s} \\ \hat{N}_3 \cdot \underline{s} \end{bmatrix} = \begin{bmatrix} \hat{N}_1^T \\ \hat{N}_2^T \\ \hat{N}_3^T \end{bmatrix} \underline{s}$$

from which the Sun direction vector $\underline{s}$ can be computed.

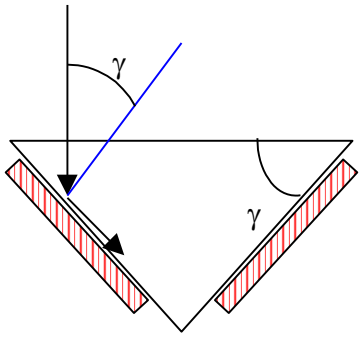
Sun presence sensors

Sun presence sensors provide a sort of binary information: the Sun is detected or it is not detected in the sensor field of view, which is usually restricted. The cosine law is not applicable in this case, due to the presence of shields or support structures. It is common practice to adopt configurations with more than one photocell, as in the following picture.



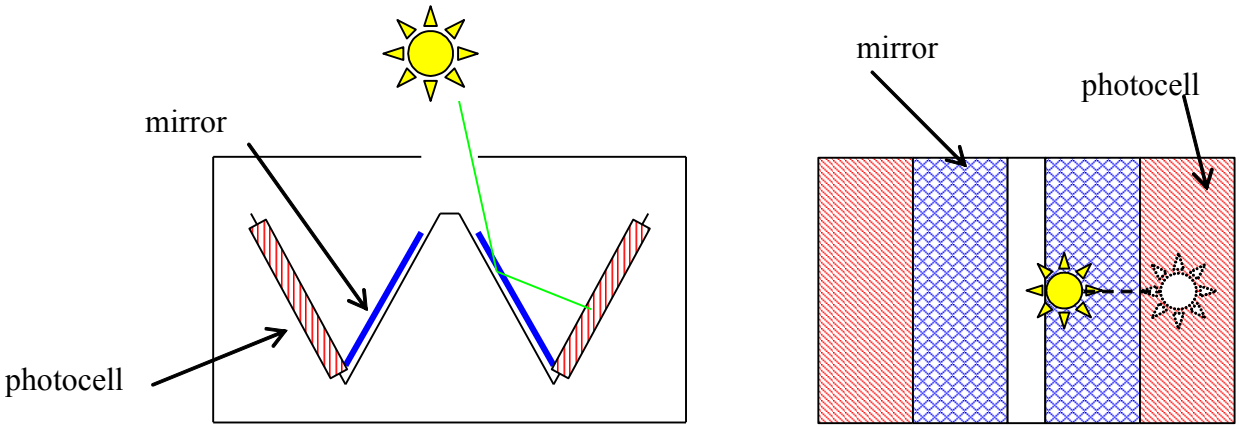
When the Sun moves, either one, both or no photocell is illuminated. Comparing the output signal of the two photocells we can then have the information of the Sun presence and a rough indication of the position.

One alternative geometry uses the principle of refraction in a prismatic configuration.



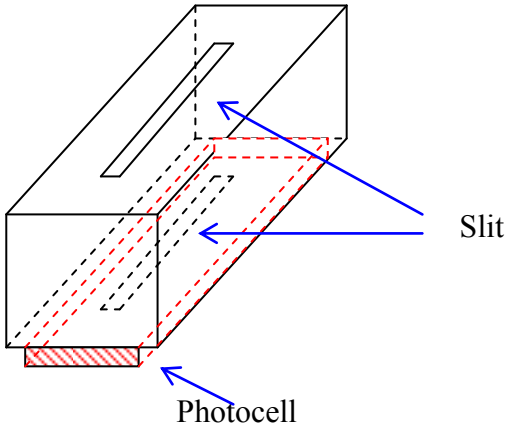
If $n \cdot \sin\gamma = 1$, with n indicating the refraction index of the prism, in the condition of the above figure radiation does not hit the photocell since it is refracted. As the angle γ varies, on one photocell the incidence will be greater than γ and on the other smaller, therefore on one photocell radiation will be totally reflected and on the other partially absorbed, providing an electric output.

A further geometry is the following:

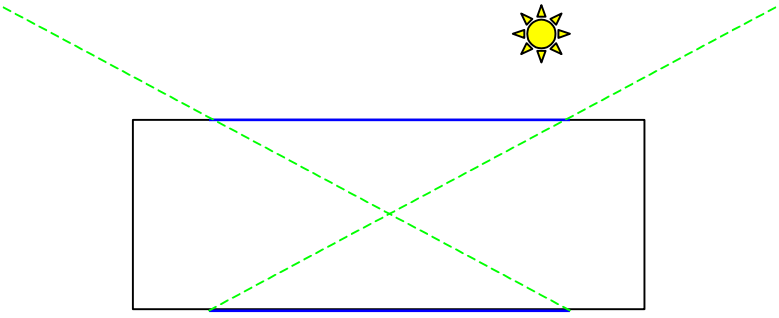


As the Sun moves from the sensor axis, the reflecting surface is more and more illuminated up to the point when the Sun image is completely on the reflecting surface. From this condition the output would be constant even for further Sun inclination, the sensor is saturated.

We can now look at a binary sensor:



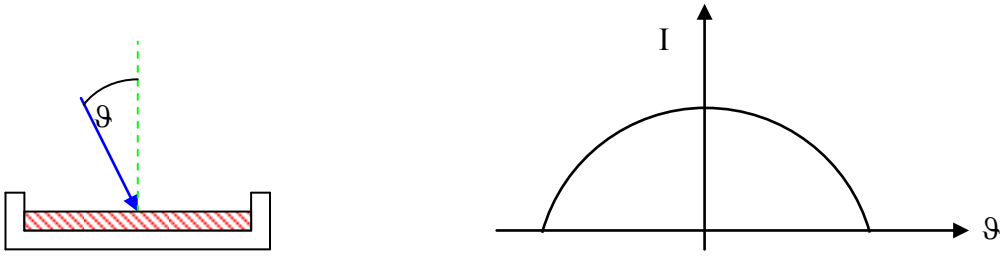
When the Sun ray passes through both slits, it means the Sun is in the plane of the slits and in a definite region of that plane, as seen in the following figure:



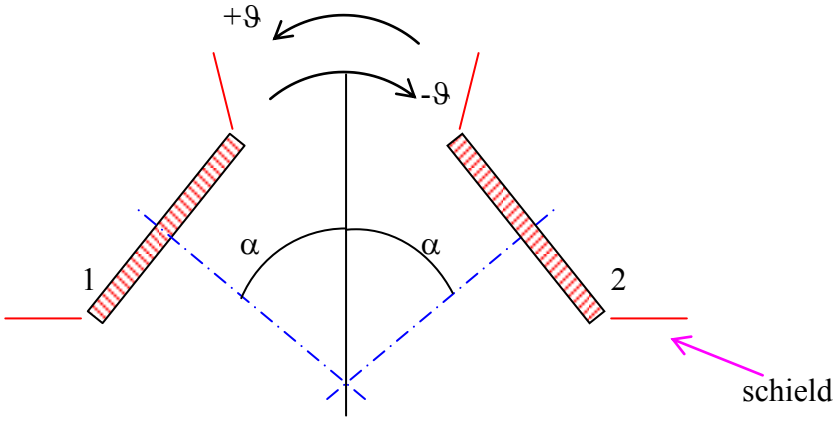
Analog and digital Sun sensors

Sensors providing the position of the Sun can be divided into two main categories: analog sensors and digital sensors. We start by considering the analog sensors.

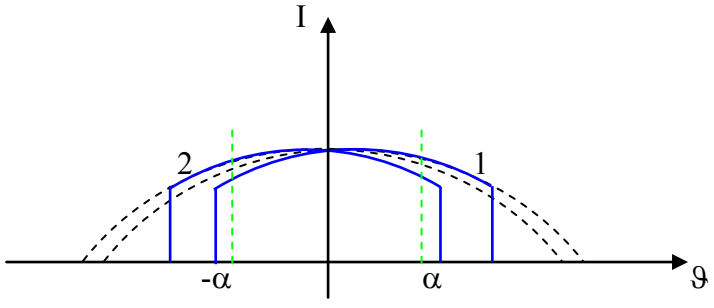
Using the cosine law it is not possible to distinguish positive angles from negative ones. Furthermore, small errors in the measurement close to the sensor zero (vertical) will deliver large errors in the evaluation of the Sun angle, since the variation of the cosine is small close to zero angles.



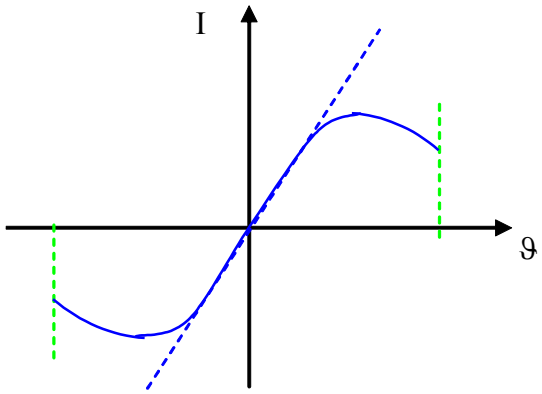
The problem can be limited by adopting two photocells inclined with respect to the zero-axis of the sensor:



Looking now at the intensity of the output current of the two photocells, within the limits guaranteed by the presence of the shields, we have:



Taking now the difference of the two:

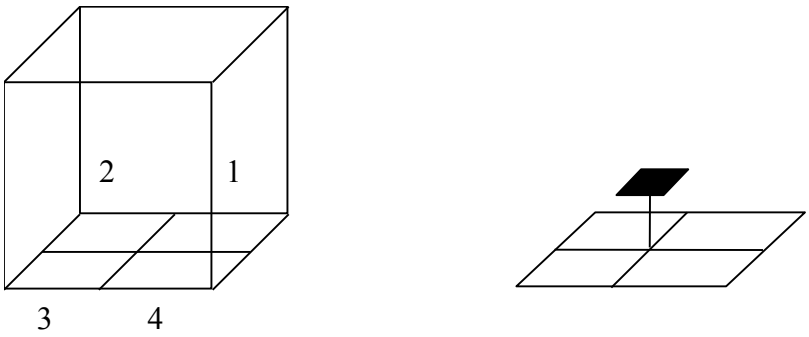


The combined output curve is expressed as:

$$cos(\vartheta - \alpha) - cos(\vartheta + \alpha) = cos\,\vartheta\,cos\,\alpha + sin\,\vartheta\,sin\,\alpha - cos\,\vartheta\,cos\,\alpha + sin\,\vartheta\,sin\,\alpha$$
$$= 2\,sin\,\vartheta\,sin\,\alpha = k \cdot sin\,\vartheta$$

where the constant 2sinα depends on the sensor geometry. The output signal is then sinusoidal, and for small angles it can be approximated by a linear function. If the angle α becomes large, then the useful operating field of view becomes smaller, since the output can be considered reliable only when both photocells are hit by Sun radiation.

With a two-cell sensor we can retrieve the information on the Sun position in one plane, while to obtain the position in space a four-cell configuration has to be adopted, as depicted in the following figure.



The sensor box or shield produces a shadow on each photocell, and the sensor will provide reliable information only when all 4 cells are illuminated. For each cell we have:

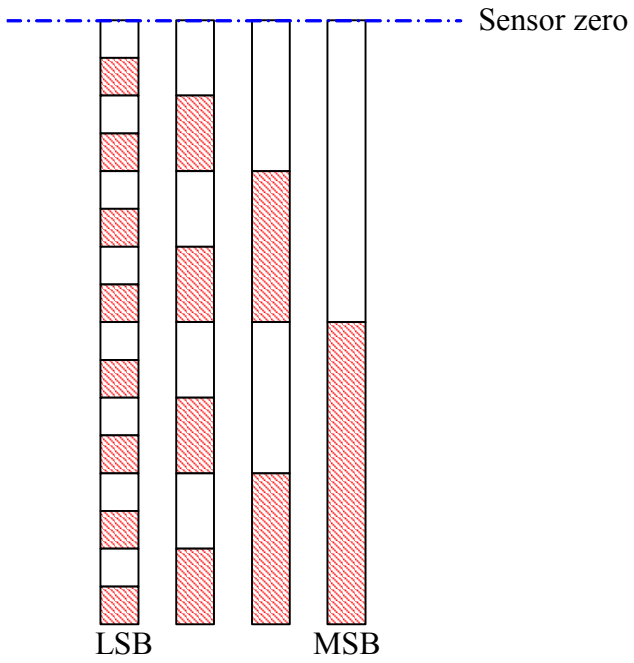
$$I = \mathrm{cA}_0 \cdot \underline{i} \cdot \underline{n}$$
$$A_0 = f(\vartheta, \phi)$$

A_0 is the illuminated area and it is a function of the Sun direction and the sensor geometry. In fact, the product $\underline{i} \cdot \underline{n}$ is known from the Sun distance, so that each output will be proportional to A_0 which in turn can be evaluated as a function of the Sun position.

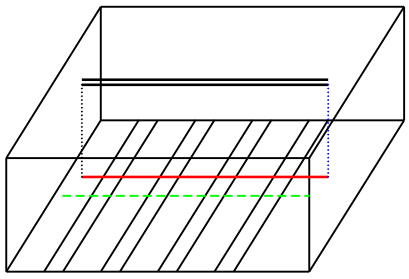
There exist also digital Sun sensors, which provide directly binary information dependent on the Sun position. To understand the sensor principle of operation, start looking at a classical binary table:

Analog	Binary	Analog	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111

The corresponding sensor can be built by using four sensor strips with appropriate mask. In the following figure the red zones correspond to bit set to 1.

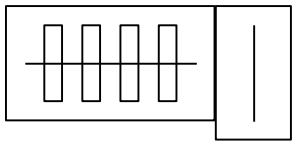


The photocells have all the same properties but their masks are different. When the cells are placed at the bottom of a box with one slit, the Sun produces a line of light that hits each photocell in the same position, generating a signal according to the mask of the series of photocells. This signal can be interpreted as a binary signal. The cell with the finest grid is the least significant bit (LSB), while the cell with the coarse grid is the most significant bit (MSB).



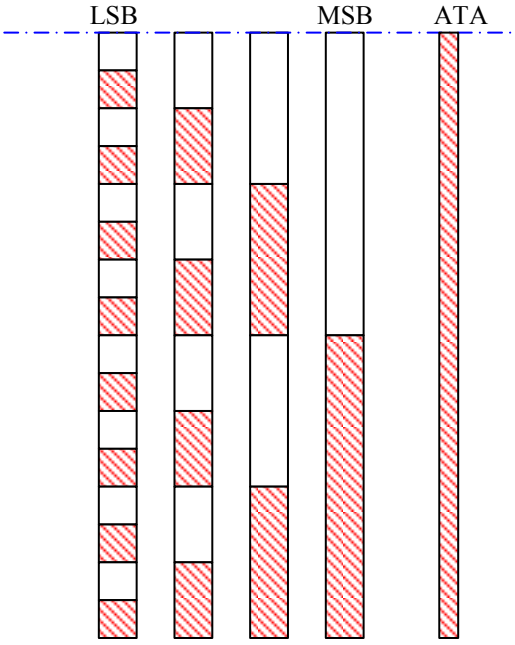
This sensor provides a meaningful signal only when the Sun is in a plane orthogonal to the top slit, otherwise some photocells might not be hit by radiation and their output would become not dependent on the Sun angle. To avoid this, the sensor includes always a Sun presence sensor with disposition

orthogonal to the top slit, as in the figure below, and the output of the Sun presence sensor triggers the decoding of the sensor signal.



At 1 AU distance, the Sun is seen under an apparent angle of approximately 0.5 degrees, so this is the thickness of the light line hitting the photocells. The finest grid should then have a 0.5 degrees sequence. The LSB is then providing a change in the output at every 0.5 degrees change in the Sun position. The overall sensor field of view (FOV) is then a function of the number of strips (bits) of the sensor, since each additional bit doubles the field of view. With 7 bits we have 64 degrees FOV, with 8 bits 128 degrees FOV, that is the maximum since it is useless to go beyond 180 degrees.

To decode the output signal, a threshold has to be defined above which the signal is interpreted as 1, below which it is interpreted as 0. The threshold can be made automatic and position-independent by including a further photocell strip, of width half the width of the other strips and with no cover grid, called Automatic Threshold Adjust (ATA).



Comparing the output signal of the ATA with the other photocells, each bit is set to 1 if the ATA has a lower output; it is set to 0 if the ATA has a higher output. This because the output of the ATA is always half the maximum output of any other photocell, due to the half width of the ATA, irrespective of the position of the Sun. In fact, calling j the grid spacing of the LSB, k the width of the LSB and k/2 the width of the ATA, we have:

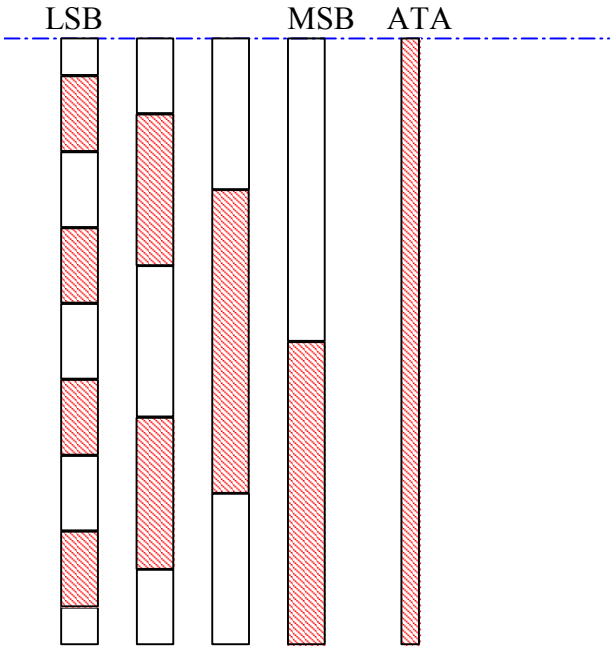
$$I_{ata} = \alpha \frac{jk}{2} \cos\theta$$
$$I_{lsb} = \alpha x k \cos\theta$$

where x is the portion of the LSB illuminated by the radiation. If $I_{ata}>I_{bit}$, then k/2>x and the bit is set to 0 because the LSB is illuminated by less than half. On the contrary, if $I_{ata}<I_{bit}$, then k/2<x and the bit is set to 1 because the LSB is illuminated by more than half. This procedure does not suffer from the changes of the output due to the cosine law and any variation in the intensity of the incident radiation. In a similar way, all bits are compared to the ATA to decode them and set them to 0 or to 1.

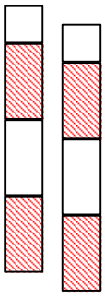
When any bit strip is illuminated exactly for half the maximum, then any measurement error can trigger the detected output to 1 or to 0, in a rather random way. This is a great problem when a series of bits are in this condition, such as between the decimal value 7 and 8. In this case, any measurement noise can trigger any bit to 1 or to 0, so the actual decoded signal would not look like 0111 (decimal 7) or 1111 (decimal 8), but it could be any combination of 0 and 1, including 0000 (decimal 0) and 1111 (decimal 15). This is obviously a huge error, and to avoid this situation the binary code has to be modified in order to avoid having to consecutive integer numbers differing by more than 1 bit. This appropriate code is called Gray code.

Analog	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100

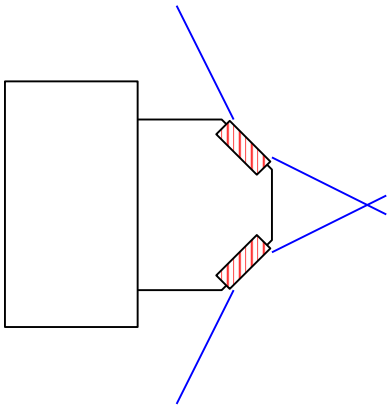
The grid spacing of the Gray code is twice as large as the classical binary code, but is has a symmetric configuration so that the resolution is not changed. With this code, random errors due to misinterpretation of bits are limited to 1 sensor unit, in this case 0,5 degrees.



By placing one additional grid besides the LSB, but shifted by half unit, the sensor resolution can be improved. Once the basic bits are interpreted, a further analysis of the LSB and the additional bits provides a further half unit of resolution (one quarter of degree) if only one further bit is added, and a quarter unit resolution in the case of two added bits. This has the effect of providing resolutions up to approximately 1/8 of degree. For higher resolution, specific and mission dependent solutions must be implemented.

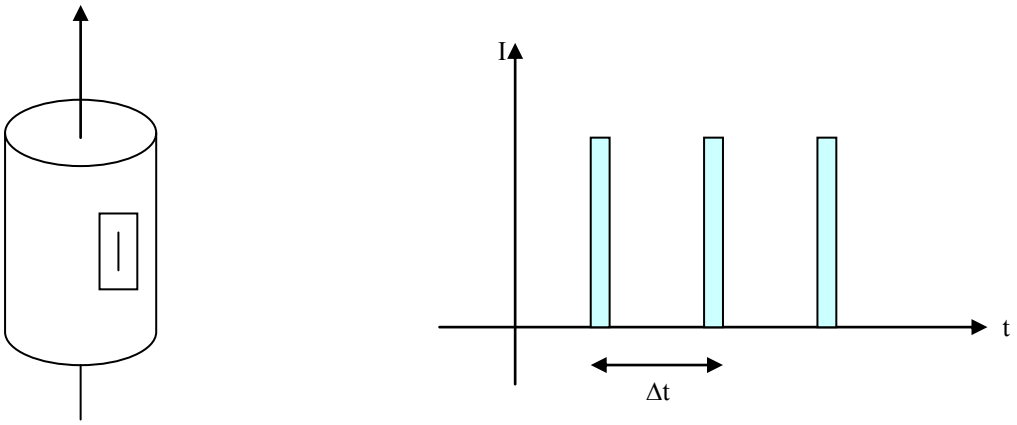


In order to have a full 180 degrees FOV, at least two sensors must be used with overlap of their individual FOVs.

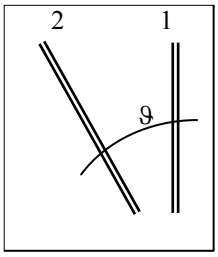


Use of Sun presence sensors on board simple spin satellites

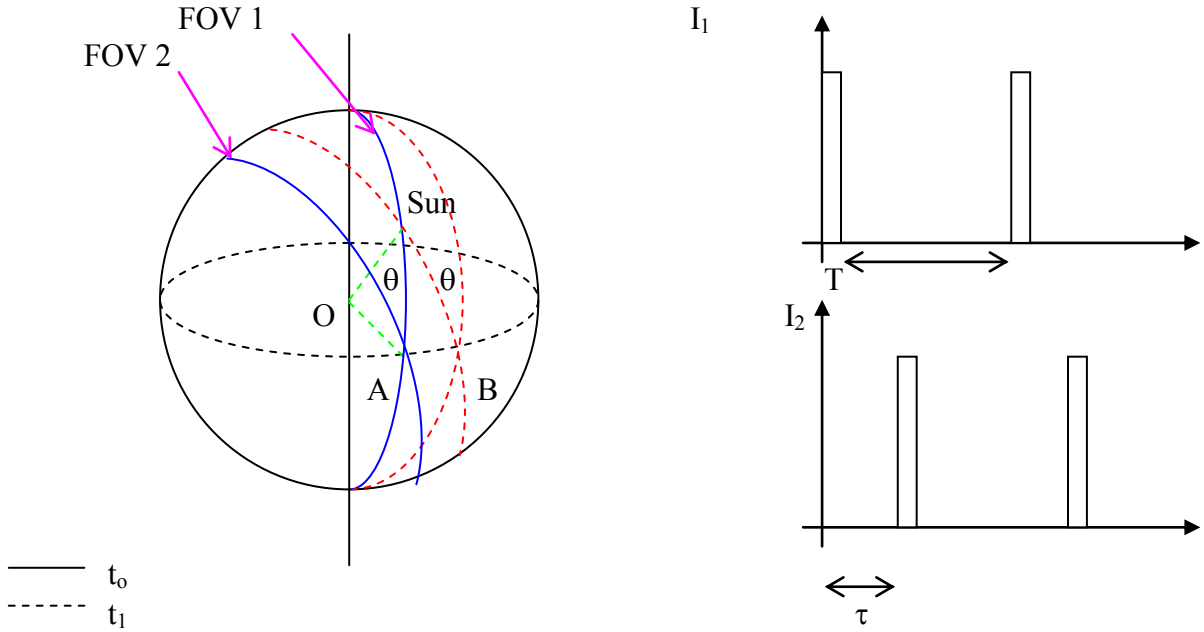
The following example shows the possibility to use a Sun presence sensor to infer the Sun angle when mounted on a simple spin satellite. Consider the simple spin satellite with a two-slit Sun presence sensor mounted on the lateral surface. At every rotation, a signal is produced by the sensor if the Sun is in the sensor FOV.



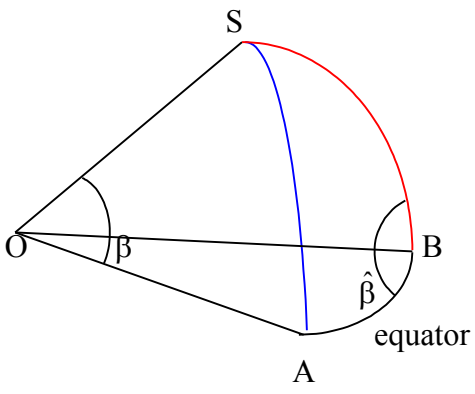
The Sun angle can be calculated by the analysis of the time interval between two consecutive electric pulses produced by a pair of two-slit sensors, with relative inclination ϑ.



Looking at the celestial sphere centered on the satellite, we can draw the following geometry of the problem:



The reference system is centered on the sensor and the equator is orthogonal to the sensor itself. Notice that the line of sight of sensor 1 is orthogonal to the equator. The interval T is connected to the spin angular velocity of the satellite, while the interval τ depends on the spin angular velocity and on the slit relative inclination ϑ. Consider the spherical triangle OABS:



We have:

$$\omega = 2\pi/T$$
$$\mathrm{AB} = \omega\tau$$
$$\hat{B} = \frac{\pi}{2} - \vartheta$$
$$\hat{A} = \frac{\pi}{2}$$

Using the cosine theorem for spherical triangles we evaluate β. Hence, measuring τ and ω it is possible to evaluate the Sun angle β, considering that some limitations to the general FOV of the sensor exist due to the relative inclination between the two slits (angle ϑ).

Starting from the measurement τ, the equations to solve the problem are:

$$\mathrm{tg}\left(\frac{\pi}{2} - \beta\right) = \frac{\mathrm{tg}\vartheta}{\sin(\omega\tau)}$$

$$\mathrm{cotg}\ \beta = \frac{\mathrm{tg}\ \vartheta}{\sin(\omega\tau)}$$

In a simulation environment, it is also possible to analyze and predict the sensor output, so evaluating the time interval τ as a function of the problem geometry, with the following equation

$$\tau = \frac{1}{\omega}\sin^{-1}(\mathrm{tg}\beta\ \mathrm{tg}\ \vartheta)$$

This is useful to define the minimum sampling time of the signal, sufficiently lower than the interval τ. Considering also some possible construction and mounting errors for the sensor, the above equation should be modified as:

$$\mathrm{cotg}^2\beta = \left[\frac{\mathrm{tg}(\vartheta + \Delta\vartheta + \varepsilon) - \mathrm{tg}\varepsilon\ \cos(\omega\tau - \delta)}{\sin(\omega\tau - \delta)}\right]^2 + \mathrm{tg}^2\varepsilon$$

where:

$\Delta\vartheta$	error on the relative angle between slits (not known)
δ	alignment error of slit 1 and spin axis
ε	in-plane rotation angle error of slit 1

In general, the simplest equation is used as a solution method, while the more complete model is used only to model the sensor and infer the possible operating errors, in connection also with the sampling frequency of the signals.

Horizon sensors (Earth sensors)

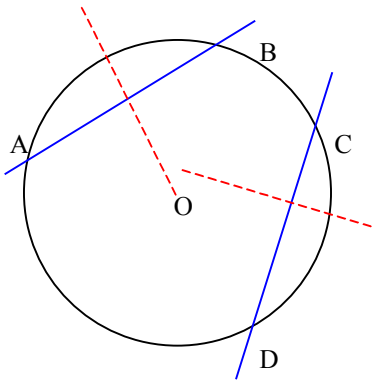
If the spacecraft is orbiting close to an extended body, such as a planet, the information on the presence of the planet within the FOV of a sensor is not sufficient to have information on the satellite attitude. To do so, the precise position of the center of the planet has to be evaluated. Horizon sensors work by analyzing the areas of the FOV that are not illuminated (deep space) compared to the illuminated areas (planet), and in general do not work in the visible spectrum to reduce the effects of interferences. It is common practice to design infrared sensors, for which the ratio of the Earth radiation compared to Sun radiation is "only" 1 to 400 (it is 1 to 30.000 in the visible spectrum). For Low Earth Orbits, the Earth can occupy 40% of the celestial sphere.

Earth horizon sensors use thermopile detectors to compute the direction vector of the Earth with respect to the spacecraft. A thermopile is a device for measuring the power of incident electromagnetic radiation via the heating of a material with a temperature-dependent electrical resistance. This sensor operates in one of two principal modes. The first is based on scanning the Earth's horizon to locate the centre of the Earth using a narrow field of view sensor (2 – 7 degrees); the second mode is based on static determination of the location of the Earth's contour inside the instrument's field of view (FOV is slightly larger than the Earth). Both kinds of sensors will be briefly explained.

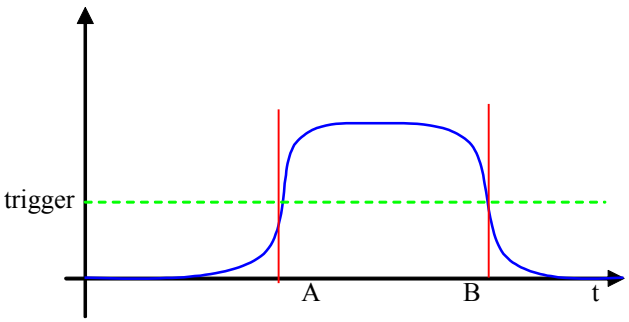
Scanning sensor

Scanning Earth horizon sensors use a spinning mirror or prism and focus a narrow beam of light onto a sensing element usually called a thermopile. The spinning causes the device to sweep out the area of a cone, and electronics inside the sensor detect when the infrared signal from Earth is first received and then lost. A factor that plays into the accuracy of such sensors is the fact the Earth is not perfectly circular.

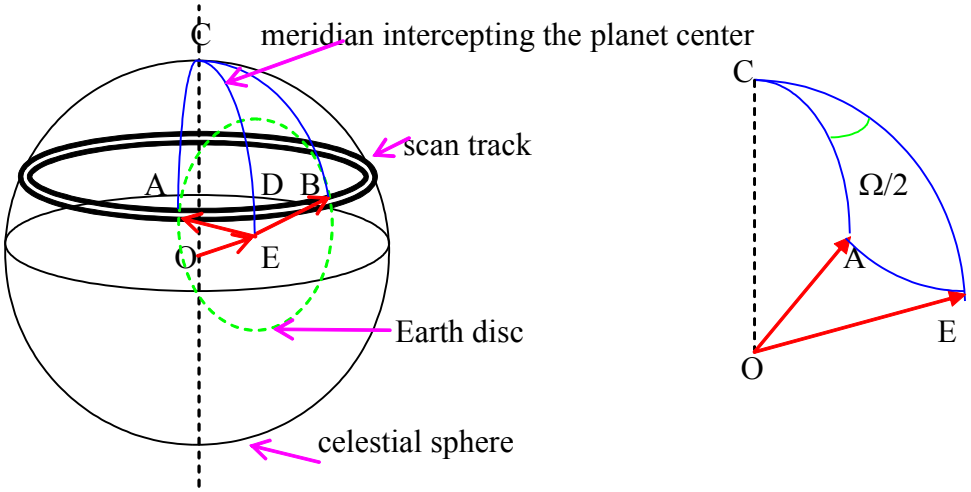
One option to calculate the planet center position is to detect two planet chords, assuming the planet is a sphere.



The sensor must be capable of detecting 4 points on the horizon, and associate the points to build the two chords. In general the sensor has a narrow field of view (2 degrees), but it can scan the horizon to produce the required output. When the sensor FOV is on the planet, the output is non-zero, when the sensor FOV is in deep space the output is zero.



Given a reference frame on the satellite, position of points A and B is known, and in similar way positions of points C and D and then the planet center. There are obviously some critical issues, since the boundary of the planet (points A and B) might not be so neat due to the transient nature of the sensor output induced by planet atmosphere. This can be solved by choosing an appropriate trigger level of the signal, above which we consider the signal as correct and below which we assume there is no signal. Points A and B will then correspond to the adopted trigger value. This value should be not fixed, but allowed to vary as a function of the peak signal or on the average of the signal itself, in order to consider also seasonal changes in the radiation emitted or day/night transitions. It is also to notice that the intensity of the sensor signal is not constant, due to change in the distance from ground as the sensor scans the horizon.
The center of the Earth can be detected using spherical geometry. Assuming the Earth is a perfect sphere, knowledge of the orbit altitude provides the information of the Earth apparent radius, and so that only one chord becomes necessary. In the following picture the case of scanning above the equatorial plane is illustrated.



For the spherical triangle CAEO we know $\overline{CA}$ and angle $\hat{C}$. Arc $\overline{AE}$ is the apparent radius of the Earth, known from the altitude, and calling θ the inclination of the scan direction with respect to the satellite horizon we have:

$$\overline{\mathrm{CA}} = \gamma \quad (\text{latitude}) = \frac{\pi}{2} - \theta$$
$$\hat{C} = \frac{\Omega}{2} = \frac{\omega \Delta t}{2} \text{ detected bysensor} = \text{arcAD}$$
$$\overline{\mathrm{AE}} = \rho$$

From spherical triangle geometry we then have:

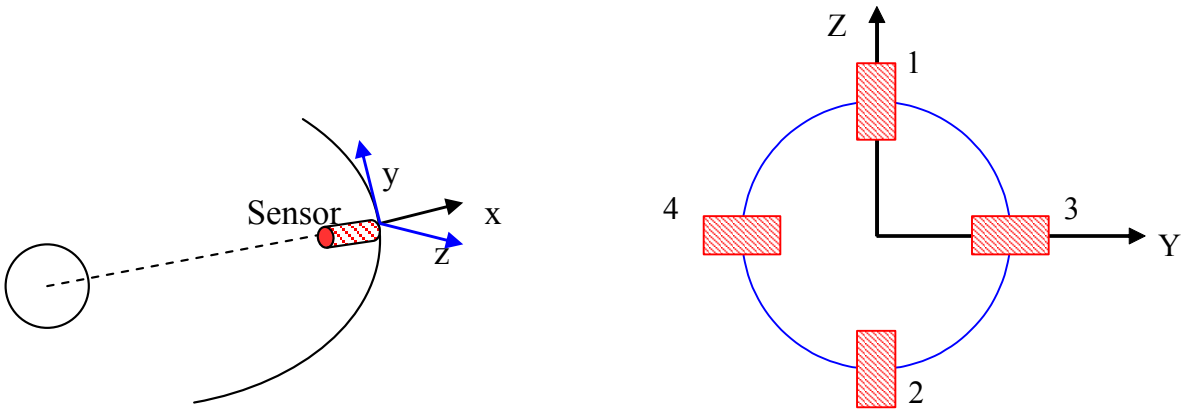
$$\frac{\Omega}{2} = \cos^{-1}\left(\frac{\cos\rho - \cos\gamma\cos\eta}{\sin\gamma\,\sin\eta}\right)$$

from which $\eta = \overline{CE}$

This type of sensor is rather heavy due to the presence of precise scan mechanism, that must guarantee a scan velocity much higher that the spin velocity of satellite.

Static sensor

Static Earth horizon sensors contain a number of sensors and sense infrared radiation from the Earth's surface with a field of view slightly larger than the Earth. The accuracy of determining the geocenter is 0.1 degrees in near-Earth orbit to 0.01 degrees at GEO. Their use is generally restricted to spacecraft with a circular orbit. An example configuration of 4 thermopile sensors is given below.

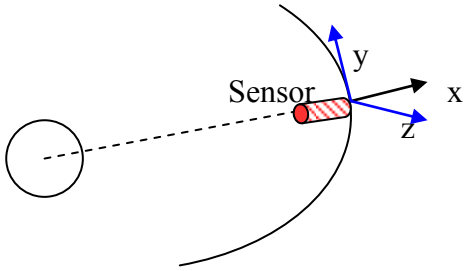


When the Earth is in the center of the FOV, the four detectors have all the same output, so that:

$$(1) - (2) = 0$$
$$(3) - (4) = 0$$

When the image of the Earth is not centered, the four detectors have different outputs and taking appropriate differences we have direct information of the misalignment angles:

$$(1) - (2) = \Delta z \text{ proportional to } \alpha_y$$
$$(3) - (4) = \Delta y \text{ proportional to } \alpha_z$$



Given these two angles we can then compute the Earth horizon vector using spherical coordinates with $\hat{\underline{h}}_B = [cos\,\alpha_y\,cos\,\alpha_z \quad -sin\,\alpha_z\,cos\,\alpha_y \quad sin\,\alpha_y]^T$

Given that we can also have an orbital model on-board or from our orbit determination we also have the direction vector of the Earth with respect to the spacecraft measured in the inertial frame $\hat{\underline{h}}_N$. Therefore, we have

$$\hat{\underline{h}}_B = A_{B/N}\hat{\underline{h}}_N$$

We could parameterize $A_{B/N}$with Euler angles so that there are only three unknowns and then we would have three nonlinear equations and three unknowns. Since it is nonlinear we can try to solve for the Euler angles using a numerical optimization scheme that minimizes the function:

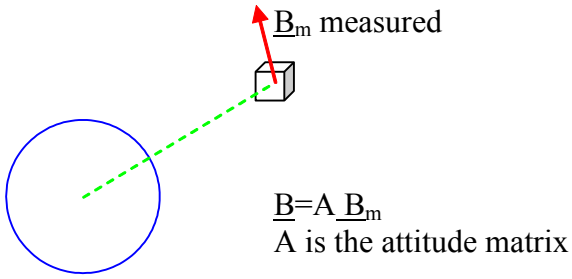
$$J = \left\|\hat{\underline{h}}_B - A_{B/N}\hat{\underline{h}}_N\right\|$$

However, since $\hat{\underline{h}}_B$and $\hat{\underline{h}}_N$ are unit vectors the problem is underdetermined, so it cannot be solved. This is intuitive since with a direction vector only, we cannot know the angle of the roll axis. So we cannot retrieve the attitude matrix just with an Earth horizon sensor.

For simple spin or dual spin satellites, with nominal angular momentum aligned with the z axis, measurement of α_z is always required since the z equation is decoupled from x and y equations. The x and y equations are instead coupled, so that measuring only α_x or α_y we can in any case reconstruct the satellite attitude with a state observer. Therefore this sensor, that measures the two angles α_y and α_z, is suitable for a full state observation in the linear satellite dynamics model.
This sensor is not capable of detecting rotations around the yaw axis (x), and furthermore the optical head must be designed in order to have a FOV that is twice the Earth apparent radius. Therefore this sensor is commonly used only for circular orbits, at constant altitude, and on Earth pointing satellites.

Magnetic field sensor

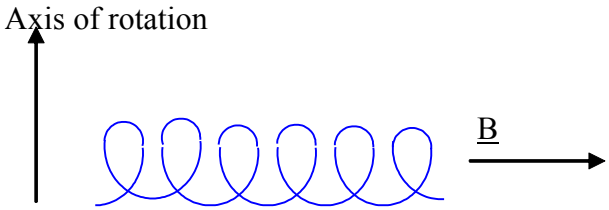
This sensor can provide the measurement of one vector. It has to be coupled with a mathematical model of the magnetic field and to the knowledge of the position of the satellite in order to provide attitude information.



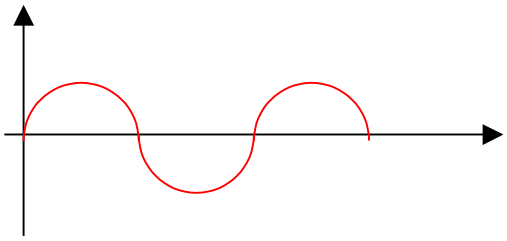
It is not simple to measure a constant magnetic field; in fact typically the rate of change of the flux of the magnetic field vector can be measured by a coil. Indicating with dove ϕ_B the flux of the magnetic field and μ the magnetic permeability, we have

$$I = \frac{d\varphi_B}{dt}\mu$$

This could be in principle obtained by having one coil rotating around a defined axis

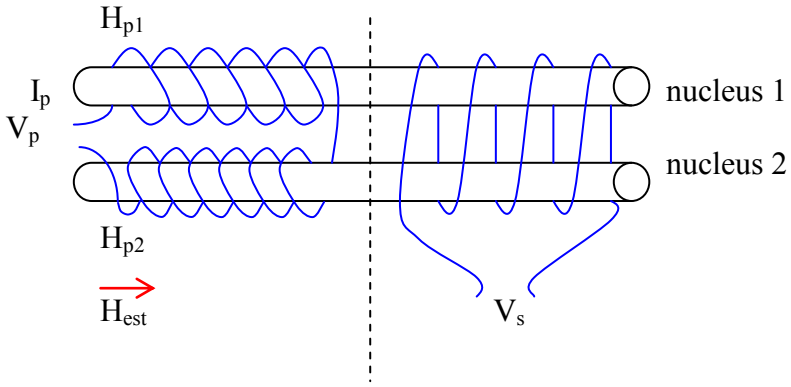


For a constant B vector the measurement would be sinusoidal



and the derivative of the measured signal is the projection of the B vector onto the coil axis, rotating inside the satellite. This procedure is complicated and provides a measurement not directly useful for attitude determination purpose, therefore alternative sensors are built, called fluxgate magnetometers.

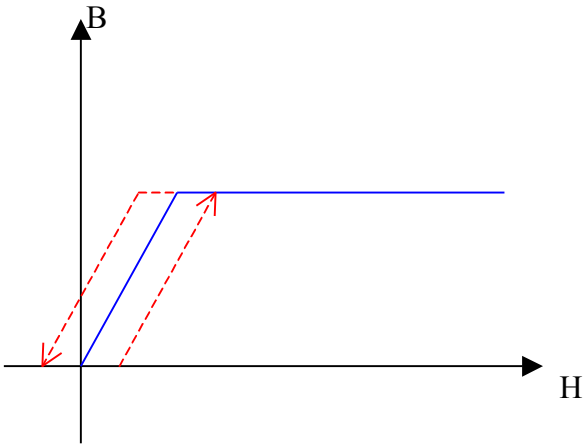
Consider two nuclei of ferromagnetic material, with two coils winded as shown in the figure below



On the left side the coil, called primary coil, is winded in such a way to produce on the two nuclei exactly the same magnetic induction but with opposite sign. Since the voltage at the secondary coil (on the right) is

$$V_s = -\frac{d\varphi(B_1 + B_2)}{dt}$$

In the case of zero external magnetic field the voltage would be zero. If the external magnetic field is non-zero, it is added to the induction due to the primary coil and the voltage at the secondary coil would be non-zero.
The coils create a magnetic induction field that generates the magnetic field B. This is normally a nonlinear mapping, which for sake of clarity will be modeled as linear with saturation, neglecting nonlinearity and hysteresis.

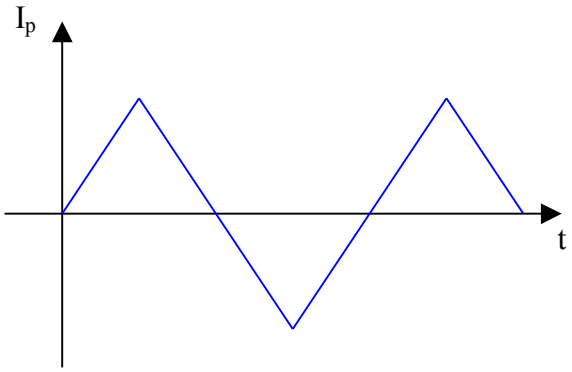


The current flowing in the primary coil is usually generated with a triangular wave form, so that the generated magnetic induction will also have a triangular waveform.

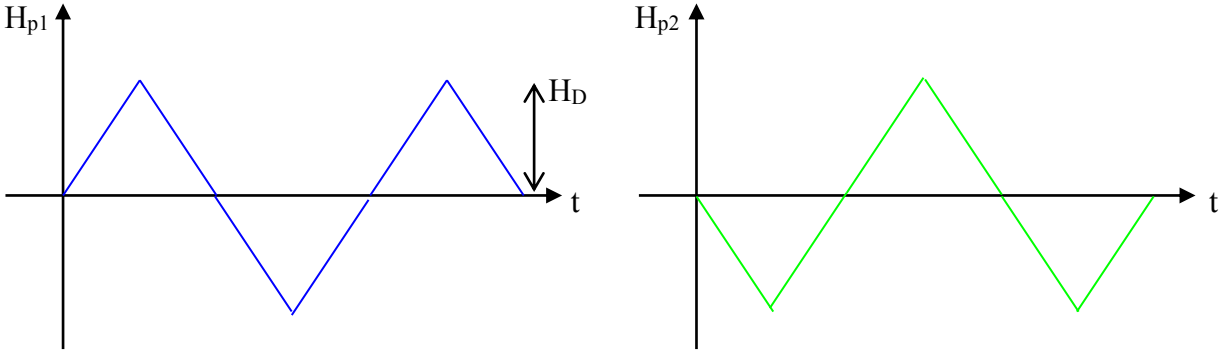
$$V = L\frac{di}{dt} = -\frac{dH}{dt}$$

It will be clear later on that the triangular waveform as input generates a pulsed output, easy to convert into a magnetic field measurement.

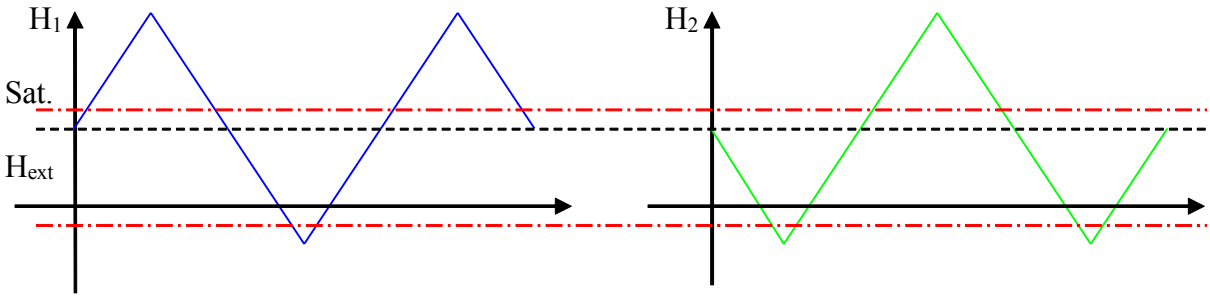
Define the following input to the primary coil



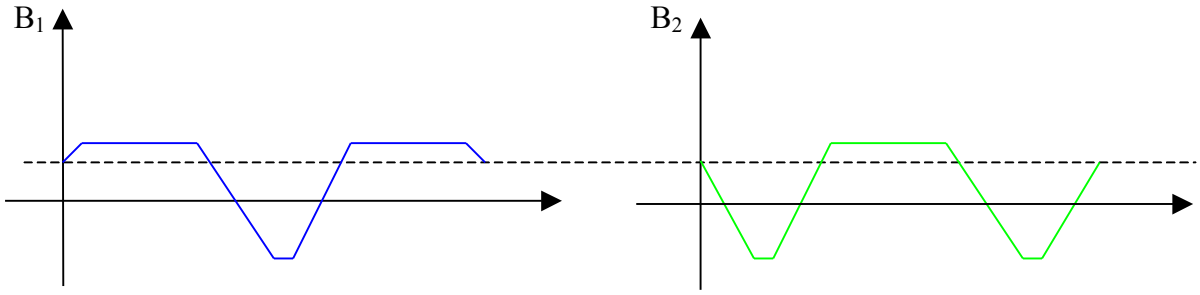
The two magnetic induction in the two nuclei are



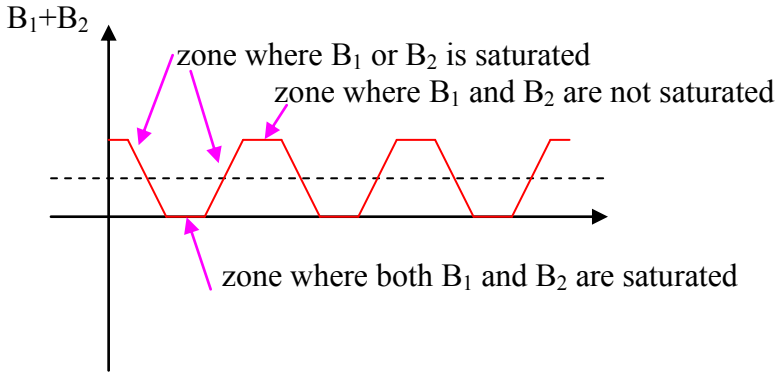
Adding the effect of the external magnetic field we get to



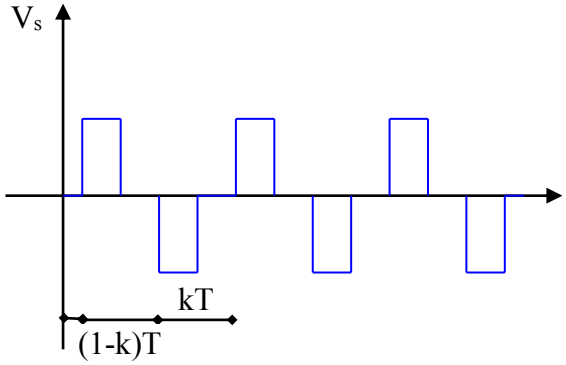
The magnetic field in the two nuclei becomes, considering nucleus saturation



We can now add the two magnetic fields generated, to analyze what is included in the secondary coil on the right part of the sensor.



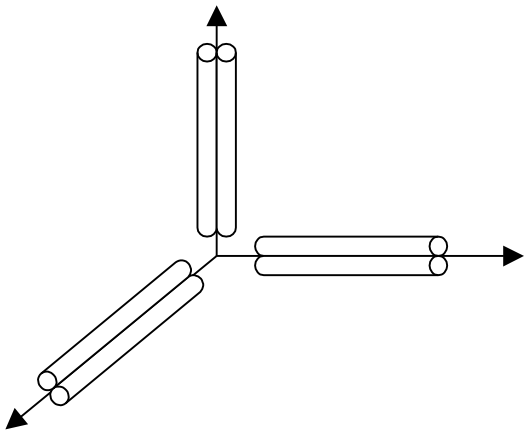
The voltage at the secondary coil is the derivative of the flux of the magnetic field vector induced



The spacing between the impulses is related to the component of the external magnetic field in the direction of the coil axis. The amplitude of the impulse is instead depending on the peak current in the primary coil.

$$H_{\text{ext}} = (1 - 2k)H_D$$

Since the principle of operation of this sensor relies on the saturation of the two nuclei, the value of the current in the primary coil must consider the expected value of the external magnetic field. In case no saturation of the two nuclei occurs, or if one nucleus is permanently saturated and the other never saturated, then the sensor would produce no meaningful output.
Using three pairs of nuclei, as shown below, we can measure the components of the magnetic field along three orthogonal directions, so the B vector.



They have an accuracy of about 300 arc minutes (5 degree) at 200km altitude to about 1 degree at 5000km. They are the least accurate attitude reference sensors as there is a lot of uncertainty in the modelling and measurement of the magnetic field. The magnetometers are also subject to noisy measurements and interference from on-board electronics. Variations in the solar wind, solar storms can lead to large inaccuracies in the on-board model.
Note that this measurement is taken in a body axes frame ($\underline{b}_B$), while the mathematical model of the magnetic field produces a vector expressed in a local horizontal frame ($\underline{b}_H$ or $\underline{b}_{r,\varphi,\vartheta}$). Detecting the rotation that maps one vector onto the other leads to the attitude determination in the local horizontal frame. In principle, we can think about looking for the solution to this problem

$$\left\| \underline{b}_B - A_{B/H}\underline{b}_H \right\| = 0$$

This can be solved using a numerical optimization procedure that seeks to find A that minimises the function

$$J = \left\| \underline{b}_B - A_{B/H}\underline{b}_H \right\|^2$$

If we are using DCM then to guarantee that the solution is orthonormal, a constrained optimal control problem must be solved, such that *J* is minimized subject to the constraints:

$$A_{B/H}^T A_{B/H} = I, det(A_{B/H}) = 1$$

As for the horizon sensor, there is however always uncertainty in the rotation around the direction of the measured $\underline{b}_B$. The problem is underdetermined, so it cannot be solved. This is the reason why the sensor is always coupled to another measurement in order to provide reliable attitude determination.

Star sensors

The best precision attainable with Sun sensors is approximately 1/8 of degree, and in addition the Sun might not be always visible if the orbit has a phase in eclipse. Star sensors allow improving the accuracy of attitude determination and, with appropriate choice of the stars observed, no eclipse problem arises. On the other hand, star sensors are extremely expensive, in terms of constructions and in terms of operations to be performed on board for data processing.

Star sensors have to perform the following operations:

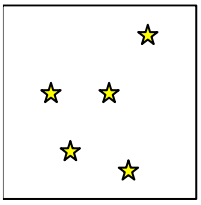
Phase 1: data acquisition
Phase 2: correct positioning in space of the acquired data
Phase 3: interpretation of the data acquired (star identification) that requires analysis of a star catalogue
Phase 4: attitude determination

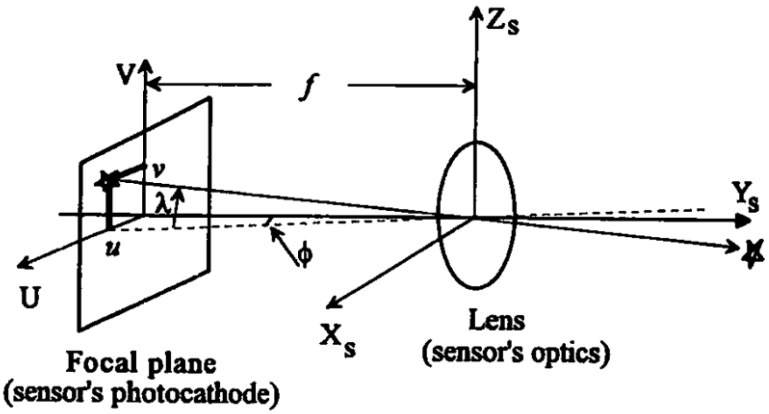
The interpretation of the data is an extremely consuming activity in terms of computation resources. It requires the comparison of the sensor data with a map of the stars, available on board the sensor, in terms of light intensity (magnitude of the star) and radiation spectrum.

Star sensors are intrinsically precise sensors, but to achieve maximum accuracy the sensor has to be also extremely sensible, so that spurious radiation can deteriorate the performances and the operating

lifetime of the sensor. A Sun presence sensor can be used to switch off the star sensor when its FOV is too close to the Sun, and light shields can also be used to limit interference with other sources of light (even Earth radiation can be source of disturbances).

In the case of star-sensors the number of measurements available are often much greater than 3 (typically 20 spacecraft-star measurements are given). Each star direction vector can be computed by detecting the point at which it intersects the focal plane. Modern sensors are built with a matrix sensor (CCD), on which the star image is projected.





The direction of the star is computed from the projection of the image on the focal plane

$$\tan\phi = \frac{u}{f} \quad ; \quad \tan\lambda = \frac{v}{f}\cos\phi \quad ; \quad \hat{O} = [-sin\phi\, cos\lambda \quad cos\phi\, cos\lambda \quad -sin\lambda]^T$$

It is pointed out that, even with narrow FOV, the stars captured by the sensor might be numerous. If the sensor has tilting capacity it can detect one star and tack it so that it is always in the center of the FOV. This sensor is called star tracker.

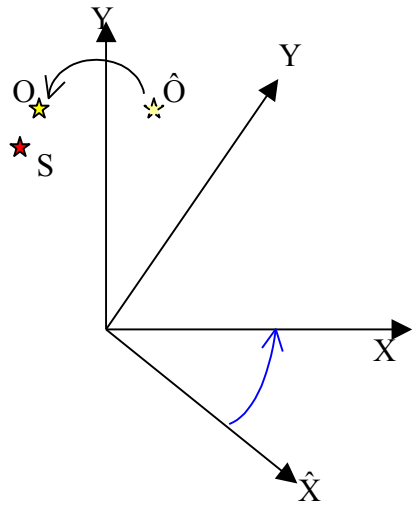
A different possibility is to keep the sensor fixed on the satellite and to reconstruct the map of the observed sky, in what is called a star mapper sensor. In this case the problem is to have a sufficiently fine CCD sensor.

Attitude determination requires first the identification of the stars observed; now we look at how this process can be completed.

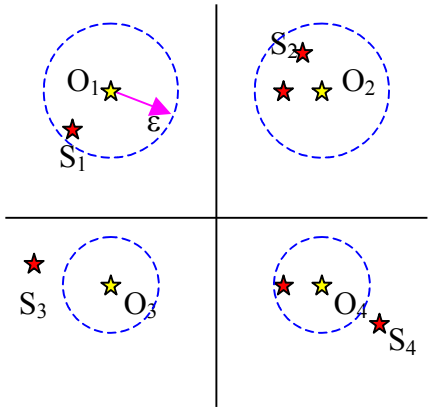
Assume to have S stars in the catalogue and Ô observations available, the goal is to associate each observation (in the sensor reference frame) with one star of the catalogue (in the inertial frame).

$$\begin{array}{ccc} \hat{O}_1 \overset{A}{\rightarrow} O_1 \rightarrow S_1 & & \\ \hat{O}_2 \overset{A}{\rightarrow} O_2 \rightarrow S_2 & & \\ \hat{O}_3 \overset{A}{\rightarrow} O_3 \rightarrow S_3 & & \\ \vdots \quad \vdots \quad \vdots & & \\ \hat{O}_n \overset{A}{\rightarrow} O_n \rightarrow S_n & & \hat{O}_i \quad \text{local reference} \\ & & O_i \quad \text{inertial reference} \end{array}$$

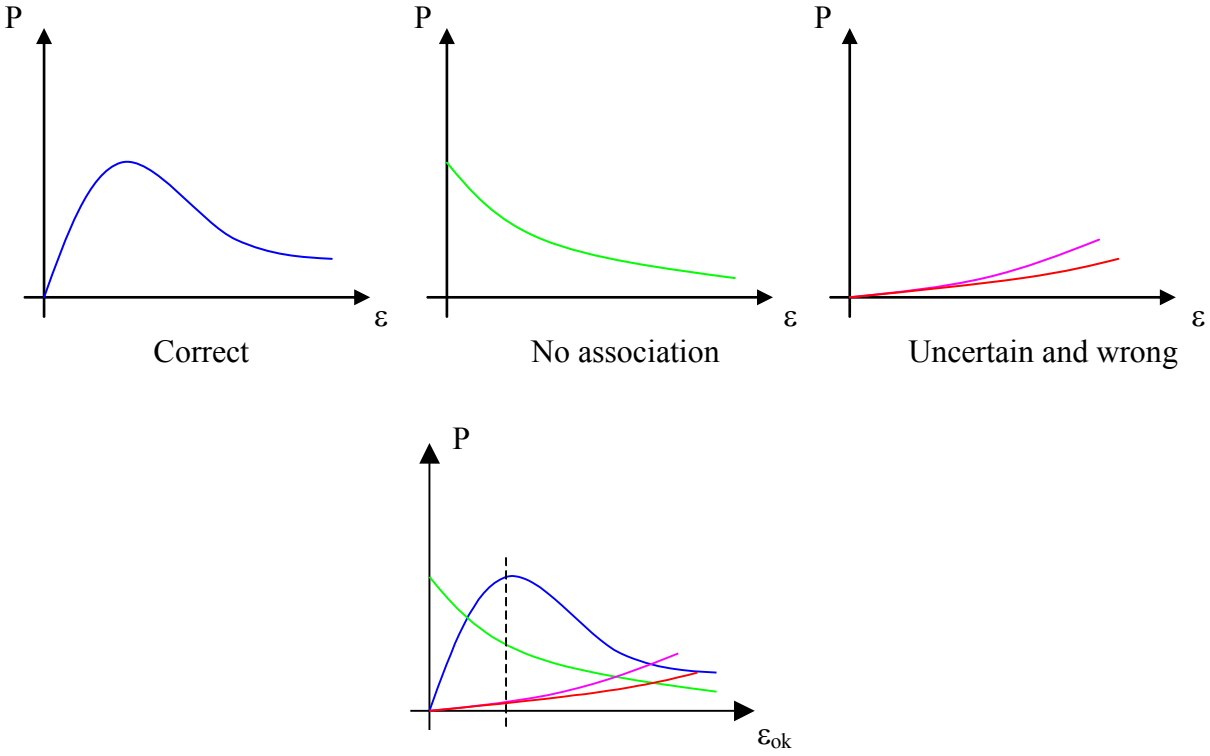
The process requires a first guess estimation of the attitude matrix A, to refine after the correct association of the observed stars with the catalogue.



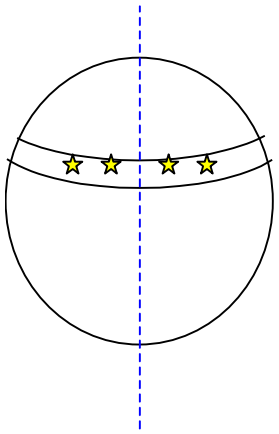
Once the observed star Ô is rotated in the catalogue system (O), the distance between O and S is used to correct the estimation of the attitude A, but this operation is correct only if we are sure that O is the correct image of S and not of a different star. To have a higher probability of success, the identification can be based also on relative geometry of a set of stars.



First, the star closer to O_1 is searched until one is found. If the association is correct, the remaining image stars should be immediately associated to the respective catalogue stars. This is called direct matching of stars. Some problem might arise in the case represented in the case O_2 in the above figure, where a multiplicity of stars can be associated to the observation, or the case O_3 in the above figure, where no stars in the catalogue are sufficiently close to the observation (no identification). The worst case is however the case 4 above, where the wrong star in the catalogue is closer to the observation, leading to an erroneous association. If the search area is enlarged (increase of neighborhood ε), the situation of O_3 is improved, leading to possible correct association, but as a general rule the number of uncertain associations or wrong associations might increase as ε increases. The optimal search radius ε maximizes the probability to have, for each observed star, a unique and correct association with the catalogue.

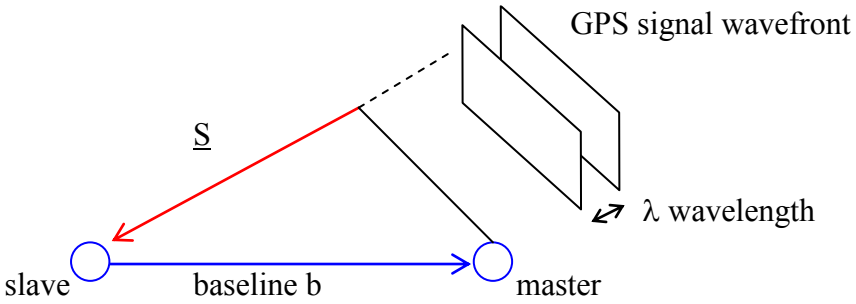


The optimal value of ε depends on the completeness of the catalogue and on the density of stars of the same magnitude in the observed area. Sky areas with a diffuse radiation can create problems since the sensor can erroneously associate one star to a portion of image, even if this is only diffuse background radiation.
A different method can be used if a circular scan of a portion of sky is performed, and if this image strip is compared to the catalogue. The correct attitude would match the highest number of image stars to the stars in the catalogue. This method is useful only if stars on the strip are not uniformly distributed, and if at least the altitude of the scan is known to reduce the search space within the catalogue.



Use of GPS sensors for attitude determination

GPS sensors can be used to determine attitude, provided at least 3 antennas are available.



Slave and master are two antennas mounted on the satellite. If their distance b is known we can detect its orientation in space. Taking the projection of $\underline{b}$ on the direction $\underline{S}$ of the incoming GPS signal, we have the path difference of the signal received by the two antennas.

$$\underline{S}^T b$$

The position of the GPS satellite is known, so that direction $\underline{S}$ is known in geocentric coordinates if we know the position of the master antenna. Measuring the path difference

$$\Delta r = \underline{S}^T b$$

and transforming the vector $\underline{S}$ from geocentric to body frame:

$$\underline{S} = AS$$
$$\Delta r = S^T A^T b$$

the unknown is the rotation matrix A. Notice that most of the times instead of measuring Δr we measure a phase difference:

$$\Delta\varphi = \Delta r \, 2\pi/\lambda$$

The problem can be solved if we have at least three independent measurements, that requires the use of at least two different baselines (3 antennas) receiving signals from at least two GPS satellites. In this case the available measurements are:

$$\Delta r_{11} = S_1^T A^T b_1 \quad \text{baseline 1 GPS satellite 1}$$
$$\Delta r_{21} = S_2^T A^T b_1 \quad \text{baseline 1 GPS satellite2}$$
$$\Delta r_{12} = S_1^T A^T b_2 \quad \text{baseline 2 GPS satellite 1}$$
$$\Delta r_{22} = S_2^T A^T b_2 \quad \text{baseline 2 GPS satellite2}$$

Note that two baselines are required; having three GPS satellites in view of one single baseline would not allow to determine the rotation around the baseline. To determine the attitude, the following function can be minimized:

$$J = \sum_{i=1}^{N.S} \sum_{j=1}^{N.b} \left(\Delta r_{ij} - S_i^T A^T b_J \right)$$

Note that when considering phase differences, the integer number of wavelengths K has to be correctly determined:

$$\Delta\phi = (\Delta r + K\lambda) \frac{2\pi}{\lambda} + \nu$$

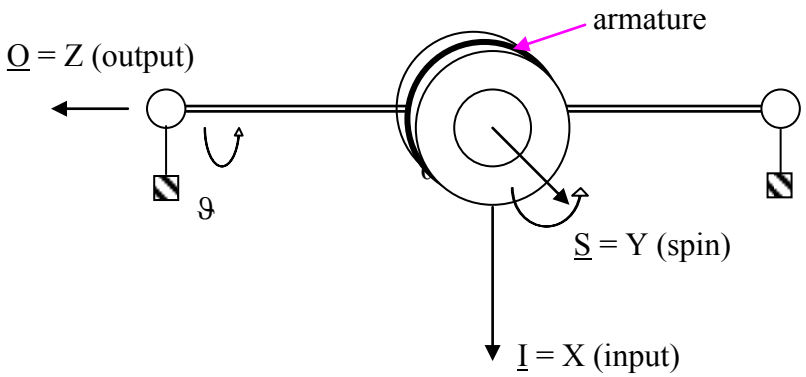
In addition, proper account of the measurement noise ν must be taken. This noise can have various sources, among which reflection of incoming signal is predominant.

Experiments performed to date have guaranteed a precision in the order of some fraction of degree.

Gyroscopes

Mechanical gyroscopes

Gyroscopes, or gyros, are used to measure angular velocities. The classical gyro is a mechanical sensor, with a spinning rotor and one degree of freedom in the rotor support mechanism.



The rotor spins around the S axis, while the support mechanism and the armature are free to rotate around axis O. The angular momentum in the XYZ body axes is then:

$$h_T = I_R \omega_R \underline{y} + J_z \dot{\vartheta} \underline{z}$$

I_R rotor

J_Z rotor + armature + support

We can write the Euler equation of the system, as:

$$\underline{\dot{h}} + \underline{\omega} \wedge \underline{h} = \underline{M}$$

where $\underline{\omega}$ is the composition of the angular velocity of the body on which the sensor is mounted, typically the satellite, and the relative angular velocity between gyro and satellite:

$$\underline{\omega} = \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z + \dot{\vartheta} \end{Bmatrix}$$

The rotor angular velocity ω_r does not appear since the rotor is totally free inside the armature.

The Euler equations are then written as:

$$\begin{cases} \omega_y J_z \dot{\vartheta} - I_R \omega_R \left(\omega_z + \dot{\vartheta}\right) = M_x \\ I_R \dot{\omega}_R - \omega_x J_z \dot{\vartheta} = M_y \\ J_z \ddot{\vartheta} + I_R \omega_R \omega_x = M_z \end{cases}$$

The third equation is used to evaluate the angular velocity ω_x while the first two equations can be used to determine the support reaction forces. If M_z is zero, from the third equation we evaluate ω_x by measuring the second derivative of the output angle ϑ. In general, to avoid having a constant acceleration at the output axis, a restrain torque is present. Assuming this has the general viscoelastic form, we have:

$$M_z = -k\vartheta - c\dot{\vartheta}$$

The third equation then becomes:

$$J_z\ddot{\vartheta} + c\dot{\vartheta} + k\vartheta = -I_R\omega_R\omega_x$$

The steady state solution is then:

$$\overline{\vartheta} = -\frac{I_R\omega_R\omega_x}{k} \qquad\qquad \omega_x = -\frac{k\overline{\vartheta}}{I_r\omega_r}$$

Measuring the steady state rotation of the output axis we measure the angular velocity component along the input axis. This instrument is known as RATE GYRO (RG). In a similar way, if we avoid using an elastic support and have only a viscous torque at the output axis, the steady state output will be a constant angular rate, which can still be used to measure the angular velocity along the input axis:

$$\overline{\dot{\vartheta}} = -\frac{I_R\omega_R\omega_x}{c} \qquad\qquad \omega_x = -\frac{c\overline{\dot{\vartheta}}}{I_r\omega_r}$$

This sensor is known as RATE INTEGRATING GYRO (RIG).
The transient terms in the output equation have an influence on the response time of the sensor, since the assumption is made that measures are taken at steady state.

Three gyros can be used to measure the three components of the angular velocity. It is pointed out that, since the measure implies a rotation of the output axis and therefore also of the input axis, it is almost impossible to measure directly the angular velocity in principal axes. To avoid this, and measure the angular velocity along a fixed reference frame, a control torque M_z can be applied at the sensor output axis such that the output angle ϑ is zero. In this condition, the angular velocity is measured by measuring the control torque M_z, and the measure is always corresponding to the same input axis.

$$M_z(\vartheta) = I_R\omega_R\omega_x$$
$$\omega_x = \frac{M_z(\vartheta)}{I_R\omega_R}$$

An alternative solution to keep the input axis I fixed is to rotate the whole support platform, which in this case becomes an inertial platform. Measuring the rotation angles of the platform with respect to the satellite we would directly retrieve the information of the inertial attitude of the satellite.

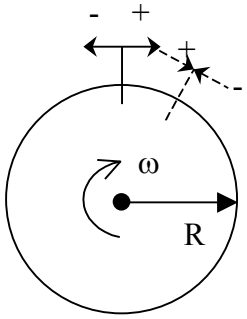
The precision of the gyro depends on the term $I_R\omega_R$; the larger it is, the smaller is the full scale of the sensor but the sensor would be more precise. Furthermore, the bandwidth of the sensor depends on J_z, since

$$\lambda = \sqrt{\frac{k}{J_z}}$$

To have a fast sensor we need a high bandwidth and so to reduce J_z and as a consequence also I_R. Also, ω_R can be increased, but this can be a problem above a certain limit due to friction and dissipation.
The mechanical gyro is suitable to measure relatively high angular velocities, for low velocities the nonlinearities and the internal friction make the measurement highly uncertain.

Laser gyroscopes

A conceptual scheme of a laser gyroscope can be reduced to an optical path in which a signal is injected by a transmitter. The same signal is then retrieved by a receiver at the end of the optical path. For simplicity, the description will assume a circular optical path.



The transmitter produces two signals (+ and -) in two opposite directions of the optical path. Since the gyroscope is rotating, the receiver, collocated with the transmitter, will receive the two signals at different time instants since the two signals will have travelled for different lengths.

$$ct^- = (2\pi - \omega t^-)R$$
$$ct^+ = (2\pi + \omega t^+)R$$

The instants in which the signals are received are then:

$$t^- = \frac{2\pi R}{c + \omega R}$$
$$t^+ = \frac{2\pi R}{c - \omega R}$$

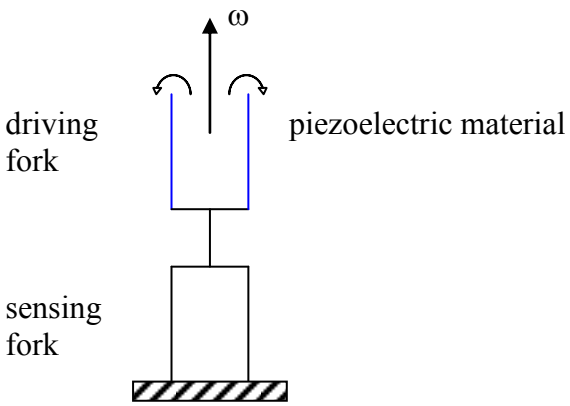
Measuring the time difference we compute the angular velocity:

$$\Delta t = (t^+ - t^-) = \frac{2\pi R(c+\omega R) - 2\pi R(c-\omega R)}{(c^2 - \omega^2 R^2)} = \frac{4\pi R^2 \omega}{(c^2 - \omega^2 R^2)} \cong \frac{4\pi R^2 \omega}{c^2} = \frac{4A\omega}{c^2}$$
$$\omega = \frac{\Delta t c^2}{4A}$$

This instrument is known as RING LASER GYRO (RLG). If the optical path is made of a fiber optic, it is possible to wind the fiber optic for more than one winding, improving the instrument sensitivity. This instrument is known as FIBER OPTIC GYRO (FOG). These gyroscopes measure the angular velocity in a reference frame fixed with the satellite, and are not subject to friction and motion problems.

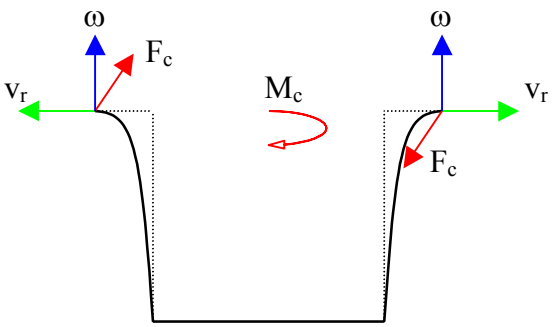
Piezoelectric gyroscopes

One further scheme of angular velocity sensor is based on the use of piezoelectric materials. Two forks made of piezoelectric materials are used. If subject to an electric field, the fork is deformed, while if deformed it produces an electric signal.



If the sensor rotates with velocity ω, and the two prongs are electrically driven in phase opposition, the Coriolis force, perpendicular to ω and the tip velocity, generates a torque. The torque is transmitted to the base prongs that, due to their deformation, generate a second electric signal that can be measured.

$$F_c = -2\underline{\omega} \wedge \underline{v_r}$$



Also this sensor measures the angular velocity along a fixed direction, and has no moving parts. However, as in part also true for laser gyros, it is extremely sensitive to temperature oscillations, which cause a thermal deformation that has an effect on the scale factor of the sensor.

Simple Gyro noise modelling

Solid-state gyroscopes output noisy measurements, that can be modelled as

$$\underline{\omega}_i^M = \underline{\omega}_i + \underline{n} + \underline{b}$$

Where $\underline{b}$ is the gyro bias

$$\underline{n} = \sigma_n \underline{\zeta}_n$$
$$\dot{\underline{b}} = \sigma_b \underline{\zeta}_b$$

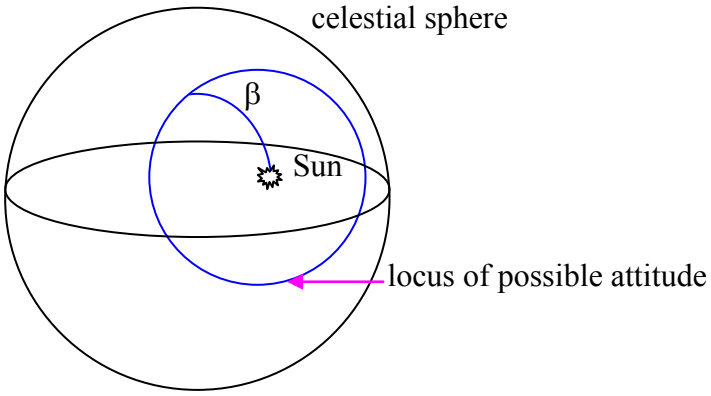
$\sigma_n \underline{\zeta}_n$ is white Gaussian noise with zero mean and standard deviation σ_n . This noise is known as angular random walk (ARW) attributed to thermo mechanical noise of the system and $\sigma_b \underline{\zeta}_b$ is white Gaussian noise with zero mean and standard deviation σ_b known as rate random walk. (RRW) attributed to electronic noise (flicker). These are the main contributions of noise in a gyro and manufacturers will often provide values for ARW and RRW.

Attitude determination

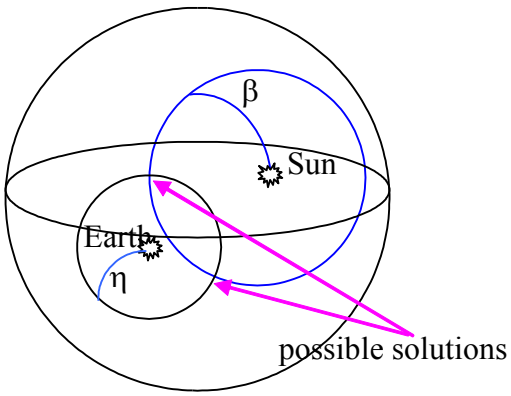
The sensors in general provide either a measure of one angle or of a direction in space, which must be converted into an attitude determination of one axis, for some simple spin satellites, or of three axes. Attitude determination methods are classified as geometric methods, algebraic methods or statistical methods.

Geometrical methods

Let suppose that one angle of the Sun or of Earth is known in the principal inertia reference, then the orientation in space of the sensor axis can be determined as a circle on the celestial sphere. This means the optical axis of the sensor has been placed on a geometrical locus.

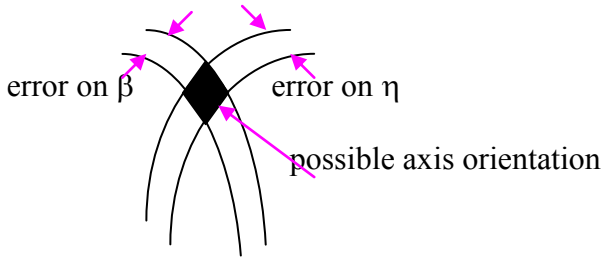


This because any rotation by the angle β starting from the great circle would rotate the optical axis of the sensor to match the direction of the Sun. If a second measurement of the same kind, referring to a different celestial body, is available, the intersection of the two cones provides the attitude of one axis of the satellite in inertial reference. Two solutions are obtained.

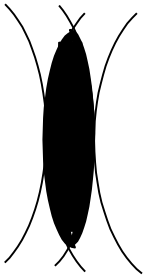


If the satellite is spinning, taking two sets of measurements with a given time separation it is expected that the orientation of the spin axis will not change. The real solution will then be the one that moves less in the inertial space, while the two circles will rotate around this point, since the relative geometry of the satellite with respect to Sun and Earth will change. The solution depends in the end on sensor geometry.

In the real situation, the intersection is not between two lines, but between two narrow strips with amplitude given by the measurement errors. The exact orientation of the spin axis will then be determined inside a small square surface.

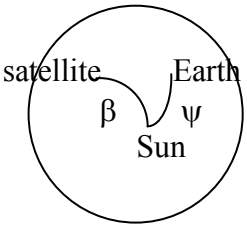


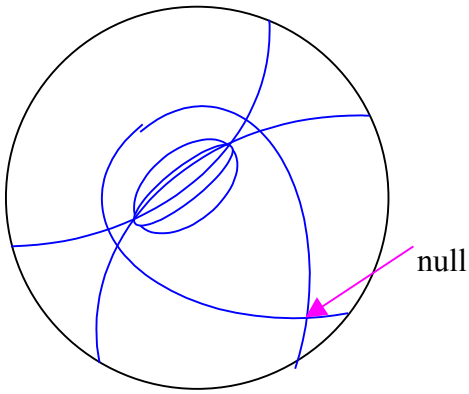
The error on the identified position of the spin axis is then higher than the error on a single measurement. If in addition the two strips do not intersect orthogonally, along one direction the error can be greatly amplified:



This situation depends on the relative position of the two detected celestial bodies, and the worst situation corresponds to the two bodies in the equatorial plane of the satellite. Notice that this event is totally independent from the sensor precision, and to minimize it the only solution is the appropriate selection of the celestial bodies to detect.

A second alternative is based on the measurement of one absolute angle and one relative angle between two celestial bodies. The shape of one locus will be still a circle on the celestial sphere, while the second locus is rather complex.





The actual shape of the second locus depends on ψ. The null point is a singular condition that appears when the angle ψ is equal to the angle β.

Algebraic methods

Knowledge of two angles (two scalar measurements) allows detecting the orientation of one axis in space. The complete knowledge of the attitude is yet undetermined, and requires at least one further scalar measurement to fix the phase of the reference frame around the detected axis.

With the exception of the GPS sensor, that can provide directly the attitude in the GPS reference system, or the fixed Earth sensor that provides the two small angles α_y and α_z, from which a state observer can reconstruct the complete attitude, the general case requires measures of two or more directions in space (celestial bodies) to correctly determine the attitude.

Calling s_i the unit vectors measured by the sensors, in body-fixed frame, v_i the unit vectors of the same celestial bodies but referred to an inertial frame or in any case a given reference frame, the following holds:

$$s_i = Av_i$$

The solution in terms of A requires at least two independent measurements, and the method of solution can differ for the cases with two measurements (s_1 and s_2), three measurements (s_1, s_2 and s_3) or more than three measurements. The simplest case refers to 3 measurements:

$$[s_1 \quad s_2 \quad s_3] = A[v_1 \quad v_2 \quad v_3]$$

If the measurements are independent, V is non-singular so that:

$$S = AV$$
$$A = SV^{-1}$$

V is invertible provided that v_i are independent, which means that the 3 measurements v_i refer to non-coinciding directions in space. This means that the selection of the target celestial bodies must be appropriate, and some degrees of freedom can also be exploited in order to improve the numerical accuracy of the solution. The best condition refers to 3 orthogonal measurements.

In case the number of measurements available is greater than 3:

$$[s_1 \quad s_2 \quad s_3 \quad s_4] = A[v_1 \quad v_2 \quad v_3 \quad v_4]$$

The system is redundant, so that if we want to make use of all available measurements the solution requires the evaluation of the pseudo inverse of matrix V:

$$SV^T = AVV^T$$
$$SV^T(VV^T)^{-1} = A$$
$$V^* = V^T(VV^T)^{-1}$$
$$A = SV^*$$

An alternative method would require the selection of the best three measurements and revert to the previous method; nevertheless using all available measurements can provide a better filtering of random errors.

In case only two measurements are available, the solution can be obtained with the following algorithm. Call the two measurements p and q, and call a and b their corresponding directions in inertial space. Associate the measurement p to unit vector s_1 and the corresponding direction a to unit vector v_1. Two orthogonal frames can then be associated to the measurements and to the reference directions as follows:

$$s_1 = p$$
$$s_2 = \frac{p \wedge q}{|p \wedge q|}$$
$$s_3 = p \wedge s_2$$

$$v_1 = a$$
$$v_2 = \frac{a \wedge b}{|a \wedge b|}$$
$$v_3 = a \wedge v_2$$

Since by construction the 3 unit vectors s_1, s_2, s_3 and v_1,v_2, v_3 are orthogonal, we can write:

$$V^{-1} = V^T$$
$$A = SV^{-1} = SV^T$$

To minimize errors, vector p should be measured with the maximum possible precision and q should be as orthogonal to p as possible. If p and q are almost aligned, as could be the case with one Earth sensor and one magnetometer in proximity of the poles, the attitude could not be determined.

Some filtering of the errors can be provided if we take:

$$p^* = \frac{p + q}{2}$$
$$q^* = \frac{p - q}{2}$$

And then start the attitude determination algorithm. In practice, this method detects first of all the plane containing the two measurements and then the perpendicular direction to the plane.

Statistical methods

Alternative methods to determine the attitude need at least two vector measurements and knowledge of the relative precision of the sensors, in order to build and minimize a suitable weighted error function:

$$J = \frac{1}{2}\sum_{i=1}^{N} \alpha_i |s_i - \mathrm{Av}_i|^2$$

where N is arbitrary and α_i is a constant identifying the sensor precision, more precise sensors are associated to the higher α_i parameters. In the following, without loss of generality, it will be assumed that the vector of weights α is normalized to 1

$$\sum_{i=1}^{N} \alpha_i = 1$$

The real solution is the one minimizing J. In this approach, we take for granted the impossibility to solve exactly the problem when measurement errors are considered, so we try to find the optimal solution that minimizes the cumulative sum of errors. A minimum of two independent measurements are needed.

$$J = \frac{1}{2}\sum_{i=1}^{N} \alpha_i (s_i^T s_i + v_i^T A^T \mathrm{Av}_i - 2\mathrm{s}_i^T \mathrm{Av}_i)$$

Since A is a rotation matrix, A^TA is identity matrix, and since s_i and v_i are unit vectors the first two terms are equal to 1, so that:

$$J = 1 - \sum_{i=1}^{N} \alpha_i (s_i^T \mathrm{Av}_i)$$

The optimal solution minimizes J(A), and therefore maximizes

$$\tilde{J} = \sum_{i=1}^{N} \alpha_i (s_i^T \mathrm{Av}_i) = \mathrm{tr}(\mathrm{AB}^T)$$

where tr indicates the trace operator and B matrix is

$$B = \sum_{i=1}^{N} \alpha_i s_i v_i^T$$

A direct solution in terms of direction cosine matrix is computationally expensive, and is most often searched adopting singular value decomposition techniques. If the B matrix is non-singular, which means having at least 3 observed vectors, then

$$A = (B^T)^{-1}(B^T B)^{1/2} = B(B^T B)^{-1/2}$$

However, after the construction of B, this requires the tricky computation of $(B^TB)^{-1/2}$. In fact, the square root of a matrix can have multiple solutions. In a general case, matrix B can be decomposed using singular value decomposition as

$$B = U\Sigma^T V^T = \mathrm{U}\ \mathrm{diag}[\Sigma_{11} \quad \Sigma_{22} \quad \Sigma_{33}]^T V^T$$

where U and V are unitary orthogonal matrices, representing the eigenvectors of matrices BB^T and B^TB respectively, and UU^T=I and VV^T=I. The singular values Σ are the square roots of the eigenvalues of matrix B^TB or of matrix BB^T. Then

$$\Sigma = diag\,[1 \quad 1 \quad (det\ U)(det\ V)]$$
$$A = \mathrm{U}\ \mathrm{diag}\,[1 \quad 1 \quad (\det \mathrm{U})(\det \mathrm{V})]V^T$$

A simpler solution can be sought if we replace A by its representation in terms of quaternions

$$A = \left(q_4^2 - \underline{q}^T\underline{q}\right) I + 2\underline{q}\underline{q}^T - 2\mathrm{q}_4[q\ \wedge]$$

In this case the cost function to maximize becomes

$$\tilde{J} = q^T \mathrm{Kq}$$

where the matrix K, square of order 4, is assembled as

$B = \sum_{i=1}^{N} \alpha_i s_i v_i^T = \left[b_{\mathrm{ij}}\right]$	square matrix of order 3
$S = B^T + B$	square matrix of order 3
$z = [b_{23} - b_{32} \ , \ \mathrm{b}_{31} - b_{13} \ , \ \mathrm{b}_{12} - b_{21}]^T$	vector of order 3
$\sigma = \mathrm{tr}(B)$	scalar
$K = \begin{bmatrix} S - \sigma\mathrm{I} & z \\ z^T & \sigma \end{bmatrix}$	square matrix of order 4

To maximize function $\tilde{J}$, the constraint $q^Tq = 1$ must be added with the associated Lagrange multiplier λ, and then the function $\tilde{G} = q^T\mathrm{Kq}\text{-}\lambda(q^T\mathrm{q}\text{-}1)$ is maximized with no constraints. Evaluating the gradient of $\tilde{G}$ and equating the gradient to zero we have

$$\mathrm{Kq} = \lambda\mathrm{q}$$

so that, back substituting into the expression of $\tilde{J}$ we have

$$\tilde{J}(q) = q^T\mathrm{Kq} = q^T\lambda\mathrm{q} = \lambda$$

The maximum value of the cost function is then associated with the maximum eigenvalue of the K matrix. As a consequence the optimal attitude, from a statistical point of view, is the eigenvector associated to the maximum eigenvalue of matrix K.
In a normal operating condition, with all sensors operational, it should be expected that the errors in attitude determination are rather small, which means that we expect the cost function J to be close to zero. In this condition, function $\tilde{J}$ will be close to one, so that we expect that the maximum eigenvalue of K is close to 1. With this assumption, we can avoid the evaluation of eigenvalues and eigenvectors of matrix K, and try to solve an approximate problem that leads to the evaluation of the Gibbs vector

giving the satellite attitude. To do so, rewrite the equation $Kq=\lambda q$ in partitioned form, considering the form of matrix K. We have the following two equations

$$(S - \sigma I)\underline{q} + z q_4 = \lambda_{MAX}\underline{q}$$
$$z^T \underline{q} + \sigma q_4 = \lambda_{MAX} q_4$$

Dividing the first equation by q_4 and taking the approximation $\lambda_{MAX}=1$, we have

$$(S - \sigma I - I)\underline{g} + z = 0$$

that represents a linear system with unknown vector $\underline{g}$ and coefficient matrix $(S-\sigma I-I)$. Solving this linear system we evaluate the approximate Gibbs vector that provides the satellite attitude.

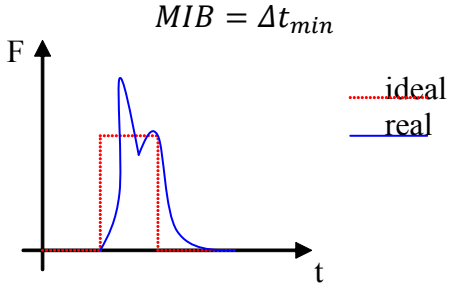
Actuators for spacecraft attitude control

Thrusters for attitude control

The simplest way to create torques is to create a set of forces with direction not aligned with the center of mass, and this can be obtained by mass expulsion techniques. Jet thrusters pose some operational problems due to the ignition transient, and besides it is not simple to finely control the magnitude of the force, so these devices are not used for fine attitude control. In addition, for control quite often the force needed is rather small (milli Newton-meters), while chemical thrusters produce forces in the order of at least some Newton. To make them compatible with attitude control, they are switched on and off with a given modulation, but this enhances the problems due to ignition transients and can cause mechanical ware of the thruster.

These problems can be solved by adopting electric propulsion thrusters, based on electrodynamic acceleration of a suitable ionized propellant, that need de-ionization immediately after expulsion to avoid charging electrically the spacecraft. These thrusters can be easily modulated in amplitude, have a high specific impulse (over 3000) that allows a reduced propellant consumption. The thrust produced can be in the order of a few Newton down to 10^{-6} Newton, so they are well suited to fine control actions. Unfortunately, electric thrusters are extremely power consuming, more or less 90% is devoted simply to keep it ready to use and only 10% is due to the thrust produced, therefore electric propulsion units are often coupled to extremely large solar panels.

With conventional (chemical) thrusters it is not possible to control the amplitude of the thrust; they are either switched on or off. The transient delay and the presence of hydraulic circuits make the actual thrust profile quite different from the ideal one, requiring a careful calibration for proper command selection.
An important characteristic of the thruster for attitude control is the minimum impulse bit (MIB) which is a measure of the accuracy and related to the precision it can point.



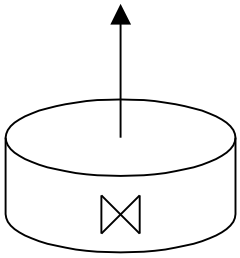
Cold-gas thrusters use a non-reactive gas, stored at high pressure (around 30 MPa), commonly use Nitrogen~SP 70s or Helium~SP 175s. Helium saves mass but is more prone to leakage and more expensive.

Monopropellant uses a propellant that is decomposed catalytically. Hydrazine~SP 242s is by far the most common. They do not require high pressure but are highly toxic.

Electric Thrusters use electric fields to accelerate the expelled ionized mass at high velocity. High SP 500-3000s.

Use of thruster on spinning satellites

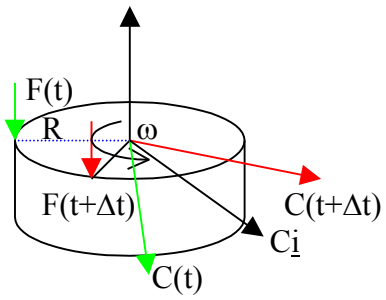
In case of spinning satellites, to control the spin velocity the thrusters must be located on the side of the satellite and the thrust direction must be orthogonal to the angular velocity:



$I\dot{\omega} = M$

Control of the direction of the spin is more critical. It requires thrust in the same direction of the angular velocity, but shifted with respect to the satellite centerline. In this case the ignition delay does not affect the magnitude of the angular velocity (or, the angular momentum), but affects the direction of the velocity. In fact, due to satellite spin, a delay in the application of the control force implies that in inertial space the force is applied with a different line of action, so the torque applied has a different direction in inertial space and hence the different final direction of $\underline{\omega}$ and $\underline{h}$.

$$\Delta\phi = \omega\Delta t$$
$$dM = FR\cos\phi\, d\phi \underline{i}$$



If the force is constant, and applied for a rotation angle equal to 2φ, also the torque will be constant, orthogonal to the average force direction and directed along the centerline of the rotation, characterized by the angleφ:

$$M_{TOT} = 2\int_0^{\phi} FR\cos\phi\, d\phi \underline{i} = 2FR\sin\phi\, \underline{i}$$

In general, to know the direction of the resulting torque the exact thrust profile must be known, in order to evaluate the average direction of the thrust.
If the thruster is switched on for one full spin rotation of the satellite, thus for φ equal to π, M_{TOT} will be zero. The net effect is null even if some propellant has been expelled. The maximum effect is obtained for φ equal to π/2, so that the thruster should be switched on and off every half spin rotation. This is however not efficient, since at the beginning and end of the thrust the effects would tend to be opposite. The efficiency of the process can be evaluated by taking the ratio of the impulse of the torque applied divided by the impulse of the force.
Since:

$$dt = \frac{d\phi}{\omega}$$

and since the maximum torque is equal to FR, the torque impulse is:

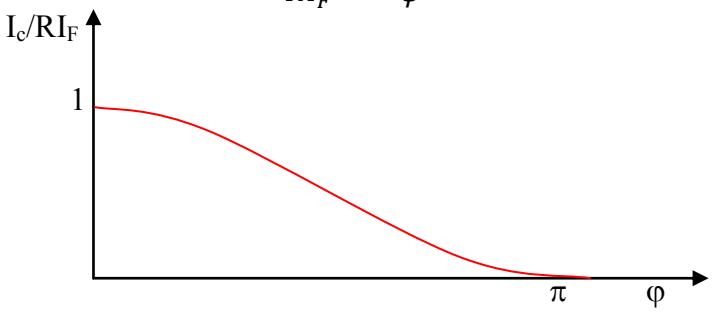
$$I_c = 2\int_0^t FR\cos\phi\, dt = 2\int_0^\phi FR\cos\phi\frac{d\phi}{\omega} = \frac{2FR}{\omega}\sin\phi$$

while the force impulse is:

$$I_F = 2\int_0^t F dt \Rightarrow$$
$$RI_F = 2R\int_0^t F dt = 2R\int_0^\phi F\frac{d\phi}{\omega} = \frac{2FR}{\omega}\phi$$

The ratio of the two gives an indication of the efficiency of the process, as:

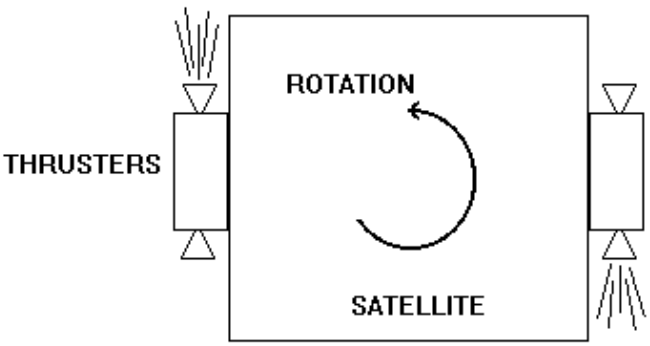
$$\frac{I_c}{RI_F} = \frac{\sin\phi}{\phi}$$



This tells us how much torque we are obtaining along the average direction compared to the total torque applied. If φ tends to zero the efficiency tends to 1. It is obvious that the torque obtained is low, but with a repeating sequence of short pulses, one per rotation, the total torque obtained can be high.

Attitude Thruster configurations

In the simplest configuration, thrusters operate in couples as displayed below:



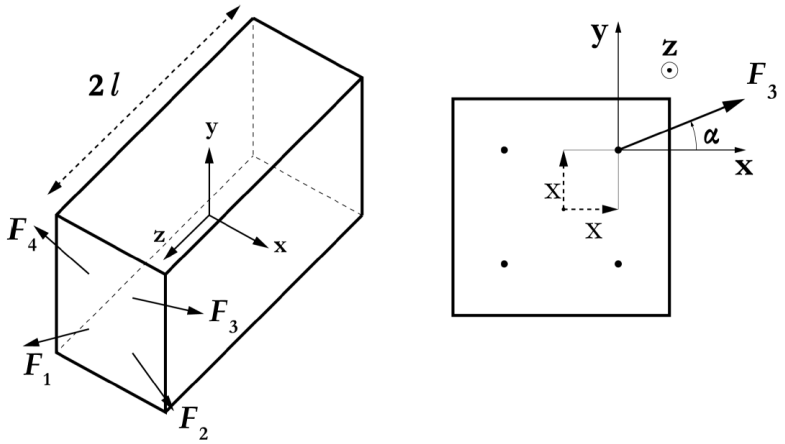
Then the torque and force are then

$$\underline{T} = \underline{r}_1 \wedge \underline{F}_m + (\underline{r}_2 \wedge -\underline{F}_m) = (\underline{r}_1 - \underline{r}_2) \wedge \underline{F}_m$$
$$\underline{F} = \underline{0}$$

In a more general configuration, with an arbitrary number of thrusters, the total torque generated becomes

$$\underline{T}_{tot} = [\hat{R}]\underline{F}$$

where matrix $\hat{R}$ maps the force generated by the set of thrusters into the torque with respect to the body axes. The example below shows the map of the forces generated by 4 thrusters.



$$[\hat{R}] = \begin{bmatrix} l\sin\alpha & 0 & 0 \\ 0 & l\cos\alpha & 0 \\ 0 & 0 & x\sin\alpha - x\cos\alpha \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

At first glance the problem of solving for $\underline{F}$ seems straightforward since we have seemingly to invert the equation $\underline{u}_{ideal} = [\hat{R}]\underline{F}$. In general, $[\hat{R}]$ is not a square matrix so we can think to apply the Moore-Penrose pseudo inverse. Then we can write $\underline{F} = [\hat{R}]^* \underline{u}_{ideal}$. A general pseudo inverse can be defined that also satisfies this constraint:

$$\underline{F} = [\hat{R}]^* \underline{u}_{ideal} + \gamma w$$

where γ is a scalar defined by

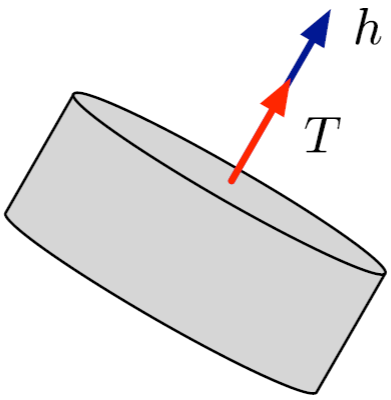
$$\gamma = max_{i=1,..,N}([\hat{R}]^* \underline{u}_{ideal})_i / w_i$$

where $w > 0$ is the null space vector of $\hat{R}$. Therefore, we can apply this method if the null space $w > 0$ of $\hat{R}$ exists. To compute the nullspace *w* we need to solve $\hat{R}w=0$. For the example given above, with 4 thrusters, $w = [1 \quad 1 \quad 1 \quad 1]^T$. Thus, the generalized pseudo inverse can be applied to this configuration.

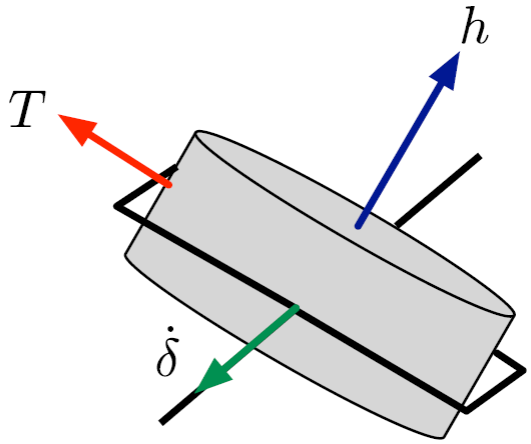
Inertia and reaction wheels

These actuators are based on acceleration and deceleration of spinning rotors. Normally, when the nominal condition is with zero angular velocity the actuator is called Reaction Wheel (RW); those that have a non-zero nominal velocity are called Inertia Wheels (IW). In some cases the distinction between the two categories is not evident, and all actuators can be classified as RW. A further class of actuators is based on wheels that spin at a constant rate but can be tilted around an axis orthogonal to the spin axis, and their principle of operation is that of gyroscopes. These actuators are called

Control Moment Gyroscopes (CMG). All these actuators can be considered as one family of actuators; all can only exchange angular momentum with the satellite but not produce a net torque.



RW and IW torque diagram

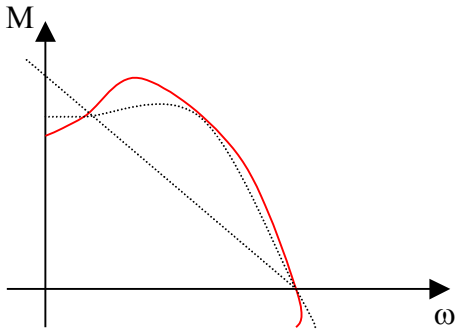


GMG torque diagram

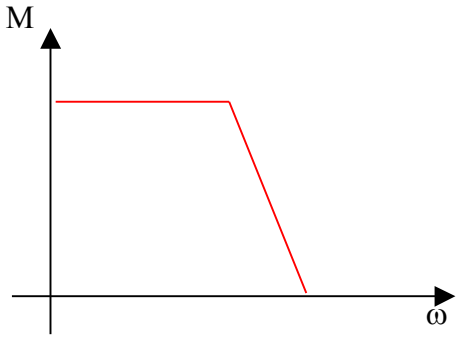
Looking at the equations for a dual spin satellite with no external torques applied, in fact, we can write:

$$I_r \dot{\omega}_r = M_r$$
$$I_z \dot{\omega}_z = -I_r \dot{\omega}_r$$

The torque M_r is in general provided by an electric motor that has its typical operational curve. One possible curve is the following:



Despite the variability of electric motor characteristic, it is commonly accepted that there is a velocity for which the torque falls to zero, this is called saturation speed. When the wheel reaches this speed it can provide no additional torque, either positive or negative, and requires a special desaturation mechanism to make it again operational. An approximation of the characteristic curve can be the following:



In general, actuators are operated in order to keep them within the speed range for which the torque is constant, so that the actuator can be somehow modeled as a linear device if commanded by a current in the armature.
If the torque required by the control is more or less periodic, then appropriate sizing of the actuator can be made in order to have it always within the linear functioning regime. If the torque required is the combination of a periodic component and a secular component, it will be inevitable to reach the saturation speed. Desaturation mechanisms must apply a net torque to the satellite; therefore they can be based on thrusters or on magnetic torquers. In case the secular component of disturbances is persistent, several desaturation maneuvers will be required.
Considering that desaturation maneuvers are studied apart with specific models, the general problem of control with RW, IW or CMG can be modeled by writing the appropriate set of Euler equations, including the effects of the n actuators in the expression of the angular momentum

$$\underline{h} = I\underline{\omega} + A\underline{h}_r$$

Matrix A is composed by 3 rows and as many columns as number of actuators. Each column represents the direction of the axis of rotation of the wheel. Vector $\underline{h}_r$ has as many elements as actuators, and represents the angular momentum of each actuator around its spin axis.
Assuming as usual that the term $I\underline{\omega}$ considers the presence of the rotors with no relative velocity, and that $\underline{T}$ represents the disturbance torque, the dynamics becomes:

$$I\dot{\underline{\omega}} + \underline{\omega} \wedge I\underline{\omega} + \dot{I}\underline{\omega} + \dot{A}\underline{h}_r + A\dot{\underline{h}}_r + \underline{\omega} \wedge A\underline{h}_r = \underline{T}$$

Additional equations for the relative dynamics of the rotors must be added to the system:

$$\dot{A}\underline{h}_r + A\dot{\underline{h}}_r = \underline{M}_r$$

We can now analyze the individual terms of the equation:

$\dot{I}\underline{\omega}$	exists only for CMG, and can be neglected if these are small compared to the satellite
$\dot{A}\underline{h}_r$	exists only for CMG, and indicates the relative attitude rotation in the spacecraft body frame
$A\dot{\underline{h}}_r$	in general it exists for RW and IW, not for CMG since these have often constant spin velocity
$\underline{\omega} \wedge A\underline{h}_r$	exists for any actuator, could be neglected for RW since they are characterized by nominal $\underline{h}_r = 0$

To simplify the solution of the control problem, the four terms can be grouped together in one term. Upon the determination of the control torque required, a separate equation can be solved including the actuator terms to infer the effective control command for the actuator. We can consider one example in which we have only RW.

$$\begin{cases} I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega} + \underline{\omega} \wedge A\underline{h}_r + A\underline{\dot{h}}_r = \underline{T} \\ M_c = -\underline{\omega} \wedge A\underline{h}_r - A\underline{\dot{h}}_r \\ I_r\underline{\dot{\omega}}_r = \underline{M}_r \end{cases}$$

We can write the equation for the evaluation of the control law as:

$$I\underline{\dot{\omega}} + \underline{\omega} \wedge I\underline{\omega} = \underline{T} + \underline{M}_c$$

Any linear control design technique can be used to calculate $\underline{M}_c$ after which $\underline{M}_r$ can be evaluated, i.e., $\underline{\dot{h}}_r = I_r\underline{\dot{\omega}}_r$ with the following procedure:

$$A\underline{\dot{h}}_r = -\underline{M}_c - \underline{\omega} \wedge A\underline{h}_r$$
$$\underline{\dot{h}}_r = -A^* \left(\underline{M}_c + \underline{\omega} \wedge A\underline{h}_r\right)$$

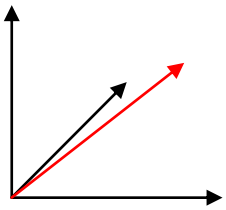
where A^* is the pseudo inverse matrix of A, in the general case, and becomes the inverse only if 3 actuators are used. Notice that the pseudo inverse can be calculated as

$$A_{3,n}x_{n,1} = y_{3,1}$$
$$A_{n,3}^T A_{3,n}x_{n,1} = A_{n,3}^T y_{3,1}$$
$$\left(A_{n,3}^T A_{3,n}\right)_{n,n} x_{n,1} = A_{n,3}^T y_{3,1}$$

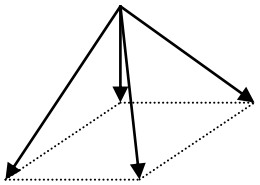
Assuming $\left(A_{n,3}^T A_{3,n}\right)_{n,n}$ is invertible:

$$\left(A_{n,3}^T A_{3,n}\right)_{n,n}^{-1} \left(A_{n,3}^T A_{3,n}\right)_{n,n} x_{n,1} = \left(A_{n,3}^T A_{3,n}\right)_{n,n}^{-1} A_{n,3}^T y_{3,1}$$
$$x_{n,1} = \left(A_{n,3}^T A_{3,n}\right)_{n,n}^{-1} A_{n,3}^T y_{3,1}$$
$$x_{n,1} = A_{n,3}^* y_{3,1}$$

In general 3 rotors are enough to solve exactly the control problem, in practice a fourth rotor is added for redundancy with a geometry that allows a full control capacity even with one rotor failed. This means that the 4 rotors have distinct rotation axes, and any combination of 3 rotors would allow the invertibility of the rotor A matrix.
The two most popular configurations are with 3 rotors aligned with the principal axes and the fourth with equal components along the three axes, or a full pyramid configuration with no actuator aligned with any of the principal axes. In this second configuration no direction is privileged, and the pyramid parameters can be tuned to have different control capacity along the three axes, this can be useful if the disturbances are not uniform along the principal axes.



Three axes and diagonal



pyramid

Analyze now the configuration with 3 rotors aligned with the principal axes and the fourth with equal components along the three axes:

$$A = \begin{bmatrix} 1 & 0 & 0 & 1/\sqrt{3} \\ 0 & 1 & 0 & 1/\sqrt{3} \\ 0 & 0 & 1 & 1/\sqrt{3} \end{bmatrix}$$

We can extract any of the actuators, and the remaining three will guarantee the invertibility of A. The pseudo-inverse is:

$$A^* = \begin{bmatrix} 5/6 & -1/6 & -1/6 \\ -1/6 & 5/6 & -1/6 \\ -1/6 & -1/6 & 5/6 \\ 1/2\sqrt{3} & 1/2\sqrt{3} & 1/2\sqrt{3} \end{bmatrix}$$

And the system is solved as:

$$\dot{\underline{h}}_r = \underline{M}_r = -A^* \left(\underline{M}_c + \underline{\omega} \wedge A\underline{h}_r \right) = A^* \underline{M}_e$$

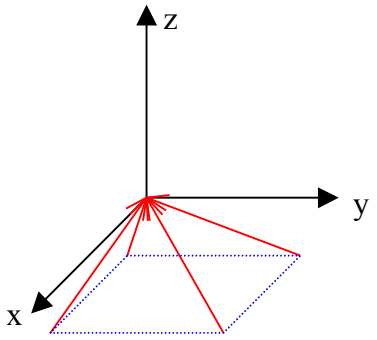
Assume now that the control torque must be aligned with the x-axis, we see that multiplying A^* with the required torque $\{1,0,0\}^T$ we have a control command for each actuator in the system, not simply the actuator aligned with x. In this case, the norm of the wheel control vector $\underline{M}_r$ is

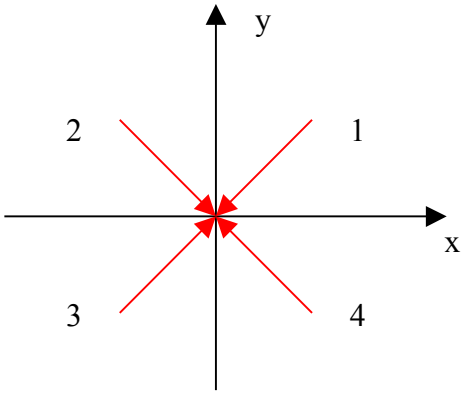
$$\underline{M}_c = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$\underline{M}_r = \begin{Bmatrix} 5/6 \\ -1/6 \\ -1/6 \\ 1/2\sqrt{3} \end{Bmatrix}$$

$$\left\| \underline{M}_r \right\| = \sqrt{\frac{27}{36} + \frac{1}{12}} = \sqrt{\frac{30}{36}}$$

Using the pyramid configuration, with all actuators with equal components along the principal axes we have instead:





$$A = \begin{bmatrix} -a & a & a & -a \\ -a & -a & a & a \\ a & a & a & a \end{bmatrix}$$

$$a = \frac{1}{\sqrt{3}}$$

The pseudo-inverse is:

$$A^* = \begin{bmatrix} -b & -b & b \\ b & -b & b \\ b & b & b \\ -b & b & b \end{bmatrix}$$

$$b = \frac{\sqrt{3}}{4}$$

Assume now that the control torque must be aligned with the x-axis, we see that multiplying A^* with the required torque $\{1,0,0\}^T$ we have the same control command on all actuators, with different signs. The norm of the wheel control vector $\underline{M}_r$ is:

$$\|\underline{M}_r\| = \sqrt{\frac{3}{4}}$$

The norm is lower than the previous case, so this configuration is preferred. In case one rotor fails, as example rotor number 4, we still have control capability since:

$$A_{1,2,3}^{-1} = \begin{bmatrix} -\frac{\sqrt{3}}{2} & 0 & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

Any torque component is provided by activating two rotors with equal acceleration.
As a general rule, adoption of IW can be preferred to RW since there are minor friction problems. In addition, a suitable configuration of IW can provide a non-zero nominal angular momentum, useful for passive attitude stability. In this case, it must be remembered that the overall rotor angular momentum must be aligned with the satellite angular momentum, so for some geometries of the actuator set the nominal angular velocities of the actuators might not be the same.

Considering the pyramid configuration seen above, if all rotors have the same velocity then the only nonzero component of $\underline{h}$ is along axis z, and we can control the satellite exactly as if we had one IW along axis z and 2 RW along axes x and y. Accelerating rotors 3 and 4 and slowing rotors 1 and 2 the net effect is equivalent to a RW along axis y, slowing rotors 2 and 3 and accelerating rotors 1 and 4 the net effect is equivalent to a RW along axis x. Accelerating all rotors together the effect is equivalent to IW along axis z.
Notice that, due to dynamic coupling, it is not always required to have one actuator for each axis, since 2 actuators can guarantee controllability of the system and a third actuator can be used for redundancy.

Note that RW will have a maximum momentum storage or in other words a saturation limit on their angular velocity. When the RW reaches the saturation limit no additional exchange of momentum may occur in the same direction and therefore the RW must be de-saturated. For example, for a RW with dynamics $\dot{\underline{h}}_{rw} = \underline{T}_{rw}$ where $\underline{T}_{rw}$ is the torque applied to the RW we can use a simple feedback control to de-celerate the wheel $\underline{T}_{rw} = -\underline{T}sign\left(\underline{h}_r\right)$. However, de-saturating the wheel will result in a generally undesired momentum exchange with the spacecraft e.g a de-tumbled spacecraft will begin to move when the RW is de-saturated. Thus, during the de-saturation of the RW a torque must be applied to the spacecraft using another actuator such as a magnetic torquer or thruster to maintain the spacecraft orientation, for example $\underline{T}_{sc} = -\underline{T}sign\left(\underline{h}_r\right)$. When the RWs velocities fall within a prescribed tolerance the applied torque to them can be switched off while $\underline{T}_{sc} = -\underline{T}sign\left(\underline{h}_r\right)$is applied to de-tumble the spacecraft due to angular velocities induced by the mismatch of the applied reaction wheel torques and the other actuator used.

Control moment gyroscopes

Control moment gyroscopes (CMG) generate a torque by gyroscopic effect. A rotating disc is tilted around one axis orthogonal to the spin axis of the disc, obtaining a torque long the third axis of an orthogonal frame. The angular momentum is then exchanged with the satellite not by a variation in the spin of the actuator, but by changing its relative orientation within the satellite. This effect allows on one side to obtain torques in general higher than those obtained with RW, but on the other side the evolution of the actuator configuration can create some singularity problem for specific configurations.
Following the same steps as shown for the RW, we can formulate the general control problem with CMG actuators as

$$I\dot{\underline{\omega}} + \underline{\omega} \wedge I\underline{\omega} + \dot{A}\underline{h}_r + \underline{\omega} \wedge A\underline{h}_r = \underline{T}$$

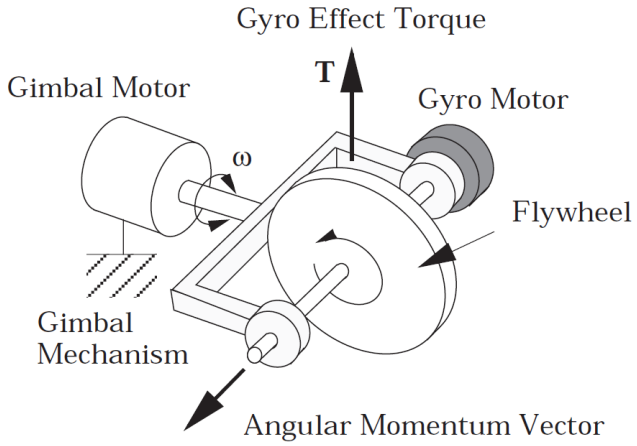
$$\begin{cases} I\dot{\underline{\omega}} + \underline{\omega} \wedge I\underline{\omega} + \dot{A}\underline{h}_r + \underline{\omega} \wedge A\underline{h}_r = \underline{T} \\ M_c = -\underline{\omega} \wedge A\underline{h}_r - \dot{A}\underline{h}_r \end{cases}$$

$$I\dot{\underline{\omega}} + \underline{\omega} \wedge I\underline{\omega} = \underline{T} + \underline{M}_c$$

$$\dot{A}\underline{h}_r = -\underline{M}_c - \underline{\omega} \wedge A\underline{h}_r$$

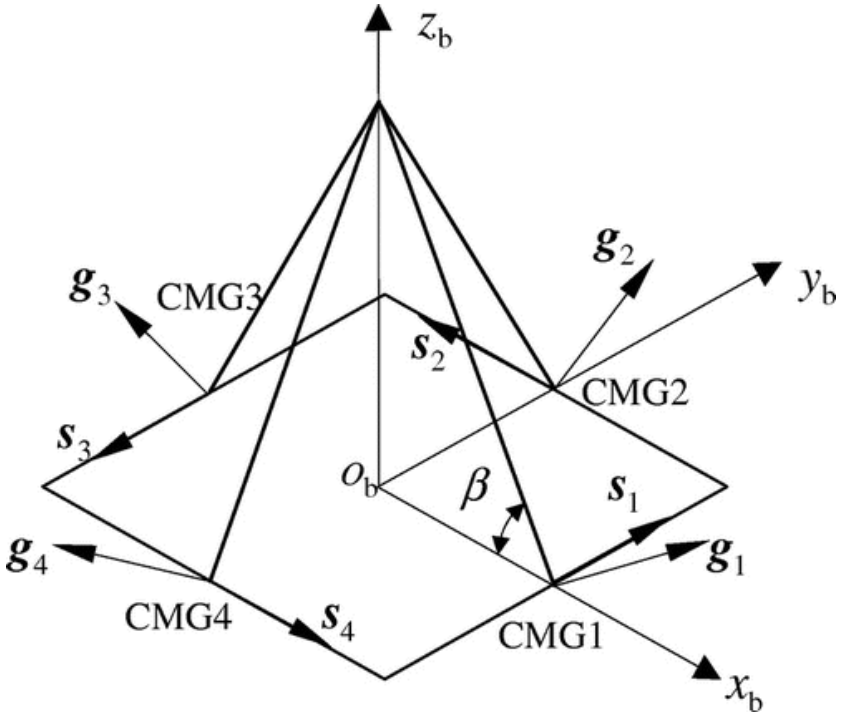
The problem is solved when the matrix $\dot{A}$ is evaluated. With the aid of one example we will show that the evaluation of $\dot{A}$ can be reformulated as the evaluation of a vector of tilt angle rates, which is easier to solve.

Assuming to have only 1 single gimbal GMG, the command becomes $\dot{\delta} = \omega$



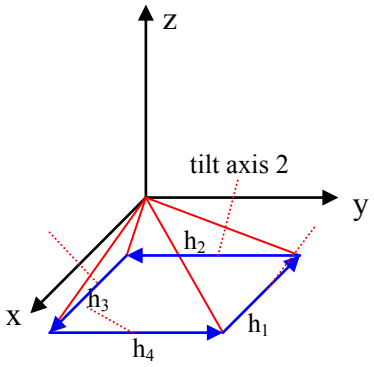
$$h_{\mathrm{CMG}} = A\underline{h}_r = h_r \begin{bmatrix} -\sin\delta \\ \cos\delta \\ 0 \end{bmatrix} \quad \dot{h}_{\mathrm{CMG}} = h_r \begin{bmatrix} -\cos\delta \\ -\sin\delta \\ 0 \end{bmatrix} \dot{\delta}$$

If the gimbal motor is also inclined by a constant angle as in the figure below



$$h_{\mathrm{CMG}} = A\underline{h}_r = h_r \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \begin{bmatrix} -\sin\delta \\ \cos\delta \\ 0 \end{bmatrix} = h_r \begin{bmatrix} -\cos\beta\sin\delta \\ \cos\delta \\ \sin\beta\sin\delta \end{bmatrix}$$

Assume a configuration with 4 CMG, whose nominal orientation makes the nominal angular momentum zero and whose tilt directions are along the 4 sides of a pyramid, as in figure



Calling δ the tilt angle of the 4 CMG and β the inclination of the pyramid, assuming all CMG have the same spin speed and therefore the same angular momentum h_r we can write

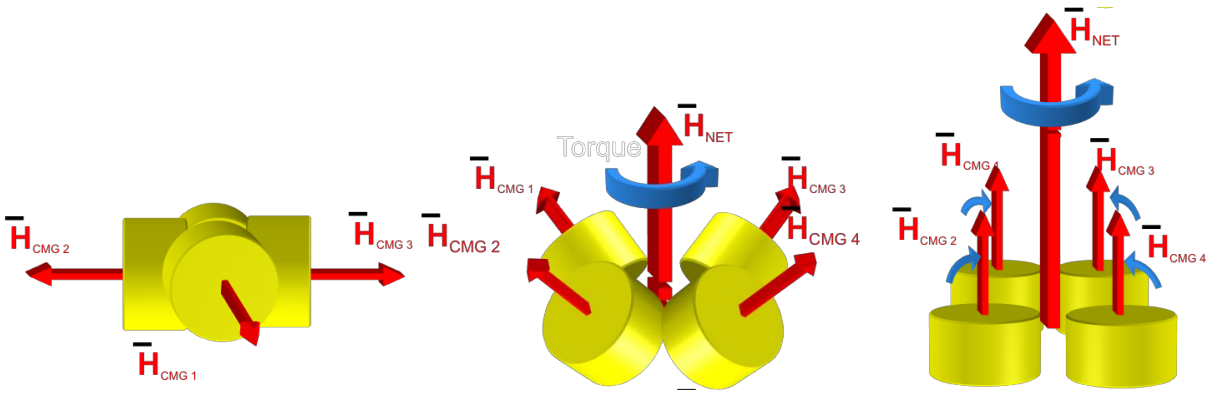
$$h_{\text{CMG}} = A\underline{h}_r = \begin{bmatrix} sin\,\beta\,sin\,\delta_1 & sin\,\beta\,sin\,\delta_2 & sin\,\beta\,sin\,\delta_3 & sin\,\beta\,sin\,\delta_4 \\ -cos\,\delta_1 & cos\,\beta\,sin\,\delta_2 & cos\,\delta_3 & -cos\,\beta\,sin\,\delta_4 \\ -cos\,\beta\,sin\,\delta_1 & -cos\,\delta_2 & cos\,\beta\,sin\,\delta_3 & cos\,\delta_4 \end{bmatrix} \begin{Bmatrix} h_r \\ h_r \\ h_r \\ h_r \end{Bmatrix}$$

$$\dot{h}_{\text{CMG}} = \dot{A}\underline{h}_r = \text{B}\dot{\delta} = h_r \begin{bmatrix} sin\,\beta\,cos\,\delta_1 & sin\,\beta\,cos\,\delta_2 & sin\,\beta\,cos\,\delta_3 & sin\,\beta\,cos\,\delta_4 \\ sin\,\delta_1 & cos\,\beta\,cos\,\delta_2 & -sin\,\delta_3 & -cos\,\beta\,cos\,\delta_4 \\ -cos\,\beta\,cos\,\delta_1 & sin\,\delta_2 & cos\,\beta\,cos\,\delta_3 & sin\,\delta_4 \end{bmatrix} \begin{Bmatrix} \dot{\delta}_1 \\ \dot{\delta}_2 \\ \dot{\delta}_3 \\ \dot{\delta}_4 \end{Bmatrix}$$

The control equation can then be reformulated as

$$\dot{A}\underline{h}_r = -\underline{M}_c - \underline{\omega} \wedge A\underline{h}_r = B\underline{\dot{\delta}} \quad \rightarrow \quad \underline{\dot{\delta}} = -B^* \left[\underline{M}_c + \underline{\omega} \wedge A\underline{h}_r\right]$$

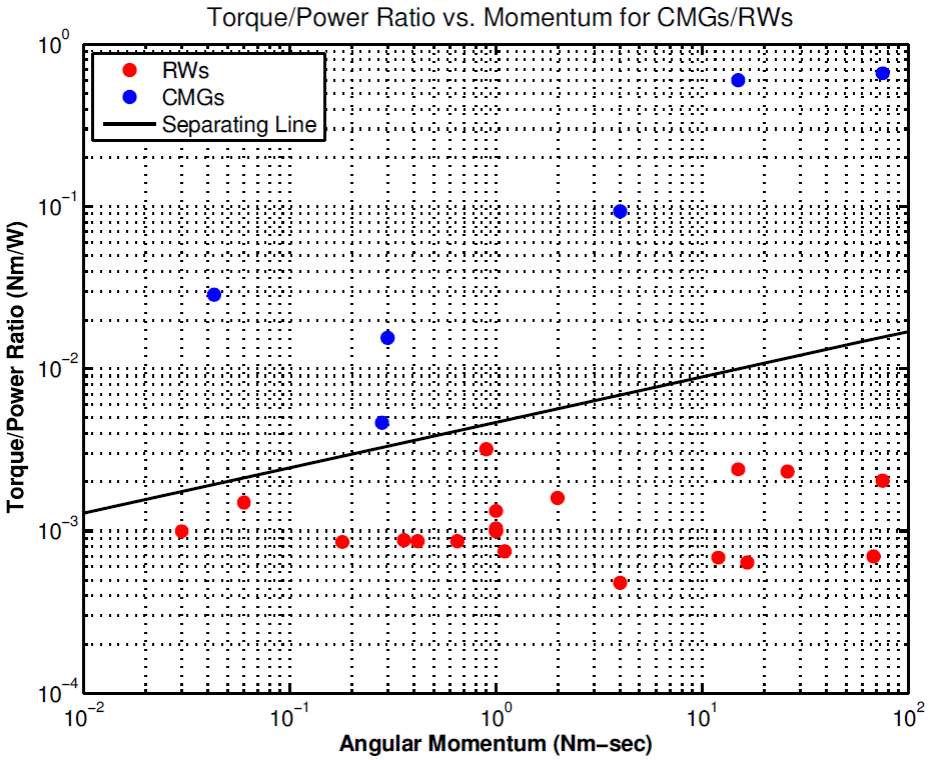
Note that CMGs are subject to saturation, as RWs, if $\underline{\dot{\delta}}$ exceeds the maximum capacity of the drive motor. In addition, CMGs can assume configurations that become singular, which means that the overall control torque can not be arbitrary. This singularity arises from tha fact that the configuration of the set of CMGs changes with time and could bring some of the CMGs to have the same direction of the applicable torque, as shown in the following set of figures.

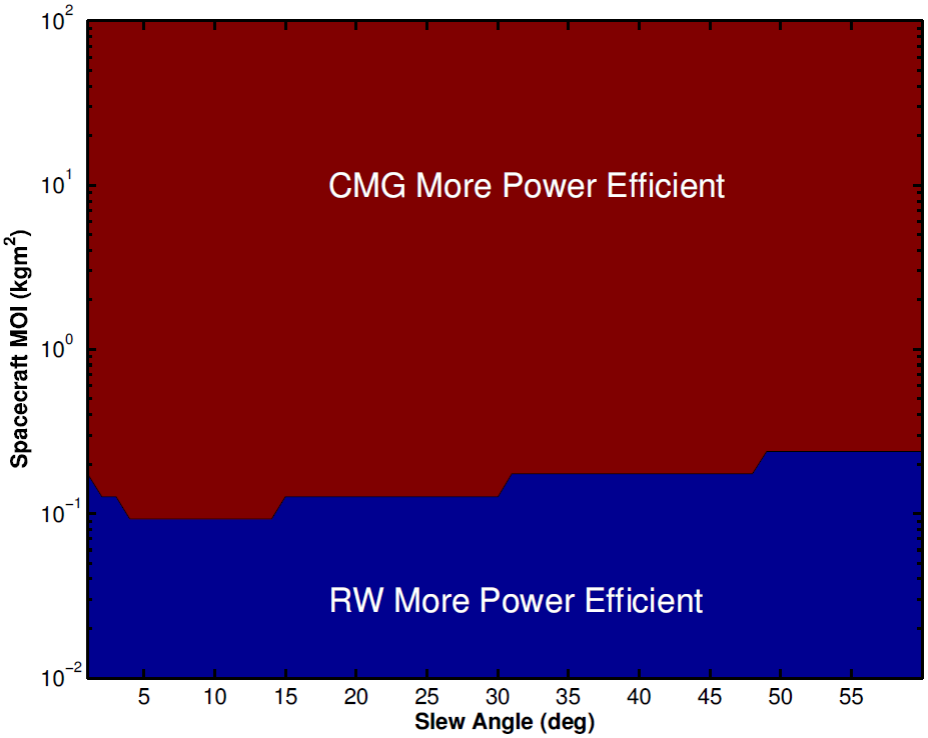


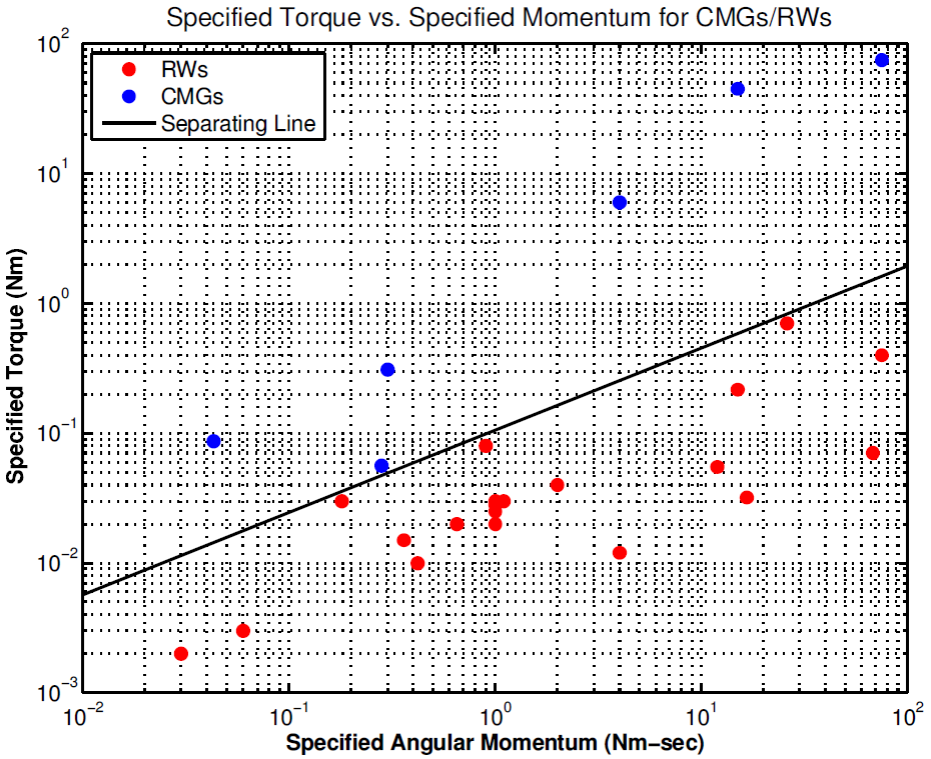
Singular configurations can be avoided by adding a command that generates rotation of the gimbal axes but overall a null torque. This is called null motion command $\underline{\dot{\delta}}_n$, it is the null space of matrix B and can be computed as

$$B\underline{\dot{\delta}_n} = 0 \quad ; \quad \underline{\dot{\delta}} = -B^*\left[\underline{M_c} + \underline{\omega} \wedge A\underline{h_r}\right] + k\underline{\dot{\delta}_n}$$

As a general rule, CMGs have a higher torque capacity compared to RWs and are more efficient for larger spacecraft.






Specified Torque vs. Specified Momentum for CMGs/RWs
RWs
CMGs
Separating Line
Specified Torque (Nm)
Specified Angular Momentum (Nm–sec)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}

Magnetic actuators

Magnetic actuators generate a torque by inducing a magnetic dipole in a coil that is surrounded by the Earth's magnetic field. The magnetic dipole generated can be modeled as:

$$\underline{D} = \mu n S \underline{I}$$

$\mu =$ magnetic permeability
$n =$ number of windings
$S =$ area of coil
$\underline{I} =$ current intensity

Torque is then given by the vector product of this dipole moment and the external magnetic field:

$$\underline{M} = \underline{D} \wedge \underline{B}$$

The effectiveness of these actuators varies with the orbit height, due to the change in the external magnetic field, and can typically provide torques in the range $10^{-3} \div 10^{-6}$ Nm. They can be customized by changing the number of windings and/or the area of the coils.

The torque generated is perpendicular to the magnetic dipole generated $\underline{D}$ and to the external magnetic field $\underline{B}$, so that it is never possible to generate three independent components of the control torque. However, if the satellite dynamics is coupled on two axes, typically x and y, the availability of control along z and along a second axis guarantees complete controllability.

$$\underline{M} = \underline{D} \wedge \underline{B} = -\underline{B} \wedge \underline{D}$$

Expanding matrix notation we have:

$$\begin{Bmatrix} M_x \\ M_y \\ M_z \end{Bmatrix} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{Bmatrix} D_x \\ D_y \\ D_z \end{Bmatrix}$$

The control command is $\underline{D}$ so that we should write:

$$\underline{D} = [-B \wedge]^{-1} \underline{M}$$

Unfortunately $[-B \wedge]$ is a singular matrix, so $\underline{D}$ cannot be evaluated through this procedure. This reflects the fact that it is not possible to provide three independent components of the control torque. A solution can be found if the dynamics is written in a reference system in which $\underline{B}$ is aligned with axis z, so that it can be made explicit that the control torque is in the x-y plane of this new reference. This method is quite tedious since it requires a continuous change of reference, and a further transformation of the commands from the rotated reference to the principal axes, in which the actuators are physically located. We can instead reformulate the problem in order to require only 2 components of control, along z and y for example, and write:

$$\begin{Bmatrix} 0 \\ M_y \\ M_z \end{Bmatrix} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{Bmatrix} D_x \\ D_y \\ D_z \end{Bmatrix}$$

Assuming furthermore that $D_x = 0$ we get to:

$$M_y = B_x D_z \quad \Rightarrow \quad D_z = \frac{M_y}{B_x}$$
$$M_z = -B_x D_y \quad \Rightarrow \quad D_y = -\frac{M_z}{B_x}$$

Now, we notice that the first equation produces still a residual torque M_x:

$$M_{xr} = -M_z \frac{B_z}{B_x} - M_y \frac{B_y}{B_x}$$

This residual is not required by the control, it is due to the interaction between $\underline{B}$ and $\underline{D}$ necessary to produce M_y and M_z. In general, this control structure is used driving to zero the component M_z and using the pitch component of the magnetic field. It is then necessary to add one additional actuator to provide control along the pitch axis z, decoupled from the x and y:

$$D_y = -\frac{M_x}{B_z}$$
$$D_x = -\frac{M_y}{B_z}$$

Passive stability along z can be provided by gravity gradient or by a RW or IW, in such a case the control can be modeled as:

$$\begin{Bmatrix} M_x \\ M_y \\ M_z \end{Bmatrix} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{Bmatrix} D_x \\ D_y \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ -\dot{h}_z \end{Bmatrix}$$

or:

$$\begin{Bmatrix} M_x \\ M_y \\ M_z \end{Bmatrix} = \begin{bmatrix} 0 & B_z & 0 \\ -B_z & 0 & 0 \\ B_y & -B_x & 1 \end{bmatrix} \begin{Bmatrix} D_x \\ D_y \\ -\dot{h}_z \end{Bmatrix}$$

The control allocation matrix is now nonsingular, so that the actuator commands can be calculated in order to have three independent components of control torque:

$$\begin{Bmatrix} D_x \\ D_y \\ -\dot{h}_z \end{Bmatrix} = \frac{1}{B_z} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ B_x & B_y & B_z \end{bmatrix} \begin{Bmatrix} M_x \\ M_y \\ M_z \end{Bmatrix}$$

If the magnetic field component B_z is nonzero the problem can be solved.

A different alternative to evaluate $\underline{D}$ leads to an approximation of the control problem. We can write:

$$\underline{M} = \underline{D} \wedge \underline{B}$$

$$\underline{B} \wedge \underline{M} = \underline{B} \wedge (\underline{D} \wedge \underline{B})$$
$$\underline{B} \wedge \underline{M} = (\underline{B} \cdot \underline{B})\underline{D} - (\underline{D} \cdot \underline{B})\underline{B}$$

Now, we can assume that D is generated in such a way to be always perpendicular to B so that:

$$\underline{D} \cdot \underline{B} = 0$$

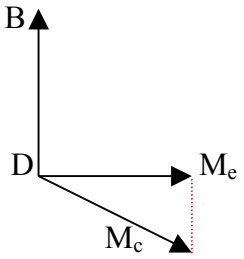
Therefore we can evaluate D:

$$\underline{B} \wedge \underline{M} = B^2 \underline{D} \quad \rightarrow \quad \underline{D} = \frac{\underline{B} \wedge \underline{M_c}}{B^2}$$

However we also have:

$$\underline{M_{\text{eff}}} = \frac{1}{B^2}\left(\underline{B} \wedge \underline{M_c}\right) \wedge \underline{B}$$

and this effective control is equal to the desired control only if the desired control $\underline{M_c}$ is really orthogonal to B, which also means that if D is orthogonal to B we can only calculate the component of control torque M that is orthogonal to B and to D:



This because, having used the vector product to find the solution, this is limited to the plane perpendicular to B, so the method is effective only when the control torques required are by their nature almost perpendicular to the magnetic field.

The above control law can be used to relate an ideal control law to the magnetic dipole. In practice, due to the approximate nature of the control law and the fact that magnetic torquers are an under-actuated controller, the control has only been used for de-saturation and coarse Sun-pointing. For de-saturation of momentum exchange devices, for example, when the RW reaches its saturation limit no additional exchange of momentum may occur in the same direction and therefore the RW must be de-saturated. For example, for a RW with dynamics $\dot{\underline{h}}_{rw} = \underline{T}_{rw}$ where $\underline{T}_{rw}$ is the torque applied to the RW can use a simple feedback control to decelerate the wheel $\underline{T}_{rw} = -k\, sign(\underline{h}_{rw})$. However, de-saturating the wheel will result in a generally undesired momentum exchange with the spacecraft e.g a de-tumbled spacecraft will begin to move when the RW is de-saturated. Thus, during the de-saturation of the RW a magnetic torque can be applied with the control $\underline{M_c} = -M_{max}\, sign(\underline{h}_{rw})$ where $M_{max}$is the maximum possible torque that can be produced by a RW. The torque induced on the spacecraft by decelerating the reaction wheel should be as equal in magnitude as possible. The tuning parameter $k$should be tuned to approximatley $k \approx M_{max}$. This type of control law will only be useful when the angular velocities of the spacecraft are small, since otherwise there is a gyroscopic couling term between the angular velocities of the spacecraft and the wheels that could be non-negligible. In summary we have the control law for a de-saturation maneuver:

Torque applied to the reaction wheel:

$$\underline{T}_{rw} = -M_{max}\, sign(\underline{h}_{rw})$$

And the magnetic dipole feedback:

$$\underline{D} = \frac{M_{max}\, sign(\underline{h}_{rw})}{\underline{b}^2}$$